Category Theory

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

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Introduction

The mathematics a student meets first is a mathematics of elements. A group is a set with a multiplication, a topological space a set with a notion of nearness, and one studies such an object by looking inside it. Category theory begins by setting that habit aside. An object is understood not through its internal constitution but through its *morphisms*, the structure-preserving maps into and out of it, and two objects indistinguishable by their morphisms are, categorically, the same. The slogan that category theory is ten percent objects and ninety percent maps is made a theorem by the *Yoneda lemma* (chapter 2), the first landmark of the book: an object is determined, up to unique isomorphism, by the maps into it.

Chapter 1 installs the language that makes this precise: categories, the functors between them, and the natural transformations between those. From there the subject unfolds as one line of thought. The constructions worth naming are those fixed by a *universal property*, a defining condition on maps in or maps out, and chapter 3 collects them as limits and colimits. A universal property relating two whole categories is an *adjunction* (chapter 4), the notion Mac Lane placed at the center of the subject; a *monad* (chapter 5) is what an adjunction leaves behind on one side; and *Kan extensions* (chapter 6) subsume the rest, in the sense of the dictum that all concepts are Kan extensions. The last two chapters enrich the maps themselves: a *monoidal* category equips them with a tensor product (chapter 7), and letting hom-sets become objects of such a category, or acquire morphisms of their own, yields the *enriched* and *2-categorical* theories of chapter 8.

The book is a graduate introduction and assumes the algebra of *Core Algebra* [4]. Groups, rings, modules, and fields are the running examples, and their isomorphism theorems, free constructions, and Galois correspondence prove to be one categorical statement viewed from several sides. The language reaches well past algebra, so topology, order theory, and set theory supply examples throughout; a reader at home in those subjects will meet the vocabulary already at work there, while a reader who is not can follow the algebraic thread alone. Everything here is strict category theory, where composition is associative on the nose and sameness means isomorphism. Chapter 8 ends at the edge of that setting, where equality relaxes to coherent isomorphism and the subject passes into higher category theory, the concern of *Higher Category Theory* [1].

The viewpoint of this book, and much of its selection and order of material, is owed to Emily Riehl's *Category Theory in Context* [8], whose insistence on universal properties and the Yoneda perspective shaped the arc above. Mac Lane's *Categories for the Working Mathematician* [6] remains the standard reference and the source of the telling that centers on adjunctions and Kan extensions, and Leinster's *Basic Category Theory* [5] is the most inviting way in. Any chapter here is developed at greater length in one of the three.

1

Categories, Functors, and Natural Transformations

Category theory begins from a change of viewpoint: an object is understood not through its internal constitution but through its relationships to every other object of its kind. A group is a set carrying a multiplication; but what a group *is*, categorically, is recorded in the homomorphisms into and out of it, not in the elements it happens to contain. This chapter installs the three notions that make the viewpoint precise and usable. A *category* packages a collection of objects together with the morphisms between them and a law for composing those morphisms. A *functor* is the matching notion of map between categories, carrying objects to objects and morphisms to morphisms while respecting composition and identities. A *natural transformation*, in turn, is a map between functors; with it the word "canonical", used informally for a construction that rests on no arbitrary choice, acquires an exact meaning.

The reward for the abstraction is uniformity. The isomorphism theorems for groups, rings, and modules; the sense in which a group action is a group of a special shape, or a free module a direct sum of copies of its ring; the Galois correspondence between intermediate fields and subgroups: each is an instance of one categorical pattern rather than a coincidence restated in three notations. The examples in this chapter are drawn deliberately from across mathematics and not from algebra alone, so a reader who knows some topology or order theory will meet the language already at work in familiar disguises. The thread running through the chapter, which the Yoneda lemma will later promote to a theorem, is that an object is pinned down completely by the morphisms it receives from every other object, or dually by those it sends to them.

1.1 Categories, Morphisms, and Size

A category is a bookkeeping device. It records a collection of mathematical objects together with the transformations allowed between them, and it insists on almost nothing else: that transformations compose, that the composition is associative, and that each object carries a transformation doing nothing. From this austere data an entire discipline grows, because the axioms are met not only by sets and functions but by groups and homomorphisms, spaces and continuous maps, and by structures in which the "transformations" are not maps of any underlying set at all. The definition below is the one on which every later chapter rests, so it is worth reading with the care its brevity conceals.

Definition 1.1.1: Category

A *category* C consists of

- a collection of *objects*; and
- a collection of *morphisms* (also called *arrows*), each equipped with a specified *domain* object and a specified *codomain* object. That f is a morphism with domain X and codomain Y is written $f\colon X \to Y$;

subject to the following data. Each object X carries an *identity morphism* $1_X\colon X \to X$. For each pair of morphisms $f\colon X \to Y$ and $g\colon Y \to Z$, in which the codomain of f is the domain of g, there is a specified *composite morphism* $gf\colon X \to Z$. These are required to satisfy two axioms.

1. *Unitality.* For every $f\colon X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f.
2. *Associativity.* For every composable triple $f\colon W \to X$, $g\colon X \to Y$, $h\colon Y \to Z$, the composites $h(gf)$ and $(hg)f$ are equal, so that the triple composite may be written hgf without ambiguity.

Two points in this definition carry most of its weight. The first is the choice of the word *collection* rather than *set*, deliberate and returned to at the end of this section: a category may have so many objects or morphisms that they do not form a set, and the definition is engineered to tolerate this. The second is that a morphism $f\colon X \to Y$ *need not be a function.* Nothing in the axioms mentions elements, images, or preimages; a morphism is a primitive arrow from one object to another, and its only obligation is to compose. Several of the examples below realize morphisms as functions with extra structure, but others, matrices, order relations, homotopy classes, are morphisms that are not functions on any underlying set, and the theory treats them on exactly the same footing.

A word on the order of composition, fixed once and used throughout. The composite gf means "g after f": one applies $f\colon X \to Y$ first and $g\colon Y \to Z$ second, and the notation is read right to left, matching the classical convention $(g \circ f)(x) = g(f(x))$ for functions. The equivalent notation $g \circ f$ is used interchangeably when the circle aids legibility. The requirement that the codomain of f agree with the domain of g before gf is defined is precisely what makes the arrows chain: composition is a partial operation, defined exactly when the arrows line up head to tail.

The identity morphisms deserve a remark, because they let one dispense with objects entirely. Since 1_X is the unique morphism $e\colon X \to X$ with $ef = f$ and $ge = g$ for all composable f, g, distinct objects have distinct identities, and objects stand in bijection with their identity morphisms. Some treatments exploit this and define a category using morphisms alone, recovering each object as its identity arrow. The object/morphism formulation is retained here as the more transparent one, but the observation underlies the slogan that category theory is about the arrows: even the objects are secretly morphisms.

The definition matters only through its instances, and the supply is enormous. The first family of examples is the algebraic one, where objects are sets equipped with operations and morphisms are the maps respecting those operations. These are the categories a reader of [4] has been working inside all along, now named as such.

Example 1.1.2: Algebraic Categories

Each of the following is a category; in each case the identity morphism on an object is the identity function on its underlying set, and composition is composition of functions, so the two axioms of definition 1.1.1 are inherited from the associativity and unitality of function composition.

1. **Set** has sets as objects and functions (each with a specified domain and codomain) as morphisms. This is the ambient category of classical mathematics, and many of the categories below are built by equipping its objects with structure and cutting down its morphisms to those preserving that structure.
2. **Grp** has groups as objects and group homomorphisms as morphisms. That a composite of homomorphisms is a homomorphism, and that the identity function on a group is a homomorphism, are the two facts needed; both are immediate from the definition of a homomorphism (see [4]). The word *morphism* itself descends etymologically from *homomorphism*.
3. **Ring** has (unital) rings as objects and ring homomorphisms as morphisms.
4. **Field** has fields as objects and ring homomorphisms between them as morphisms. A field is a commutative unital division ring, so a field homomorphism is nothing but a ring homomorphism whose domain and codomain happen to be fields; changing which objects are admitted, while keeping the same notion of morphism, produces a different category.
5. $\mathbf{Mod}_R$, for a fixed ring R, has left R-modules as objects and R-module homomorphisms as morphisms. Two cases are frequent enough to earn their own names. When $R = k$ is a field, an R-module is a k-vector space and the category is written $\mathbf{Vect}_k$. When $R = \mathbb{Z}$, a $\mathbb{Z}$-module is precisely an abelian group, and the category is written **Ab**.

In every algebraic category the pattern is the same: an object is a set carrying structure, a morphism is a function preserving that structure, and the category axioms reduce to the observation that structure-preservation is closed under composition and holds for identity functions. This is worth verifying once, as above for **Grp**, and thereafter taken for granted. A category assembled in this way, one whose objects have underlying sets and whose morphisms are the structure-preserving functions between them, is called *concrete*; the examples of example 1.1.2 are all concrete. The reach of the definition, however, extends well past algebra and well past concreteness.

The next family shows that concreteness has nothing to do with algebra: these categories carry topological, measure-theoretic, differential, or order-theoretic structure, yet each is still a set with structure and a structure-preserving function.

Example 1.1.3: Non-Algebraic Concrete Categories

Each of the following is a concrete category; as before, identities are identity functions and composition is function composition, so the axioms of definition 1.1.1 hold automatically once one checks that the chosen class of functions is closed under composition and contains the identities.

1. **Top** has topological spaces as objects and continuous functions as morphisms. A composite of continuous functions is continuous and every identity function is continuous, which is all the axioms require.

2. $\mathbf{Top}_*$ has pointed topological spaces, that is, pairs (X, x_0) of a space with a chosen basepoint, as objects, and basepoint-preserving continuous functions as morphisms. The analogous construction on **Set** gives $\mathbf{Set}_*$, the category of pointed sets.

3. **Meas** has measurable spaces as objects and measurable functions as morphisms.

4. **Man** has smooth manifolds as objects and smooth maps as morphisms.

5. **Poset** has partially ordered sets as objects and order-preserving (monotone) functions as morphisms: a function $f\colon (P, \le) \to (Q, \le)$ with $x \le y$ implying $f(x) \le f(y)$.

Every category met so far is concrete, and the uniformity can mislead. It would be easy to conclude that a morphism is always a function of some kind, decorated with a preservation property, and that a category is always a collection of structured sets. Both conclusions are false, and the examples that break them are not exotic. In the categories below the morphisms are matrices, or comparisons in an order, or homotopy classes of maps, none of which is a function from the domain object to the codomain object, indeed in some of them the objects are not sets at all.

Example 1.1.4: Categories Whose Morphisms Are Not Functions

1. Fix a unital ring R. The category $\mathbf{Mat}_R$ has the natural numbers $0, 1, 2, \dots$ as objects, and a morphism $n \to m$ is an $m \times n$ matrix with entries in R. Composition is matrix multiplication: given $A\colon n \to m$ (an $m \times n$ matrix) and $B\colon m \to k$ (a $k \times m$ matrix), the composite $BA\colon n \to k$ is the $k \times n$ product matrix, formed along the arrow chain

$$n \xrightarrow{A} m \xrightarrow{B} k$$

The identity morphism on n is the $n \times n$ identity matrix. Associativity of composition is associativity of matrix multiplication, and unitality is the fact that the identity matrix acts trivially; both are standard, so $\mathbf{Mat}_R$ is a category. A morphism here is not a function from n to m: the objects n and m are natural numbers, not sets, and it would make no sense to feed an "element of n" to the matrix A. The arrows are matrices, full stop.

2. Any partially ordered set $(P, \le)$ is itself a category. The objects are the elements of P, and between elements x and y there is a *unique* morphism $x \to y$ when $x \le y$ and no morphism otherwise. Reflexivity, $x \le x$, supplies the identity morphism on each object; transitivity, $x \le y \le z$ implies $x \le z$, supplies the composite of $x \to y$ and $y \to z$. Because there is at most one morphism between any two objects, the associativity and unitality axioms hold trivially: any two morphisms with matching domain and codomain are equal. This is a category whose morphisms are comparisons, not functions, and it should be contrasted with **Poset**, whose *objects* are posets and whose morphisms are monotone functions; here a single poset *is* a category.

3. The homotopy category **hTop** has topological spaces as objects, but a morphism $X \to Y$ is a *homotopy class* of continuous maps, two maps being identified when one deforms continuously into the other. Composition is well defined because composing homotopic maps yields homotopic composites. A morphism is now an equivalence class of maps rather than a single map, so **hTop** is not concrete in the naive way; it is the natural home of much of algebraic topology, where a space matters only up to homotopy.

4. There is a category **Cat** whose objects are (small, in the sense defined below) categories and whose morphisms are the functors between them, the structure-preserving maps of categories taken up later in this chapter. That the objects are categories rather than sets is exactly the kind of self-reference that must be handled with care, and the size restriction to *small* categories, deferred to the size discussion that closes this section, is what keeps **Cat** from collapsing into paradox.

These last examples make precise a slogan already stated: the objects of a category are almost beside the point, and it is the morphisms that carry the content. A category is named for its objects only by convention and convenience, because in most cases the intended morphisms are clear once the objects are fixed, one says "the category of groups" and the reader supplies "and group homomorphisms". The convention is harmless but occasionally misleading, for the same collection of objects can support wildly different categories depending on the morphisms chosen. The natural numbers are the objects both of the discrete category with only identity arrows and of $\mathbf{Mat}_R$ with all matrices as arrows, and these are not the same category; the underlying objects settle nothing. What a category theorist studies is how objects relate, and relation is the business of arrows.

The word *collection* in definition 1.1.1 has been flagged twice as deliberate, and it is time to explain it. The difficulty is that the objects of **Set** are all sets, and there are too many sets to form a set. Attempting to gather them into one leads directly to contradiction, a warning that naive set formation is unsafe.

1.1.5 Russell's Paradox Suppose there were a set S of all sets. Consider

$$R = \{x \in S \big| x \notin x\}$$

the set of all sets that are not members of themselves. Since S contains every set, R is a set, so $R \in S$, and one may ask whether $R \in R$. If $R \in R$, then R satisfies the defining condition $R \notin R$, a contradiction. If $R \notin R$, then R satisfies the condition for membership in R, so $R \in R$, again a contradiction. Either horn is absurd, so no set S of all sets can exist.

The paradox is not a defect of category theory but of unrestricted set formation, and the standard remedy, adopted throughout the series, is to distinguish two sizes of aggregate. A *set* is an aggregate small enough to be handled by the usual axioms without contradiction; a *proper class*, or informally a *collection*, may be larger, large enough to hold all sets, or all groups, or all topological spaces, and it is precisely not required to be a set. The objects of **Set** form a proper class, not a set, and definition 1.1.1 was written with "collection" so that **Set** counts as a category despite this. The same care applies to **Cat**: were its objects allowed to include all categories, **Cat** would contain itself and the paradox would return, which is why its objects are restricted to *small* categories.

This forces a distinction between categories that are set-sized and those that have only set-sized pieces. The strictest condition asks that all of a category's arrows form a set.

Definition 1.1.6: Small Category

A category is *small* if its collection of morphisms is a set (rather than a proper class).

Since each object has its own identity morphism, and distinct objects have distinct identities, a small category also has only a set's worth of objects: the objects inject into the morphisms via $X \mapsto 1_X$. Smallness is therefore a single condition controlling the whole category at once. A poset viewed as a category (the second item of example 1.1.4) is small, as is any finite category and any group regarded as a one-object category. But **Set**, **Grp**, and their relatives are not small: already their objects form a proper class, so a fortiori their morphisms do.

Smallness is often too much to ask, and a weaker condition suffices for nearly everything the theory needs. It requires only that the arrows between each *fixed* pair of objects form a set, while allowing the objects, and hence all the arrows together, to be a proper class.

Definition 1.1.7: Locally Small Category

A category C is *locally small* if for every ordered pair of objects (X, Y) the morphisms from X to Y form a set. This set is the *hom-set* from X to Y, written

$$\mathrm{Hom}(X, Y) \qquad \text{or} \qquad \mathrm{Hom}_C(X, Y) \qquad \text{or} \qquad C(X, Y)$$

when the ambient category C must be named.

The name *hom-set* is a fossil of the algebraic origin of the notion, where $\mathrm{Hom}(X, Y)$ was literally a set of homomorphisms; in a general category its elements are morphisms, which need not be homomorphisms or even functions, but the name has stuck. The notation is convenient enough that $\mathrm{Hom}(X, Y)$ and $C(X, Y)$ are written even for categories that are not locally small, with the understanding that the aggregate in question is then a proper class rather than a set. Local smallness is the standing assumption behind almost every construction in this book, and for good reason: it is exactly the hypothesis that lets one apply set-level tools, functions, bijections, cardinalities, to the arrows between two objects, and the Yoneda lemma of the next chapter will turn each hom-set into a probe that reads off an object from its arrows.

The distinction between small and locally small is exactly the distinction between **Cat** and its examples. **Set** is locally small but not small: its objects form a proper class, so it is not small, yet for any two sets X and Y the functions from X to Y form a genuine set (a subset of the power set $\mathcal{P}(X \times Y)$), so it is locally small. The same holds for **Grp**, **Ring**, **Top**, and every other category of example 1.1.2 and example 1.1.3: each has a proper class of objects but only a set of morphisms between any fixed pair. These are the categories one works in daily, and they are locally small without exception; smallness, by contrast, is a special property enjoyed only by categories one can in principle write out.

Example 1.1.8: The Hom-Set of a Group

Hom-set notation names constructions the reader already knows. Let G and H be groups, regarded as objects of **Grp**. Then $\mathrm{Hom}_{\mathbf{Grp}}(G, H)$ is the set of all group homomorphisms $G \to H$, exactly the object studied under that name in [4]. A concrete instance illuminates a familiar idea: take $H = S_A$, the symmetric group of a set A. A group homomorphism $G \to S_A$ is by definition an action of G on A by permutations, so the hom-set

$$\mathrm{Hom}_{\mathbf{Grp}}(G, S_A)$$

is precisely the set of G-actions on A (see [4] for the action-homomorphism correspondence). A single piece of categorical notation thus repackages a definition from group theory: an action is a morphism into a symmetric group, and the collection of actions is a hom-set. The same phenomenon recurs throughout the book, and recognizing prior constructions as hom-sets is the first step toward the representability viewpoint developed in later chapters, where an object is recovered from the hom-sets into or out of it.

1.2 Isomorphisms, Monomorphisms, and Epimorphisms

The previous section built the vocabulary of a category: objects, morphisms, composition, identities. What that vocabulary conspicuously lacks is a notion of two objects being "the same". In the category **Set** two sets ought to count as interchangeable when they have the same cardinality; in **Grp** two groups when they are isomorphic; in **Top** two spaces when they are homeomorphic. Each of these is a bespoke definition phrased in terms of the internal anatomy of the objects. The thread of this book is that such internal descriptions are avoidable: sameness, injectivity, and surjectivity can all be read off the morphisms alone. This section makes good on that promise for the first three, defining an isomorphism, a monomorphism, and an epimorphism purely by how a morphism composes with others, and then harvesting the definitions for a categorical account of groups and of the invertible part of any category.

Two objects should count as the same when there is a morphism between them that can be undone. In **Set** a bijection $f\colon X \to Y$ has a two-sided inverse function; in **Grp** an isomorphism has an inverse homomorphism. The common feature, visible without ever mentioning elements, is the existence of a morphism running the other way that composes to the identity on both sides.

Definition 1.2.1: Isomorphism

Let C be a category. A morphism $f\colon X \to Y$ in C is an *isomorphism* if there exists a morphism $g\colon Y \to X$ with

$$gf = 1_X \qquad \text{and} \qquad fg = 1_Y$$

Such a g is called the *inverse* of f; two objects X and Y are *isomorphic*, written $X \cong Y$, when there exists an isomorphism between them.
An *endomorphism* of X (a morphism $X \to X$) that is an isomorphism is called an *automorphism* of X. The automorphisms of X form a set (in a locally small category) denoted $\mathrm{Aut}(X)$.

The inverse deserves the definite article: if g and g' are both inverses of f, then

$$g = g\,1_Y = g(fg') = (gf)g' = 1_X\,g' = g'$$

so an inverse, when it exists, is unique. This little computation is the categorical form of the familiar uniqueness of inverses in a group, and it uses nothing but associativity and the identity laws. The notation $g = f^{-1}$ is therefore unambiguous. The relation $X \cong Y$ is an equivalence relation: reflexivity comes from 1_X, symmetry from the fact that f^{-1} is again an isomorphism with inverse f, and transitivity from the observation that a composite of isomorphisms is an isomorphism, which is worked out in lemma 1.2.7 below.

The point of the definition is that it is category-relative. The morphism f carries no elements and no structure of its own; whether it is an isomorphism is a question about the category it lives in, and the answer can change dramatically as the category changes even when the underlying function stays fixed. The following computation of isomorphisms across the standing cast makes this concrete, and already exhibits the phenomenon that will recur throughout the book: a bijective structure-preserving map need not be an isomorphism.

Example 1.2.2: Isomorphisms in the Standing Cast

The isomorphisms of a category are computed by unwinding what a two-sided inverse must be.

1. In **Set** the isomorphisms are exactly the bijections. A function $f\colon X \to Y$ with a two-sided inverse function is injective (if $f(x) = f(x')$ then $x = f^{-1}f(x) = f^{-1}f(x') = x'$) and surjective (every $y = f(f^{-1}(y))$ is a value), and conversely a bijection has the inverse function as its two-sided inverse. Consequently the only invariant a set carries up to isomorphism is its cardinality: the particular names of the elements are invisible to the category.

2. In **Grp**, **Ring**, and $\mathbf{Mod}_R$ the isomorphisms are exactly the bijective homomorphisms. If a homomorphism f is bijective, its set-theoretic inverse f^{-1} is automatically a homomorphism: for $y, y' \in Y$, writing $x = f^{-1}(y)$ and $x' = f^{-1}(y')$, the identity $f(xx') = f(x)f(x') = yy'$ gives $f^{-1}(yy') = xx' = f^{-1}(y)f^{-1}(y')$, and similarly for the additive and scalar operations, see [4]. So f^{-1} is a morphism in the same category and f is an isomorphism. Here a bijective morphism *is* an isomorphism.

3. In **Top** the isomorphisms are the *homeomorphisms*: continuous bijections whose inverse is again continuous. This is strictly stronger than being a continuous bijection. The map from the half-open interval $[0, 1)$ to the circle S^1 sending t to $(\cos 2\pi t, \sin 2\pi t)$ is a continuous bijection, but its inverse is discontinuous at the point $(1, 0)$, where the circle is torn open. There is no isomorphism $[0, 1) \cong S^1$ in **Top**, even though the two underlying sets are isomorphic in **Set**. This is the first place where a bijective "structure-preserving" map fails to be an isomorphism, and it is the reason the categorical definition, which requires an inverse morphism rather than a bijective morphism, is the correct one.

4. In a poset $(P, \leq)$ regarded as a category, the only isomorphisms are the identities. A morphism $x \to y$ exists precisely when $x \leq y$, so an isomorphism $x \to y$ forces both $x \leq y$ and $y \leq x$, and antisymmetry gives $x = y$. A poset therefore has no nontrivial isomorphisms at all.

The contrast between the second and third items is the lesson to carry forward. In an algebraic category a bijective morphism inverts to a morphism for free, because the algebraic operations are equations that transport across the bijection; in **Top** continuity is not an equation but a genuine extra constraint on the inverse, and it can fail. Defining "sameness" as the existence of an inverse morphism, rather than as bijectivity of the morphism, is what makes the notion track the right equivalence in every category at once.

The poset computation, where every isomorphism is trivial, is one extreme. The opposite extreme, where *every* morphism is an isomorphism, is important enough to name, because it turns out to be the categorical home of group theory.

Definition 1.2.3: Groupoid

A *groupoid* is a category in which every morphism is an isomorphism.

A groupoid is the categorical distillation of "a system of reversible processes". Any category has, sitting inside it, a largest groupoid, obtained by throwing away every non-invertible morphism; that construction is lemma 1.2.7 below. But the single-object groupoids already recover an entire algebraic theory, and identifying which one is the first genuine payoff of the categorical language: a group is nothing but a groupoid with one object.

Example 1.2.4: A Group as a One-Object Groupoid

Let $\mathcal{G}$ be a groupoid with a single object, call it $*$. All the data of $\mathcal{G}$ lives in the single hom-set

$$G := \mathrm{Hom}_{\mathcal{G}}(*, *) = \mathrm{Aut}(*)$$

where the second equality holds because $\mathcal{G}$ is a groupoid, so every endomorphism of $*$ is an isomorphism, hence an automorphism. Composition of morphisms gives a binary operation

$$G \times G \to G \qquad (h, g) \mapsto hg$$

and the claim is that $(G, \cdot)$ is a group. Each group axiom is exactly one of the category axioms specialized to the one object.

Associativity. For any three morphisms $f, g, h \in G$, composition is defined (all domains and codomains equal $*$) and the associativity axiom of a category gives $h(gf) = (hg)f$. This is the associativity of the group operation.

Identity. The object $*$ carries an identity morphism $1_* \in G$, and the identity axiom of a category gives $1_* g = g = g\, 1_*$ for every $g \in G$. So 1_* is a two-sided identity element, and it is the unique such element by the general uniqueness of identities in a category, which is the same argument as the uniqueness of a group identity, see [4].

Inverses. Because $\mathcal{G}$ is a groupoid, every $g \in G$ is an isomorphism, so there is a morphism $g^{-1} \in G$ with $gg^{-1} = 1_*$ and $g^{-1}g = 1_*$. Since there is only one object, 1_* is the only identity available, so this g^{-1} is a two-sided inverse of g in the group sense.

The three category axioms have produced exactly the three group axioms, and no data was left over: a one-object groupoid *is* a group, and conversely a group G gives a one-object groupoid $\mathbf{B}G$ with $\mathrm{Hom}(*, *) = G$ and composition the group multiplication (every element is invertible, so it is a groupoid). This is not an analogy but an identification, and it is the first sign of the book's thread at work. The group axioms are not an arbitrary list to be memorized; they are what the category axioms say when there is a single object and every arrow is reversible.

The one-object-groupoid picture reframes group theory as the study of the automorphisms of a single point, and this vantage point pays dividends later: a group action becomes a functor out of $\mathbf{B}G$, and the homomorphisms $G \to H$ become the functors $\mathbf{B}G \to \mathbf{B}H$. For now the identification stands as evidence that categorical definitions, far from being empty generalities, can reconstruct familiar structures exactly.

Groupoids and one-object groupoids are examples of a general move: carving a smaller category out of a larger one by restricting the objects and morphisms. When the restriction is compatible with the categorical structure, the result is again a category.

Definition 1.2.5: Subcategory

A *subcategory* D of a category C consists of a subcollection of the objects of C and a subcollection of the morphisms of C, such that

1. the domain and codomain of every morphism in D are objects of D;
2. the identity morphism 1_X of every object X of D lies in D; and
3. the composite gf of any composable pair of morphisms f, g of D lies in D.

A subcategory D is *full* if for every pair of objects X, Y of D one has $\mathrm{Hom}_D(X, Y) = \mathrm{Hom}_C(X, Y)$, that is, D omits objects but no morphisms between the objects it keeps.

The three conditions are exactly what is needed for D, equipped with the composition and identities inherited from C, to satisfy the category axioms in its own right: they are the closure requirements guaranteeing that composition never leads outside D and that every object has its identity. Associativity and the unit laws are inherited from C for free, since D's morphisms are C's morphisms and are composed the same way. A subcategory is thus the categorical analogue of a subgroup or a subspace: a sub-collection closed under the ambient operations.

Example 1.2.6: Rings with and without Identity

The categories of rings sit in a chain of subcategories

$$\mathbf{CRing} \subseteq \mathbf{Ring} \subseteq \mathbf{Rng}$$

where **Rng** has as objects the (not necessarily unital) rings with morphisms the ring homomorphisms not required to preserve a unit, **Ring** has as objects the unital rings with morphisms the unit-preserving homomorphisms, and **CRing** restricts **Ring** to the commutative unital rings. Each inclusion satisfies the three conditions: a homomorphism between unital rings has unital rings as domain and codomain, the identity homomorphism of a unital ring is unit-preserving, and a composite of unit-preserving homomorphisms is unit-preserving, so **Ring** is a subcategory of **Rng**; the same checks with commutativity added give $\mathbf{CRing} \subseteq \mathbf{Ring}$. The inclusion $\mathbf{CRing} \subseteq \mathbf{Ring}$ is *full*, since a ring homomorphism between commutative rings is automatically a morphism of **Ring** with no further condition. The inclusion $\mathbf{Ring} \subseteq \mathbf{Rng}$ is *not* full: between two unital rings there are non-unital homomorphisms (for instance the zero map $\mathbb{Z} \to \mathbb{Z}$, which sends 1 to 0) that live in **Rng** but not in **Ring**.

The distinction between full and non-full subcategories, illustrated cleanly by the rings example, matters throughout: a full subcategory is determined by its objects alone, whereas a non-full one records a genuine choice of which morphisms to keep. The one construction that packages the isomorphisms of a category always produces a subcategory, and identifying it settles the transitivity of $\cong$ left open after definition 1.2.1.

Every category, no matter how few of its morphisms are invertible, has a canonical groupoid inside it: keep all the objects but only the isomorphisms. That this collection really is a subcategory, and a groupoid, is the content of the next lemma. The proof is a matter of checking that identities are isomorphisms and that isomorphisms are closed under composition, but the second point carries the useful formula for the inverse of a composite.

Lemma 1.2.7: The Maximal Groupoid

Every category C contains a *maximal groupoid* $C^{\cong}$: the subcategory whose objects are all the objects of C and whose morphisms are exactly the isomorphisms of C. It is a groupoid, and it is maximal in the sense that any subcategory of C that is a groupoid is contained in $C^{\cong}$.

***Proof*:**
Write $C^{\cong}$ for the collection consisting of all objects of C together with those morphisms of C that are isomorphisms. The verification proceeds in three steps: that $C^{\cong}$ is a subcategory, that it is a groupoid, and that it is maximal.

Step 1: $C^{\cong}$ is a subcategory. The three closure conditions of definition 1.2.5 must hold. Every isomorphism of C has its domain and codomain among the objects of C, which are all present in $C^{\cong}$, so condition (1) holds. For condition (2), each identity 1_X is an isomorphism, being its own two-sided inverse ($1_X 1_X = 1_X$), so every identity of C lies in $C^{\cong}$. For condition (3), let $f\colon X \to Y$ and $g\colon Y \to Z$ be composable isomorphisms, with inverses f^{-1} and g^{-1}. Then $f^{-1}g^{-1}$ is a two-sided inverse of gf:

$$(f^{-1}g^{-1})(gf) = f^{-1}(g^{-1}g)f = f^{-1}1_Y f = f^{-1}f = 1_X$$

and symmetrically $(gf)(f^{-1}g^{-1}) = g(ff^{-1})g^{-1} = g1_Y g^{-1} = gg^{-1} = 1_Z$. Hence gf is an isomorphism, with inverse the reversed composite $(gf)^{-1} = f^{-1}g^{-1}$, and condition (3) holds. So $C^{\cong}$ is a subcategory of C.

Step 2: $C^{\cong}$ is a groupoid. Every morphism of $C^{\cong}$ is by construction an isomorphism of C; its inverse is again an isomorphism of C, hence again a morphism of $C^{\cong}$. So every morphism of $C^{\cong}$ is invertible *within* $C^{\cong}$, which is precisely the condition in definition 1.2.3 for a groupoid.

Step 3: maximality. Let D be any subcategory of C that is a groupoid. Every morphism f of D is invertible in D, so it has an inverse g in D satisfying $gf = 1$ and $fg = 1$; since g is in particular a morphism of C, this exhibits f as an isomorphism of C. Thus every morphism of D is a morphism of $C^{\cong}$, and D is contained in $C^{\cong}$, completing the proof.

The inverse formula extracted in Step 1, $(gf)^{-1} = f^{-1}g^{-1}$, is the categorical version of the "socks and shoes" rule familiar from group theory: to undo two operations performed in order, undo them in the reverse order. It also closes the loose end from earlier: transitivity of the isomorphism relation is exactly the statement that a composite of isomorphisms is an isomorphism, which Step 1 establishes. The maximal groupoid is a functor of C in a sense to be made precise later, and it is the object one names when speaking of "the isomorphisms of C" as a category in their own right; the one-object case recovers example 1.2.4, since the maximal groupoid of a one-object category with all endomorphisms invertible is that group.

Isomorphisms capture sameness. The finer information carried by a morphism, whether it identifies distinct inputs or misses part of its target, is captured by two weaker conditions, each phrased entirely through composition. The clue is the elementary characterization of injections and surjections of sets by cancellation. A function f is injective exactly when it can be cancelled on the left ($fh = fk$ forces $h = k$), and surjective exactly when it can be cancelled on the right ($hf = kf$ forces $h = k$). These cancellation properties make no reference to elements and so define morphisms in any category.

Definition 1.2.8: Monomorphism and Epimorphism

Let $f\colon X \to Y$ be a morphism in a category C.

1. f is a *monomorphism* (or is *monic*) if it is left-cancellable: for every pair of morphisms $h, k\colon W \to X$,
$$fh = fk \quad\Longrightarrow\quad h = k$$
A monomorphism is often written $f\colon X \rightarrowtail Y$.

2. f is an *epimorphism* (or is *epic*) if it is right-cancellable: for every pair of morphisms $h, k\colon Y \to Z$,
$$hf = kf \quad\Longrightarrow\quad h = k$$
An epimorphism is often written $f\colon X \twoheadrightarrow Y$.

The two definitions are word-for-word the same with the composition order reversed, which is the first hint of a pervasive symmetry: a monomorphism in C is the same thing as an epimorphism in the category C^{op} with all arrows reversed, and vice versa. Reversing the arrows turns left-cancellation into right-cancellation, so every statement about monomorphisms has a mirror statement about epimorphisms obtained by passing to the opposite category. This is the duality principle, the organizing tool of the next section; here it already halves the work, since anything proved for monomorphisms holds for epimorphisms by reversing arrows.

Every isomorphism is both monic and epic: if f has an inverse g, then $fh = fk$ gives $h = gfh = gfk = k$, and $hf = kf$ gives $h = hfg = kfg = k$. The converse fails in general, and the failure is exactly what makes mono and epi worth naming separately from iso. In the concrete algebraic categories the two conditions usually coincide with injectivity and surjectivity, but the coincidence is not automatic, and epimorphisms are where it breaks first.

Example 1.2.9: Epic Need Not Mean Surjective

In many concrete categories a monomorphism is the same as an injective map and an epimorphism the same as a surjective map. In **Set** this holds on both counts: a function is monic exactly when it is injective and epic exactly when it is surjective, since one can probe it with maps out of and into a one-point set. In **Grp**, **Ab**, and $\mathbf{Mod}_R$ the monomorphisms are again the injective homomorphisms, and in these categories the epimorphisms are also exactly the surjective homomorphisms. The agreement can fail, however, and the standard witness is the inclusion of the integers into the rationals.

Consider **Ring** (unital rings, unit-preserving homomorphisms) and the inclusion $\iota\colon \mathbb{Z} \to \mathbb{Q}$. This map is not surjective, since $\frac{1}{2}$ is not in its image. It is nonetheless an epimorphism. To see this, let R be any unital ring and let $h, k\colon \mathbb{Q} \to R$ be ring homomorphisms with $h\iota = k\iota$, that is, h and k agree on every integer. For a rational number $\frac{a}{b}$ with $b \neq 0$, the element $h(b) = k(b)$ is invertible in R (it is the image of the unit $b \in \mathbb{Q}$, and a unit-preserving homomorphism sends units to units), and
$$h\left(\tfrac{a}{b}\right) = h(a)\,h(b)^{-1} = k(a)\,k(b)^{-1} = k\left(\tfrac{a}{b}\right)$$
because $h(a) = k(a)$ and $h(b) = k(b)$ force $h(b)^{-1} = k(b)^{-1}$. Thus $h = k$ on all of $\mathbb{Q}$, so ι is right-cancellable and hence epic. The homomorphism out of $\mathbb{Q}$ is pinned down by its values on the integers because the rationals are generated from the integers by inverting nonzero elements, and inverses are forced once the elements they invert are known. A ring epimorphism can therefore fail

to be surjective; indeed there is no nonzero ring homomorphism $\mathbb{Q} \to \mathbb{Z}$ at all, so ι has no hope of being "undone" set-theoretically.

The lesson is that mono and epi are categorical conditions: they test a morphism against all the other morphisms of the category rather than inspecting its effect on elements, and this test can diverge from injectivity or surjectivity precisely when the category's morphisms are constrained in a way that lets a non-surjective map still be cancellable. In **Ring** the constraint is that a homomorphism is determined by less data than its full graph, because it must respect inverses; the inclusion $\mathbb{Z} \to \mathbb{Q}$ exploits exactly that. Keeping the categorical definitions rather than the element-wise ones is what allows a single notion of "injective-like" and "surjective-like" morphism to make sense in every category, including those such as **Poset** or the arrow-only categories where elements are not available to inspect at all.

1.3 Duality and Opposite Categories

The philosophy of category theory is that an object is known by its morphisms, and the morphisms carry a direction: an arrow $f\colon X \to Y$ is not the same datum as an arrow $Y \to X$. This directedness is why morphisms are drawn as arrows rather than undirected lines. It also raises a question that turns out to organize a large part of the subject. What happens if every arrow is reversed? If reversal always produced the same theory, the direction would be inessential and $X - Y$ would be adequate notation. It does not: in most categories reversal yields a different category, and comparing a category with its reverse is one of the most productive moves available. The reversal is packaged into a single construction, the opposite category, and the payoff is a metatheorem, the duality principle, that halves the labor of category theory by turning every proved statement into a second proved statement for free.

The construction reverses each arrow while leaving the objects untouched. Composition must be reversed with it, since the composite of $X \to Y \to Z$ should become an arrow $Z \to X$ built from the reversed pieces.

Definition 1.3.1: Opposite Category

Let C be a category (definition 1.1.1). The *opposite category* (or *dual category*) C^{op} is the category with

- the same objects as C;
- for each morphism $f\colon X \to Y$ of C, a morphism $f^{\mathrm{op}}\colon Y \to X$ of C^{op}, with domain and codomain interchanged; every morphism of C^{op} arises this way, so that

$$\mathrm{Hom}_{C^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_C(Y, X)$$

  as sets;

subject to the requirements

1. for each object X, the identity in C^{op} is $1_X^{\mathrm{op}} = 1_X$;
2. for morphisms $f\colon X \to Y$ and $g\colon Y \to Z$ of C, whose composite $gf\colon X \to Z$ is defined, the reversed morphisms $g^{\mathrm{op}}\colon Z \to Y$ and $f^{\mathrm{op}}\colon Y \to X$ are composable in C^{op}, with composite

$$f^{\mathrm{op}}g^{\mathrm{op}} = (gf)^{\mathrm{op}}$$

The composition rule is forced by the direction of the arrows and is worth reading slowly. In C the composite gf runs $X \to Y \to Z$. Reversing each arrow turns f into $f^{\mathrm{op}}\colon Y \to X$ and g into $g^{\mathrm{op}}\colon Z \to Y$, and the only way to compose these two is $Z \to Y \to X$, that is $f^{\mathrm{op}}g^{\mathrm{op}}$. The reversed composite therefore takes the label $(gf)^{\mathrm{op}}$, the reverse of the original composite. The rule $(gf)^{\mathrm{op}} = f^{\mathrm{op}}g^{\mathrm{op}}$ says exactly that reversal turns the order of composition around, the same order-reversal seen in the transpose of matrices $(AB)^{\top} = B^{\top}A^{\top}$ or the inverse of a product $(gh)^{-1} = h^{-1}g^{-1}$ in a group. The following square, the two ways of reading the composable pair $X \xrightarrow{f} Y \xrightarrow{g} Z$ and its reverse, records the construction.

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[rr, bend left, "gf"] & Y \arrow[r, "g"] & Z \\
X & Y \arrow[l, "f^{\mathrm{op}}"'] & Z \arrow[l, "g^{\mathrm{op}}"'] \arrow[ll, bend right, "(gf)^{\mathrm{op}} = f^{\mathrm{op}}g^{\mathrm{op}}"']
\end{tikzcd}$$

The top row lives in C and the bottom row in C^{op}; the two rows carry the same objects and the same underlying arrows, only the directions and the composition order differ.

The construction is an involution: reversing twice restores the original category. Interchanging domain and codomain twice returns each arrow to its starting orientation, $1_X^{\mathrm{opop}} = 1_X$, and the composition rule applied twice gives $(f^{\mathrm{op}}g^{\mathrm{op}})^{\mathrm{op}} = g^{\mathrm{opop}}f^{\mathrm{opop}} = gf$, so $(C^{\mathrm{op}})^{\mathrm{op}} = C$ on the nose. This is what makes duality a tool rather than a mere relabeling: a category and its opposite stand on equal footing, and any construction available in one is available in the other by symmetry. There is no distinguished "forward" direction, only a chosen one.

The definition is uniform across all categories, but its content varies sharply with the category. The next example runs it through three of the standard cast to show that the opposite can be isomorphic to the original, isomorphic in a less obvious way, or a category of an entirely different kind.

Example 1.3.2: Opposites of Posets, Groups, and Sets

1. *Posets.* A partially ordered set $(P, \leq)$ is a category with one object per element and a unique arrow $x \to y$ exactly when $x \leq y$ (definition 1.1.1). Reversing the arrows reverses the order: P^{op} has a unique arrow $x \to y$ exactly when $y \leq x$, so P^{op} is the poset $(P, \geq)$ with the order turned upside down. For the chain $(\mathbb{Z}, \leq)$ the covering arrows run
$$\cdots \longrightarrow -1 \longrightarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$$
and in $(\mathbb{Z}, \leq)^{\mathrm{op}} = (\mathbb{Z}, \geq)$ they run the other way,
$$\cdots \longleftarrow -1 \longleftarrow 0 \longleftarrow 1 \longleftarrow 2 \longleftarrow \cdots$$
The opposite of a poset is again a poset, but usually a different one: a least element becomes a greatest element, an infimum becomes a supremum. This is the categorical root of the fact that every order-theoretic notion comes in a matched pair.

2. *Groups.* A group G is a one-object groupoid with $\mathrm{Hom}(*, *) = G$, composition given by the group multiplication. Its opposite G^{op} is again a one-object groupoid, hence again a group: the underlying set is the same, but the product is $g^{\mathrm{op}} h^{\mathrm{op}} = (hg)^{\mathrm{op}}$, so multiplication in G^{op} is the multiplication of G with the factors in reversed order. This G^{op} is the *opposite group*. It is isomorphic to G, but the isomorphism is not the identity map on the underlying set unless G is abelian; the natural isomorphism is inversion,
$$\varphi\colon G \to G^{\mathrm{op}}, \qquad \varphi(g) = g^{-1}$$
which is a homomorphism precisely because inversion reverses products,
$$\varphi(gh) = (gh)^{-1} = h^{-1} g^{-1}$$
the right-hand side being the product $\varphi(g)\varphi(h)$ computed in G^{op}. Passing to the opposite group is the same operation as passing between left and right actions. A left action of G on a set X is a rule with $(gh) \cdot x = g \cdot (h \cdot x)$; a right action satisfies $x \cdot (gh) = (x \cdot g) \cdot h$. Setting $g \cdot x := x \cdot g^{-1}$, or equivalently reading a right action of G as a left action of G^{op}, converts one into the other, and the order-reversal in G^{op} is exactly what makes the two associativity laws match. Dropping the requirement of inverses leaves a one-object category rather than a one-object groupoid, that is, a monoid; the opposite monoid M^{op} is defined by the same reversed product, and here M^{op} need not be isomorphic to M.

3. *Sets.* The opposite of **Set** is not **Set**. A morphism $X \to Y$ in $\mathbf{Set}^{\mathrm{op}}$ is, by definition, a function $Y \to X$, so composition in $\mathbf{Set}^{\mathrm{op}}$ is composition of functions read backwards. One is tempted to picture a morphism $f\colon X \to Y$ of $\mathbf{Set}^{\mathrm{op}}$ as "f run in reverse", a map $Y \to X$, but a function need not be invertible, so there is no such reverse function in general. What the reversed arrows track is not functions but relations. A function $f\colon A \to B$ is a relation that is single-valued (each $a \in A$ has exactly one image) and total (every $a \in A$ has an image); its reversed graph $\{(f(a), a) : a \in A\} \subseteq B \times A$ is a relation that is injective-on-the-nose (distinct a give distinct pairs) and surjective (every a appears), but is single-valued and total only when f is a bijection. So the arrows of $\mathbf{Set}^{\mathrm{op}}$ are reversed graphs of functions, which

are relations of a constrained type, not functions. The precise statement is that $\mathbf{Set}^{\mathrm{op}}$ is equivalent to the category of complete atomic Boolean algebras (the powersets PX with their function-induced maps), not to **Set** itself. The takeaway needs no such machinery: reversing arrows in **Set** leaves the world of sets and functions entirely, so "the opposite" is a strictly new category, not a repackaging of the old one.

The three cases exhibit the full range. For a poset the opposite is another object of the same kind, obtained by an evident flip. For a group the opposite is again of the same kind and even isomorphic to the original, though by a map (inversion) that is invisible if one looks only at underlying sets. For **Set** the opposite is a category of a different nature altogether. In every case the objects are unchanged and only the direction of the arrows moves, yet the categories that result can be as close as isomorphic or as far apart as sets and Boolean algebras. This variability is what gives the next principle its force: a single argument, reinterpreted in the opposite category, can say something new precisely because the opposite category is different.

Everything so far concerns one category at a time. The reason opposite categories are indispensable is a metatheorem about the whole theory. Because the axioms of a category are self-dual, reversing every arrow in a valid argument produces another valid argument, so any statement proved for all categories automatically holds for all opposite categories, which is to say for all categories again with the arrows reversed. Stating this cleanly halves the work of the subject.

1.3.3 The Duality Principle Let S be a statement about categories, expressed in the language of objects, morphisms, domains, codomains, composites, and identities. Let S^{op} be the *dual statement*, obtained from S by reversing the direction of every morphism and the order of every composite (and leaving the objects, identities, and logical structure unchanged). Then S holds in a category C if and only if S^{op} holds in C^{op}. In particular, if S holds in *every* category, then so does S^{op}: each theorem of category theory yields a second theorem, its *dual*, for free.

The justification is immediate from the involution $(C^{\mathrm{op}})^{\mathrm{op}} = C$. A proof of S valid in every category is valid in particular in C^{op}; unwinding the definition of C^{op} (each arrow reversed, each composite reordered) turns the assertion "S holds in C^{op}" into the assertion "S^{op} holds in C". Since C was arbitrary, S^{op} holds in every category. There is nothing to check beyond the bookkeeping of reversal, and that bookkeeping is exactly the passage to the opposite category. The practical consequence is a discipline: a construction and its dual, conventionally distinguished by the prefix "co-", are literally the same construction performed in the opposite category. A colimit is a limit in C^{op}, a coproduct is a product in C^{op}, an initial object is a terminal object in C^{op}. To prove a fact about colimits, prove the corresponding fact about limits and read it in C^{op}; no separate argument is ever needed for the dual.

The cleanest illustration available at this stage is the pairing of monomorphisms with epimorphisms. Recall the definitions (definition 1.2.8): a morphism $f\colon X \to Y$ is a *monomorphism* if $fh = fk$ implies $h = k$ for all h, k into X, and an *epimorphism* if $hf = kf$ implies $h = k$ for all h, k out of Y. Written side by side, the epimorphism condition is the monomorphism condition with every composite reordered, which is to say read in the opposite category.

Proposition 1.3.4: Monomorphisms are Epimorphisms in the Opposite

Let C be a category and $f\colon X \to Y$ a morphism of C, with corresponding morphism $f^{\mathrm{op}}\colon Y \to X$ of C^{op}. Then f is a monomorphism in C if and only if f^{op} is an epimorphism in C^{op}. Dually, f is an epimorphism in C if and only if f^{op} is a monomorphism in C^{op}.

***Proof*:**
Fix $f\colon X \to Y$ in C. By definition 1.2.8, f is a monomorphism in C precisely when, for every object W and every pair of morphisms $h, k\colon W \to X$ of C,

$$fh = fk \implies h = k$$

Passing to C^{op}, the morphisms $h, k\colon W \to X$ of C become morphisms $h^{\mathrm{op}}, k^{\mathrm{op}}\colon X \to W$ of C^{op}, and $f^{\mathrm{op}}\colon Y \to X$ of C^{op} is composable before them. Applying the composition rule of definition 1.3.1 to the hypothesis and conclusion,

$$fh = fk \iff (fh)^{\mathrm{op}} = (fk)^{\mathrm{op}} \iff h^{\mathrm{op}} f^{\mathrm{op}} = k^{\mathrm{op}} f^{\mathrm{op}}$$

and $h = k \iff h^{\mathrm{op}} = k^{\mathrm{op}}$, since reversal is a bijection on morphisms. Substituting, the defining implication of a monomorphism for f in C reads

$$h^{\mathrm{op}} f^{\mathrm{op}} = k^{\mathrm{op}} f^{\mathrm{op}} \implies h^{\mathrm{op}} = k^{\mathrm{op}}$$

for all $h^{\mathrm{op}}, k^{\mathrm{op}}\colon X \to W$ of C^{op}. As W ranges over all objects and $h^{\mathrm{op}}, k^{\mathrm{op}}$ over all such pairs, this is exactly the condition that f^{op} be an epimorphism in C^{op} (definition 1.2.8). Hence f is a monomorphism in C if and only if f^{op} is an epimorphism in C^{op}. The second equivalence follows by applying the first to C^{op} in place of C and using $(C^{\mathrm{op}})^{\mathrm{op}} = C$, so that f^{op} is a monomorphism in C^{op} if and only if f is an epimorphism in C, as we sought to show.

The proposition is duality in miniature. Monomorphism and epimorphism are not two unrelated conditions that happen to resemble each other; they are a single condition read in the two directions, one the dual of the other. Anything proved about monomorphisms in every category, by the duality principle (box 1.3.3), holds verbatim for epimorphisms once every arrow is reversed, and conversely. This is why the two notions appear throughout the subject as an inseparable pair, and why the prefix “co-” recurs so insistently: it is the marker of a statement obtained by passing to the opposite category.

Duality is not confined to category theory as an internal bookkeeping device. Across mathematics one meets theorems asserting that two apparently unrelated classes of objects are opposite to each other, in the precise sense that one category is equivalent to the opposite of another. Each such theorem lets questions about one class be translated into questions about the other, often turning a hard geometric or analytic problem into a tractable algebraic one, and each is evidence that the reversal of arrows is a pervasive phenomenon. Three named examples indicate the range. *Gelfand duality* identifies the category of compact Hausdorff spaces with the opposite of the category of commutative unital C^*-algebras, sending a space X to its algebra $C(X)$ of continuous complex functions, so that the topology of X is recovered entirely from an algebra of functions on it. *Stone duality* identifies the category of Boolean algebras with the opposite of the category of compact Hausdorff totally disconnected spaces (Stone spaces), the algebraic and the topological faces of the same data. *Pontryagin duality* sends a locally compact abelian group G to its dual group $\hat{G} = \mathrm{Hom}(G, S^1)$ of continuous characters and exhibits the category of locally compact abelian groups as equivalent to its own opposite, with the double dual $\hat{\hat{G}}$ canonically G again; it is the structural home of the Fourier transform. The proofs of these three belong to functional analysis, general topology, and harmonic analysis respectively and lie well beyond this chapter; they are cited here only as evidence that the opposite category, introduced above as a formal reversal of arrows, names a structure that recurs at the center of several mathematical disciplines.

1.4 Functors

Categories were built to record how objects of one kind relate to one another through their morphisms. The next question is the one that always follows the introduction of a new mathematical structure: what are the structure-preserving maps between two such structures? A map from one category to another must carry objects to objects and morphisms to morphisms, and it must respect the two pieces of data a category carries beyond its raw collections, composition and identities. Such a map is a *functor*, and it is the vehicle by which one part of mathematics is transported into another: the fundamental group turns spaces into groups, the spectrum turns rings into spaces, and the derivative turns smooth maps into linear ones. Each of these is a functor, and in each case the price of admission is exactly that composition be respected.

Definition 1.4.1: Functor

A *functor* $F\colon C \to D$ between categories C and D consists of

- an object Fc of D for each object c of C, and
- a morphism $Ff\colon Fc \to Fc'$ of D for each morphism $f\colon c \to c'$ of C, so that the domain and codomain of Ff are, respectively, F applied to the domain and codomain of f,

subject to the two axioms

1. (functoriality) $F(gf) = Fg\,Ff$ for every composable pair $f\colon c \to c'$, $g\colon c' \to c''$ of morphisms of C, and
2. (identities) $F(1_c) = 1_{Fc}$ for every object c of C.

A functor is thus a map of the underlying data of a category that preserves everything a category is made of: the assignment of domains and codomains, composition, and identities. The functoriality axiom is the substantive one. It says that F cannot see a composite gf as anything other than the composite $Fg\,Ff$ of the images, so a functor can never manufacture a relation between images that was not already present between the sources, nor destroy one that was. This is what makes a functor a faithful conduit of structure rather than an arbitrary reassignment: a commutative triangle in C is sent to a commutative triangle in D, so any equation between composites that holds in the source survives into the target. When $C = D$ the functor is an *endofunctor* of C.

The examples below are the standing cast the rest of the book returns to. Several are functors the reader has already met under other names; the point is that a single word now covers the powerset operation, the act of forgetting algebraic structure, the fundamental group, the derivative, and the notion of a group action all at once.

Example 1.4.2: Functors

1. *The covariant powerset endofunctor.* There is an endofunctor $P\colon \mathbf{Set} \to \mathbf{Set}$ sending a set A to its powerset PA, the set of all subsets of A, and sending a function $f\colon A \to B$ to the *direct image* function
$$Pf\colon PA \to PB \qquad A' \mapsto f(A')$$
where $f(A') = \{f(a)\big| a \in A'\}$. This is a functor: for $A' \subseteq A$ one has $(Pg\,Pf)(A') = g(f(A')) = (gf)(A') = P(gf)(A')$ because the image of a set under gf is the image under g of its image under f, and $P(1_A)(A') = 1_A(A') = A' = 1_{PA}(A')$, so P carries identities to identities.

2. *Forgetful functors.* There is a functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ carrying each group to its underlying set and each group homomorphism to its underlying function. Functoriality and preservation of identities are immediate, since composition and identities of homomorphisms are computed as composition and identities of the underlying functions. A functor that discards part of the structure of its source is a *forgetful functor*. The same pattern gives $\mathbf{Mod}_R \to \mathbf{Ab}$ (forget the R-action, remember the underlying abelian group) and $\mathbf{Ring} \to \mathbf{Ab}$ (forget the multiplication, remember the additive group). These need not be inclusions: a single abelian group underlies many different rings, so $\mathbf{Ring} \to \mathbf{Ab}$ is far from injective on objects.

3. *The free-group functor.* In the opposite direction there is the *free functor* $F\colon \mathbf{Set} \to \mathbf{Grp}$ sending a set X to the free group FX on X and a function $f\colon X \to Y$ to the unique homomorphism $Ff\colon FX \to FY$ extending $x \mapsto f(x)$ on generators, see [4]. Uniqueness of the extending homomorphism forces $F(gf)$ and $Fg\,Ff$ to agree, since both extend $x \mapsto g(f(x))$, and $F(1_X)$ and 1_{FX} both extend $x \mapsto x$, so F is a functor.

4. *The fundamental group.* The fundamental group is the functor $\pi_1\colon \mathbf{Top}_* \to \mathbf{Grp}$ from pointed spaces and basepoint-preserving continuous maps to groups: it sends a pointed space (X, x_0) to its fundamental group $\pi_1(X, x_0)$ of homotopy classes of loops at x_0, and a pointed map $f\colon (X, x_0) \to (Y, y_0)$ to the induced homomorphism $f_*\colon \pi_1(X, x_0) \to \pi_1(Y, y_0)$, $[\gamma] \mapsto [f \circ \gamma]$. That $(g \circ f)_* = g_*\,f_*$ and $(1_X)_* = 1_{\pi_1(X,x_0)}$ is exactly the functoriality of π_1, developed in [7].

5. *The derivative and the chain rule.* Let $\mathbf{Euclid}_*$ be the category whose objects are pointed finite-dimensional Euclidean spaces $(\mathbb{R}^n, a)$ and whose morphisms $(\mathbb{R}^n, a) \to (\mathbb{R}^m, b)$ are differentiable maps f with $f(a) = b$. The derivative is a functor
$$D\colon \mathbf{Euclid}_* \to \mathbf{Mat}_{\mathbb{R}} \qquad D(\mathbb{R}^n, a) = n, \qquad Df = \text{the Jacobian of } f \text{ at } a$$
sending a pointed Euclidean space to its dimension (an object of $\mathbf{Mat}_{\mathbb{R}}$) and a pointed differentiable map to its Jacobian matrix at the marked point. For composable $f\colon (\mathbb{R}^n, a) \to (\mathbb{R}^m, b)$ and $g\colon (\mathbb{R}^m, b) \to (\mathbb{R}^k, c)$, functoriality reads
$$D(gf) = Dg\,Df$$
the Jacobian of the composite equals the product of the Jacobian of g at b with the Jacobian of f at a. This equation is the chain rule; that D preserves identities is the statement that the derivative of an identity map is the identity matrix. The chain rule is the functoriality of the derivative.

6. *A group as a one-object category.* A group G is a category $\mathbf{B}G$ with a single object $*$, one morphism $* \to *$ for each element of G, composition given by the group operation, and 1_* the identity element. A functor $X \colon \mathbf{B}G \to C$ is then precisely an *action of G on an object of C*: it chooses an object $X*$ of C and, for each $g \in G$, an endomorphism $g_* \colon X* \to X*$, subject to

$$(hg)_* = h_* \, g_* \qquad \text{and} \qquad e_* = 1_{X*}$$

which are the functoriality and identity axioms written out. These are exactly the axioms of an action. Specializing the target recovers the familiar notions: a functor $\mathbf{B}G \to \mathbf{Set}$ is a G-set, a functor $\mathbf{B}G \to \mathbf{Vect}_k$ is a k-linear representation of G, and a functor $\mathbf{B}G \to \mathbf{Top}$ is a G-space. The three classical settings in which a group acts are three choices of target for one functor.

The last two examples repay a second look. The chain rule, which is usually presented as a formula to be memorized, is nothing more than the assertion that one particular assignment respects composition; once the derivative is recognized as a functor, the chain rule is not a theorem to be proved separately but the definition of what it means for D to be one. The one-object viewpoint is the same economy applied to group actions: rather than three definitions of a G-set, a G-representation, and a G-space, there is a single definition, a functor out of $\mathbf{B}G$, whose three instances differ only in where the arrows land. This is the recurring dividend of the categorical language, that constructions which look unrelated turn out to be one construction viewed through different targets.

Duality (definition 1.3.1) supplies a second kind of functor. Reversing the arrows of the source category before applying a functor produces a map that turns composition around, and this arrow-reversing behavior is so common that it is given its own name.

Definition 1.4.3: Contravariant Functor

A *contravariant functor* F from C to D is a functor $F \colon C^{\mathrm{op}} \to D$. Unwinding the definition of the opposite category (definition 1.3.1), a contravariant functor assigns

- an object Fc of D to each object c of C, and
- a morphism $Ff \colon Fc' \to Fc$ of D to each morphism $f \colon c \to c'$ of C, reversing domain and codomain,

subject to

1. $F(gf) = Ff \, Fg$ for every composable pair $f \colon c \to c'$, $g \colon c' \to c''$ of C, and
2. $F(1_c) = 1_{Fc}$ for each object c of C.

For emphasis, a functor in the sense of definition 1.4.1 is called *covariant*.

The single point of difference is the reversal in the first axiom: a covariant functor sends gf to $Fg\,Ff$, whereas a contravariant functor sends it to $Ff\,Fg$, varying *contrary* to the direction of composition. This is not a new species of object but a bookkeeping convenience: a contravariant functor $C \to D$ is literally a covariant functor $C^{\mathrm{op}} \to D$, so every theorem about functors applies to contravariant ones after the source is replaced by its opposite. The examples are as ubiquitous as the covariant ones, and each is built by the same move, sending a map to a map that goes the other way.

Example 1.4.4: Contravariant Functors

1. *The linear dual.* The dual-space construction is a contravariant functor $(-)^*\colon \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$ carrying a vector space V to its dual $V^* = \mathrm{Hom}_k(V, k)$ and a linear map $f\colon V \to W$ to the transpose $f^*\colon W^* \to V^*$ given by precomposition,

$$f^*(\varphi) = \varphi \circ f \qquad (\varphi\colon W \to k)$$

The reversal is forced by precomposition: since f^* eats a functional on W and returns one on V, it points from W^* to V^*. For composable $f\colon U \to V$ and $g\colon V \to W$ one computes $(gf)^*(\varphi) = \varphi \circ g \circ f = f^*(g^*(\varphi))$, so $(gf)^* = f^*g^*$, and $(1_V)^*(\varphi) = \varphi = 1_{V^*}(\varphi)$; the arrows reverse exactly as the definition requires.

2. *The contravariant powerset.* There is a contravariant functor $P\colon \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ sending a set A to PA and a function $f\colon A \to B$ to the *preimage* function

$$f^{-1}\colon PB \to PA \qquad B' \mapsto f^{-1}(B') = \{a \in A \big| f(a) \in B'\}$$

In contrast to the covariant powerset of example 1.4.2, which uses direct image and preserves the direction of f, the preimage points the other way. Functoriality holds because $(gf)^{-1}(C') = f^{-1}(g^{-1}(C'))$ for $C' \subseteq C$, which is the identity $(gf)^{-1} = f^{-1}g^{-1}$, and $(1_A)^{-1} = 1_{PA}$.

3. *The prime spectrum.* The spectrum is a contravariant functor $\mathrm{Spec}: \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Top}$ sending a commutative ring R to its prime spectrum $\mathrm{Spec}\, R$ and a ring homomorphism $\varphi\colon R \to S$ to the map

$$\varphi^*\colon \mathrm{Spec}\, S \to \mathrm{Spec}\, R \qquad \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$$

taking a prime $\mathfrak{p}$ of S to its preimage $\varphi^{-1}(\mathfrak{p})$, which is a prime of R because the preimage of a prime ideal under a ring homomorphism is prime. The contravariance is again a matter of preimages: φ runs from R to S, but φ^* runs from $\mathrm{Spec}\, S$ to $\mathrm{Spec}\, R$. Functoriality follows from $(\psi\varphi)^{-1}(\mathfrak{p}) = \varphi^{-1}(\psi^{-1}(\mathfrak{p}))$.

4. *The opens of a space.* For a topological space X, let $\mathcal{O}(X)$ be the set of open subsets of X, partially ordered by inclusion and so a poset. This is a contravariant functor $\mathcal{O}\colon \mathbf{Top}^{\mathrm{op}} \to \mathbf{Poset}$: a continuous map $f\colon X \to Y$ is sent to the preimage map

$$f^{-1}\colon \mathcal{O}(Y) \to \mathcal{O}(X) \qquad U \mapsto f^{-1}(U)$$

which lands in $\mathcal{O}(X)$ precisely because f is continuous, and which preserves inclusions since $U \subseteq V$ gives $f^{-1}(U) \subseteq f^{-1}(V)$. Functoriality is once more the composite-of-preimages identity $(gf)^{-1}(U) = f^{-1}(g^{-1}(U))$.

The four contravariant examples are variations on a single theme: each sends a map to its associated preimage or precomposition, and preimage inverts the direction of the arrow. The dual space, the contravariant powerset, the spectrum, and the opens of a space are the same movement performed in four categories. Set against the covariant powerset, which uses direct image and so preserves direction, they display in the sharpest possible terms the difference between forward transport by image and backward transport by preimage.

One property holds for every functor, covariant or contravariant, and it is the first indication that a functor is a genuine conduit of categorical structure rather than an arbitrary map: a functor cannot tell isomorphic objects apart, because it carries the very equations that witness an isomorphism.

Lemma 1.4.5: Functors Preserve Isomorphisms

Let $F: C \to D$ be a functor. If $f: x \to y$ is an isomorphism in C, then $Ff: Fx \to Fy$ is an isomorphism in D, with inverse $F(f^{-1})$. In particular, $x \cong y$ in C implies $Fx \cong Fy$ in D.

***Proof*:**
Let $f: x \to y$ be an isomorphism in C with inverse $g = f^{-1}: y \to x$, so that $gf = 1_x$ and $fg = 1_y$ (definition 1.2.1). Applying F and using the functoriality and identity axioms of definition 1.4.1,

$$Fg\,Ff = F(gf) = F(1_x) = 1_{Fx}$$

so $Fg: Fy \to Fx$ is a left inverse of Ff. The same computation with the roles of f and g interchanged gives

$$Ff\,Fg = F(fg) = F(1_y) = 1_{Fy}$$

so Fg is also a right inverse of Ff. Thus Ff is an isomorphism with two-sided inverse $Fg = F(f^{-1})$, and $Fx \cong Fy$, as we sought to show.

The proof used only the two functor axioms, and it applies verbatim to a contravariant functor: there the images of f and g compose in the opposite order, but the two products $Ff\,Fg$ and $Fg\,Ff$ still equal 1_{Fx} and 1_{Fy} in some order, so Ff remains invertible. Every functor therefore respects the categorical notion of sameness, which is why a functorial invariant, one whose value is a functor, can be used to distinguish objects: if $Fx \ncong Fy$ then $x \ncong y$. This is the mechanism behind the fundamental group as a tool in topology, since π_1, being a functor, sends homeomorphic pointed spaces to isomorphic groups, so non-isomorphic fundamental groups certify that two spaces are not homeomorphic.

Isomorphisms are the only class of maps a functor is guaranteed to preserve. Functors need not preserve monomorphisms or epimorphisms (definition 1.2.8). The spectrum makes the failure concrete: the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism in **CRing** (example 1.2.9), yet Spec reverses it, and its image $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a monomorphism rather than an epimorphism, so the epi property is not carried across. What a functor does preserve, by the identical one-line argument that proves lemma 1.4.5, is any mono or epi that *splits*: if f has a one-sided inverse s with $sf = 1$, then applying F gives $Fs\,Ff = 1$, so Ff retains that one-sided inverse and remains a split monomorphism, and dually for a split epimorphism. The equations a splitting provides survive because a functor preserves exactly the equations between composites, and nothing more.

1.5 Natural Transformations

Functors compare categories; natural transformations compare functors. This third layer is what makes the categorical vocabulary powerful enough to say precisely what mathematicians have always meant by a construction being *canonical*, or *basis-free*, or *independent of choices*. The guiding example is one every reader has met in linear algebra. A finite-dimensional vector space V is isomorphic to its dual V^*, but only after a basis is chosen; change the basis and the isomorphism changes. Yet V is also isomorphic to its double dual V^{**}, and this time no choice is needed at all: a single formula, valid for every V and compatible with every linear map, produces the isomorphism. The distinction between these two isomorphisms, one requiring a choice and one not, is not a vague matter of taste. It is captured exactly by whether a family of maps assembles into a natural transformation. The section develops the double-dual example first, extracts the general definition, packages functors and their transformations into a category of their own, and closes with the notion of sameness that naturality makes available for categories themselves: equivalence.

The dual space is the starting point. For a field k and a vector space V, the linear dual is $V^* = \mathrm{Hom}_k(V, k)$, the space of linear functionals on V; this is the object developed in Core Algebra, and its basic properties are cited rather than reproved here (see [4]). When V is finite-dimensional, V^* has the same dimension as V, so the two are isomorphic. An isomorphism is built by choosing a basis $e_1, \dots, e_n$ of V and sending e_i to the dual basis vector e_i^*, the functional determined by $e_i^*(e_j) = \delta_{ij}$. This map is a linear isomorphism, but it depends on the basis: a different basis of V yields a different isomorphism $V \to V^*$, and there is no way to single one out. The double dual behaves differently, and pinning down that difference is the object of the example.

Example 1.5.1: The Double Dual of a Vector Space

Let k be a field and V a vector space over k. Its *double dual* is $V^{**} = \mathrm{Hom}_k(V^*, k) = \mathrm{Hom}_k(\mathrm{Hom}_k(V, k), k)$. For each vector $v \in V$ define the *evaluation map* $\mathrm{ev}_V(v) \colon V^* \to k$ by

$$\mathrm{ev}_V(v)(\varphi) = \varphi(v) \qquad (\varphi \in V^*)$$

so $\mathrm{ev}_V(v)$ is the functional on V^* that reads off the value of a given functional at v. This assembles into a single map $\mathrm{ev}_V : V \to V^{**}$, $v \mapsto \mathrm{ev}_V(v)$, defined without any reference to a basis.

Step 1: $\mathrm{ev}_V(v)$ is linear for each v. For $\varphi, \psi \in V^*$ and $c \in k$,

$$\mathrm{ev}_V(v)(\varphi + \psi) = (\varphi + \psi)(v) = \varphi(v) + \psi(v) = \mathrm{ev}_V(v)(\varphi) + \mathrm{ev}_V(v)(\psi)$$

and $\mathrm{ev}_V(v)(c\varphi) = (c\varphi)(v) = c\,\varphi(v) = c\,\mathrm{ev}_V(v)(\varphi)$, using the vector-space structure on V^*. So $\mathrm{ev}_V(v) \in V^{**}$.

Step 2: ev_V is linear in v. For $v, w \in V$ and $c \in k$, the identity of functionals $\mathrm{ev}_V(v + w) = \mathrm{ev}_V(v) + \mathrm{ev}_V(w)$ is checked by evaluating both sides at an arbitrary $\varphi \in V^*$:

$$\mathrm{ev}_V(v + w)(\varphi) = \varphi(v + w) = \varphi(v) + \varphi(w) = \mathrm{ev}_V(v)(\varphi) + \mathrm{ev}_V(w)(\varphi)$$

and likewise $\mathrm{ev}_V(cv)(\varphi) = \varphi(cv) = c\,\varphi(v) = c\,\mathrm{ev}_V(v)(\varphi)$, so $\mathrm{ev}_V(cv) = c\,\mathrm{ev}_V(v)$. The linearity of ev_V is inherited from the linearity of the functionals it evaluates against.

Step 3: ev_V is injective. Suppose $v \neq 0$. Extend v to a basis of V and let $\varphi \in V^*$ be the coordinate functional with $\varphi(v) = 1$ (see [4] for the existence of such a functional). Then $\mathrm{ev}_V(v)(\varphi) = \varphi(v) = 1 \neq 0$, so $\mathrm{ev}_V(v) \neq 0$. Thus $\ker \mathrm{ev}_V = 0$.

Step 4: in finite dimension ev_V is an isomorphism. If $\dim V = n < \infty$ then $\dim V^* = n$ and $\dim V^{**} = n$, so ev_V is an injective linear map between spaces of equal finite dimension and is therefore an isomorphism. Hence $V \cong V^{**}$ with no choice of basis.

Step 5: the evaluation maps are compatible with every linear map. Given a linear map $f \colon V \to W$, the dual map $f^* \colon W^* \to V^*$ is $f^*(\psi) = \psi \circ f$, and the double dual map $f^{**} \colon V^{**} \to W^{**}$ is

$f^{**} = (f^*)^*$, so $f^{**}(\Phi) = \Phi \circ f^*$ for $\Phi \in V^{**}$. The square

$$\begin{tikzcd} V \arrow[r, "\mathrm{ev}_V"] \arrow[d, "f"'] & V^{**} \arrow[d, "f^{**}"] \\ W \arrow[r, "\mathrm{ev}_W"'] & W^{**} \end{tikzcd}$$

commutes. Indeed, for $v \in V$ and any $\psi \in W^*$,

$$f^{**}(\mathrm{ev}_V(v))(\psi) = \mathrm{ev}_V(v)(f^*\psi) = (f^*\psi)(v) = \psi(f(v)) = \mathrm{ev}_W(f(v))(\psi)$$

so $f^{**} \circ \mathrm{ev}_V = \mathrm{ev}_W \circ f$ as maps $V \to W^{**}$.

The five steps isolate exactly what is meant by canonicity. The map ev_V is defined by one formula that mentions only the data present in every vector space, so it exists for all V simultaneously (Steps 1 and 2), it is an isomorphism whenever V is finite-dimensional (Steps 3 and 4), and Step 5 shows that the whole family $\{\mathrm{ev}_V\}_V$ is compatible with the maps between spaces: transporting a vector along f and then evaluating gives the same answer as evaluating and then transporting along f^{**}. The isomorphism $V \cong V^*$ has no analogue of Step 5, precisely because it is built from a basis rather than from a formula uniform in V. The commuting square of Step 5 is the whole content of naturality, and abstracting it gives the central definition.

Definition 1.5.2: Natural Transformation

Let C and D be categories and let $F, G\colon C \to D$ be functors. A *natural transformation* $\alpha\colon F \Rightarrow G$ assigns to each object c of C a morphism

$$\alpha_c\colon Fc \to Gc$$

in D, called the *component* of α at c, such that for every morphism $f\colon c \to c'$ in C the *naturality square*

$$\begin{tikzcd} Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"'] & Gc \arrow[d, "Gf"] \\ Fc' \arrow[r, "\alpha_{c'}"'] & Gc' \end{tikzcd}$$

commutes, that is,

$$Gf \circ \alpha_c = \alpha_{c'} \circ Ff$$

A natural transformation $\alpha\colon F \Rightarrow G$ is a *natural isomorphism* if every component α_c is an isomorphism in D. When a natural isomorphism $F \Rightarrow G$ exists, the functors are *naturally isomorphic*, written $F \cong G$.

A natural transformation is a family of morphisms, one per object of the source category, wired together by the naturality condition so that the family respects every morphism of the source. The double arrow $\Rightarrow$ marks the new altitude: an ordinary morphism $\to$ connects objects, a functor $\to$ connects categories, and a natural transformation $\Rightarrow$ connects functors. It is often drawn as a 2-cell filling the region between two parallel functors,

$$\begin{tikzcd} C \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & D \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$$

which records the shape of the data: two functors $C \to D$ and a transformation between them. The naturality square is the substantive clause. Without it a family $\{\alpha_c\}$ is just an unstructured collection of morphisms; with it, α interacts coherently with all of C, and this coherence is what earns the word "natural". A useful reading of the square is that α commutes with change of object: whether one applies α before or after transporting along f, the result is the same.

The double-dual example is now an instance of the definition, and naming the two functors involved makes the fit exact.

Example 1.5.3: Evaluation as a Natural Isomorphism

Take $C = D = \mathbf{Vect}_k$. The identity functor $\mathrm{Id}\colon \mathbf{Vect}_k \to \mathbf{Vect}_k$ sends each space to itself and each linear map to itself. The *double dual functor* $(-)^{**}\colon \mathbf{Vect}_k \to \mathbf{Vect}_k$ sends V to V^{**} and $f\colon V \to W$ to $f^{**}\colon V^{**} \to W^{**}$; it is a covariant functor because dualizing twice restores the original variance (each application of $(-)^*$ is contravariant, and the composite of two contravariant functors is covariant, cf. definition 1.4.3).

The evaluation maps of example 1.5.1 are the components of a natural transformation

$$\mathrm{ev}\colon \mathrm{Id} \Rightarrow (-)^{**}$$

with component $\mathrm{ev}_V\colon V \to V^{**}$ at each space V. The naturality square at a linear map $f\colon V \to W$ is

$$\begin{tikzcd} V \arrow[r, "\mathrm{ev}_V"] \arrow[d, "f"'] & V^{**} \arrow[d, "f^{**}"] \\ W \arrow[r, "\mathrm{ev}_W"'] & W^{**} \end{tikzcd}$$

which is exactly the commuting square proved in Step 5 of example 1.5.1. Restricting to the full subcategory of finite-dimensional spaces, every component ev_V is an isomorphism (Step 4), so ev is a natural isomorphism there and $\mathrm{Id} \cong (-)^{**}$. This is the precise sense in which a finite-dimensional space is canonically its own double dual: not only that each V is isomorphic to V^{**}, but that the isomorphisms are the components of a single natural transformation.

The single dual admits no such transformation, and the reason is worth isolating carefully, since it is a matter of variance. Each application of $(-)^*$ is contravariant, so the single dual is a functor $(-)^*\colon \mathbf{Vect}_k^{\mathrm{op}} \to \mathbf{Vect}_k$, not an endofunctor of $\mathbf{Vect}_k$; the symbol $\mathrm{Id} \Rightarrow (-)^*$ does not even typecheck, because a natural transformation compares two functors of the same variance and the naturality square would need its right edge $f^*\colon W^* \to V^*$ to point the same way as $f\colon V \to W$, which it does not. The comparison is made on the groupoid $\mathbf{Vect}_k^{\mathrm{fd},\cong}$ of finite-dimensional spaces and their isomorphisms, where every morphism is invertible and variance is immaterial; there both Id and $(-)^*$ are genuine endofunctors, so a natural isomorphism $\mathrm{Id} \cong (-)^*$ is a meaningful thing to ask for. None exists: the family of basis-dependent isomorphisms $V \to V^*$ does *not* assemble into a natural transformation, because the naturality square fails for the isomorphisms of V that rescale the chosen basis. Naturality is exactly the property the basis-free construction has and the basis-dependent one lacks.

The contrast in the example is worth stating as a principle. A construction is canonical precisely when it is natural, that is, when its instances at the various objects are the components of a natural transformation. Any construction that requires an arbitrary choice, a basis, a splitting, a lift, fails to be natural, because different choices produce components that violate the naturality square. The vocabulary of natural transformations turns the informal "no choices were made" into a checkable condition on a commuting diagram.

Once natural transformations are in hand, functors between two fixed categories become the objects of a category in their own right, with natural transformations as the morphisms. This packaging is not a formality: it is the setting in which diagrams, presheaves, and much of the rest of the subject live, and it is where the double-dual isomorphism becomes an isomorphism in the literal categorical sense. To build it, one needs to know how natural transformations compose. Given $\alpha\colon F \Rightarrow G$ and $\beta\colon G \Rightarrow H$ between functors $F, G, H\colon C \to D$, the composite is defined componentwise by $(\beta\alpha)_c = \beta_c \circ \alpha_c$, and the identity transformation $1_F\colon F \Rightarrow F$ has components $(1_F)_c = 1_{Fc}$.

Definition 1.5.4: Functor Category

Let C and D be categories. The *functor category* $[C, D]$ (also written D^C) has

- as objects, the functors $F\colon C \to D$;
- as morphisms $F \to G$, the natural transformations $\alpha\colon F \Rightarrow G$.

The identity morphism on a functor F is the natural transformation 1_F with components $(1_F)_c = 1_{Fc}$, and the composite of $\alpha\colon F \Rightarrow G$ with $\beta\colon G \Rightarrow H$ is the natural transformation $\beta\alpha\colon F \Rightarrow H$ with components $(\beta\alpha)_c = \beta_c \circ \alpha_c$.

The definition asserts that $[C, D]$ is a category, and this must be verified: the proposed composite and identities have to be natural transformations, and they have to satisfy the category axioms. The verification is short, and doing it in full shows how the naturality squares of the factors combine.

***Proof*:**
Two things require checking: that $\beta\alpha$ and 1_F are natural transformations, and that composition is associative and unital.

Step 1: the componentwise composite is natural. Let $\alpha\colon F \Rightarrow G$ and $\beta\colon G \Rightarrow H$ be natural, and let $f\colon c \to c'$ be a morphism in C. Naturality of α gives $Gf \circ \alpha_c = \alpha_{c'} \circ Ff$, and naturality of β gives $Hf \circ \beta_c = \beta_{c'} \circ Gf$. Then

$$Hf \circ (\beta\alpha)_c = Hf \circ \beta_c \circ \alpha_c = \beta_{c'} \circ Gf \circ \alpha_c = \beta_{c'} \circ \alpha_{c'} \circ Ff = (\beta\alpha)_{c'} \circ Ff$$

so the naturality square for $\beta\alpha$ commutes. Diagrammatically, the two naturality squares stack vertically and the outer rectangle is what commutes:

$$\begin{tikzcd} Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"'] & Gc \arrow[r, "\beta_c"] \arrow[d, "Gf"] & Hc \arrow[d, "Hf"] \\ Fc' \arrow[r, "\alpha_{c'}"'] & Gc' \arrow[r, "\beta_{c'}"'] & Hc' \end{tikzcd}$$

Step 2: the identity is natural. For 1_F with components 1_{Fc} and any $f\colon c \to c'$, both routes around the square equal Ff, since $Ff \circ 1_{Fc} = Ff = 1_{Fc'} \circ Ff$. So 1_F is a natural transformation.

Step 3: the category axioms. Composition and identities in $[C, D]$ are defined componentwise, and in each component they are ordinary composition and identities in D. Associativity and the unit laws therefore hold component by component, inherited from the corresponding axioms in D. For a third composable transformation $\gamma\colon H \Rightarrow K$ and each object c,

$$((\gamma\beta)\alpha)_c = \gamma_c\beta_c\alpha_c = (\gamma(\beta\alpha))_c, \qquad (1_G\,\alpha)_c = \alpha_c = (\alpha\,1_F)_c$$

A pair of natural transformations agreeing in every component are equal, so these componentwise identities give the required equalities of natural transformations. Thus $[C, D]$ satisfies the category axioms, completing the proof.

The functor category is the natural home for the double-dual isomorphism. In $[\mathbf{Vect}_k, \mathbf{Vect}_k]$ the transformation $\mathrm{ev}\colon \mathrm{Id} \Rightarrow (-)^{**}$ of example 1.5.3 is a morphism, and because every component is an isomorphism (in finite dimensions) it is an isomorphism in the functor category: the componentwise inverse ev^{-1}, with components $(\mathrm{ev}_V)^{-1}$, is again natural and is a two-sided inverse to ev. So the statement "V is canonically isomorphic to V^{**}" becomes the statement "Id and $(-)^{**}$ are isomorphic objects of the functor category". The composition just defined, taking place along the objects of C, is the *vertical* composition of natural transformations; there is a second, *horizontal* composition that composes transformations between functors that themselves compose, developed when it is needed. Functor categories are already familiar in disguise: when C is a small category and $D = \mathbf{Set}$, an object of $[C, \mathbf{Set}]$ is a diagram of sets shaped like C, and morphisms of diagrams are exactly natural transformations.

Natural isomorphism is also the ingredient needed to fix the notion of sameness for categories themselves. The obvious candidate, isomorphism of categories, asks too much. Two categories C and D would be isomorphic if there were functors $F\colon C \to D$ and $G\colon D \to C$ with $GF = \mathrm{Id}_C$ and $FG = \mathrm{Id}_D$ on the nose, meaning $GFc = c$ for every object c and $GFf = f$ for every morphism. In practice this equality-on-objects requirement is almost never met by the functors one cares about, because the objects of C and D are typically built from different raw material even when the categories carry the same mathematical content. What one wants is that GFc be canonically isomorphic to c, not equal to it, uniformly in c; and "canonically isomorphic, uniformly in c" is precisely a natural isomorphism $GF \cong \mathrm{Id}_C$. Weakening equality to natural isomorphism gives the right notion.

Definition 1.5.5: Equivalence of Categories

An *equivalence of categories* between C and D consists of functors $F\colon C \to D$ and $G\colon D \to C$ together with natural isomorphisms

$$\eta\colon \mathrm{Id}_C \Rightarrow GF \qquad \text{and} \qquad \varepsilon\colon FG \Rightarrow \mathrm{Id}_D$$

(each component an isomorphism). When such data exist, F is an *equivalence*, G is a *quasi-inverse* of F, and the categories C and D are *equivalent*, written $C \simeq D$.

An equivalence relaxes the two rigid equations $GF = \mathrm{Id}_C$ and $FG = \mathrm{Id}_D$ of an isomorphism to the two natural isomorphisms $GF \cong \mathrm{Id}_C$ and $FG \cong \mathrm{Id}_D$. The functor F need not hit every object of D on the nose, nor establish a bijection between objects; it need only do so up to isomorphism and in a way compatible with all morphisms. This is the correct notion of sameness for categories: it identifies categories that carry the same mathematical information even when their objects are packaged differently, and essentially every "two descriptions of the same theory" result in mathematics is an equivalence rather than an isomorphism of categories.

There is a second characterization of equivalences that is usually easier to check than exhibiting a quasi-inverse, phrased entirely in terms of the single functor F. Recall from definition 1.4.1 that F acts on hom-sets by $f \mapsto Ff$; call F *full* if each of these maps $\mathrm{Hom}_C(c, c') \to \mathrm{Hom}_D(Fc, Fc')$ is surjective, *faithful* if each is injective, and *fully faithful* if each is a bijection. Call F *essentially surjective* if every object d of D is isomorphic to Fc for some object c of C. The two conditions together say that F is a bijection on morphisms between any two objects and hits every object up to isomorphism.

Proposition 1.5.6: Characterization of Equivalences

A functor $F: C \to D$ is an equivalence of categories if and only if F is full, faithful, and essentially surjective.

The forward direction is elementary and can be given now; the converse requires building a quasi-inverse object by object, and its construction is deferred to the chapter on the Yoneda lemma, where the tools for choosing the components coherently are in place.

***Proof*:**
Only the forward implication is proved here; for the converse, the construction of a quasi-inverse from full faithfulness and essential surjectivity is carried out once the necessary machinery is available, and is taken up later in the book.

Suppose F is an equivalence, with quasi-inverse G and natural isomorphisms $\eta: \mathrm{Id}_C \Rightarrow GF$ and $\varepsilon: FG \Rightarrow \mathrm{Id}_D$.

Step 1: F is essentially surjective. For any object d of D, the component $\varepsilon_d: FGd \to d$ is an isomorphism, so $d \cong Fc$ for $c = Gd$. Every object of D is thus isomorphic to one in the image of F.

Step 2: F is faithful. Let $f, g: c \to c'$ satisfy $Ff = Fg$. Applying G gives $GFf = GFg$. Naturality of η at the morphism f gives $GFf \circ \eta_c = \eta_{c'} \circ f$, and likewise for g, so

$$\eta_{c'} \circ f = GFf \circ \eta_c = GFg \circ \eta_c = \eta_{c'} \circ g$$

Since $\eta_{c'}$ is an isomorphism it is monic, and cancelling it gives $f = g$. Thus F is faithful. The identical argument applied to G (with the natural isomorphism ε in place of η) shows G is faithful as well.

Step 3: F is full. Let $h: Fc \to Fc'$ be any morphism in D. Set $f = \eta_{c'}^{-1} \circ Gh \circ \eta_c : c \to c'$, a morphism of C built from $Gh: GFc \to GFc'$ and the isomorphisms η_c, $\eta_{c'}$. Naturality of η at f reads $GFf \circ \eta_c = \eta_{c'} \circ f$, and substituting the definition of f gives $\eta_{c'} \circ f = Gh \circ \eta_c$, whence $GFf = Gh$. Because G is faithful (Step 2), $GFf = Gh$ forces $Ff = h$. So every $h: Fc \to Fc'$ is Ff for some f, and F is full, as we sought to show.

The two descriptions of an equivalence serve different purposes. The definition in terms of a quasi-inverse and two natural isomorphisms is the one to use when the inverse construction is at hand and one wants to transport structure back and forth. The characterization by full faithfulness and essential surjectivity is the one to use when only F is given and building an explicit G would be laborious: three conditions on a single functor replace the search for a partner. A concrete instance shows the second criterion at work.

Example 1.5.7: Matrices as an Equivalence

Let k be a field, let $\mathbf{Vect}_k^{\mathrm{fd}}$ be the category of finite-dimensional k-vector spaces and linear maps, and let $\mathbf{Mat}_k$ be the category whose objects are the natural numbers $0, 1, 2, \ldots$ and whose morphisms $m \to n$ are the $n \times m$ matrices over k, with composition given by matrix multiplication and identities the identity matrices (this is the matrix category $\mathbf{Mat}_k$ of example 1.1.4, specialized to $R = k$). Define $F \colon \mathbf{Mat}_k \to \mathbf{Vect}_k^{\mathrm{fd}}$ by $F(n) = k^n$ on objects and by sending an $n \times m$ matrix A to the linear map $k^m \to k^n$, $x \mapsto Ax$.

The claim is that F is an equivalence, verified through proposition 1.5.6.

- *Fully faithful.* A linear map $k^m \to k^n$ is represented, with respect to the standard bases, by a unique $n \times m$ matrix, and every such matrix arises this way. So $\mathbf{Mat}_k(m, n) \to \mathrm{Hom}_k(k^m, k^n)$, $A \mapsto (x \mapsto Ax)$, is a bijection: F is full and faithful.
- *Essentially surjective.* Every finite-dimensional space V has a basis of some size n, and choosing one gives an isomorphism $V \cong k^n = F(n)$ (see [4]). So every object of $\mathbf{Vect}_k^{\mathrm{fd}}$ is isomorphic to one in the image of F.

Hence F is an equivalence, and $\mathbf{Mat}_k \simeq \mathbf{Vect}_k^{\mathrm{fd}}$.

The equivalence in the example is not an isomorphism of categories, and the reason is instructive. The objects of $\mathbf{Mat}_k$ are natural numbers, while the objects of $\mathbf{Vect}_k^{\mathrm{fd}}$ are vector spaces; there are a proper class of two-dimensional spaces but only one object 2 in $\mathbf{Mat}_k$, so no functor can be a bijection on objects and $GF = \mathrm{Id}$ cannot hold on the nose. Yet the two categories carry identical information: linear algebra on finite-dimensional spaces is the theory of matrices up to change of basis, and the equivalence is the exact expression of that identity. This is the recurring pattern. Equivalence, not isomorphism, is the notion of sameness for categories, because it records that two categories have the same objects and morphisms up to canonical isomorphism, which is all that the categorical viewpoint, where an object is known only through its morphisms, can or should ask.

2

The Yoneda Lemma and Representability

Chapter 1 built the categorical language and returned to one thread: an object is determined by its morphisms. This chapter makes it a theorem. The *representable functor* $\mathrm{Hom}(c, -)$ records how an object c maps to every other; the *Yoneda lemma* computes its natural transformations into any set-valued functor, and the *Yoneda embedding* recovers c from the presheaf it represents. Along the way we settle questions left open in Chapter 1, among them the converse of the equivalence characterization (proposition 1.5.6).

2.1 Representable Functors

The first step toward the Yoneda lemma is to package the morphisms out of a fixed object as a functor. Fixing an object c of a locally small category and letting the second slot of the hom-set vary produces a set-valued functor; the functors that arise this way, and the objects that produce them, are the subject of this section. A functor isomorphic to one of these is *representable*, and the isomorphism is pinned down by a single element, the universal element.

We work throughout in a locally small category C (definition 1.1.7), so that each collection $\mathrm{Hom}_C(x, y)$ is a set.

Definition 2.1.1: Representable Functor

Let C be locally small and c an object of C. The *covariant hom-functor*

$$\mathrm{Hom}(c,-)\colon C \to \mathbf{Set}$$

sends an object x to the set $\mathrm{Hom}(c,x)$ and a morphism $f\colon x \to y$ to *post-composition*

$$f_*\colon \mathrm{Hom}(c,x) \to \mathrm{Hom}(c,y), \qquad f_*(h) = fh$$

The *contravariant hom-functor*

$$\mathrm{Hom}(-,c)\colon C^{\mathrm{op}} \to \mathbf{Set}$$

sends x to $\mathrm{Hom}(x,c)$ and $f\colon x \to y$ to *pre-composition*

$$f^*\colon \mathrm{Hom}(y,c) \to \mathrm{Hom}(x,c), \qquad f^*(h) = hf$$

A functor $F\colon C \to \mathbf{Set}$ is *representable* if there is an object c and a natural isomorphism $F \cong \mathrm{Hom}(c,-)$; the object c is a *representing object* for F. Dually, a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ is *representable* if $F \cong \mathrm{Hom}(-,c)$ for some c.

Functoriality of $\mathrm{Hom}(c,-)$ is the two composition laws of C: for $f\colon x \to y$ and $g\colon y \to z$ one has $(gf)_*(h) = gfh = g_*(f_*(h))$, so $(gf)_* = g_*f_*$, and $(1_x)_*(h) = h$, so $(1_x)_* = 1_{\mathrm{Hom}(c,x)}$. The same identities read backwards give $(gf)^* = f^*g^*$, the order-reversal that makes $\mathrm{Hom}(-,c)$ contravariant. A representing object, if one exists, is a solution to the problem posed by F: it is an object whose maps out of it (or into it) reproduce the sets Fx compatibly with all the morphisms of C. Chapter 1 already showed how much a hom-set encodes, since a morphism is an isomorphism exactly when post-composition with it is a bijection of hom-sets for every test object (example 1.1.8 recorded the first instances); here the hom-set is promoted from a single set to a functor.

Example 2.1.2: The Underlying Set of a Group

The forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$, sending a group to its underlying set and a homomorphism to its underlying function, is representable, and the integers $\mathbb{Z}$ represent it. A homomorphism $\mathbb{Z} \to G$ is determined by the image of the generator 1, and any element of G is a legitimate image, so

$$\mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z},G) \cong UG, \qquad \varphi \mapsto \varphi(1)$$

This bijection is natural in G: for a homomorphism $\psi\colon G \to H$ the square comparing $\varphi \mapsto \varphi(1)$ on either side commutes, since $(\psi\varphi)(1) = \psi(\varphi(1))$. So $U \cong \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z},-)$. That $\mathbb{Z}$ is the free group on one generator ([4], free groups) is exactly the statement that it represents the underlying-set functor.

A representing object is a compressed description of a whole functor, and the examples of most interest name a construction the reader already knows. Before collecting more of them it is worth isolating the contravariant case, where the objects of the represented functor are the presheaves.

Definition 2.1.3: Presheaf

A *presheaf* on a category C is a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$. Presheaves on C, together with natural transformations between them, form the *presheaf category*

$$\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$$

the functor category (definition 1.5.4) of contravariant set-valued functors on C.

For each object c the contravariant hom-functor $\mathrm{Hom}(-, c)$ is a presheaf, the *presheaf represented by* c; a presheaf isomorphic to some $\mathrm{Hom}(-, c)$ is representable in the sense of definition 2.1.1. The two variances are best seen together. Holding neither slot of the hom-set fixed gives a functor of both arguments at once.

Definition 2.1.4: The Two-Sided Hom-Bifunctor

Let C be locally small. The *hom-bifunctor*

$$\mathrm{Hom}(-, -)\colon C^{\mathrm{op}} \times C \to \mathbf{Set}$$

sends an object (x, y) to $\mathrm{Hom}(x, y)$ and a morphism $(f, g)\colon (x, y) \to (x', y')$, i.e. a pair $f\colon x' \to x$ in C and $g\colon y \to y'$ in C, to

$$\mathrm{Hom}(x, y) \to \mathrm{Hom}(x', y'), \qquad h \mapsto ghf$$

Functoriality is one line: given a second pair $(f', g')\colon (x', y') \to (x'', y'')$, composing the induced maps sends h to $g'(ghf)f' = (g'g)h(ff')$, which is the map induced by the composite $(f', g')(f, g) = (ff', g'g)$ in $C^{\mathrm{op}} \times C$; and the identity pair $(1_x, 1_y)$ induces $h \mapsto h$. Fixing the first slot at c recovers $\mathrm{Hom}(c, -)$ and fixing the second recovers $\mathrm{Hom}(-, c)$, so the bifunctor is the common source of both hom-functors.

The definition of representability quantifies over all objects and all elements at once, through the natural isomorphism. It has a pointwise reformulation carried by a single piece of data. An isomorphism $F \cong \mathrm{Hom}(c, -)$ is pinned down by where the identity 1_c goes, and the element of Fc it corresponds to satisfies a universal property that says exactly "every element of every Fd factors through it uniquely."

Definition 2.1.5: Universal Element

Let $F\colon C \to \mathbf{Set}$. A *universal element* of F is a pair (c, x) with c an object of C and $x \in Fc$ such that for every object d and every $y \in Fd$ there is a unique morphism $f\colon c \to d$ with $(Ff)(x) = y$.

The pair (c, x) is universal in that every element of the functor is obtained from x by transport along a unique morphism. The following makes precise that a universal element is the same data as a representation.

Proposition 2.1.6: Representability by a Universal Element

A functor $F\colon C \to \mathbf{Set}$ is representable if and only if it has a universal element. Explicitly, a natural isomorphism $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ corresponds to the universal element (c,x) with $x = \alpha_c(1_c)$, and conversely every universal element arises this way from a unique natural isomorphism.

***Proof*:**
Step 1: A representation gives a universal element. Suppose $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is a natural isomorphism, and set $x = \alpha_c(1_c) \in Fc$. Fix an object d and an element $y \in Fd$. Naturality of α at a morphism $f\colon c \to d$ is the commuting square

$$\begin{tikzcd} \mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "f_*"'] & Fc \arrow[d, "Ff"] \\ \mathrm{Hom}(c,d) \arrow[r, "\alpha_d"'] & Fd \end{tikzcd}$$

Chasing 1_c around it gives $\alpha_d(f) = \alpha_d(f_*(1_c)) = (Ff)(\alpha_c(1_c)) = (Ff)(x)$. Thus for every $f \in \mathrm{Hom}(c,d)$,

$$\alpha_d(f) = (Ff)(x) \tag{2.1}$$

Since α_d is a bijection, for the given $y \in Fd$ there is exactly one $f\colon c \to d$ with $\alpha_d(f) = y$, that is, exactly one f with $(Ff)(x) = y$. So (c,x) is a universal element.

Step 2: A universal element gives a representation. Conversely, let (c,x) be a universal element of F. Define, for each object d, a map

$$\alpha_d\colon \mathrm{Hom}(c,d) \to Fd, \qquad \alpha_d(f) = (Ff)(x)$$

The universal property says precisely that for each $y \in Fd$ there is a unique $f\colon c \to d$ with $(Ff)(x) = y$, which is the statement that α_d is a bijection. The maps α_d are natural: for $g\colon d \to d'$ and $f\colon c \to d$,

$$\alpha_{d'}(g_*(f)) = \alpha_{d'}(gf) = (F(gf))(x) = (Fg)((Ff)(x)) = (Fg)(\alpha_d(f))$$

so the naturality square for g commutes. Hence $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is a natural isomorphism, and F is representable.

Step 3: The two constructions are inverse. A representation α produces $x = \alpha_c(1_c)$, and (2.1) shows that the representation rebuilt from x in Step 2 has components $f \mapsto (Ff)(x) = \alpha_d(f)$, recovering α. In the other direction, the representation built from (c,x) in Step 2 sends 1_c to $(F1_c)(x) = x$, recovering the element. So the correspondence is a bijection between representations and universal elements, completing the proof.

The proposition converts a question about a natural isomorphism, which involves a component at every object, into a question about one element and its universal property. Verifying representability then reduces to exhibiting a single x and checking that every y factors through it uniquely, as the following examples show.

Example 2.1.7: Representable Functors

1. The identity functor on **Set** is representable, with representing object a one-point set $1 = \{*\}$. A function $1 \to X$ is a choice of a point of X, so $\mathrm{Hom}_{\mathbf{Set}}(1, X) \cong X$ by $\varphi \mapsto \varphi(*)$, naturally in X. The universal element is the pair $(1, *)$: every element y of every set X is $(Ff)(*) = f(*)$ for the unique $f\colon 1 \to X$ with $f(*) = y$.

2. The contravariant powerset $P\colon \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$, sending a set X to its powerset PX and a function $f\colon X \to Y$ to the preimage map $f^{-1}\colon PY \to PX$, is a representable presheaf. Let $2 = \{0, 1\}$. A subset $A \subseteq X$ is the same data as its indicator function $\chi_A\colon X \to 2$, with $\chi_A(a) = 1$ exactly when $a \in A$, and this correspondence is a bijection

$$PX \cong \mathrm{Hom}_{\mathbf{Set}}(X, 2), \qquad A \mapsto \chi_A$$

It is natural in X: for $f\colon X \to Y$ and $B \subseteq Y$, the indicator of $f^{-1}(B)$ is $\chi_B \circ f$, which is exactly $f^*(\chi_B)$. So $P \cong \mathrm{Hom}_{\mathbf{Set}}(-, 2)$, and 2 is the representing object. The universal element is the pair $(2, U)$ with $U = \{1\} \subseteq 2$: every subset $A \subseteq X$ is the preimage $\chi_A^{-1}(U)$ under a unique map $X \to 2$.

2.2 The Yoneda Lemma

The representable functor $\mathrm{Hom}(c, -)$ of definition 2.1.1 records how c maps to every object of C. This section computes the natural transformations out of it. The answer, the Yoneda lemma, is that a natural transformation from $\mathrm{Hom}(c, -)$ to a set-valued functor F is the same thing as a single element of Fc: the whole transformation is pinned down by where it sends the identity 1_c. The embedding of the next section and everything that follows rest on this computation.

The mechanism is visible before any proof. A natural transformation $\alpha\colon \mathrm{Hom}(c, -) \Rightarrow F$ has, at each object d, a component $\alpha_d\colon \mathrm{Hom}(c, d) \to Fd$. The set $\mathrm{Hom}(c, c)$ holds a distinguished element, the identity 1_c, and α_c sends it to some $\alpha_c(1_c) \in Fc$. Naturality forces every other value of every component to be determined by this one element: for $f\colon c \to d$, naturality applied to f and evaluated at 1_c expresses $\alpha_d(f)$ through $\alpha_c(1_c)$. Reading off $\alpha_c(1_c)$ is thus a map from natural transformations to Fc, and the lemma is that it is a bijection.

Theorem 2.2.1: The Yoneda Lemma

Let C be a locally small category, c an object of C, and $F\colon C \to \mathbf{Set}$ a functor. Evaluation at the identity,

$$\Phi\colon\ \mathrm{Nat}(\mathrm{Hom}(c, -), F) \to Fc, \qquad \Phi(\alpha) = \alpha_c(1_c)$$

is a bijection, natural in both c and F.

The proof produces an explicit inverse and checks that the two maps compose to the identity in each order. The inverse takes an element $x \in Fc$ and builds a natural transformation from it; the functoriality of F is what makes that transformation natural.

***Proof*:**
Step 1: the inverse map. Fix $x \in Fc$. Define a family $\Psi(x)$ by giving, at each object d, the component

$$\Psi(x)_d \colon \mathrm{Hom}(c,d) \to Fd, \qquad \Psi(x)_d(f) = (Ff)(x)$$

for $f \colon c \to d$, where $Ff \colon Fc \to Fd$ is the action of F on f. This is well-defined: Ff is a function $Fc \to Fd$ and $x \in Fc$, so $(Ff)(x) \in Fd$.

The claim is that $\Psi(x)$ is a natural transformation $\mathrm{Hom}(c,-) \Rightarrow F$. Let $g \colon d \to d'$ be a morphism of C. The functor $\mathrm{Hom}(c,-)$ sends g to post-composition $g_* \colon \mathrm{Hom}(c,d) \to \mathrm{Hom}(c,d')$, $g_*(f) = gf$. The naturality square to check is

$$\begin{tikzcd}
\mathrm{Hom}(c,d) \arrow[r, "\Psi(x)_d"] \arrow[d, "g_*"'] & Fd \arrow[d, "Fg"] \\
\mathrm{Hom}(c,d') \arrow[r, "\Psi(x)_{d'}"'] & Fd'
\end{tikzcd}$$

Chase $f \colon c \to d$ around both routes. Down then across gives $\Psi(x)_{d'}(g_*(f)) = \Psi(x)_{d'}(gf) = (F(gf))(x)$; across then down gives $(Fg)(\Psi(x)_d(f)) = (Fg)((Ff)(x))$. These agree because F is a functor, $F(gf) = Fg \circ Ff$, so

$$(F(gf))(x) = (Fg \circ Ff)(x) = (Fg)((Ff)(x))$$

The square commutes for every g, so $\Psi(x)$ is natural, and $\Psi \colon Fc \to \mathrm{Nat}(\mathrm{Hom}(c,-),F)$, $x \mapsto \Psi(x)$, is a well-defined candidate inverse to Φ.

Step 2: Φ and Ψ are mutually inverse. There are two composites to compute.

First, $\Phi\Psi = 1_{Fc}$. For $x \in Fc$,

$$\Phi(\Psi(x)) = \Psi(x)_c(1_c) = (F1_c)(x) = 1_{Fc}(x) = x$$

using that F preserves identities, $F1_c = 1_{Fc}$. This direction is immediate.

Second, $\Psi\Phi$ is the identity on $\mathrm{Nat}(\mathrm{Hom}(c,-),F)$. Let $\alpha \colon \mathrm{Hom}(c,-) \Rightarrow F$ be given and put $x = \Phi(\alpha) = \alpha_c(1_c)$. Two natural transformations agree exactly when all components agree, so it suffices to show $\Psi(x)_d = \alpha_d$ for every d. Fix d and $f \colon c \to d$. Naturality of α at f is the commuting square

$$\begin{tikzcd}
\mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "f_*"'] & Fc \arrow[d, "Ff"] \\
\mathrm{Hom}(c,d) \arrow[r, "\alpha_d"'] & Fd
\end{tikzcd}$$

Evaluate it at $1_c \in \mathrm{Hom}(c,c)$. Down then across gives $\alpha_d(f_*(1_c)) = \alpha_d(f1_c) = \alpha_d(f)$; across then down gives $(Ff)(\alpha_c(1_c)) = (Ff)(x)$. Equating the routes,

$$\alpha_d(f) = (Ff)(x) = \Psi(x)_d(f)$$

Since f and d were arbitrary, $\alpha_d = \Psi(x)_d$ for all d, so $\alpha = \Psi(\Phi(\alpha))$. Both composites are identities, so Φ is a bijection with inverse Ψ.

Step 3: naturality of the bijection. The source and target of Φ each depend on the pair (c,F), and the claim is that Φ commutes with change of F and change of c. Consider F first, with c

fixed. A natural transformation $\theta\colon F \Rightarrow G$ acts on the target by its component $\theta_c\colon Fc \to Gc$ and on the source by post-composition $\alpha \mapsto \theta\alpha$, carrying $\mathrm{Hom}(c,-) \Rightarrow F$ to $\mathrm{Hom}(c,-) \Rightarrow G$. Naturality in F is the square

$$\begin{tikzcd}
\mathrm{Nat}(\mathrm{Hom}(c,-),F) \arrow[r, "\Phi_F"] \arrow[d, "\theta\circ(-)"'] & Fc \arrow[d, "\theta_c"] \\
\mathrm{Nat}(\mathrm{Hom}(c,-),G) \arrow[r, "\Phi_G"'] & Gc
\end{tikzcd}$$

For $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$, down then across gives $\Phi_G(\theta\alpha) = (\theta\alpha)_c(1_c) = \theta_c(\alpha_c(1_c))$, since composition of natural transformations is componentwise (definition 1.5.4); across then down gives $\theta_c(\Phi_F(\alpha)) = \theta_c(\alpha_c(1_c))$. The two agree, so Φ is natural in F.

Now fix F and vary c. A morphism $h\colon c \to c'$ acts on the target by $Fh\colon Fc \to Fc'$. On the source it acts by precomposition with the representable's own functoriality: h induces $\mathrm{Hom}(h,-)\colon \mathrm{Hom}(c',-) \Rightarrow \mathrm{Hom}(c,-)$, whose component at d is $\mathrm{Hom}(h,d)\colon \mathrm{Hom}(c',d) \to \mathrm{Hom}(c,d)$, $k \mapsto kh$, and a transformation $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$ is carried to $\alpha\circ\mathrm{Hom}(h,-)\colon \mathrm{Hom}(c',-) \Rightarrow F$. Naturality in c is the square

$$\begin{tikzcd}
\mathrm{Nat}(\mathrm{Hom}(c,-),F) \arrow[r, "\Phi_c"] \arrow[d, "(-)\circ\mathrm{Hom}(h,-)"'] & Fc \arrow[d, "Fh"] \\
\mathrm{Nat}(\mathrm{Hom}(c',-),F) \arrow[r, "\Phi_{c'}"'] & Fc'
\end{tikzcd}$$

Take $\alpha\colon \mathrm{Hom}(c,-) \Rightarrow F$. Across then down gives $Fh(\Phi_c(\alpha)) = (Fh)(\alpha_c(1_c))$. Down then across gives

$$\Phi_{c'}(\alpha \circ \mathrm{Hom}(h,-)) = \big(\alpha \circ \mathrm{Hom}(h,-)\big)_{c'}(1_{c'}) = \alpha_{c'}\big(\mathrm{Hom}(h,c')(1_{c'})\big) = \alpha_{c'}(1_{c'}\,h) = \alpha_{c'}(h)$$

To match the two, apply naturality of α (a transformation $\mathrm{Hom}(c,-) \Rightarrow F$) at the morphism $h\colon c \to c'$: its square is

$$\begin{tikzcd}
\mathrm{Hom}(c,c) \arrow[r, "\alpha_c"] \arrow[d, "h_*"'] & Fc \arrow[d, "Fh"] \\
\mathrm{Hom}(c,c') \arrow[r, "\alpha_{c'}"'] & Fc'
\end{tikzcd}$$

Evaluated at $1_c \in \mathrm{Hom}(c,c)$, down then across gives $\alpha_{c'}(h_*(1_c)) = \alpha_{c'}(h1_c) = \alpha_{c'}(h)$; across then down gives $(Fh)(\alpha_c(1_c))$. Hence $\alpha_{c'}(h) = (Fh)(\alpha_c(1_c))$, which is exactly the equality of the two routes around the outer square. So Φ is natural in c, completing the proof.

Out of a lone element $x \in Fc$ the map Ψ manufactures an entire natural family $\{(Ff)(x)\}_f$, one function per object, coherent across every morphism. Nothing beyond the functoriality of F went into the construction, which is why the correspondence is as tight as it is: the data of a natural transformation, a priori a large amount of information spread over all objects, collapses to one point of a single set.

A first payoff sets $F = \mathrm{Hom}(c',-)$ for a second object c'. Then $Fc = \mathrm{Hom}(c',c)$, and the lemma reads $\mathrm{Nat}(\mathrm{Hom}(c,-),\mathrm{Hom}(c',-)) \cong \mathrm{Hom}(c',c)$: natural transformations between two representables are exactly the morphisms of C in the reverse direction. This is the computation on which the next section builds the Yoneda embedding.

Example 2.2.2: The Underlying Set as a Representable

Let $F\colon \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor, sending a group to its underlying set. It is represented by the free group $\mathbb{Z}$ on one generator: a homomorphism $\mathbb{Z} \to G$ is determined by, and freely choosable as, the image of the generator 1, so evaluation at 1 gives a bijection $\mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, G) \cong FG$ natural in G (definition 2.1.1). The Yoneda lemma names the element of $F\mathbb{Z} = \mathbb{Z}$ that encodes this: it is

$$\Phi(\alpha) = \alpha_{\mathbb{Z}}(1_{\mathbb{Z}}) \in F\mathbb{Z} = \mathbb{Z}$$

the image, under the natural isomorphism $\alpha\colon \mathrm{Hom}(\mathbb{Z}, -) \Rightarrow F$, of the identity homomorphism $1_{\mathbb{Z}}$. Tracing the definitions, $\alpha_{\mathbb{Z}}(1_{\mathbb{Z}})$ is the generator $1 \in \mathbb{Z}$, and the whole isomorphism α is recovered from it by $\alpha_G(\varphi) = (F\varphi)(1) = \varphi(1)$, the image of the generator under $\varphi\colon \mathbb{Z} \to G$. The single element $1 \in \mathbb{Z}$ carries the entire natural family.

The presheaf version is the same statement read in C^{op}. Recall the duality principle (box 1.3.3): a statement provable for every locally small category holds equally for every opposite. Applying the Yoneda lemma in C^{op} turns the covariant representable $\mathrm{Hom}_{C^{\mathrm{op}}}(c, -)$ into the contravariant $\mathrm{Hom}_C(-, c)$, and a covariant functor $C^{\mathrm{op}} \to \mathbf{Set}$ is a presheaf on C (definition 2.1.3). No new argument is needed.

Corollary 2.2.3: The Yoneda Lemma, Presheaf Form

Let C be a locally small category, c an object of C, and $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ a presheaf. Evaluation at the identity,

$$\mathrm{Nat}(\mathrm{Hom}(-, c), F) \to Fc, \qquad \alpha \mapsto \alpha_c(1_c)$$

is a bijection, natural in c and F.

***Proof*:**

The Yoneda lemma holds in every locally small category, and by the duality principle it applies in particular to C^{op}, which is again locally small (box 1.3.3). Applied there to the object c and the functor F, the covariant representable of C^{op} at c is $\mathrm{Hom}_{C^{\mathrm{op}}}(c, -) = \mathrm{Hom}_C(-, c)$ read in the free variable, and a functor $F\colon C^{\mathrm{op}} \to \mathbf{Set}$ is precisely a covariant functor on C^{op}. So theorem 2.2.1 yields the bijection $\mathrm{Nat}(\mathrm{Hom}_C(-, c), F) \cong Fc$, with the same evaluation-at-1_c map and the same naturality in c and F, as required.

The two forms are one theorem seen from two sides, and both will be used: the covariant form for functors out of C, the presheaf form for the embedding $c \mapsto \mathrm{Hom}(-, c)$ into presheaves.

The lemma also closes the loop with the universal element of section 2.1. An element $x \in Fc$ is called universal when every $y \in Fd$ arises as $(Ff)(x)$ for a unique $f\colon c \to d$ (definition 2.1.5), and F is representable exactly when such an x exists. Under the Yoneda bijection x corresponds to the transformation $\Psi(x)\colon \mathrm{Hom}(c, -) \Rightarrow F$ with $\Psi(x)_d(f) = (Ff)(x)$, and x is universal precisely when every $\Psi(x)_d$ is a bijection, that is, when $\Psi(x)$ is a natural isomorphism. So the representable-iff-universal-element statement is the Yoneda lemma specialized to a representable F: the universal element is the point of Fc that Yoneda hands back, and for $F = \mathrm{Hom}(c, -)$ itself it is the identity 1_c, the element from which the whole representable is generated.

2.3 The Yoneda Embedding

The Yoneda lemma of the previous section computes the natural transformations out of a representable functor. Read one way, it is a calculation; read another, it says that the assignment $c \mapsto \mathrm{Hom}(-, c)$ loses no information. This section makes that reading precise. Packaging the contravariant representables into a single functor $\mathbf{y}\colon C \to \mathbf{PSh}(C)$, the presheaf form of the lemma becomes the statement that $\mathbf{y}$ is fully faithful, and full faithfulness in turn says that an object is recovered, up to isomorphism, from the presheaf it represents.

Recall from definition 2.1.3 that $\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$ is the category of presheaves on C, whose objects are the contravariant set-valued functors and whose morphisms are natural transformations. Each object c of C gives such a presheaf, the represented $\mathrm{Hom}(-, c)$ of definition 2.1.1; each morphism of C gives a natural transformation between two of them. Assembling these into a functor is the content of the first definition.

Definition 2.3.1: The Yoneda Embedding

Let C be a locally small category (definition 1.1.7). The *Yoneda embedding* is the functor

$$\mathbf{y}\colon C \to \mathbf{PSh}(C)$$

defined as follows. On an object c, it is the represented presheaf

$$\mathbf{y}(c) = \mathrm{Hom}(-, c) \in \mathbf{PSh}(C)$$

On a morphism $f\colon c \to d$, it is the natural transformation

$$\mathbf{y}(f) = \mathrm{Hom}(-, f)\colon \mathrm{Hom}(-, c) \Rightarrow \mathrm{Hom}(-, d)$$

whose component at an object x is post-composition with f,

$$\mathrm{Hom}(-, f)_x\colon \mathrm{Hom}(x, c) \to \mathrm{Hom}(x, d), \qquad g \mapsto fg$$

That $\mathbf{y}$ is a functor is exactly the statement that its two clauses respect composition and identities, and both follow from the category axioms of C. The verification is short and worth doing once, because it exhibits how the unit and associativity laws of composition become the functoriality of $\mathbf{y}$.

***Proof*:**
Two checks are needed: that each $\mathbf{y}(f)$ is a natural transformation, and that $\mathbf{y}$ preserves composites and identities.

Step 1: $\mathbf{y}(f)$ is natural. Fix $f\colon c \to d$ and a morphism $h\colon x' \to x$ in C. The presheaf $\mathrm{Hom}(-, c)$ sends h to pre-composition $\mathrm{Hom}(h, c)\colon \mathrm{Hom}(x, c) \to \mathrm{Hom}(x', c)$, $g \mapsto gh$, and similarly for $\mathrm{Hom}(-, d)$. Naturality of $\mathbf{y}(f) = \mathrm{Hom}(-, f)$ at h is the commuting of

$$\begin{tikzcd}
\mathrm{Hom}(x, c) \arrow[r, "g \mapsto fg"] \arrow[d, "g \mapsto gh"'] & \mathrm{Hom}(x, d) \arrow[d, "g \mapsto gh"] \\
\mathrm{Hom}(x', c) \arrow[r, "g \mapsto fg"'] & \mathrm{Hom}(x', d)
\end{tikzcd}$$

Both routes send $g\colon x \to c$ to $f(gh)$: down-then-across gives $f(gh)$, across-then-down gives $(fg)h$, and these agree by associativity of composition in C. So $\mathbf{y}(f)$ is natural.

Step 2: $\mathbf{y}$ preserves composition. Let $f\colon c \to d$ and $f'\colon d \to e$. Both $\mathbf{y}(f'f)$ and $\mathbf{y}(f')\mathbf{y}(f)$ are natural transformations $\mathrm{Hom}(-,c) \Rightarrow \mathrm{Hom}(-,e)$, so it suffices to compare their components at each object x. On $g\colon x \to c$,

$$\mathbf{y}(f'f)_x(g) = (f'f)g, \qquad \big(\mathbf{y}(f')\mathbf{y}(f)\big)_x(g) = f'(fg)$$

and $(f'f)g = f'(fg)$ by associativity. Hence $\mathbf{y}(f'f) = \mathbf{y}(f')\mathbf{y}(f)$.

Step 3: $\mathbf{y}$ preserves identities. The component of $\mathbf{y}(1_c)$ at x sends $g\colon x \to c$ to $1_c g = g$, so it is the identity function on $\mathrm{Hom}(x,c)$. As this holds for every x, $\mathbf{y}(1_c) = 1_{\mathrm{Hom}(-,c)}$, the identity natural transformation on $\mathbf{y}(c)$.

Naturality of each $\mathbf{y}(f)$, preservation of composites, and preservation of identities together make $\mathbf{y}$ a functor, completing the proof.

The proof isolates a small dictionary: associativity of composition in C becomes both the naturality of each $\mathbf{y}(f)$ and the functoriality of $\mathbf{y}$, and the left unit law becomes the preservation of identities. The embedding carries an object to the record of all maps into it and a morphism to post-composition with that morphism, and this record transforms functorially because composition does. A concrete category makes the abstract assignment tangible.

Example 2.3.2: The Yoneda Embedding of a Poset

Let P be a poset, regarded as a category with a single morphism $x \to y$ whenever $x \le y$ and none otherwise (definition 1.1.1). For objects x, y the hom-set $\mathrm{Hom}(x,y)$ is a one-element set when $x \le y$ and empty otherwise, so a presheaf $\mathrm{Hom}(-,c) = \mathbf{y}(c)$ on P is the indicator of the down-set of c: for each object x,

$$\mathrm{Hom}(x,c) = \begin{cases} \{*\} & x \le c \\ \varnothing & x \not\le c \end{cases}$$

Thus $\mathbf{y}(c)$ records precisely which elements lie below c, that is, the principal down-set ${\downarrow}c = \{x \in P \,|\, x \le c\}$. A morphism $c \le d$ in P gives the natural transformation $\mathbf{y}(c \le d)\colon \mathrm{Hom}(-,c) \Rightarrow \mathrm{Hom}(-,d)$, which at each x is the unique function from $\mathrm{Hom}(x,c)$ to $\mathrm{Hom}(x,d)$; such a function exists as a morphism of presheaves exactly because $x \le c \le d$ forces $x \le d$, so the down-set of c is contained in that of d. The embedding $\mathbf{y}\colon P \to \mathbf{PSh}(P)$ is therefore the assignment $c \mapsto {\downarrow}c$ sending each element to its principal down-set, ordered by inclusion. Since $c \le d$ if and only if ${\downarrow}c \subseteq {\downarrow}d$, distinct elements have distinct down-sets and the order is faithfully reproduced: $\mathbf{y}$ is the poset embedding of P into its lattice of down-sets. This is the order-theoretic case of the full faithfulness proved next.

The example already displays the pattern of the general theorem. The embedding of a poset is injective on objects and reflects the order, so the whole of P can be read off from its image in $\mathbf{PSh}(P)$. For a general category the statement is not that $\mathbf{y}$ is injective on objects, which is too rigid, but that it is a bijection on morphisms between any two objects, full and faithful in the sense recalled from proposition 1.5.6. This is where the Yoneda lemma does the work: its presheaf form, applied to a represented presheaf, is precisely the assertion that $\mathbf{y}$ is fully faithful.

Theorem 2.3.3: The Yoneda Embedding is Fully Faithful

Let C be a locally small category. For all objects c, d of C the function

$$\mathrm{Hom}_C(c,d) \to \mathrm{Hom}_{\mathbf{PSh}(C)}\big(\mathbf{y}(c), \mathbf{y}(d)\big) = \mathrm{Nat}\big(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)\big), \qquad f \mapsto \mathbf{y}(f)$$

is a bijection. Equivalently, the Yoneda embedding $\mathbf{y}\colon C \to \mathbf{PSh}(C)$ is full and faithful.

***Proof*:**
Apply the presheaf form of the Yoneda lemma, corollary 2.2.3, to the presheaf $F = \mathrm{Hom}(-,d)$ and the object c. It supplies a bijection

$$\mathrm{Nat}\big(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)\big) \xrightarrow{\cong} \big(\mathrm{Hom}(-,d)\big)(c) = \mathrm{Hom}_C(c,d) \tag{2.2}$$

natural in c and d. The bijection of (2.2) sends a natural transformation $\alpha\colon \mathrm{Hom}(-,c) \Rightarrow \mathrm{Hom}(-,d)$ to the element $\alpha_c(1_c) \in \mathrm{Hom}_C(c,d)$, the image of the identity 1_c under the component of α at c. The claimed map $f \mapsto \mathbf{y}(f)$ runs the other way, so it suffices to check that the two are mutually inverse. In fact $f \mapsto \mathbf{y}(f)$ is exactly the inverse produced by corollary 2.2.3, and identifying it is the whole content of the theorem.

Step 1: the Yoneda bijection sends $\mathbf{y}(f)$ to f. Take $\alpha = \mathbf{y}(f) = \mathrm{Hom}(-,f)$ for a morphism $f\colon c \to d$. Its component at c is post-composition with f, so it sends the identity to

$$\big(\mathbf{y}(f)\big)_c(1_c) = f \circ 1_c = f$$

Thus the composite $\mathrm{Hom}_C(c,d) \to \mathrm{Nat}(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d)) \to \mathrm{Hom}_C(c,d)$, first $f \mapsto \mathbf{y}(f)$ and then the bijection of (2.2), is the identity on $\mathrm{Hom}_C(c,d)$.

Step 2: the other composite is the identity. The map of (2.2) is already a bijection by corollary 2.2.3, and Step 1 shows $f \mapsto \mathbf{y}(f)$ is a right inverse to it. A right inverse to a bijection is its two-sided inverse, so $f \mapsto \mathbf{y}(f)$ is itself a bijection, inverse to the Yoneda bijection. Concretely, starting from a natural transformation α, its image $f = \alpha_c(1_c)$ satisfies $\mathbf{y}(f) = \alpha$: for any object x and any $g\colon x \to c$, naturality of α at g applied to 1_c gives

$$\alpha_x(g) = \alpha_x\big(\mathrm{Hom}(g,c)(1_c)\big) = \mathrm{Hom}(g,d)\big(\alpha_c(1_c)\big) = \alpha_c(1_c) \circ g = fg = \big(\mathbf{y}(f)\big)_x(g)$$

so $\alpha = \mathbf{y}(f)$, recovering the second half of the inverse relation.

The two composites being identities, $f \mapsto \mathbf{y}(f)$ is a bijection $\mathrm{Hom}_C(c,d) \xrightarrow{\cong} \mathrm{Nat}(\mathrm{Hom}(-,c), \mathrm{Hom}(-,d))$ for every pair c, d. This is exactly the statement that $\mathbf{y}$ is full (the map is surjective, so every natural transformation between representables is $\mathbf{y}(f)$ for some f) and faithful (the map is injective, so $\mathbf{y}(f)$ determines f), as we sought to show.

Full faithfulness is the sharpest possible statement that $\mathbf{y}$ loses no information at the level of morphisms. Every natural transformation between the presheaves $\mathrm{Hom}(-,c)$ and $\mathrm{Hom}(-,d)$ is post-composition with a unique morphism $c \to d$: there are no natural transformations between representables beyond the ones the category already supplies. The functor $\mathbf{y}$ therefore identifies the hom-set $\mathrm{Hom}_C(c,d)$ with the hom-set $\mathrm{Hom}_{\mathbf{PSh}(C)}(\mathbf{y}(c), \mathbf{y}(d))$ without collapsing or inventing morphisms, so C sits inside $\mathbf{PSh}(C)$ as a full subcategory (up to the identification of objects with the presheaves they represent). The step from this to objects is one general fact about fully faithful functors: they detect isomorphisms as well as reflect the morphisms that witness them.

Before stating the corollary, it is worth pinning the one property of fully faithful functors it rests on. A functor always preserves isomorphisms (lemma 1.4.5): if f is invertible, so is Ff. A fully faithful functor also *reflects* them, and this is the new content. Suppose $Ff\colon Fc \to Fd$ is an isomorphism with inverse $k\colon Fd \to Fc$. Because F is full, $k = Fg$ for some $g\colon d \to c$; because F is faithful, the equations $Fg \circ Ff = 1_{Fc} = F(1_c)$ and $Ff \circ Fg = 1_{Fd} = F(1_d)$, which say $F(gf) = F(1_c)$ and $F(fg) = F(1_d)$, force $gf = 1_c$ and $fg = 1_d$. So f is an isomorphism (definition 1.2.1) with inverse g. A morphism whose image under a fully faithful functor is invertible is therefore already invertible.

Corollary 2.3.4: Objects are Determined by their Points

Let C be a locally small category. Two objects c, d of C are isomorphic in C if and only if the represented presheaves $\mathbf{y}(c) = \mathrm{Hom}(-, c)$ and $\mathbf{y}(d) = \mathrm{Hom}(-, d)$ are isomorphic in $\mathbf{PSh}(C)$. An object of C is determined, up to isomorphism, by the presheaf it represents, its *functor of points*.

***Proof*:**
The Yoneda embedding $\mathbf{y}$ is full and faithful by theorem 2.3.3. If $c \cong d$ in C, then $\mathbf{y}$ preserves the isomorphism (lemma 1.4.5), giving $\mathbf{y}(c) \cong \mathbf{y}(d)$ in $\mathbf{PSh}(C)$. Conversely, suppose $\varphi\colon \mathbf{y}(c) \to \mathbf{y}(d)$ is an isomorphism of presheaves. By fullness $\varphi = \mathbf{y}(f)$ for some $f\colon c \to d$, and $\mathbf{y}(f)$ is an isomorphism, so by the reflection of isomorphisms established above f is itself an isomorphism. Hence $c \cong d$ in C, as we sought to show.

This is the precise form of the chapter's thread: an object is known, up to isomorphism, by how everything maps into it. The functor of points $\mathrm{Hom}(-, c)$ carries the full isomorphism type of c, so two objects with naturally isomorphic points are the same object seen twice.

2.4 Applications

The embedding of section 2.3 converts three recurring questions into questions about hom-functors, where they become easy. This section collects the payoffs. First, a representing object is pinned down not just up to isomorphism but up to a *unique* isomorphism, so "the" representing object is a legitimate phrase. Second, the deferred converse from Chapter 1 falls out: a full, faithful, essentially surjective functor is an equivalence (proposition 1.5.6). Third, a concrete instance of corollary 2.3.4: an object of a category is recovered, up to isomorphism, from its functor of points.

A representing object for a set-valued functor F is any object c together with a natural isomorphism $F \cong \mathrm{Hom}(c, -)$. Two such objects are isomorphic by the general principle that represented functors determine their representing object (corollary 2.3.4), but more is true: the isomorphism between them is forced, with no freedom of choice. This is what licenses treating the representing object as canonical data attached to F.

Corollary 2.4.1: Uniqueness of Representing Objects

Let C be a locally small category and $F\colon C \to \mathbf{Set}$ a functor. If $\varphi\colon F \Rightarrow \mathrm{Hom}(c,-)$ and $\varphi'\colon F \Rightarrow \mathrm{Hom}(c',-)$ are natural isomorphisms, then there is a unique isomorphism $u\colon c' \to c$ in C for which the induced natural isomorphism $\mathrm{Hom}(u,-)\colon \mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$ equals $\varphi'\varphi^{-1}$. In particular the representing object of F, when it exists, is unique up to a unique isomorphism.

***Proof*:**
Write ψ for the composite $\varphi'\varphi^{-1}\colon \mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$, a natural isomorphism, being a composite of natural isomorphisms.

The covariant hom-functors are the objects in the image of the covariant embedding $c \mapsto \mathrm{Hom}(c,-)$, which sends C^{op} fully faithfully into $[C,\mathbf{Set}]$. This is the covariant analog of theorem 2.3.3 and is read off the Yoneda lemma directly: applying theorem 2.2.1 to the functor $F = \mathrm{Hom}(c',-)$ gives a bijection

$$\mathrm{Nat}\big(\mathrm{Hom}(c,-), \mathrm{Hom}(c',-)\big) \;\cong\; \mathrm{Hom}(c',-)(c) \;=\; \mathrm{Hom}(c',c)$$

natural in both variables. Tracing the Yoneda bijection, a morphism $v\colon c' \to c$ corresponds to the natural transformation $\mathrm{Hom}(v,-)$ whose component at x is $g \mapsto gv$; and every natural transformation $\mathrm{Hom}(c,-) \Rightarrow \mathrm{Hom}(c',-)$ arises this way from a unique such v. Applying this to ψ yields a unique $u\colon c' \to c$ with $\mathrm{Hom}(u,-) = \psi = \varphi'\varphi^{-1}$.

It remains to check u is an isomorphism. Applying the same bijection to $\psi^{-1}\colon \mathrm{Hom}(c',-) \Rightarrow \mathrm{Hom}(c,-)$ produces a unique $w\colon c \to c'$ with $\mathrm{Hom}(w,-) = \psi^{-1}$. Since $c \mapsto \mathrm{Hom}(c,-)$ is a functor $C^{\mathrm{op}} \to [C,\mathbf{Set}]$, precomposition is sent to composition of natural transformations, so $\mathrm{Hom}(uw,-) = \mathrm{Hom}(w,-)\mathrm{Hom}(u,-) = \psi^{-1}\psi = \mathrm{id}$ and likewise $\mathrm{Hom}(wu,-) = \mathrm{id}$. Faithfulness of the embedding (again theorem 2.2.1, whose bijection is injective) forces $uw = \mathrm{id}_c$ and $wu = \mathrm{id}_{c'}$, so u is an isomorphism with inverse w. Uniqueness of u is the uniqueness already supplied by the Yoneda bijection, completing the proof.

The statement isolates the mechanism behind the familiar slogan that universal objects are unique "up to canonical isomorphism". The isomorphism u is not merely one of possibly several: it is singled out by the requirement that it intertwine the two representations φ, φ', and full faithfulness of the embedding says there is exactly one morphism doing so. Every construction recognized by a universal property in the chapters to come (products, kernels, tensor products) inherits this uniqueness for free, which is why one speaks of *the* product rather than a product.

The second application discharges a debt from Chapter 1. There the notion of an equivalence of categories was introduced (definition 1.5.5), and proposition 1.5.6 showed that an equivalence is necessarily full, faithful, and essentially surjective. The converse was deferred, because constructing the quasi-inverse requires choosing, for each object of the target, an object of the source mapping to it, and then transporting morphisms along those choices coherently. The bookkeeping is exactly the naturality that the Yoneda circle of ideas trains one to handle, so this is the natural place to settle it.

Proposition 2.4.2: Full, Faithful, Essentially Surjective Implies Equivalence

Let $F\colon C \to D$ be a functor that is full, faithful, and essentially surjective. Then F is an equivalence of categories: there exist a functor $G\colon D \to C$ and natural isomorphisms $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ and $\eta\colon \mathrm{id}_C \Rightarrow GF$.

***Proof*:**
Step 1: Define G on objects. For each object d of D, essential surjectivity supplies an object of C mapping isomorphically to d. Choose one such object, call it Gd, together with an isomorphism $\varepsilon_d\colon F(Gd) \to d$ in D. This is one simultaneous selection over all objects of D, the only appeal to choice in the argument; it is harmless, and when D is small it is a choice from a set.

Step 2: Define G on morphisms. Given $g\colon d \to d'$ in D, consider the morphism

$$\varepsilon_{d'}^{-1}\, g\, \varepsilon_d\colon F(Gd) \to F(Gd')$$

Because F is full and faithful, the map $\mathrm{Hom}_C(Gd, Gd') \to \mathrm{Hom}_D(F(Gd), F(Gd'))$ induced by F is a bijection. Hence there is a unique morphism $Gg\colon Gd \to Gd'$ in C with

$$F(Gg) \;=\; \varepsilon_{d'}^{-1}\, g\, \varepsilon_d \tag{2.3}$$

Existence gives G a value; uniqueness is what makes the value well-defined and, as we now see, functorial.

Step 3: G is a functor. For the identity id_d, the morphism $\varepsilon_d^{-1}\,\mathrm{id}_d\,\varepsilon_d = \mathrm{id}_{F(Gd)} = F(\mathrm{id}_{Gd})$, so by the uniqueness in (2.3) the unique preimage is id_{Gd}, giving $G(\mathrm{id}_d) = \mathrm{id}_{Gd}$. For composability $d \xrightarrow{g} d' \xrightarrow{g'} d''$, faithfulness lets us compare $G(g'g)$ and $(Gg')(Gg)$ by comparing their images under F. Applying F and using (2.3) termwise,

$$F\big((Gg')(Gg)\big) = F(Gg')\,F(Gg) = \big(\varepsilon_{d''}^{-1}\, g'\, \varepsilon_{d'}\big)\big(\varepsilon_{d'}^{-1}\, g\, \varepsilon_d\big) = \varepsilon_{d''}^{-1}\, g' g\, \varepsilon_d = F\big(G(g'g)\big)$$

where the middle equality cancels $\varepsilon_{d'}\varepsilon_{d'}^{-1} = \mathrm{id}$. Faithfulness then gives $(Gg')(Gg) = G(g'g)$, so G is a functor.

Step 4: $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ is a natural isomorphism. Each component $\varepsilon_d\colon F(Gd) \to d$ is an isomorphism by construction, so it suffices to check naturality. For $g\colon d \to d'$, rewriting (2.3) as $g\,\varepsilon_d = \varepsilon_{d'}\,F(Gg)$ is exactly the statement that the square

$$\begin{tikzcd}
F(Gd) \arrow[r, "\varepsilon_d"] \arrow[d, "F(Gg)"'] & d \arrow[d, "g"] \\
F(Gd') \arrow[r, "\varepsilon_{d'}"'] & d'
\end{tikzcd}$$

commutes. Thus ε is natural, and being componentwise an isomorphism it is a natural isomorphism.

Step 5: Construct $\eta\colon \mathrm{id}_C \Rightarrow GF$. For an object c of C, the object $GF(c)$ comes with the chosen isomorphism $\varepsilon_{Fc}\colon F(GFc) \to Fc$. Since F is full and faithful, the isomorphism $\varepsilon_{Fc}^{-1}\colon Fc \to F(GFc)$ has a unique preimage $\eta_c\colon c \to GF(c)$ with

$$F(\eta_c) \;=\; \varepsilon_{Fc}^{-1} \tag{2.4}$$

Each η_c is an isomorphism: its inverse is the preimage of ε_{Fc} under the same bijection, and faithfulness turns the identities $F(\eta_c)\,\varepsilon_{Fc} = \mathrm{id}$ and vice versa into identities in C, as in the argument of corollary 2.4.1. For naturality, let $f\colon c \to c'$. Applying F to both candidate composites and using (2.4) together with (2.3) for the morphism Ff,

$$F\big(GF(f)\big)\,F(\eta_c) = \big(\varepsilon_{Fc'}^{-1}\,(Ff)\,\varepsilon_{Fc}\big)\,\varepsilon_{Fc}^{-1} = \varepsilon_{Fc'}^{-1}\,(Ff) = F(\eta_{c'})\,F(f)$$

so $F\big(GF(f)\,\eta_c\big) = F\big(\eta_{c'}\,f\big)$, and faithfulness gives $GF(f)\,\eta_c = \eta_{c'}\,f$. Hence η is a natural isomorphism $\mathrm{id}_C \Rightarrow GF$.

With G a functor and ε, η natural isomorphisms, F and G are quasi-inverse, so F is an equivalence of categories, as we sought to show.

Together with proposition 1.5.6 this completes the characterization: a functor is an equivalence if and only if it is full, faithful, and essentially surjective. The "if" direction just proved is the useful one in practice, since it lets one certify an equivalence without ever exhibiting the inverse functor. One checks three local conditions, two of them (fullness, faithfulness) about hom-sets and one (essential surjectivity) about the image on objects, and the quasi-inverse is then guaranteed to exist. The proof also shows the quasi-inverse is far from unique as bare data, since it depends on the choices in Step 1, yet any two such choices yield equivalent functors, matching the uniqueness pattern of corollary 2.4.1.

The last application makes corollary 2.3.4 tangible. That corollary says an object is determined up to isomorphism by the functor it represents, its functor of points; two objects with naturally isomorphic points are isomorphic. On a category of algebraic objects this recovers the object from the ways of mapping it to arbitrary test objects, which is the content of the "functor of points" viewpoint that pervades algebraic geometry.

Example 2.4.3: Recovering a Ring from Its Points

Work in $C = \mathbf{Ring}$ (commutative rings with unit). Each ring R represents the covariant functor of points $\mathrm{Hom}(R,-)\colon \mathbf{Ring} \to \mathbf{Set}$, sending a ring A to the set $\mathrm{Hom}(R,A)$ of ring homomorphisms out of R. Suppose R and S are rings and

$$\Phi\colon \mathrm{Hom}(R,-) \Rightarrow \mathrm{Hom}(S,-)$$

is a natural isomorphism of these functors. Then $R \cong S$ as rings, and the isomorphism is canonical.

The point is the covariant form of corollary 2.3.4: represented functors determine their representing object, and here the representation is covariant, so the roles are dual to the presheaf statement. Concretely, apply corollary 2.4.1 with $F = \mathrm{Hom}(R,-)$. The two natural isomorphisms $\mathrm{id}\colon F \Rightarrow \mathrm{Hom}(R,-)$ and $\Phi\colon F \Rightarrow \mathrm{Hom}(S,-)$ represent the same F, so there is a unique isomorphism $u\colon S \to R$ with $\mathrm{Hom}(u,-) = \Phi$.

The isomorphism u can be named by hand, which shows the abstraction costs nothing. Chase the identity through the components. At $A = R$ the map $\Phi_R\colon \mathrm{Hom}(R,R) \to \mathrm{Hom}(S,R)$ carries id_R to a homomorphism $u\colon S \to R$, that is, set $u = \Phi_R(\mathrm{id}_R)$. For any A and any $f\colon R \to A$, naturality of Φ along f gives the square

$$\begin{tikzcd}
\mathrm{Hom}(R,R) \arrow[r, "\Phi_R"] \arrow[d, "f\circ(-)"'] & \mathrm{Hom}(S,R) \arrow[d, "f\circ(-)"] \\
\mathrm{Hom}(R,A) \arrow[r, "\Phi_A"'] & \mathrm{Hom}(S,A)
\end{tikzcd}$$

Evaluating both routes at id_R gives $\Phi_A(f) = fu$, so Φ_A is precomposition with u, that is, $\Phi = \mathrm{Hom}(u, -)$. Running the same chase on Φ^{-1} produces $v = (\Phi^{-1})_S(\mathrm{id}_S)\colon R \to S$ with $\Phi^{-1} = \mathrm{Hom}(v, -)$. Then $\mathrm{Hom}(uv, -) = \mathrm{Hom}(v, -)\mathrm{Hom}(u, -) = \Phi^{-1}\Phi = \mathrm{id}$ forces $uv = \mathrm{id}_R$, and symmetrically $vu = \mathrm{id}_S$, so $u\colon S \to R$ is an isomorphism of rings with inverse v. The functor of points has recovered the ring.

The computation is the whole Yoneda lemma in miniature: the natural transformation Φ is completely determined by its value on a single element, the identity id_R, and that value is the morphism representing it. The same argument works verbatim in any locally small category, with **Ring** replaced by **Grp**, **Top**, or the category of schemes, and it is the reason the functor of points is a faithful record of the object rather than a lossy one.

3

Universal Properties, Limits, and Colimits

Chapters 1 and 2 built the categorical language and its central theorem; this chapter turns to construction. Products, quotients, kernels, unions, gluings: most objects mathematics builds are pinned down not by their elements but by a mapping property. This chapter isolates that pattern as a *universal property*, organizes the constructions it governs into *limits* and *colimits*, and computes them in the categories at hand.

3.1 Universal Properties and Universal Arrows

An object is often pinned down not by naming its elements but by a rule describing how it maps to, or receives maps from, every other object. The empty set is the set with a unique map to every set; the free group on a set X is the group into which every function from X extends uniquely. This section isolates the two building blocks of such descriptions. The first is the pair of extremal objects, *initial* and *terminal*, which sit at the ends of a category. The second, the *universal arrow*, relativizes them along a functor and turns out to be the general form of the universal element from definition 2.1.5.

Definition 3.1.1: Initial and Terminal Objects

Let C be a category. An object 0 of C is *initial* if for every object X there is exactly one morphism $0 \to X$. An object 1 of C is *terminal* if for every object X there is exactly one morphism $X \to 1$. An object that is both initial and terminal is a *zero object*.

The two notions are dual: a terminal object of C is an initial object of C^{op}, so every statement about one specializes to a statement about the other by box 1.3.3. Neither need exist, and when one does it carries no further data beyond the bare mapping property; that austerity is what makes it useful.

Example 3.1.2: Initial and Terminal Objects in the Standing Cast
The following run through the standing cast.

1. In **Set** the empty set $\varnothing$ is initial: the empty function is the unique map $\varnothing \to X$ for every X. Every singleton $\{*\}$ is terminal, since a map $X \to \{*\}$ has no choice in where to send its elements. There is no zero object: $\varnothing$ is not terminal (no map $\{*\} \to \varnothing$ exists).

2. In **Grp** and in $\mathbf{Mod}_R$ the trivial object (the group $\{e\}$, the zero module 0) is both initial and terminal, hence a zero object. Every homomorphism to or from it is forced, since a homomorphism preserves the identity; this is the algebraic fact recorded in [4] (trivial group, zero module).

3. In **Ring** (unital rings, $1 \neq 0$ not required) the ring $\mathbb{Z}$ is initial: for any ring R the unique homomorphism $\mathbb{Z} \to R$ sends n to $n \cdot 1_R$, forced because a ring map preserves 1. The zero ring, in which $1 = 0$, is terminal. Here the two extremes are distinct, so **Ring** has no zero object.

4. A poset $(P, \leq)$, viewed as a category with a unique arrow $x \to y$ when $x \leq y$, has an initial object exactly when P has a least element and a terminal object exactly when it has a greatest element.

The **Ring** example already exhibits the pattern the rest of the chapter formalizes: the map $\mathbb{Z} \to R$ is not just *some* distinguished morphism, it is the *unique* one, and that uniqueness is the whole content. A universal property is always a uniqueness statement of this kind.

The defining clause of an initial object, "exactly one morphism", has an immediate consequence: an initial object is determined by the property, not only up to isomorphism but up to a *unique* isomorphism. This is the prototype every universal-property uniqueness argument imitates, so it is worth seeing once in full.

Proposition 3.1.3: Initial and Terminal Objects are Unique up to Unique Isomorphism

If 0 and $0'$ are both initial objects of a category C, then there is a unique isomorphism $0 \to 0'$. Dually, any two terminal objects are related by a unique isomorphism.

***Proof*:**
Since 0 is initial, there is a unique morphism $u\colon 0 \to 0'$; since $0'$ is initial, there is a unique morphism $v\colon 0' \to 0$. The composite $vu\colon 0 \to 0$ is a morphism from 0 to itself, and 0 is initial, so there is only one such morphism. The identity id_0 is one, hence $vu = \mathrm{id}_0$. The same argument in C with the roles of 0 and $0'$ exchanged gives $uv = \mathrm{id}_{0'}$. Thus u is an isomorphism with inverse v, and it is the only morphism $0 \to 0'$ by initiality of 0, so it is the unique isomorphism. The terminal statement is the initial statement read in C^{op}, completing the proof.

The argument uses nothing about 0 except that maps out of it are forced. Every "unique up to unique isomorphism" claim for a universal construction, for products, kernels, free objects, limits, runs the same way: two solutions each map uniquely to the other, and the round trips are identities because they compete with id for a slot the property allows only one morphism to fill. The reader may reduce any such uniqueness proof to proposition 3.1.3 by exhibiting the construction as an initial or terminal object of a suitable category, which is precisely the reduction the next two definitions make available.

Initial and terminal objects describe an object by its maps to or from *everything*. Most constructions of interest carry data along a functor: the free group on X is built from the set X through the forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$. The universal arrow relativizes the extremal notion to this setting.

Definition 3.1.4: Universal Arrow

Let $U\colon C \to D$ be a functor and d an object of D. A *universal arrow from d to U* is a pair (c, η) consisting of an object c of C and a morphism $\eta\colon d \to Uc$ in D with the following property: for every object x of C and every morphism $f\colon d \to Ux$ in D, there is a unique morphism $g\colon c \to x$ in C such that $f = (Ug) \circ \eta$. The defining equation is the commuting triangle

$$\begin{tikzcd} d \arrow[r, "\eta"] \arrow[dr, "f"'] & Uc \arrow[d, "Ug"] \\ & Ux \end{tikzcd}$$

Dually, a *universal arrow from U to d* is a pair (c, ε) with $\varepsilon\colon Uc \to d$ such that every $f\colon Ux \to d$ factors as $f = \varepsilon \circ (Ug)$ for a unique $g\colon x \to c$.

The pair (c, η) is universal in that η is the most efficient map from d into the image of U: any other map f from d into U is recovered from η by post-composing with a C-morphism, and in exactly one way. Setting $U = \mathrm{id}_D$ recovers the extremal case, an initial object of the coslice under d; the functor U is what makes the notion carry structure across categories.

This connects directly to the representability material of the previous chapter. A universal element of a functor $F\colon C \to \mathbf{Set}$ (definition 2.1.5) is an element $u \in Fc$ such that every $x \in Fa$ equals $Fg(u)$ for a unique $g\colon c \to a$; that is the special case of definition 3.1.4 in which $D = \mathbf{Set}$, the object d is a singleton $\{*\}$, and a map $\{*\} \to Fc$ is named by the element it picks out. Both say the same thing: the datum η (or u) generates every map into U (or every element of F) freely and uniquely. Equivalently, a universal arrow from d to U makes the functor $D(d, U-)$ representable (definition 2.1.1), represented by c with the element η corresponding under Yoneda; the connection to preservation of limits is taken up later in the chapter.

Example 3.1.5: The Universal Arrow of the Free Group

Take $U\colon \mathbf{Grp} \to \mathbf{Set}$ the forgetful functor and X a set. The free group FX carries the inclusion of generators $\eta\colon X \to UFX$. Given any group H and any function $f\colon X \to UH$, the universal property of the free group ([4]) supplies a unique homomorphism $g\colon FX \to H$ extending f on generators, so $f = (Ug) \circ \eta$. Thus (FX, η) is a universal arrow from X to U: the free group is exactly the solution to this universal problem, with η marking the generators.

The example is the template for "free" constructions in general, and the next definition packages the search for a universal arrow into a single object of a single category, so that finding one becomes finding an initial object.

Definition 3.1.6: Comma Category

Let $F\colon A \to C$ and $G\colon B \to C$ be functors into a common category C. The *comma category* $F \downarrow G$ has as objects the triples (a, b, h) with a an object of A, b an object of B, and $h\colon Fa \to Gb$ a morphism of C. A morphism $(a, b, h) \to (a', b', h')$ is a pair $(p\colon a \to a',\ q\colon b \to b')$ making the square

$$\begin{tikzcd} Fa \arrow[r, "h"] \arrow[d, "Fp"'] & Gb \arrow[d, "Gq"] \\ Fa' \arrow[r, "h'"'] & Gb' \end{tikzcd}$$

commute; composition and identities are computed componentwise in A and B.

Two special cases recur. When $B = C$ and $G = \mathrm{id}_C$, an object is a map $Fa \to c$ with c fixed, giving the *slice category* C/c of objects over c; dually $F = \mathrm{id}_C$ and a fixed c give the *coslice* c/C of objects under c. The unifying point is a single reformulation of definition 3.1.4.

Proposition 3.1.7: Universal Arrows are Initial Objects of a Comma Category

Let $U\colon C \to D$ be a functor and d an object of D, viewed as a functor $d\colon \mathbf{1} \to D$ from the one-object one-morphism category. A universal arrow from d to U is the same thing as an initial object of the comma category $d \downarrow U$.

***Proof*:**
An object of $d \downarrow U$ is a pair $(c, \eta\colon d \to Uc)$ with c an object of C, and a morphism $(c, \eta) \to (x, f)$ is a C-morphism $g\colon c \to x$ with $(Ug) \circ \eta = f$. To say (c, η) is initial is to say that for every (x, f) there is exactly one such g, which is verbatim the defining property of a universal arrow from d to U in definition 3.1.4, as we sought to show.

proposition 3.1.7 discharges the uniqueness of universal arrows into proposition 3.1.3: a universal arrow, being an initial object, is determined up to a unique isomorphism, so the free group, the initial ring, and every other solution of a universal problem is unique in the strongest sense available. The comma category is the "neighbouring category" whose initial object is the construction one is after, and the sections that follow build limits and colimits by finding exactly such extremal objects.

Example 3.1.8: Universal Arrows Across the Standing Cast

Each construction below is a universal arrow, read off from its factorization property.

1. *The integers as initial ring.* Let $U\colon \mathbf{Ring} \to \mathbf{Set}$ be the forgetful functor and $\{*\}$ a one-point set. A universal arrow from $\{*\}$ to U is a ring c together with a chosen element $\eta(*) \in Uc$ such that every pointed ring (R, r), with $r = f(*) \in UR$, receives a unique ring map $g\colon c \to R$ sending $\eta(*)$ to r. Take $c = \mathbb{Z}[x]$ with $\eta(*) = x$: a ring map $\mathbb{Z}[x] \to R$ is determined by, and freely specifies, the image of x ([4], universal property of the polynomial ring), so $(\mathbb{Z}[x], x)$ is the universal arrow. Forgetting the marked point instead, and taking U with no chosen element, the empty universal problem is solved by $\mathbb{Z}$ itself: $\mathbb{Z}$ is the initial object of $\mathbf{Ring}$ (example 3.1.2), the universal arrow from the empty set to U.
2. *Abelianization.* Let $U\colon \mathbf{Ab} \to \mathbf{Grp}$ be the inclusion of abelian groups into groups and let G

be a group. The abelianization $G^{\mathrm{ab}} = G/[G,G]$ carries the quotient map $\eta\colon G \to U(G^{\mathrm{ab}})$. Every homomorphism $f\colon G \to UA$ into an abelian group A kills commutators, so it factors as $f = (Ug) \circ \eta$ through a unique homomorphism $g\colon G^{\mathrm{ab}} \to A$ ([4]). Thus (G^{ab}, η) is a universal arrow from G to U, the best abelian approximation of G from the left.

The two items sit on opposite sides of a forgetful (or inclusion) functor. The polynomial ring is universal *from* a set *to* U, a free construction; abelianization is likewise a universal arrow from an object to an inclusion, a reflection into a subcategory. Both are computed the same way, by recognizing that a map out of the constructed object is forced by its effect on the universal datum η. The organizing principle behind such matched pairs of universal arrows, one for each object, is the subject of the chapter on adjunctions.

3.2 Limits

An initial object receives a unique map from nothing, a terminal object receives one from everything; both are universal against the empty diagram. A limit asks for more: a universal way to map compatibly into a whole configuration of objects and morphisms at once. Products, intersections, kernels, and fibered products are all this one requirement read against different shapes. This section isolates the shape as a small category J, packages a configuration as a functor out of it, and defines the limit as the terminal object among the compatible maps into it. The atomic cases, products and equalizers, are then shown to generate every limit.

We first fix what "a configuration" means. A configuration of objects and morphisms in $\mathbf{C}$ is exactly a functor into $\mathbf{C}$ whose domain records the shape; the domain is small so that the configuration has only a set's worth of pieces.

Definition 3.2.1: Diagram, Cone, and Limit

Let $\mathbf{C}$ be a category and J a small category, the *index category*. A *diagram* of shape J in $\mathbf{C}$ is a functor $D\colon J \to \mathbf{C}$ (definition 1.4.1); the object $D(j)$ is written Dj.
A *cone* over D with *apex* X is an object X of $\mathbf{C}$ together with a family of morphisms $\lambda_j\colon X \to Dj$, one for each object j of J, such that for every morphism $u\colon j \to j'$ in J the triangle

$$\begin{tikzcd} & X \arrow[dl, "\lambda_j"'] \arrow[dr, "\lambda_{j'}"] & \\ Dj \arrow[rr, "Du"'] & & Dj' \end{tikzcd}$$

commutes, that is $Du \circ \lambda_j = \lambda_{j'}$. A morphism of cones $(X, \{\lambda_j\}) \to (Y, \{\mu_j\})$ is a morphism $f\colon X \to Y$ with $\mu_j \circ f = \lambda_j$ for every j. Cones over D and their morphisms form a category $\mathrm{Cone}(D)$.
A *limit* of D is a terminal object of $\mathrm{Cone}(D)$: a cone $(L, \{\pi_j\})$, the *limit cone*, such that every cone $(X, \{\lambda_j\})$ factors through it by a unique morphism $\langle \lambda_j \rangle\colon X \to L$ with $\pi_j \circ \langle \lambda_j \rangle = \lambda_j$ for all j. The apex L is written $\lim D$ or $\lim_J D$, and the π_j are its *projections*.

The commuting condition is the whole content of "compatible": a cone is not just a map into each Dj but a family that respects every morphism the shape imposes among the Dj. The limit is the cone through which all others factor, so a map into L is precisely a compatible family of maps into the diagram. This is the universal-arrow pattern of definition 3.1.4 for the diagonal functor $\Delta\colon \mathbf{C} \to [J, \mathbf{C}]$ sending X to the constant diagram at X: a cone with apex X is a natural transformation $\Delta X \Rightarrow D$ (definition 1.5.2), and the limit is the universal such X.

Being terminal in $\mathrm{Cone}(D)$ settles uniqueness at once.

Proposition 3.2.2: Uniqueness of Limits

If $(L, \{\pi_j\})$ and $(L', \{\pi'_j\})$ are both limits of $D\colon J \to \mathbf{C}$, there is a unique isomorphism $L \to L'$ commuting with the projections.

***Proof*:**
Both are terminal objects of $\mathrm{Cone}(D)$, and terminal objects are unique up to a unique isomorphism by proposition 3.1.3. The isomorphism is a morphism of cones, so it commutes with the projections by definition of $\mathrm{Cone}(D)$, completing the proof.

The limit is thus determined by D alone, and one speaks of "the" limit. The rest of the section reads specific shapes J off this single definition, then shows two of them suffice to build every other.

The simplest shapes carry no morphisms beyond identities, so the cone condition is vacuous and a cone is just a chosen map into each object.

Definition 3.2.3: Product

Let J be a *discrete* category (only identity morphisms) with objects indexed by a set I, and let D pick out a family $\{X_i\}_{i\in I}$. The limit of D, when it exists, is the *product* $\prod_{i\in I} X_i$, with projections $\pi_i\colon \prod_{i\in I} X_i \to X_i$. Its universal property: for every object X and every family $f_i\colon X \to X_i$ there is a unique $\langle f_i\rangle\colon X \to \prod_{i\in I} X_i$ with $\pi_i \circ \langle f_i\rangle = f_i$ for all i. When $I = \{1, 2\}$ the product is written $X_1 \times X_2$; when $I = \varnothing$ the limit over the empty diagram is a terminal object 1, the *empty product*.

A product is a single object out of which one can name any tuple of maps to the factors and nothing more; the projections read the coordinates back. There are no compatibility constraints because a discrete shape imposes none, so a map into $\prod_i X_i$ is exactly an unconstrained family of maps into the factors. The empty case is worth pausing on: with no factors to hit, the universal property says every object admits a unique map to the limit, which is the definition of a terminal object (definition 3.1.1).

Next introduce the shape with two parallel arrows between two objects. A cone must equalize the two composites, so it selects the part of the source on which they agree.

Definition 3.2.4: Equalizer

Let J be the category $\{\bullet \rightrightarrows \bullet\}$ with two objects and two parallel non-identity morphisms, and let D send it to a parallel pair $f, g\colon A \to B$ in $\mathbf{C}$. A cone over D is an object E with maps to A and to B making both triangles commute; the map to B is forced to be the common composite, so a cone is a morphism $e\colon E \to A$ with $fe = ge$. The limit is the *equalizer* of f and g: a morphism $e\colon E \to A$ with $fe = ge$ such that every $h\colon X \to A$ with $fh = gh$ factors as $h = e \circ \bar{h}$ for a unique $\bar{h}\colon X \to E$.

The equalizer is the categorical "locus where two maps agree": e is the universal way to map into A so that f and g become equal after the map. That universality forces e to be injective on maps.

Proposition 3.2.5: Equalizers are Monic

The equalizer $e\colon E \to A$ of a parallel pair $f, g\colon A \to B$ is a monomorphism.

***Proof*:**
Let $p, q\colon X \to E$ satisfy $ep = eq$, and set $h = ep$. Then $fh = fep = gep = gh$, so $h\colon X \to A$ satisfies $fh = gh$ and factors uniquely through e. Both p and q are factorizations of h through e: indeed $e \circ p = h$ and $e \circ q = eq = ep = h$. Uniqueness of the factorization gives $p = q$, so e is monic, as required.

That equalizers are monic recovers the intuition that "where f and g agree" is a subobject of the source A, and it is the ingredient that later lets equalizers cut a limit out of a product.

The third shape is the cospan: two arrows into a common target. Its limit combines a product with an agreement condition.

Definition 3.2.6: Pullback

Let J be the *cospan* category $\{\bullet \to \bullet \leftarrow \bullet\}$, and let D send it to a cospan $f\colon X \to Z$, $g\colon Y \to Z$ in $\mathbf{C}$. The limit is the *pullback* $X \times_Z Y$, with projections $p\colon X \times_Z Y \to X$ and $q\colon X \times_Z Y \to Y$ satisfying $fp = gq$, universal among such squares:

$$\begin{tikzcd} X \times_Z Y \arrow[r, "q"] \arrow[d, "p"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

For every object W with maps $a\colon W \to X$, $b\colon W \to Y$ such that $fa = gb$, there is a unique $\langle a, b\rangle\colon W \to X \times_Z Y$ with $p\langle a, b\rangle = a$ and $q\langle a, b\rangle = b$.

A pullback is a product "constrained to agree over Z": it is the subobject of $X \times Y$ on which the two ways to reach Z coincide. When $Z = 1$ is terminal the agreement condition is vacuous and the pullback collapses to the product $X \times Y$, so the pullback refines the product. This one shape specializes to intersections of subobjects, preimages, fibers of a map, and fibered products of schemes, which is why it recurs throughout the series.

Each of these shapes should be seen at work in the category the reader knows best.

Example 3.2.7: Limits in Set
In **Set** each of the three limits above has a concrete description, and the universal property is verified by hand.

1. *Products.* The product $\prod_{i \in I} X_i$ is the cartesian product, the set of tuples $(x_i)_{i \in I}$ with $x_i \in X_i$, and π_i is the i-th coordinate. Given $f_i\colon X \to X_i$, the map $\langle f_i\rangle(x) = (f_i(x))_{i \in I}$ satisfies $\pi_i\langle f_i\rangle = f_i$, and any g with $\pi_i g = f_i$ has i-th coordinate f_i, so $g = \langle f_i\rangle$. The empty product is a one-element set $\{*\}$, terminal in **Set**. In **Grp**, **Ab**, and $\mathbf{Mod}_R$ the

same underlying cartesian product carries the pointwise (coordinatewise) operations, and the projections are homomorphisms.

2. *Equalizers.* The equalizer of $f, g\colon A \to B$ is the subset

$$E = \{\, a \in A \,|\, f(a) = g(a) \,\}$$

with e the inclusion. If $h\colon X \to A$ satisfies $fh = gh$ then $f(h(x)) = g(h(x))$ for all x, so $h(x) \in E$; corestricting h to E gives the unique $\bar{h}$ with $e\bar{h} = h$. The inclusion is injective, illustrating proposition 3.2.5.

3. *Pullbacks.* The pullback of $f\colon X \to Z$ and $g\colon Y \to Z$ is the subset of $X \times Y$

$$X \times_Z Y = \{\, (x, y) \in X \times Y \,|\, f(x) = g(y) \,\}$$

with p, q the coordinate maps. Given $a\colon W \to X$ and $b\colon W \to Y$ with $fa = gb$, the map $w \mapsto (a(w), b(w))$ lands in $X \times_Z Y$ and is the unique factorization. Taking $g\colon \{z\} \hookrightarrow Z$ the inclusion of a point recovers the fiber $f^{-1}(z)$, and taking $X, Y \subseteq Z$ with f, g the inclusions recovers the intersection $X \cap Y$.

The three examples share a pattern: the limit is a subset of a product cut out by the equations the shape requires. The next theorem promotes that pattern to a construction, showing that products and equalizers alone build every limit.

Products handle the objects of a diagram, equalizers handle the constraints its morphisms impose. Stack every Dj into one product, stack every constraint into a second, and the two maps that express "constraint satisfied versus not" have an equalizer that is exactly the limit. This reduces the existence of all limits to two special cases.

Theorem 3.2.8: Limits from Products and Equalizers

Let $\mathbf{C}$ have all small products and all equalizers. Then $\mathbf{C}$ has all small limits. Explicitly, for a diagram $D\colon J \to \mathbf{C}$ with J small, form the products

$$P = \prod_{j \in J} Dj, \qquad Q = \prod_{u\colon j \to j' \text{ in } J} D(\operatorname{cod} u)$$

indexed by the objects and by the morphisms of J respectively. Let $s, t\colon P \to Q$ be the morphisms determined on the u-component $(u\colon j \to j')$ by

$$\rho_u \circ s = D(u) \circ \mathrm{pr}_j, \qquad \rho_u \circ t = \mathrm{pr}_{j'}$$

where $\mathrm{pr}_j\colon P \to Dj$ and $\rho_u\colon Q \to D(\operatorname{cod} u) = Dj'$ are the product projections. Then the equalizer $e\colon L \to P$ of s and t is a limit of D, with projections $\pi_j = \mathrm{pr}_j \circ e$.

***Proof*:**
The proof runs in three stages: define the two maps whose disagreement measures failure of the cone condition, check that the equalizer carries a cone, and check that this cone is terminal.

Step 1: The maps s and t. By the universal property of the product Q (definition 3.2.3), a morphism into Q is determined by its composite with each projection ρ_u. So the two displayed families of composites define s and t uniquely. Reading them off: for a point of P, thought of as an arbitrary family $(x_j)_j$ with $x_j : T \to Dj$ for a test object T, the u-component of s is $D(u) \circ x_j$ (push the source coordinate along $D(u)$), while the u-component of t is $x_{j'}$ (read off the target coordinate). A family (x_j) satisfies $s = t$ on the u-component precisely when $D(u) \circ x_j = x_{j'}$, which is the cone condition for the morphism u. Thus s and t agree on (x_j) if and only if (x_j) is a cone over D.

Step 2: The equalizer is a cone. Let $e\colon L \to P$ equalize s and t, so $se = te$. Set $\pi_j = \mathrm{pr}_j \circ e\colon L \to Dj$. For any $u\colon j \to j'$, compose $se = te$ with ρ_u:

$$\rho_u \circ s \circ e = \rho_u \circ t \circ e$$

The left side is $D(u) \circ \mathrm{pr}_j \circ e = D(u) \circ \pi_j$ and the right side is $\mathrm{pr}_{j'} \circ e = \pi_{j'}$. Hence $D(u) \circ \pi_j = \pi_{j'}$ for every u, so $(L, \{\pi_j\})$ is a cone over D.

Step 3: The cone is terminal. Let $(X, \{\lambda_j\})$ be any cone over D. By the universal property of P the family $\{\lambda_j\}$ determines a unique morphism $\ell\colon X \to P$ with $\mathrm{pr}_j \circ \ell = \lambda_j$ for all j. The cone condition $D(u) \circ \lambda_j = \lambda_{j'}$ says, on each u-component,

$$\rho_u \circ s \circ \ell = D(u) \circ \mathrm{pr}_j \circ \ell = D(u) \circ \lambda_j = \lambda_{j'} = \mathrm{pr}_{j'} \circ \ell = \rho_u \circ t \circ \ell$$

Since these hold for every u, and morphisms into Q are determined by their ρ_u-components, $s\ell = t\ell$. So ℓ equalizes s and t and factors uniquely as $\ell = e \circ \bar{\ell}$ for a unique $\bar{\ell}\colon X \to L$. Then

$$\pi_j \circ \bar{\ell} = \mathrm{pr}_j \circ e \circ \bar{\ell} = \mathrm{pr}_j \circ \ell = \lambda_j$$

so $\bar{\ell}$ is a morphism of cones $(X, \{\lambda_j\}) \to (L, \{\pi_j\})$. For uniqueness, suppose $\varphi\colon X \to L$ is any cone morphism, so $\pi_j \circ \varphi = \lambda_j$ for all j. Then $\mathrm{pr}_j \circ (e \circ \varphi) = \pi_j \circ \varphi = \lambda_j = \mathrm{pr}_j \circ \ell$ for every j, and the uniqueness clause of P's universal property forces $e \circ \varphi = \ell$. Since e is monic (proposition 3.2.5), $e \circ \varphi = e \circ \bar{\ell}$ gives $\varphi = \bar{\ell}$. Thus $\bar{\ell}$ is the unique cone morphism $X \to L$, so $(L, \{\pi_j\})$ is terminal in $\mathrm{Cone}(D)$ and is a limit of D, as we sought to show.

The construction is the abstract form of the **Set** computation in example 3.2.7: P is the product of the objects, s and t encode the two sides of each morphism-constraint, and the equalizer $L \subseteq P$ is the sub-configuration on which every constraint holds. What was "a subset of a product cut out by equations" in **Set** is now "an equalizer of two maps out of a product" in any category with those two limits, and the same argument gives the universal property. A category with all products and equalizers is therefore complete for all small diagrams, and to check completeness one need only supply products and equalizers.

A limit is the universal way to map compatibly into an entire diagram at once, so that a single morphism into the apex is the same data as a family of morphisms into the diagram respecting its shape. Products and equalizers are the atomic cases, the first handling objects without constraint and the second handling a constraint without extra objects, and theorem 3.2.8 shows these two generate all the rest.

3.3 Colimits

Every construction of the previous section has a mirror image obtained by reversing all arrows. Where a limit sits at the tip of an incoming cone, its dual sits at the tip of an outgoing one; where a limit is a terminal cone, its dual is an initial cocone. This section develops that mirror image. The payoff is not only a second family of constructions but a method: because a colimit in C is exactly a limit in C^{op}, every theorem about limits transports across the duality principle (box 1.3.3) with no new proof, and the constructions themselves become recognizable as the coproducts, quotients, and gluings the reader already meets throughout algebra and topology.

The starting point is the dual of a cone. A cone over a diagram receives a compatible family of maps from a single apex; a cocone under a diagram emits a compatible family of maps into a single nadir.

Definition 3.3.1: Cocone and Colimit

Let $D\colon J \to C$ be a diagram. A *cocone* under D with *nadir* X is an object X of C together with a family of morphisms

$$\iota_j\colon Dj \to X, \qquad j \in J$$

one for each object of J, such that for every morphism $u\colon j \to k$ of J the triangle

$$\iota_k \circ Du = \iota_j$$

commutes. A *morphism of cocones* $(X, \{\iota_j\}) \to (Y, \{\kappa_j\})$ is a morphism $f\colon X \to Y$ of C with $f\iota_j = \kappa_j$ for every j.

A *colimit* of D is an *initial* cocone: a cocone $(L, \{\iota_j\})$ such that for every cocone $(X, \{\kappa_j\})$ under D there is a unique morphism $f\colon L \to X$ with $f\iota_j = \kappa_j$ for all j. The nadir L is written $\operatorname{colim}_J D$, and the maps ι_j are the *structure maps* (or *coprojections*) of the colimit.

The definition is definition 3.2.1 read backwards. A cocone under D is precisely a cone over the diagram $D^{\text{op}}\colon J^{\text{op}} \to C^{\text{op}}$ obtained by reversing every arrow, and an initial cocone is a terminal cone in C^{op}. This is the organizing fact of the section.

Proposition 3.3.2: Colimits are Limits in the Opposite Category

Let $D\colon J \to C$ be a diagram and write $D^{\text{op}}\colon J^{\text{op}} \to C^{\text{op}}$ for the same assignment on objects, regarded as a diagram in C^{op}. An object L with maps $\iota_j\colon Dj \to L$ is a colimit of D in C if and only if L with the same maps, now read as $\iota_j\colon L \to Dj$ in C^{op}, is a limit of D^{op} in C^{op}. Thus

$$\operatorname*{colim}_J D \text{ in } C \quad = \quad \lim_{J^{\text{op}}} D^{\text{op}} \text{ in } C^{\text{op}}$$

***Proof*:**
Reversing arrows sends the compatibility condition $\iota_k \circ Du = \iota_j$ for a cocone under D to the condition $D^{\text{op}}u \circ \iota_k = \iota_j$ for a cone over D^{op} in C^{op}, since composition in C^{op} is composition in C in the opposite order and each Du becomes $D^{\text{op}}u$ pointing the other way. So cocones under D in C are the same data as cones over D^{op} in C^{op}, and a morphism of cocones $f\colon X \to Y$ in C is a morphism of cones $f\colon Y \to X$ in C^{op}. An initial object among cocones is therefore a terminal object among cones, so L is a colimit of D exactly when it is a limit of D^{op} in C^{op}, completing the proof.

This proposition licenses a mechanical translation. Every notion of the previous section has a dual with the prefix "co": the terminal cone becomes the initial cocone, and each specific limit shape becomes a colimit shape by reversing the diagram J. The three shapes worth naming are the duals of the product, the equalizer, and the pullback. Their definitions are stated directly, but each one could equally be obtained by applying proposition 3.3.2 to the corresponding definition of section 3.2.

The first is the dual of the product (definition 3.2.3): a discrete diagram $\{X_i\}_{i \in I}$ with no nontrivial morphisms, whose colimit assembles the objects side by side rather than cutting out their common part.

Definition 3.3.3: Coproduct

Let $\{X_i\}_{i \in I}$ be a family of objects of C, viewed as a diagram on the discrete index category I. A *coproduct* of the family is a colimit: an object $\coprod_{i \in I} X_i$ together with *injections*

$$\iota_i \colon X_i \to \coprod_{i \in I} X_i$$

such that for every object Y and every family of morphisms $f_i \colon X_i \to Y$ there is a unique morphism $f \colon \coprod_{i \in I} X_i \to Y$ with $f\iota_i = f_i$ for all i. A binary coproduct is written $X_1 \sqcup X_2$.

A coproduct is the object into which every X_i maps and out of which maps are prescribed by naming one map from each summand independently. Where a product is determined by projections and its universal property governs *maps in*, a coproduct is determined by injections and governs *maps out*. The empty coproduct, indexed by $I = \varnothing$, is an object with a unique map to every Y and no data to specify: it is an initial object 0 (definition 3.1.1).

Example 3.3.4: Coproducts in the Standing Cast

The coproduct instantiates as a familiar assembly in each concrete category.

1. In **Set** the coproduct is the *disjoint union* $\coprod_i X_i$, the set of pairs (i, x) with $x \in X_i$, and $\iota_i(x) = (i, x)$. A family of functions $f_i \colon X_i \to Y$ assembles into the single function $(i, x) \mapsto f_i(x)$, which is the only function restricting to each f_i on the i-th copy. For two sets this is $X_1 \sqcup X_2$.

2. In **Grp** the coproduct is the *free product* $G_1 * G_2$: its elements are reduced words alternating between nonidentity elements of G_1 and of G_2, with ι_1, ι_2 the inclusions of the one-letter words. A pair of homomorphisms $f_i \colon G_i \to H$ extends to the unique homomorphism sending a word to the product of the images of its letters, each letter mapped by the f_i of the factor it comes from, and this assignment is forced ([4], free products). The free group on a set S is the coproduct of $|S|$ copies of $\mathbb{Z}$.

3. In **Ab** and in $\mathbf{Mod}_R$ the coproduct of a family is the *direct sum* $\bigoplus_i M_i$, the submodule of the product $\prod_i M_i$ consisting of tuples with finitely many nonzero entries, with ι_i the insertion into the i-th coordinate. A family $f_i \colon M_i \to N$ assembles into $(m_i)_i \mapsto \sum_i f_i(m_i)$, a finite sum because only finitely many m_i are nonzero ([4], direct sums). For finitely many summands the direct sum and the direct product coincide, so the biproduct $M_1 \oplus M_2$ is at once a product and a coproduct; this coincidence is a feature of additive categories, not a general phenomenon.

The contrast with products is sharpest in **Grp**: the product $G_1 \times G_2$ forces the two factors to commute, while the coproduct $G_1 * G_2$ imposes no relations at all between them.

The direct-sum example already shows the divergence between limits and colimits concretely: in **Set** the product is a Cartesian product and the coproduct a disjoint union, unrelated constructions, and in **Grp** the gap between $G_1 \times G_2$ and $G_1 * G_2$ is the gap between imposing all cross-relations and imposing none. The next shape is the dual of the equalizer (definition 3.2.4). Where an equalizer selects the sub-object on which two maps agree, its dual forms the quotient on which two maps are forced to agree.

Definition 3.3.5: Coequalizer

Let $f, g\colon A \to B$ be a parallel pair of morphisms in C. A *coequalizer* of f and g is a morphism $q\colon B \to Q$ with $qf = qg$ that is universal with this property: for every morphism $h\colon B \to Y$ satisfying $hf = hg$ there is a unique morphism $\bar{h}\colon Q \to Y$ with $\bar{h}q = h$.

The coequalizer is the colimit of the two-object diagram $A \rightrightarrows B$ whose only nontrivial morphisms are f and g: the condition $qf = qg$ is the cocone compatibility for that diagram, and universality is initiality among such cocones. The map q is the largest quotient of B on which f and g become equal, the categorical quotient. It is always epic, the dual of the fact that an equalizer is always monic.

Lemma 3.3.6: A Coequalizer is Epic

The coequalizer map $q\colon B \to Q$ of any parallel pair $f, g\colon A \to B$ is an epimorphism.

***Proof*:**
Suppose $s, t\colon Q \to Y$ satisfy $sq = tq$. Set $h = sq = tq\colon B \to Y$. Then $hf = sqf = sqg = hg$ using $qf = qg$, so h satisfies the coequalizing condition and factors uniquely through q. Both s and t are such factorizations, since $sq = h$ and $tq = h$, so uniqueness forces $s = t$. Thus q is epic, as we sought to show.

Example 3.3.7: The Coequalizer in Sets is a Quotient
For functions $f, g\colon A \to B$ in **Set**, let $\sim$ be the smallest equivalence relation on B containing every pair $(f(a), g(a))$ for $a \in A$. The coequalizer is the quotient set $Q = B/{\sim}$ with $q\colon B \to Q$ the quotient map. A function $h\colon B \to Y$ satisfies $hf = hg$ exactly when $h(f(a)) = h(g(a))$ for all a, that is, when h is constant on each generating pair, hence on each $\sim$-class; such an h descends to a unique $\bar{h}\colon Q \to Y$ with $\bar{h}q = h$. The generated equivalence relation is not just the set of pairs $(f(a), g(a))$: it is their reflexive, symmetric, transitive closure, since a quotient map respecting the relations must respect everything they entail. Taking g constant recovers, as a special case, the quotient of B collapsing a subset to a point.

The equivalence closure in example 3.3.7 is the concrete face of a general fact: a coequalizer imposes the relations $f = g$ together with every consequence forced by the category's own composition, exactly as a coproduct in **Grp** imposes no relations while a quotient group imposes stated ones. The last named shape is the dual of the pullback (definition 3.2.6), which glues two objects along a shared source rather than intersecting them over a shared target.

Definition 3.3.8: Pushout

Let $f\colon Z \to X$ and $g\colon Z \to Y$ be a pair of morphisms with common *source* Z. A *pushout* of f and g is a colimit of this diagram: an object $X \sqcup_Z Y$ with morphisms $i\colon X \to X \sqcup_Z Y$ and $j\colon Y \to X \sqcup_Z Y$ satisfying $if = jg$, universal among such cocones. For every object W with maps $p\colon X \to W$ and $q\colon Y \to W$ satisfying $pf = qg$ there is a unique $h\colon X \sqcup_Z Y \to W$ with $hi = p$ and $hj = q$.

A pushout glues X and Y along the images of Z: the map f marks a copy of Z inside X, the map g marks a copy inside Y, and $X \sqcup_Z Y$ is the result of identifying the two marked copies. In **Set** it is the quotient of $X \sqcup Y$ by the relation $f(z) \sim g(z)$, which is why the pushout is the formal model of every "glue two pieces along a common part" construction. When $Z = 0$ is initial the constraint is vacuous and the pushout is the coproduct $X \sqcup Y$, exactly as a pullback over a terminal object is a product.

Example 3.3.9: Pushouts as Gluing

Two instances make the gluing concrete.

1. In **Set**, if $Z \hookrightarrow X$ and $Z \hookrightarrow Y$ are inclusions of a common subset, the pushout $X \sqcup_Z Y$ is the union of X and Y overlapping exactly along Z: form the disjoint union $X \sqcup Y$ and identify each copy of a point of Z. Explicitly it is $(X \sqcup Y)/{\sim}$ with $\iota_1 f(z) \sim \iota_2 g(z)$, the coequalizer of the two maps $Z \rightrightarrows X \sqcup Y$ given by $\iota_1 f$ and $\iota_2 g$.
2. In **Top** the same colimit glues spaces. Taking X and Y to be two closed disks D^2 and $Z = S^1$ their common boundary circle, mapping into each by the boundary inclusion, the pushout $D^2 \sqcup_{S^1} D^2$ is the 2-sphere S^2: two disks glued along their boundary. This is the mechanism behind attaching cells and building spaces from simpler pieces, the staple of the algebraic topology of CW complexes.

The two constructions from which every colimit is built are now in hand: coproducts assemble a family of objects, and coequalizers impose relations. The dual of theorem 3.2.8 states that these two suffice, and its proof is that theorem read in the opposite category. The motivation is the same as in the limit case: to check that a category has all small colimits, it is enough to check two constructions rather than one for every diagram shape.

Theorem 3.3.10: Colimits from Coproducts and Coequalizers

Let C be a category with all small coproducts and all coequalizers. Then C has all small colimits. Concretely, the colimit of a diagram $D\colon J \to C$ with J small is the coequalizer of a canonical parallel pair of maps between two coproducts,

$$\coprod_{(u\colon j\to k)\in \mathrm{mor}\, J} Dj \rightrightarrows \coprod_{j\in \mathrm{ob}\, J} Dj \longrightarrow \operatorname*{colim}_J D$$

where the first coproduct is indexed by the morphisms of J and the second by its objects.

***Proof*:**
By proposition 3.3.2, a colimit of $D\colon J \to C$ is a limit of $D^{\mathrm{op}}\colon J^{\mathrm{op}} \to C^{\mathrm{op}}$. The hypotheses transport across the duality principle (box 1.3.3): a coproduct in C is a product in C^{op} and a coequalizer in C is an equalizer in C^{op}, so C having all small coproducts and coequalizers is C^{op} having all small products and equalizers. Theorem 3.2.8 then applies in C^{op} and produces the limit of D^{op}, hence the colimit of D. This proves existence; it remains only to record what the construction of that theorem becomes when read back in C.

The construction of theorem 3.2.8 builds a limit as an equalizer of two maps between the product $\prod_j Dj$ over the objects and the product $\prod_u D(\operatorname{cod} u)$ over the morphisms. Dualizing every arrow turns each product into a coproduct and the equalizer into a coequalizer, giving the coproduct $\coprod_j Dj$ over the objects, the coproduct $\coprod_u D(\operatorname{dom} u)$ over the morphisms, and a coequalizer of two maps between them. Concretely, for a morphism $u\colon j \to k$ of J the u-th summand Dj of the source coproduct is sent by the two maps to the object coproduct in two ways: one injection includes Dj as the j-th summand directly, and the other sends it through $Du\colon Dj \to Dk$ into the k-th summand. Writing $\iota_j\colon Dj \to \coprod_j Dj$ for the coproduct injections, the two maps $s, t\colon \coprod_u Dj \to \coprod_j Dj$ are determined on the u-th summand by

$$s \circ \iota_u = \iota_j, \qquad t \circ \iota_u = \iota_k \circ Du$$

Their coequalizer $q\colon \coprod_j Dj \to Q$ forces $q\iota_j$ and $q\iota_k \circ Du$ to become equal for every u, since $qs = qt$ reads $q\iota_j = q\iota_k \circ Du$ on the u-th summand. This is exactly the cocone compatibility for the maps $q\iota_j$. So the maps $q\iota_j\colon Dj \to Q$ form a cocone under D, and the coequalizer's universal property makes it the initial one: a cocone $(Y, \{\kappa_j\})$ is a family with $\kappa_k \circ Du = \kappa_j$, which is a map $\langle \kappa_j \rangle\colon \coprod_j Dj \to Y$ coequalizing s and t, hence factoring uniquely through q. Thus $Q = \operatorname{colim}_J D$, completing the proof.

The theorem and its proof are the mirror of the limit case in the strict sense that not one line of new mathematics was needed: proposition 3.3.2 moves the problem to C^{op}, the duality principle converts the hypotheses, and theorem 3.2.8 does the work. The dual construction is worth carrying explicitly all the same, because the coequalizer of two maps between coproducts is how a colimit is computed: assemble every object of the diagram into one coproduct, then force the relations that the morphisms of J impose. The equalizer-of-products picture cut a sub-object out of an assembly; the coequalizer-of-coproducts picture builds an assembly and then collapses it.

This is the interpretive content of the whole duality. Limits and colimits are two readings of one definition, exchanged by reversing arrows, and the exchange is not a formal trick but a difference in operation. A limit intersects and cuts out: the product is the joint object seen from all its projections, the equalizer is the locus where two maps agree, the pullback is the fibered intersection over a shared target. A colimit glues and quotients: the coproduct sets objects side by side, the coequalizer imposes relations, the pushout glues along a shared source. The product cuts $\prod X_i$ down by an equalizer to enforce compatibility; the coproduct is quotiented by a coequalizer to enforce it. Each theorem, each construction, and each computation of section 3.2 therefore has a twin here, obtained for free by the duality principle, and the two families are the same mathematics viewed from opposite sides.

3.4 Computing (Co)limits

The previous sections built limits and colimits abstractly and reduced their existence to two special cases: products and equalizers on one side, coproducts and coequalizers on the other (theorem 3.2.8 and its dual). This section makes the constructions concrete: a name for the categories in which every diagram has a limit; the working principle that in the algebraic categories a limit is computed on underlying sets while a colimit usually is not; the pointwise computation in functor and presheaf categories; and the functors that carry limits to limits, the representable ones foremost. That last result closes the loop with Chapter 2: an object sees limits through its hom-functor exactly because of Yoneda.

We fix conventions once. A category has *all small limits* if every diagram $D\colon J \to C$ indexed by a small category J has a limit, and likewise for colimits.

Definition 3.4.1: Complete and Cocomplete Categories

A category C is *complete* if it has all small limits, that is, if every diagram $D\colon J \to C$ with J small has a limit $\lim D$. It is *cocomplete* if it has all small colimits, that is, if every such diagram has a colimit $\operatorname{colim}_J D$.

Completeness is a strong hypothesis stated compactly, but it need never be checked diagram by diagram. Theorem 3.2.8 constructed the limit of an arbitrary diagram from a pair of products and one equalizer, so a category that has all small products and all equalizers already has all small limits. This is the only test one runs in practice.

Proposition 3.4.2: Completeness from Products and Equalizers

A category C is complete if and only if it has all small products and all equalizers. Dually, C is cocomplete if and only if it has all small coproducts and all coequalizers.

***Proof*:**
If C is complete it has, in particular, all products (limits of discrete diagrams) and all equalizers (limits over the parallel-pair shape), so the forward direction is immediate. Conversely, suppose C has all small products and all equalizers. Given any diagram $D\colon J \to C$ with J small, the construction of theorem 3.2.8 builds $\lim D$ as the equalizer of two maps between the product $\prod_j Dj$ over the objects of J and the product $\prod_u D(\operatorname{cod} u)$ over the morphisms of J; both products are small because J is small. Hence $\lim D$ exists, and C is complete. The colimit statement is the dual, obtained by applying the same argument in C^{op} (box 1.3.3): a coproduct in C is a product in C^{op}, a coequalizer in C is an equalizer in C^{op}, and completeness of C^{op} is cocompleteness of C, completing the proof.

The categories of the standing cast are all complete and cocomplete: **Set**, **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, and **Top** each have all small limits and colimits. For **Set** this is direct, and completeness of the rest follows from the next result once the limits are located inside **Set**. Cocompleteness holds too, but its proof is not dual in content: by proposition 3.4.2 it suffices to have all small coproducts and all coequalizers, and in the algebraic categories these are the free constructions and quotients of [4] (free products in **Grp**, direct sums in **Ab** and $\mathbf{Mod}_R$, and quotients by the relations generated for coequalizers), while **Top** carries the quotient and coproduct topologies. The second half of this section does not reprove these but illustrates the point that colimits are built by their own universal properties rather than read off underlying sets.

The pattern that makes limits computable in the algebraic categories is that the forgetful functor to **Set** does more than preserve them, it *creates* them: the underlying set of a limit is the limit of the underlying sets, and the algebraic structure on it is forced. A product of groups is the product set with coordinatewise multiplication; the equalizer of two homomorphisms is the subset where they agree, which is a subgroup because the operations are computed coordinatewise. The precise statement follows; the notion of creation is made formal later in this section (definition 3.4.6), but the content here is elementary and self-contained.

Proposition 3.4.3: Limits in the Algebraic Categories

Let C be any of **Grp**, **Ab**, **Ring**, or $\mathbf{Mod}_R$, and let $U\colon C \to \mathbf{Set}$ be the forgetful functor. Then C is complete, and U *creates* all small limits: for a diagram $D\colon J \to C$, every limit cone of UD in **Set** lifts to a unique cone in C lying over it, and that cone is a limit. In particular the underlying set of $\lim D$ is $\lim(UD)$, computed in **Set**, with the algebraic operations defined coordinatewise. Concretely:

1. *Products.* The product $\prod_i X_i$ has underlying set the Cartesian product $\prod_i UX_i$, with the operations applied coordinatewise; the projections $\pi_j\colon \prod_i X_i \to X_j$ are the coordinate projections.
2. *Equalizers.* The equalizer of two morphisms $f, g\colon X \to Y$ is the subobject $E = \{x \in UX \,|\, f(x) = g(x)\}$ with the operations inherited from X; the equalizing morphism $E \to X$ is the inclusion.

The same holds for **Top**, with the limit set carrying the initial topology for the cone maps.

***Proof*:**
By proposition 3.4.2 it suffices to treat products and equalizers and to assemble the general limit from these by theorem 3.2.8, since creation is inherited stage by stage: a functor that creates products and equalizers creates every limit built from them. For each of the two shapes the argument shows more than preservation. Starting from a limit cone in **Set** (the Cartesian product with its projections, or the agreement subset with its inclusion), the algebraic operations that turn it into a cone in C are forced coordinatewise, so the lift is unique; the lifted cone is then verified to be a limit in C; and U returns it to the given cone in **Set**. That is creation (definition 3.4.6), and preservation and completeness of C follow.

Step 1: Products. Take a small family $(X_i)_{i\in I}$ in C. Let P be the Cartesian product $\prod_i UX_i$ of underlying sets, an element of which is a family $(x_i)_i$ with $x_i \in UX_i$. Define the algebraic operations coordinatewise: for **Grp**, $(x_i)(y_i) = (x_i y_i)$ with identity (e_i) and inverse $(x_i)^{-1} = (x_i^{-1})$; for a ring or module the addition and multiplication (or scalar action) are applied in each coordinate. The group, ring, or module axioms hold in P because they hold in each X_i and are tested coordinatewise. The coordinate projections $\pi_j\colon P \to X_j$, $(x_i) \mapsto x_j$, are morphisms of C because the operations on P are coordinatewise. Given any Z in C with morphisms $h_i\colon Z \to X_i$, the set map $h\colon UZ \to P$, $z \mapsto (h_i(z))_i$, is the unique function with $\pi_j h = h_j$ for all j (this is the universal property of the product in **Set**), and it is a morphism of C because each h_i is and the operations on P are coordinatewise. So P with the π_j is the product in C, and

$UP = \prod_i UX_i$ is the product in **Set**; thus U preserves it. The lift is unique: any operations on the set $\prod_i UX_i$ making the coordinate projections into morphisms of C must agree with each X_j in that coordinate, hence be the coordinatewise ones, so this is the only structure lying over the **Set** product. That is creation for the product shape.

Step 2: Equalizers. Take $f, g\colon X \to Y$ in C and set $E = \{x \in UX \,|\, f(x) = g(x)\} \subseteq UX$, which is the equalizer of Uf and Ug in **Set**. The subset E is closed under the operations of X: if $f(x) = g(x)$ and $f(x') = g(x')$ then, since f and g are morphisms, $f(xx') = f(x)f(x') = g(x)g(x') = g(xx')$, and likewise f and g agree on the identity, on inverses, and on scalar multiples. Hence E is a subgroup, subring, or submodule of X, and with these inherited operations the inclusion $e\colon E \to X$ is a morphism of C with $fe = ge$. If $h\colon Z \to X$ satisfies $fh = gh$ then h lands in E at the level of sets, so it factors uniquely as $h = eh'$ through the set map $h'\colon UZ \to E$; and h' is a morphism of C because its composite with the inclusion is and E carries the inherited operations. So E is the equalizer in C, and UE is the equalizer in **Set**. Again the lift is unique: the subset E inherits its operations from X by restriction, the only structure making the inclusion a morphism, so the cone over the **Set** equalizer is forced. That is creation for the equalizer shape.

Step 3: Topological spaces. For **Top** the same two constructions work on underlying sets, and the topology is forced: the product carries the product topology (the coarsest making every projection continuous) and the equalizer carries the subspace topology. In each case this is the coarsest topology making the cone maps continuous, so a map out of a test space is continuous exactly when its composites with the cone maps are, which is the universal property. The initial topology is the unique one making the cone maps continuous while giving the universal property, so here too the lift over the **Set** limit is unique: $U\colon \mathbf{Top} \to \mathbf{Set}$ creates these limits as well. By proposition 3.4.2 every category listed is complete with limits computed on underlying sets, completing the proof.

The proposition is what lets one compute in these categories at all: a limit is found by taking the limit of the underlying sets and then observing that the algebraic structure has nowhere else to go. The equalizer clause is the categorical form of a familiar move, that the locus where two homomorphisms agree is a subgroup. The kernel of a group or module homomorphism $f\colon G \to H$ is the special case where g is the trivial homomorphism (written $g = 0$ in the additive case), so such kernels are equalizers, and the general limit packages products with these agreement loci. In **Ring** the zero map is not a morphism, so a ring-map kernel is an ideal rather than an equalizer; the equalizer description here stays in the group and module setting.

The contrast the reader should carry forward is that colimits are not computed this way. The forgetful functor does not create colimits: the coproduct in **Grp** is the free product, whose underlying set is far larger than the disjoint union of the underlying sets.

Example 3.4.4: The Coproduct in Grp Is Not the Disjoint Union

In **Set** the coproduct is the disjoint union $X \sqcup Y$. In **Grp** the coproduct of G and H is the *free product* $G * H$, whose elements are the reduced words alternating between nontrivial elements of G and of H ([4], free products). Its universal property is the one required of a coproduct: a pair of homomorphisms $G \to K$ and $H \to K$ extends uniquely to $G * H \to K$. Take $G = H = \mathbb{Z}/2$, generated by a and b respectively. The disjoint union of the underlying sets has four elements, while $\mathbb{Z}/2 * \mathbb{Z}/2$ is infinite: it contains the distinct reduced words $a, b, ab, ba, aba, bab, \ldots$ and is isomorphic to the infinite dihedral group. So $U(G * H) \neq UG \sqcup UH$, and the forgetful functor does not preserve this coproduct. In **Ab** the coproduct of G and H is instead the direct sum $G \oplus H$, which for two groups agrees with the product, and its underlying set is $UG \times UH$, not $UG \sqcup UH$: even the abelian coproduct is not the set-level coproduct.

The lesson is that "compute on underlying sets" is a statement about limits, and that colimits in the algebraic categories must be built by their own universal property (free constructions, quotients, amalgamation) rather than read off the forgetful functor. Coequalizers, for instance, are quotients rather than subobjects.

The functor and presheaf categories behave better: there both limits and colimits are computed pointwise, one object of the indexing category at a time. This is what makes presheaf theory computable, and the reason presheaf categories inherit completeness and cocompleteness from **Set**.

Proposition 3.4.5: Pointwise Limits in Functor Categories

Let J be a small category and C a category, and let $[J, C]$ be the functor category (definition 1.5.4). Suppose C is complete. Then $[J, C]$ is complete, and limits are computed *pointwise*: for a diagram $F\colon I \to [J, C]$ indexed by a small category I, the limit $\lim_I F$ is the functor

$$\left(\lim_I F\right)(j) = \lim_I \big(F(-)(j)\big)$$

sending each object j of J to the limit in C of the diagram $i \mapsto F(i)(j)$. Dually, if C is cocomplete then $[J, C]$ is cocomplete, with colimits computed pointwise. In particular the presheaf category $\mathbf{PSh}(C) = [C^{\mathrm{op}}, \mathbf{Set}]$ (definition 2.3.1) is complete and cocomplete, and both are computed pointwise from **Set**.

***Proof*:**
Write $G_j = \lim_I \big(F(-)(j)\big)$ for the object of C obtained by taking, for a fixed j, the limit of the diagram $i \mapsto F(i)(j)$; it exists because C is complete. Let $\lambda_i^j\colon G_j \to F(i)(j)$ be the limit cone.

Step 1: The pointwise limits assemble into a functor. For a morphism $u\colon j \to j'$ in J and each i, the maps $F(i)(u)\colon F(i)(j) \to F(i)(j')$ are compatible with the diagram in I, since each $F(i)$ is a functor and the arrows of the diagram over I are natural transformations, so the family $\big(F(i)(u) \circ \lambda_i^j\big)_i$ is a cone over $i \mapsto F(i)(j')$. By the universal property of $G_{j'}$ there is a unique map $G(u)\colon G_j \to G_{j'}$ with $\lambda_i^{j'} \circ G(u) = F(i)(u) \circ \lambda_i^j$ for all i. Uniqueness makes G functorial: $G(1_j) = 1_{G_j}$ and $G(u'u) = G(u')G(u)$, because both sides satisfy the equations defining the induced map. So $G\colon J \to C$ is a functor, an object of $[J, C]$, and the maps $\lambda_i^\bullet\colon G \Rightarrow F(i)$ are natural transformations, a cone over F in $[J, C]$.

Step 2: This cone is a limit. Let H be a functor in $[J, C]$ with a cone $\mu_i\colon H \Rightarrow F(i)$. For each fixed j the components $(\mu_i)_j\colon H(j) \to F(i)(j)$ form a cone over $i \mapsto F(i)(j)$ in C, so there is a unique map $\varphi_j\colon H(j) \to G_j$ with $\lambda_i^j \circ \varphi_j = (\mu_i)_j$ for all i. The components φ_j are natural in j: for $u\colon j \to j'$, both $G(u)\varphi_j$ and $\varphi_{j'}H(u)$ are maps $H(j) \to G_{j'}$ that, after composing with each $\lambda_i^{j'}$, give $(\mu_i)_{j'}H(u) = F(i)(u)(\mu_i)_j$, so they agree by the uniqueness in the universal property of $G_{j'}$. Hence $\varphi\colon H \Rightarrow G$ is a natural transformation with $\lambda_i^\bullet \circ \varphi = \mu$, and it is the unique such because each φ_j was forced. So G with the cone $\lambda_i^\bullet$ is the limit of F in $[J, C]$. The colimit statement is the dual (box 1.3.3), and applying both to $C = \mathbf{Set}$, which is complete and cocomplete, gives the claim for $\mathbf{PSh}(C)$, completing the proof.

The proposition says that a (co)cone in $[J, C]$ is exactly an objectwise-compatible family of (co)cones in C: naturality is what glues the pointwise data into a morphism of functors, and the universal property is checked one object at a time. For presheaves this means every limit and colimit is computed in **Set** and then reassembled, so all the set-level formulas transfer verbatim. A product of presheaves is the objectwise product of sets, a colimit of presheaves is the objectwise colimit, and representable presheaves, while not closed under these operations, sit inside a category where every limit and colimit exists.

Having located limits in specific categories, the next results treat how limits behave under functors. Three conditions arrange the possibilities: a functor may carry limit cones to limit cones, may detect a limit from its image, or may lift limits from the target to the source. These are preservation, reflection, and creation.

Definition 3.4.6: Preserving, Reflecting, and Creating Limits

Let $G\colon C \to D$ be a functor and $F\colon J \to C$ a diagram.

1. G *preserves* the limit of F if, whenever $(\lim F, \lambda)$ is a limit cone in C, its image $(G(\lim F), G\lambda)$ is a limit cone over GF in D.
2. G *reflects* limits if a cone (L, λ) over F in C is a limit cone whenever its image $(GL, G\lambda)$ is a limit cone over GF in D.
3. G *creates* the limit of F if, given any limit cone (M, μ) over GF in D, there is a cone (L, λ) over F in C with $GL = M$ and $G\lambda = \mu$, this cone is a limit of F, and it is the unique cone in C (up to the induced isomorphism) lying over (M, μ).

The dual notions, for colimits and cocones, are named identically.

Creation is the strongest of the three and is what proposition 3.4.3 really established: the forgetful functor $U\colon \mathbf{Grp} \to \mathbf{Set}$ creates limits, because a limit cone downstairs in **Set** lifts to a unique cone of groups, and that cone is a limit. Creation implies preservation and reflection together. Preservation alone is the condition that appears most often, and the hom-functor is the instance developed next. A morphism into a limit is the same data as a compatible family of morphisms into the diagram, which is precisely what the hom-set of the limit computes.

Proposition 3.4.7: Representable Functors Preserve Limits

Let C be a locally small category and c an object. The covariant hom-functor $\mathrm{Hom}(c, -)\colon C \to \mathbf{Set}$ (definition 2.1.1) preserves all limits that exist in C: for any diagram $D\colon J \to C$ with a limit,

$$\mathrm{Hom}\big(c, \lim_J D\big) \;\cong\; \lim_J \mathrm{Hom}(c, D-)$$

naturally in c, where the right-hand limit is taken in **Set**. Dually, $\mathrm{Hom}(-, c)\colon C^{\mathrm{op}} \to \mathbf{Set}$ sends colimits in C to limits in **Set**.

***Proof*:**
Let $(\lim_J D, \lambda)$ be the limit cone in C, with legs $\lambda_j \colon \lim_J D \to Dj$. Apply $\mathrm{Hom}(c, -)$: the images $(\lambda_j)_* \colon \mathrm{Hom}(c, \lim_J D) \to \mathrm{Hom}(c, Dj)$ form a cone over $\mathrm{Hom}(c, D-)$ in **Set**, since post-composition is functorial and the λ_j commute with the diagram. It remains to show this cone is a limit in **Set**, that is, that $\mathrm{Hom}(c, \lim_J D) \to \lim_J \mathrm{Hom}(c, Dj)$ is a bijection.

Step 1: The target is the set of compatible families. A limit in **Set** is the set of compatible families over the diagram (the equalizer inside a product built in theorem 3.2.8), so an element of $\lim_J \mathrm{Hom}(c, Dj)$ is a family $(g_j)_j$ with $g_j \in \mathrm{Hom}(c, Dj)$ such that for every morphism $u \colon j \to j'$ in J one has $Du \circ g_j = g_{j'}$. That is exactly a cone over D with apex c: a family of maps $g_j \colon c \to Dj$ compatible with the diagram.

Step 2: Cones with apex c correspond to maps into the limit. The universal property of $\lim_J D$ says precisely that cones over D with apex c are in bijection with morphisms $c \to \lim_J D$: to each cone (g_j) there is a unique $g \colon c \to \lim_J D$ with $\lambda_j g = g_j$, and every $g \colon c \to \lim_J D$ yields the cone $(\lambda_j g)_j$. Under the identification of Step 1, this bijection is exactly the map $g \mapsto ((\lambda_j)_* g)_j = (\lambda_j g)_j$, so $(\lambda_j)_* \colon \mathrm{Hom}(c, \lim_J D) \to \lim_J \mathrm{Hom}(c, Dj)$ is a bijection. Hence the cone is a limit, and $\mathrm{Hom}(c, -)$ preserves the limit of D. Naturality in c is functoriality of the two-sided hom-bifunctor (definition 2.1.4): a morphism $c' \to c$ induces compatible pre-composition maps on both sides.

Step 3: The dual. Applying the statement in C^{op} (box 1.3.3) gives the second claim. A colimit in C is a limit in C^{op}, and $\mathrm{Hom}_C(-, c) = \mathrm{Hom}_{C^{\mathrm{op}}}(c, -)$; the preservation of C^{op}-limits by $\mathrm{Hom}_{C^{\mathrm{op}}}(c, -)$ reads in C as $\mathrm{Hom}_C(\mathrm{colim}_J D, c) \cong \lim_J \mathrm{Hom}_C(D-, c)$, so $\mathrm{Hom}(-, c)$ carries colimits to limits, completing the proof.

This is the categorical version of a fact used constantly: to map into a limit is to map compatibly into each stage, and to map out of a colimit is to map compatibly out of each stage. It is why a homomorphism into a product is a tuple of homomorphisms, and why a homomorphism out of a quotient or a pushout is a compatible pair. The result also reads from the Yoneda side. By the Yoneda lemma (theorem 2.2.1) an object is determined by the functor $\mathrm{Hom}(-, c)$ it represents, and that functor turns colimits into limits, while $\mathrm{Hom}(c, -)$ preserves limits. So an object's behaviour under limits is legible through its hom-functors, the precise sense in which Chapter 2's lens on objects extends to the constructions of this chapter.

The results combine into a single computation. In **Set** a pullback is the fibered product, and the hom-functor preservation says a map into it is a compatible pair of maps into its factors. This one construction computes intersections and preimages at once.

Example 3.4.8: Pullbacks in Set as Fibered Products
Let $f \colon X \to Z$ and $g \colon Y \to Z$ be functions. Their pullback (definition 3.2.6) in **Set** is the fibered product

$$X \times_Z Y = \{(x, y) \in X \times Y \,\big|\, f(x) = g(y)\}$$

with the two projections to X and Y. This is a limit, computed on the underlying sets exactly as proposition 3.4.3 prescribes: it is the equalizer of $f \circ \pi_X$ and $g \circ \pi_Y$ inside the product $X \times Y$, so it is the subset where the two composites agree.

Two special cases recover familiar operations.

1. *Intersection.* Let $A, B \subseteq Z$ be subsets, with inclusions $f\colon A \hookrightarrow Z$ and $g\colon B \hookrightarrow Z$. Then $f(a) = g(b)$ forces $a = b$ as elements of Z, so

$$A \times_Z B = \{(a, b) \in A \times B \big| a = b\} \cong A \cap B$$

   The pullback of two subobjects along their inclusions is their intersection.

2. *Preimage.* Let $g\colon Y \to Z$ be arbitrary and take $A \subseteq Z$ with inclusion $f\colon A \hookrightarrow Z$. Then $A \times_Z Y = \{(a, y) \big| a = g(y)\}$, and the projection to Y identifies it with

$$\{y \in Y \big| g(y) \in A\} = g^{-1}(A)$$

   the preimage. Pulling back a subobject along a map is taking its preimage.

The preservation result reads off the maps into $X \times_Z Y$: for a set W,

$$\mathrm{Hom}(W, X \times_Z Y) \cong \{(p, q) \big| p\colon W \to X,\ q\colon W \to Y,\ fp = gq\}$$

which is the equalizer of f_* and g_* inside $\mathrm{Hom}(W, X) \times \mathrm{Hom}(W, Y)$, precisely the pullback in **Set** of the induced hom-sets. So a map into the fibered product is a matching pair of maps into the factors, the concrete face of proposition 3.4.7.

4

Adjunctions

Chapters 1 through 3 built the vocabulary; adjunctions turn it into an organizing principle. An *adjunction* makes precise the sense in which one functor is the best approximate inverse of another: free constructions are adjoint to forgetful ones, colimits and limits to the diagonal, tensoring to hom. This chapter gives the definition in its three equivalent forms, works the examples, and proves the theorem every adjunction obeys, that right adjoints preserve limits, before asking when an adjoint exists at all.

4.1 Adjoint Functors

The previous chapter recognized free constructions, abelianization, and the initial ring as universal arrows, one for each object of a category (definition 3.1.4). When a functor supplies such an arrow at *every* object in a coherent way, the two functors involved stand in the relationship this section defines. We give the definition first through a natural isomorphism of hom-sets, then repackage it through a unit and a counit, and prove that these two presentations and the object-by-object universal-arrow condition are the same data. The free-monoid construction runs alongside as the guiding example.

Definition 4.1.1: Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors. An *adjunction* from F to G is a natural isomorphism

$$\Phi\colon \mathrm{Hom}_D(F-,-) \cong \mathrm{Hom}_C(-,G-)$$

of functors $C^{\mathrm{op}} \times D \to \mathbf{Set}$; componentwise, for each object c of C and d of D a bijection

$$\Phi_{c,d}\colon \mathrm{Hom}_D(Fc,d) \xrightarrow{\cong} \mathrm{Hom}_C(c,Gd)$$

natural in c and in d. When such an adjunction exists we say F is *left adjoint* to G and G is *right adjoint* to F, written $F \dashv G$. For a morphism $f\colon Fc \to d$, the corresponding morphism $\Phi_{c,d}(f)\colon c \to Gd$ is its *transpose* (or *adjunct*), and likewise $\Phi^{-1}_{c,d}(g)$ is the transpose of $g\colon c \to Gd$.

Naturality is the content of the definition. It says that transposition commutes with composition on both sides: for $u\colon c' \to c$ in C and $v\colon d \to d'$ in D, the transpose of $v \circ f \circ Fu$ is $Gv \circ \Phi_{c,d}(f) \circ u$. Concretely, the naturality square in the C-variable,

$$\begin{tikzcd}
\mathrm{Hom}_D(Fc,d) \arrow[r, "\Phi_{c,d}"] \arrow[d, "(Fu)^*"'] & \mathrm{Hom}_C(c,Gd) \arrow[d, "u^*"] \\
\mathrm{Hom}_D(Fc',d) \arrow[r, "\Phi_{c',d}"'] & \mathrm{Hom}_C(c',Gd)
\end{tikzcd}$$

commutes, where $(Fu)^*$ is precomposition with Fu and u^* is precomposition with u; the analogous square in the D-variable uses postcomposition with v and with Gv. The intuition is compact: a map out of the free object Fc is the same datum as a map into the underlying object Gd. The bijection lets one pass between the world where c lives and the world where d lives without loss, and naturality guarantees the passage respects every morphism in sight.

A first structural consequence, stated here and proved in section 4.3, is that either functor determines the other up to isomorphism: if $F \dashv G$ and $F \dashv G'$ then $G \cong G'$, and dually for left adjoints. The reason is that Gd represents the functor $c \mapsto \mathrm{Hom}_D(Fc,d)$, and a representing object is unique up to unique isomorphism (corollary 2.4.1). An adjoint is thus not extra structure to be chosen but a property that F may or may not have, pinned down when it holds.

Example 4.1.2: The Free Monoid and Underlying Set

Let $U\colon \mathbf{Mon} \to \mathbf{Set}$ send a monoid to its underlying set and a homomorphism to its underlying function, and let $F\colon \mathbf{Set} \to \mathbf{Mon}$ send a set X to the free monoid FX of finite words in the alphabet X under concatenation, with the empty word as identity ([4]). A monoid homomorphism $FX \to M$ is determined by its values on the length-one words, which are the elements of X, and any assignment of those values extends uniquely to a homomorphism because a word is a product of letters. Restriction to length-one words and free extension are mutually inverse, so

$$\mathrm{Hom}_{\mathbf{Mon}}(FX,M) \cong \mathrm{Hom}_{\mathbf{Set}}(X,UM)$$

and the bijection is natural in X and M: precomposing a homomorphism with Ff for $f\colon Y \to X$ corresponds to precomposing the associated function with f, and postcomposing with a homomorphism $M \to N$ corresponds to postcomposing with its underlying function. Thus $F \dashv U$.

The two hom-sets in example 4.1.2 are compared across a change of category, and the bijection is exactly the statement that FX is free on X. Every "free object" theorem in algebra is an instance of this pattern, and the adjunction packages all of them at once. The transpose of the identity of FX under this bijection is the map $X \to UFX$ sending each element to its length-one word, the insertion of generators; the transpose of the identity of UM is a homomorphism $FUM \to M$ evaluating a word of elements of M to their product. These two distinguished maps are the general phenomenon isolated next.

Passing to hom-sets and back is often less convenient than working with two natural transformations, one in each category, from which the whole bijection can be reconstructed. Taking transposes of identity morphisms produces them.

Definition 4.1.3: Unit and Counit

Let $F \dashv G$ with natural isomorphism Φ as in definition 4.1.1. The *unit* of the adjunction is the natural transformation

$$\eta\colon \mathrm{id}_C \Rightarrow GF$$

whose component at c is $\eta_c = \Phi_{c,Fc}(\mathrm{id}_{Fc})\colon c \to GFc$, the transpose of id_{Fc}. The *counit* is the natural transformation

$$\varepsilon\colon FG \Rightarrow \mathrm{id}_D$$

whose component at d is $\varepsilon_d = \Phi^{-1}_{Gd,d}(\mathrm{id}_{Gd})\colon FGd \to d$, the transpose of id_{Gd}.

That η and ε are natural transformations follows from the naturality of Φ: naturality in c, applied to the identity, moves η_c along any morphism $c \to c'$ as a component of GF requires, and the counit argument is dual. The unit $\eta_c\colon c \to GFc$ is a universal arrow from c to G in the sense of definition 3.1.4, as the equivalence theorem below makes precise, and the counit $\varepsilon_d\colon FGd \to d$ is a universal arrow from F to d. In example 4.1.2 the unit is the insertion of generators $X \to UFX$ and the counit $FUM \to M$ multiplies out a word of monoid elements.

The unit and counit recover the full hom-set bijection. The transpose of a morphism is obtained by whiskering with the appropriate one and then applying the functor, as the next proposition records; the formulas are the computational core of everything that follows.

Proposition 4.1.4: Transposition via the Unit and Counit

Let $F \dashv G$ with unit η and counit ε as in definition 4.1.3. For every $f\colon Fc \to d$ in D and every $g\colon c \to Gd$ in C,

$$\Phi_{c,d}(f) = Gf \circ \eta_c\colon c \to Gd \tag{4.1}$$

and

$$\Phi^{-1}_{c,d}(g) = \varepsilon_d \circ Fg\colon Fc \to d \tag{4.2}$$

These two assignments are mutually inverse.

***Proof*:**
Step 1: The formula for Φ. Fix $f\colon Fc \to d$. Naturality of Φ in the D-variable, applied to $f\colon Fc \to d$ regarded as a morphism advancing the second argument, gives a commuting square

$$\begin{tikzcd} \mathrm{Hom}_D(Fc, Fc) \arrow[r, "\Phi_{c,Fc}"] \arrow[d, "f_*"'] & \mathrm{Hom}_C(c, GFc) \arrow[d, "(Gf)_*"] \\ \mathrm{Hom}_D(Fc, d) \arrow[r, "\Phi_{c,d}"'] & \mathrm{Hom}_C(c, Gd) \end{tikzcd}$$

where f_* is postcomposition with f and $(Gf)_*$ is postcomposition with Gf. Chasing id_{Fc} around the square: down then right gives $\Phi_{c,d}(f \circ \mathrm{id}_{Fc}) = \Phi_{c,d}(f)$, while right then down gives $Gf \circ \Phi_{c,Fc}(\mathrm{id}_{Fc}) = Gf \circ \eta_c$ by the definition of the unit. The two are equal, which is (4.1).

Step 2: The formula for Φ^{-1}. The dual computation uses naturality of Φ in the C-variable. Fix $g\colon c \to Gd$ and apply naturality to g advancing the first argument:

$$\begin{tikzcd} \mathrm{Hom}_D(FGd, d) \arrow[r, "\Phi_{Gd,d}"] \arrow[d, "(Fg)^*"'] & \mathrm{Hom}_C(Gd, Gd) \arrow[d, "g^*"] \\ \mathrm{Hom}_D(Fc, d) \arrow[r, "\Phi_{c,d}"'] & \mathrm{Hom}_C(c, Gd) \end{tikzcd}$$

where $(Fg)^*$ and g^* are precomposition with Fg and g. Chase $\varepsilon_d \in \mathrm{Hom}_D(FGd, d)$ around the square. Right then down gives $g^*\big(\Phi_{Gd,d}(\varepsilon_d)\big) = g^*(\mathrm{id}_{Gd}) = g$, since $\Phi_{Gd,d}(\varepsilon_d) = \mathrm{id}_{Gd}$ by definition 4.1.3. Down then right gives $\Phi_{c,d}\big(\varepsilon_d \circ Fg\big)$. Equating the two, $\Phi_{c,d}(\varepsilon_d \circ Fg) = g$, and applying $\Phi^{-1}_{c,d}$ yields $\Phi^{-1}_{c,d}(g) = \varepsilon_d \circ Fg$, which is (4.2).

Step 3: Mutual inverseness. Because $\Phi_{c,d}$ is a bijection with inverse $\Phi^{-1}_{c,d}$, the two assignments (4.1) and (4.2) are mutually inverse by construction. This can also be checked directly from the triangle identities established in theorem 4.1.5, and the two verifications agree, completing the proof.

Formulas (4.1) and (4.2) are how one computes with an adjunction in practice: to transpose a map out of Fc, apply G and precompose with the unit; to transpose back, apply F and postcompose with the counit. Since the two are inverse, feeding one into the other must return the original morphism, and unwinding what that requires of η and ε alone gives a characterization of adjunctions with no mention of hom-sets. That characterization is the equivalence theorem.

The theorem collects three descriptions of the same situation. The first is the hom-set isomorphism of definition 4.1.1. The second replaces it by the unit and counit subject to two equations, the *triangle identities*, which require only that transposing a map and transposing it back be the identity. The third is the object-by-object universal-arrow condition, tying the chapter back to definition 3.1.4. Their equivalence is what lets one build, recognize, and use adjunctions from whichever description is at hand.

Theorem 4.1.5: Equivalent Formulations of an Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors. The following data are equivalent, each determining the others.

1. A natural isomorphism $\Phi\colon \mathrm{Hom}_D(F-,-) \cong \mathrm{Hom}_C(-,G-)$, that is, an adjunction $F \dashv G$.
2. Natural transformations $\eta\colon \mathrm{id}_C \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ satisfying the *triangle identities*
$$(G\varepsilon)(\eta G) = \mathrm{id}_G, \qquad (\varepsilon F)(F\eta) = \mathrm{id}_F \tag{4.3}$$
as natural transformations $G \Rightarrow G$ and $F \Rightarrow F$.
3. A natural transformation $\eta\colon \mathrm{id}_C \Rightarrow GF$ whose component $\eta_c\colon c \to GFc$ is a universal arrow from c to G for every object c of C (definition 3.1.4).

***Proof*:**
We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

Step 1: From the hom-set isomorphism to the triangle identities. Assume (1), and define η and ε as in definition 4.1.3. They are natural transformations, as noted after that definition. The triangle identities are the componentwise statements
$$G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}, \qquad \varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc} \tag{4.4}$$
for all c and d. For the first, apply (4.1) of proposition 4.1.4 to the morphism $\varepsilon_d\colon FGd \to d$ with source object Gd: its transpose is $\Phi_{Gd,d}(\varepsilon_d) = G\varepsilon_d \circ \eta_{Gd}$. But $\varepsilon_d = \Phi^{-1}_{Gd,d}(\mathrm{id}_{Gd})$ by definition 4.1.3, so $\Phi_{Gd,d}(\varepsilon_d) = \mathrm{id}_{Gd}$, giving $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$. For the second, apply (4.2) to $\eta_c\colon c \to GFc$ with target object Fc: its transpose is $\Phi^{-1}_{c,Fc}(\eta_c) = \varepsilon_{Fc} \circ F\eta_c$, and $\eta_c = \Phi_{c,Fc}(\mathrm{id}_{Fc})$, so this transpose is id_{Fc}. Thus (4.4) holds, which is (4.5).

Step 2: From the triangle identities to universal arrows. Assume (2). Fix an object c; we show $\eta_c\colon c \to GFc$ is a universal arrow from c to G. Given any object d of D and any morphism $g\colon c \to Gd$ in C, we must produce a unique $f\colon Fc \to d$ with $Gf \circ \eta_c = g$.

Existence. Set $f = \varepsilon_d \circ Fg\colon Fc \to d$. Then
$$Gf \circ \eta_c = G\varepsilon_d \circ GFg \circ \eta_c$$
Naturality of η applied to $g\colon c \to Gd$ gives $GFg \circ \eta_c = \eta_{Gd} \circ g$, so
$$Gf \circ \eta_c = G\varepsilon_d \circ \eta_{Gd} \circ g = (G\varepsilon_d \circ \eta_{Gd}) \circ g = \mathrm{id}_{Gd} \circ g = g$$
using the first triangle identity in (4.4). This f solves the factorization problem.

Uniqueness. Suppose $f'\colon Fc \to d$ also satisfies $Gf' \circ \eta_c = g$. Then
$$\varepsilon_d \circ F(Gf' \circ \eta_c) = \varepsilon_d \circ Fg$$
The left side equals $\varepsilon_d \circ FGf' \circ F\eta_c$. Naturality of ε applied to $f'\colon Fc \to d$ gives $\varepsilon_d \circ FGf' = f' \circ \varepsilon_{Fc}$, so the left side is $f' \circ \varepsilon_{Fc} \circ F\eta_c = f' \circ \mathrm{id}_{Fc} = f'$ by the second triangle identity. The right side is our chosen f. Hence $f' = f$, and the universal arrow is unique. Since c was arbitrary, η_c is universal for every c, which is (3).

Step 3: From universal arrows to the hom-set isomorphism. Assume (3): $\eta\colon \mathrm{id}_C \Rightarrow GF$ is a natural transformation whose every component $\eta_c\colon c \to GFc$ is a universal arrow from c to G. Fix c and d and define

$$\Phi_{c,d}\colon \mathrm{Hom}_D(Fc, d) \to \mathrm{Hom}_C(c, Gd), \qquad \Phi_{c,d}(f) = Gf \circ \eta_c$$

The universal property of η_c says exactly that for each $g\colon c \to Gd$ there is a unique $f\colon Fc \to d$ with $Gf \circ \eta_c = g$, that is, $\Phi_{c,d}$ is a bijection. It remains to check naturality in c and d. For $v\colon d \to d'$, both $\Phi_{c,d'}(v \circ f)$ and $Gv \circ \Phi_{c,d}(f)$ equal $Gv \circ Gf \circ \eta_c$, since G is a functor; this is naturality in d. For $u\colon c' \to c$, naturality of η gives $GFu \circ \eta_{c'} = \eta_c \circ u$, whence

$$\Phi_{c',d}(f \circ Fu) = G(f \circ Fu) \circ \eta_{c'} = Gf \circ GFu \circ \eta_{c'} = Gf \circ \eta_c \circ u = \Phi_{c,d}(f) \circ u$$

which is naturality in c. Thus Φ is a natural isomorphism and $F \dashv G$, returning to (1) and closing the cycle, completing the proof.

The theorem is the working definition of an adjunction: one verifies whichever of the three is easiest and gains the other two. Presentation (1) is the cleanest to state and the one that makes symmetry between F and G visible. Presentation (2) is the most convenient in proofs, since the triangle identities are two equations to check rather than a family of bijections, and it survives into contexts where hom-sets are not available. Presentation (3) is the bridge from the last chapter: an adjunction is precisely a functorial choice of universal arrow, one at each object, so the free monoid, abelianization, and the initial ring of section 3.1 were adjunctions seen one object at a time. There is a symmetric fourth form, dual to (3), asking that each counit component $\varepsilon_d\colon FGd \to d$ be a universal arrow from F to d; it follows by applying the theorem in the opposite categories, since $F \dashv G$ in C, D is $G^{\mathrm{op}} \dashv F^{\mathrm{op}}$ in $D^{\mathrm{op}}, C^{\mathrm{op}}$ (box 1.3.3).

The name "triangle identity" comes from the diagrams the equations in (4.4) assert commute. The first says the outer path in

$$\begin{tikzcd} Gd \arrow[r, "\eta_{Gd}"] \arrow[dr, "\mathrm{id}_{Gd}"'] & GFGd \arrow[d, "G\varepsilon_d"] \\ & Gd \end{tikzcd}$$

is the identity, and the second says the same of

$$\begin{tikzcd} Fc \arrow[r, "F\eta_c"] \arrow[dr, "\mathrm{id}_{Fc}"'] & FGFc \arrow[d, "\varepsilon_{Fc}"] \\ & Fc \end{tikzcd}$$

Read aloud, they say that inserting generators into Gd and then evaluating, or building the free object on Fc and then collapsing it, returns where one started. In example 4.1.2 the first identity says: take a set of generators inside a monoid M (the map η_{UM}), form the free monoid on them, multiply each word out (ε_M), and the underlying-set effect is the identity on UM. The equations look slight, but by theorem 4.1.5 they carry the entire adjunction.

4.2 Examples of Adjunctions

The definition and its three forms are in place; the work of this section is to show how much of the mathematics already met is an adjunction in disguise. Four families cover most of the recurring instances: free constructions adjoint to forgetful functors, tensoring adjoint to hom, colimits and limits adjoint to the diagonal, and Galois connections between posets. Each is presented as a hom-set bijection $\mathrm{Hom}_D(Fc, d) \cong \mathrm{Hom}_C(c, Gd)$ (definition 4.1.1), natural in both variables, with the naturality left to the one-line check it always reduces to. The through-line stated at the definition returns here in its standard form: left adjoints are free constructions.

The first family is the one the definition was built to capture. A forgetful functor discards structure; its left adjoint builds the freest structure on the bare data it was handed. The universal property of a free construction is exactly a hom-set bijection, so each free-forgetful pair is an adjunction with no extra argument to make.

Example 4.2.1: Free and Forgetful Adjunctions
Across the standing cast, each free construction is left adjoint to the forgetful functor it inverts.

1. *Free group $\dashv$ forgetful.* Let $U\colon \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor (example 1.4.2) and $F\colon \mathbf{Set} \to \mathbf{Grp}$ the free-group functor. The universal property of the free group FX says that a set function $X \to UG$ extends uniquely to a homomorphism $FX \to G$, and every homomorphism $FX \to G$ restricts to such a function on generators ([4]). These two operations are mutually inverse, giving a bijection
$$\mathrm{Hom}_{\mathbf{Grp}}(FX, G) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UG)$$
so $F \dashv U$. Naturality in X and in G is the statement that restriction to generators commutes with precomposing a function on the left and postcomposing a homomorphism on the right, which holds because both sides are "evaluate on generators, then extend."

2. *Free R-module $\dashv$ forgetful.* For a ring R, let $U\colon \mathbf{Mod}_R \to \mathbf{Set}$ forget the module structure and F send a set X to the free module $R^{(X)}$ on X. The universal property of the free module ([4]) gives, for every module M,
$$\mathrm{Hom}_{\mathbf{Mod}_R}(R^{(X)}, M) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UM)$$
a function $X \to UM$ extending uniquely to an R-linear map on the basis X, so $F \dashv U$. The same check applies with $\mathbf{Vect}_k$ in place of $\mathbf{Mod}_R$: every vector space is free, so the left adjoint here sends a set to the space with that basis.

3. *Abelianization $\dashv$ inclusion.* Let $\iota\colon \mathbf{Ab} \to \mathbf{Grp}$ be the inclusion of abelian groups among all groups and $\mathrm{ab}\colon \mathbf{Grp} \to \mathbf{Ab}$ send G to its abelianization $G^{\mathrm{ab}} = G/[G, G]$. A homomorphism from G to an abelian group A kills every commutator, hence factors uniquely through the quotient $G \to G^{\mathrm{ab}}$ ([4]), which is the bijection
$$\mathrm{Hom}_{\mathbf{Ab}}(G^{\mathrm{ab}}, A) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, \iota A)$$
so $\mathrm{ab} \dashv \iota$. Here the "structure" the right adjoint remembers is commutativity, and the left adjoint imposes it as freely as possible, by quotienting out only what commutativity forces.

4. *Discrete ⊣ forgetful ⊣ indiscrete.* Let $U\colon \mathbf{Top} \to \mathbf{Set}$ forget the topology. It has adjoints on both sides. The *discrete* functor $\mathrm{disc}\colon \mathbf{Set} \to \mathbf{Top}$ equips a set with its discrete topology, in which every subset is open; the *indiscrete* functor indisc equips it with the indiscrete topology, whose only opens are $\varnothing$ and the whole set. Every function out of a discrete space is continuous and every function into an indiscrete space is continuous, so

$$\mathrm{Hom}_{\mathbf{Top}}(\mathrm{disc}\, X, Y) \cong \mathrm{Hom}_{\mathbf{Set}}(X, UY), \qquad \mathrm{Hom}_{\mathbf{Set}}(UY, X) \cong \mathrm{Hom}_{\mathbf{Top}}(Y, \mathrm{indisc}\, X)$$

The first bijection makes $\mathrm{disc} \dashv U$; the second makes $U \dashv \mathrm{indisc}$. Thus U is at once a right adjoint and a left adjoint, so it preserves both limits and colimits, a fact the next section will read off directly.

The first three items instantiate one pattern: the forgetful functor is a right adjoint and its left adjoint is the free construction, which is why free objects are recovered by the recipe "the value of the left adjoint on a set." The last item is different. A forgetful functor need not sit on only one side of an adjunction; when the structure being forgotten can be added in a freest and a cofreest way, it acquires adjoints on both. The point to carry forward is that a left adjoint is a free construction and a right adjoint is a forgetful or cofree one, with the hom-set bijection the precise content of "free."

The second family trades one bifunctor for two others. Fixing one input of a product and asking to map out of it is the same as mapping into a function object; this is currying, and in its module form it is the tensor-hom adjunction that pervades homological algebra. The prototype is in **Set**, where the function object is a plain set of functions.

Example 4.2.2: The Tensor-Hom Adjunction

Fix a set B. Currying converts a function of two variables into a function valued in functions: a map $A \times B \to C$ is the same as a map $A \to C^B$ sending a to the function $b \mapsto f(a,b)$, where $C^B = \mathrm{Hom}_{\mathbf{Set}}(B, C)$ is the set of functions $B \to C$. This assignment is a bijection, natural in A and C,

$$\mathrm{Hom}_{\mathbf{Set}}(A \times B, C) \cong \mathrm{Hom}_{\mathbf{Set}}(A, C^B)$$

so the functor $(-) \times B$ is left adjoint to $(-)^B = \mathrm{Hom}_{\mathbf{Set}}(B, -)$. This is the *cartesian closure* of **Set**: each product functor has the corresponding function-object functor as a right adjoint.

The module version replaces the product by the tensor product. Let R be a commutative ring and fix an R-module B. The defining property of the tensor product ([4]) is that R-bilinear maps $A \times B \to C$ correspond to R-linear maps $A \otimes_R B \to C$; currying the second variable identifies a bilinear map with an R-linear map $A \to \mathrm{Hom}_R(B, C)$, giving the bijection

$$\mathrm{Hom}_R(A \otimes_R B, C) \cong \mathrm{Hom}_R(A, \mathrm{Hom}_R(B, C))$$

natural in A and C. Thus $(-) \otimes_R B$ is left adjoint to $\mathrm{Hom}_R(B, -)$.

The tensor-hom adjunction is why tensoring and hom behave as complementary halves of the same structure: the left adjoint $(-) \otimes_R B$ and the right adjoint $\mathrm{Hom}_R(B, -)$ are tied by a natural bijection, and the preservation theorem of the next section will explain from this alone why tensoring is right exact and hom left exact. The **Set** prototype is the same phenomenon with $\times$ for $\otimes$ and the bare function set for the internal hom; a category carrying such an adjunction for each fixed object is called closed, a notion the monoidal chapter returns to.

The third family exhibits limits and colimits themselves as adjoints. A single diagram-valued functor, the diagonal, has the limit on one side and the colimit on the other, so the two constructions of the previous chapter are the two adjoints of one functor.

Recall the diagonal functor $\Delta\colon \mathbf{C} \to [J, \mathbf{C}]$ sending an object X to the constant diagram at X, the diagram taking every object of J to X and every morphism to 1_X (section 3.2). A natural transformation $\Delta X \Rightarrow D$ is exactly a cone over D with apex X, and a natural transformation $D \Rightarrow \Delta X$ is exactly a cocone under D with nadir X. This identification turns the universal properties of the limit and the colimit into hom-set bijections.

Example 4.2.3: Limits and Colimits as Adjoints to the Diagonal

Let J be a small category and suppose $\mathbf{C}$ has all limits and all colimits of shape J, so that $\lim\colon [J, \mathbf{C}] \to \mathbf{C}$ and $\operatorname{colim}\colon [J, \mathbf{C}] \to \mathbf{C}$ are defined. Then

$$\operatorname{colim} \;\dashv\; \Delta \;\dashv\; \lim$$

the colimit is left adjoint to the diagonal and the limit is right adjoint to it.

The two adjunctions are dual, and each is immediate once the naturality bijection is unwound. Only the $\Delta \dashv \lim$ side is proved; the $\operatorname{colim} \dashv \Delta$ side is the same argument in $\mathbf{C}^{\mathrm{op}}$ by the duality principle (box 1.3.3), since a colimit in $\mathbf{C}$ is a limit in $\mathbf{C}^{\mathrm{op}}$ (proposition 3.3.2) and Δ is self-opposite.

***Proof*:**

Fix an object X of $\mathbf{C}$ and a diagram D in $[J, \mathbf{C}]$. A morphism in the functor category $\mathrm{Hom}_{[J,\mathbf{C}]}(\Delta X, D)$ is a natural transformation $\alpha\colon \Delta X \Rightarrow D$ (definition 1.5.4): a family of components $\alpha_j\colon X \to Dj$, one for each object j of J, such that for every morphism $u\colon j \to j'$ of J the naturality square commutes. Since ΔX sends u to 1_X, that square reads $Du \circ \alpha_j = \alpha_{j'} \circ 1_X = \alpha_{j'}$, which is exactly the cone condition (definition 3.2.1). A natural transformation $\Delta X \Rightarrow D$ is therefore the same data as a cone over D with apex X.

By the universal property of the limit, a cone over D with apex X corresponds to a unique morphism $X \to \lim D$ commuting with the projections, and every morphism $X \to \lim D$ arises this way by composing with the limit cone. This correspondence is a bijection

$$\mathrm{Hom}_{\mathbf{C}}(X, \lim D) \cong \mathrm{Hom}_{[J,\mathbf{C}]}(\Delta X, D)$$

It is natural in X: precomposing a morphism $X' \to X$ transports a cone with apex X to one with apex X' by the same precomposition on each component, and the induced map to $\lim D$ transforms accordingly. It is natural in D: a morphism of diagrams $D \Rightarrow D'$ postcomposes each cone component and the induced maps to the limits commute by the universal property. The bijection, natural in both variables, is the defining data of an adjunction (definition 4.1.1), so $\Delta \dashv \lim$. The dual identity $\operatorname{colim} \dashv \Delta$ follows by applying this to $\mathbf{C}^{\mathrm{op}}$, completing the proof.

This recovers the limits and colimits of the previous chapter as the two adjoints of a single functor, and it explains a fact used repeatedly downstream: the limit functor is a right adjoint, so it preserves limits. The general preservation theorem is deferred to the next section, but the shape of the consequence is already visible, that a limit of limits may be computed in either order because the outer limit is a right adjoint. Dually the colimit functor is a left adjoint and preserves colimits. The diagonal, sitting between its two adjoints, is the categorical device that makes "take the limit" and "take the colimit" into functors with formal properties rather than case-by-case constructions.

The last family lives one dimension down, where the categories are posets and a functor is a monotone map. An adjunction between two posets is a pair of monotone maps tied by the single inequality that a hom-set, being either empty or a point, can express.

A poset $(P, \leq)$ is a category with one morphism $p \to q$ when $p \leq q$ and none otherwise (example 1.4.2 treats such thin categories). A functor between posets is a monotone map, and a hom-set $P(p, q)$ has at most one element, present exactly when $p \leq q$. The hom-set bijection defining an adjunction therefore collapses to a biconditional between inequalities.

Definition 4.2.4: Galois Connection

Let P and Q be posets, viewed as categories. A *(monotone) Galois connection* between P and Q is an adjunction $f \dashv g$ between them: a pair of monotone maps $f\colon P \to Q$ and $g\colon Q \to P$ such that for all $p \in P$ and $q \in Q$,

$$f(p) \leq q \quad \Longleftrightarrow \quad p \leq g(q)$$

The map f is the *left adjoint* (or lower adjoint) and g the *right adjoint* (or upper adjoint).

The biconditional is the poset case of the natural bijection $\mathrm{Hom}_Q(fp, q) \cong \mathrm{Hom}_P(p, gq)$: each hom-set is either empty or a single point, so the bijection between them is precisely the equivalence of the two inequalities, and naturality is automatic because there is nothing to compare beyond presence or absence of a morphism. A Galois connection is an adjunction with the analytic content stripped away, leaving the order-theoretic skeleton visible.

A single monotone connection already produces a closure operator, and the classical Galois correspondence is the archetype once one factor is passed to its opposite.

Example 4.2.5: The Galois Correspondence

Let L/k be a finite Galois extension with Galois group $G = \mathrm{Gal}(L/k)$. Write $\mathcal{S}$ for the poset of subgroups of G ordered by inclusion and $\mathcal{F}$ for the poset of intermediate fields $k \subseteq E \subseteq L$ ordered by inclusion. The two maps

$$\mathrm{Fix}\colon \mathcal{S} \to \mathcal{F}, \quad H \mapsto L^H, \qquad \mathrm{Gal}\colon \mathcal{F} \to \mathcal{S}, \quad E \mapsto \mathrm{Gal}(L/E)$$

each reverse inclusions: a larger subgroup fixes a smaller field, and a larger field is fixed by a smaller subgroup ([4]). They satisfy

$$H \subseteq \mathrm{Gal}(L/E) \quad \Longleftrightarrow \quad E \subseteq L^H$$

both sides asserting that every element of H fixes every element of E. Because each map reverses order, the pair is an *antitone* Galois connection, which is a monotone one after replacing $\mathcal{F}$ by its opposite $\mathcal{F}^{\mathrm{op}}$: then $\mathrm{Fix}\colon \mathcal{S} \to \mathcal{F}^{\mathrm{op}}$ and $\mathrm{Gal}\colon \mathcal{F}^{\mathrm{op}} \to \mathcal{S}$ are monotone and the biconditional above is definition 4.2.4. The fundamental theorem of Galois theory ([4]) states that here both composites are the identity, so the connection is an order-reversing bijection between subgroups and intermediate fields.

The example carries two lessons about Galois connections in general. First, an antitone connection is an adjunction only after an opposite category is introduced; the raw pair of order-reversing maps becomes a genuine $f \dashv g$ once one side is flipped, which is the categorical account of why order-reversing correspondences are as natural as order-preserving ones. Second, even when the composites are not the identity, a Galois connection $f \dashv g$ always yields a *closure operator* $gf\colon P \to P$: it is monotone, satisfies $p \le gf(p)$, and is idempotent, $gfgf = gf$. The Galois correspondence is the degenerate case where the closure is trivial because gf is the identity; in a general connection the closure operator gf records the "saturation" a subset acquires, exactly as the double annihilator $(-)^{\perp\perp}$ closes a subspace W to $W^{\perp\perp}$. Every adjunction produces such a monad on the domain of its left adjoint, of which the closure operator is the poset instance, a thread the chapter on monads develops.

4.3 Units, Counits, and Preservation

The three formulations of an adjunction are on the table (theorem 4.1.5), and the examples of the previous section show how often one is in play. This section pays off the unit/counit formulation. The unit and counit are tied together by two equations, the triangle identities, and those equations are exactly the data that lets one reconstruct the hom-set bijection. Once the identities are in hand, two structural facts follow with almost no work: an adjoint, when it exists, is determined up to a unique natural isomorphism, and a right adjoint carries every limit that exists to a limit. The second of these, right adjoints preserve limits, explains at one stroke why so many limits in algebra and topology are computed on underlying sets.

Throughout, $F \dashv G$ is an adjunction with $F\colon \mathbf{C} \to \mathbf{D}$ and $G\colon \mathbf{D} \to \mathbf{C}$, unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$, in the notation of definition 4.1.3.

The bijection $\mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, Gd)$ moves a morphism across the adjunction; call the image of $g\colon Fc \to d$ its *transpose* $g^\flat\colon c \to Gd$, and the image of $f\colon c \to Gd$ its transpose $f^\sharp\colon Fc \to d$. The two operations are mutually inverse, so transposing twice returns the original morphism. In terms of unit and counit, the transpose of $g\colon Fc \to d$ is $Gg \circ \eta_c$ and the transpose of $f\colon c \to Gd$ is $\varepsilon_d \circ Ff$; this is how theorem 4.1.5 recovered the bijection from the two natural transformations. The content of the next theorem is that the round trip really is the identity, and that this fact is captured by two equations relating η and ε.

Theorem 4.3.1: Triangle Identities

Let $F \dashv G$ with unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$. Then the two whiskered composites

$$G \overset{\eta G}{\Longrightarrow} GFG \overset{G\varepsilon}{\Longrightarrow} G, \qquad F \overset{F\eta}{\Longrightarrow} FGF \overset{\varepsilon F}{\Longrightarrow} F$$

are the respective identity natural transformations:

$$(G\varepsilon)(\eta G) = \mathrm{id}_G, \qquad (\varepsilon F)(F\eta) = \mathrm{id}_F \tag{4.5}$$

Conversely, given functors F, G and natural transformations $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$, $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathbf{D}}$ satisfying (4.5), the assignment $g \mapsto Gg \circ \eta_c$ is a bijection $\mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, Gd)$ natural in c and d, so η and ε present an adjunction $F \dashv G$.

The two equations (4.5) are named for the shape of the diagrams that express them. Reading the whiskered composites ηG, $G\varepsilon$ and $F\eta$, εF as edges, each identity says that a triangle commutes:

$$\begin{tikzcd} G \arrow[r, Rightarrow, "\eta G"] \arrow[dr, Rightarrow, "\mathrm{id}_G"'] & GFG \arrow[d, Rightarrow, "G\varepsilon"] \\ & G \end{tikzcd} \qquad \begin{tikzcd} F \arrow[r, Rightarrow, "F\eta"] \arrow[dr, Rightarrow, "\mathrm{id}_F"'] & FGF \arrow[d, Rightarrow, "\varepsilon F"] \\ & F \end{tikzcd}$$

Componentwise at objects $d \in \mathbf{D}$ and $c \in \mathbf{C}$, these read $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ and $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$.

***Proof*:**
The forward direction is the statement that transposing twice returns the original morphism, read off at the two extreme objects.

Step 1: The identities are the two double-transpose calculations. Fix $c \in \mathbf{C}$. The morphism $\eta_c : c \to GFc$ is the transpose of id_{Fc}, by the description of the transpose in theorem 4.1.5. Transposing back sends $f : c \to GFc$ to $\varepsilon_{Fc} \circ Ff$; applied to $f = \eta_c$ this gives $\varepsilon_{Fc} \circ F\eta_c$. Since transposing twice is the identity, $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$. As c was arbitrary, $(\varepsilon F)(F\eta) = \mathrm{id}_F$. Dually, fix $d \in \mathbf{D}$: the morphism $\varepsilon_d : FGd \to d$ is the transpose of id_{Gd}, and transposing forward sends $g : FGd \to d$ to $Gg \circ \eta_{Gd}$; applied to $g = \varepsilon_d$ this gives $G\varepsilon_d \circ \eta_{Gd}$, equal to id_{Gd} because transposing twice is the identity. As d was arbitrary, $(G\varepsilon)(\eta G) = \mathrm{id}_G$.

Step 2: The converse. Assume now only that η, ε are natural transformations satisfying (4.5). Define

$$\Phi_{c,d} : \mathrm{Hom}_{\mathbf{D}}(Fc, d) \to \mathrm{Hom}_{\mathbf{C}}(c, Gd), \qquad g \mapsto Gg \circ \eta_c$$

and

$$\Psi_{c,d} : \mathrm{Hom}_{\mathbf{C}}(c, Gd) \to \mathrm{Hom}_{\mathbf{D}}(Fc, d), \qquad f \mapsto \varepsilon_d \circ Ff$$

Both are natural in c and d. For Φ, naturality in d is functoriality of G: for $k : d \to d'$, $\Phi_{c,d'}(kg) = G(kg)\,\eta_c = Gk\,(Gg\,\eta_c) = Gk \circ \Phi_{c,d}(g)$; naturality in c is naturality of η: for $h : c' \to c$, $\Phi_{c',d}(g\,Fh) = G(g\,Fh)\,\eta_{c'} = Gg\,(GFh\,\eta_{c'}) = Gg\,(\eta_c\,h) = \Phi_{c,d}(g) \circ h$, the middle equality by the naturality square of η at h. The naturality of Ψ is dual, using functoriality of F and naturality of ε. It remains to check that Φ and Ψ are mutually inverse, and here the triangle identities enter.

Take $g : Fc \to d$. Then

$$\Psi_{c,d}(\Phi_{c,d}(g)) = \varepsilon_d \circ F(Gg \circ \eta_c) = \varepsilon_d \circ FGg \circ F\eta_c$$

Naturality of ε applied to $g : Fc \to d$ gives $\varepsilon_d \circ FGg = g \circ \varepsilon_{Fc}$, so the right side equals $g \circ \varepsilon_{Fc} \circ F\eta_c$. By the second identity in (4.5), $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$, hence $\Psi_{c,d}(\Phi_{c,d}(g)) = g$.

Take $f : c \to Gd$. Then

$$\Phi_{c,d}(\Psi_{c,d}(f)) = G(\varepsilon_d \circ Ff) \circ \eta_c = G\varepsilon_d \circ GFf \circ \eta_c$$

Naturality of η applied to $f : c \to Gd$ gives $GFf \circ \eta_c = \eta_{Gd} \circ f$, so the right side equals $G\varepsilon_d \circ \eta_{Gd} \circ f$. By the first identity in (4.5), $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$, hence $\Phi_{c,d}(\Psi_{c,d}(f)) = f$.

Thus Φ is a natural bijection with inverse Ψ, which is the hom-set formulation of $F \dashv G$ (definition 4.1.1), completing the proof.

The identities encode a single fact about the two transpose operations: they are inverse. Everything about how F and G interact through the adjunction, including the naturality of the correspondence, is compressed into these two equations. This makes the unit/counit formulation the computational one, since checking an adjunction reduces to producing η, ε and verifying two triangle equations, with no bijection to construct by hand.

Example 4.3.2: The Free Group Triangle Identities
Let $F\colon \mathbf{Set} \to \mathbf{Grp}$ be the free-group functor and $U\colon \mathbf{Grp} \to \mathbf{Set}$ the forgetful functor, $F \dashv U$ (example 4.1.2). The unit $\eta_S\colon S \to UFS$ includes a set S as the generators of the free group FS. The counit $\varepsilon_G\colon FUG \to G$ is the homomorphism from the free group on the underlying set of G that sends each generator g to the element $g \in G$; it evaluates a formal word in the elements of G as an actual product in G.

The first identity $(U\varepsilon)(\eta U) = \mathrm{id}_U$ reads, at a group G, as follows. Start with an element $g \in UG$. The unit η_{UG} views g as a generator of the free group FUG, that is, as the length-one word g. Then $U\varepsilon_G$ evaluates that word in G, returning the element g. The composite is the identity on UG, as the identity requires.

The second identity $(\varepsilon F)(F\eta) = \mathrm{id}_F$ reads, at a set S, as follows. A reduced word $w = s_1^{\pm 1} \cdots s_n^{\pm 1}$ in FS is sent by $F\eta_S$ to the same formal word, now regarded as living in $FUFS$ (each generator s_i first viewed as a generator of FS, then as an element of UFS, then as a generator of $FUFS$). Applying ε_{FS} evaluates that word back in FS, recovering w. The composite is the identity on FS. The two identities are the two ways "generator" and "evaluate" cancel.

An adjunction determines its right adjoint completely: if a functor has a left adjoint at all, the resulting right adjoint is fixed up to a unique natural isomorphism. This is what licenses the article "the" in "the right adjoint". The proof is a direct application of the Yoneda embedding (definition 2.3.1): a functor value Gd is pinned down by the functor it represents, $\mathrm{Hom}_{\mathbf{C}}(-, Gd)$, and the adjunction identifies that representable with one built from F alone.

Proposition 4.3.3: Uniqueness of Adjoints

Let $F\colon \mathbf{C} \to \mathbf{D}$. If $F \dashv G$ and $F \dashv G'$, then there is a unique natural isomorphism $\theta\colon G \Rightarrow G'$ compatible with the adjunctions, in the sense that it identifies the two transpose bijections. Dually, if $G \dashv F$ and $G' \dashv F$, the two left adjoints G, G' are isomorphic by a unique such natural isomorphism.

***Proof*:**
Fix $d \in \mathbf{D}$. The two adjunctions give, for every $c \in \mathbf{C}$, bijections

$$\mathrm{Hom}_{\mathbf{C}}(c, Gd) \cong \mathrm{Hom}_{\mathbf{D}}(Fc, d) \cong \mathrm{Hom}_{\mathbf{C}}(c, G'd)$$

natural in c, the first from $F \dashv G$ and the second from $F \dashv G'$. Their composite is a natural isomorphism

$$\mathrm{Hom}_{\mathbf{C}}(-, Gd) \cong \mathrm{Hom}_{\mathbf{C}}(-, G'd)$$

of functors $\mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$. The Yoneda embedding $\mathbf{y}\colon \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$ is fully faithful (definition 2.3.1), so this isomorphism of representables is $\mathbf{y}(\theta_d)$ for a unique isomorphism $\theta_d\colon Gd \to G'd$ in $\mathbf{C}$.

It remains to see that the θ_d assemble into a natural transformation $\theta\colon G \Rightarrow G'$. Let $k\colon d \to d'$ in $\mathbf{D}$. Both composite bijections are natural in d as well as in c, since the transpose correspondences are natural in both variables (definition 4.1.1); consequently the square of representables

$$\begin{tikzcd} \mathrm{Hom}_{\mathbf{C}}(-, Gd) \arrow[r, "\cong"] \arrow[d, "\mathrm{Hom}(-,Gk)"'] & \mathrm{Hom}_{\mathbf{C}}(-, G'd) \arrow[d, "\mathrm{Hom}(-,G'k)"] \\ \mathrm{Hom}_{\mathbf{C}}(-, Gd') \arrow[r, "\cong"] & \mathrm{Hom}_{\mathbf{C}}(-, G'd') \end{tikzcd}$$

commutes. Applying full faithfulness of $\mathbf{y}$ to this commuting square of representables yields $\theta_{d'} \circ Gk = G'k \circ \theta_d$, which is naturality of θ. Uniqueness of each θ_d, hence of θ, is the injectivity of $\mathbf{y}$ on morphisms. The compatibility with the transpose bijections holds by construction, since θ was read off from them.

For the dual statement, if $G \dashv F$ and $G' \dashv F$, then for every c the bijections $\mathrm{Hom}_{\mathbf{D}}(Gc, -) \cong \mathrm{Hom}_{\mathbf{C}}(c, F-) \cong \mathrm{Hom}_{\mathbf{D}}(G'c, -)$ are natural, so the covariant representables $\mathrm{Hom}_{\mathbf{D}}(Gc, -)$ and $\mathrm{Hom}_{\mathbf{D}}(G'c, -)$ are isomorphic; the covariant Yoneda embedding (definition 2.3.1, applied in $\mathbf{D}^{\mathrm{op}}$) delivers a unique natural isomorphism $Gc \cong G'c$ by the same argument, as we sought to show.

Two constructions each solving the same universal problem cannot differ in any observable way, and the isomorphism between them is forced rather than chosen. This is the categorical form of a phenomenon seen repeatedly in chapter 4: the free group on a set, the tensor product of two modules, the coproduct of a family of objects are each unique up to unique isomorphism, and here the reason is uniform, since all three are adjoint (or universal) constructions and Yoneda pins down what an adjoint represents. From now on we speak of *the* left adjoint and *the* right adjoint of a functor, when one exists.

The structural payoff of the section is that right adjoints preserve limits. The mechanism is the same Yoneda argument, now run against a diagram: a right adjoint commutes with a limit because its defining bijection turns the limit-detecting property of Gd into the limit-detecting property inherited through F from representables, which already preserve limits (proposition 3.4.7). The abbreviation RAPL, "right adjoints preserve limits", is standard.

Recall from definition 3.4.6 what preservation means: a functor $G\colon \mathbf{D} \to \mathbf{C}$ *preserves* the limit of $D\colon \mathbf{J} \to \mathbf{D}$ if, whenever $\lim D$ exists with limit cone $(\lambda_j\colon \lim D \to Dj)_j$, the image cone $(G\lambda_j\colon G(\lim D) \to G(Dj))_j$ is a limit cone for $G \circ D$, so that $G(\lim D) \cong \lim(G \circ D)$ canonically.

Theorem 4.3.4: Right Adjoints Preserve Limits

Let $F \dashv G$ with $G\colon \mathbf{D} \to \mathbf{C}$, and let $D\colon \mathbf{J} \to \mathbf{D}$ be a diagram whose limit $\lim D$ exists in $\mathbf{D}$. Then G preserves this limit: the canonical comparison morphism $G(\lim D) \to \lim(G \circ D)$ is an isomorphism, so

$$G(\lim D) \cong \lim_{j \in \mathbf{J}} G(Dj)$$

Dually, a left adjoint preserves all colimits that exist in its domain.

***Proof*:**
Write $L = \lim D$ in $\mathbf{D}$. The claim is that GL, with the cone $(G\lambda_j)_j$, is a limit of $G \circ D$ in $\mathbf{C}$. By the universal property of a limit (definition 3.2.1) this is equivalent to the statement that for

every $c \in \mathbf{C}$ the map

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \to \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj)), \qquad f \mapsto (G\lambda_j \circ f)_j$$

is a bijection, where the limit on the right is the limit in **Set** of the diagram $j \mapsto \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$, that is, the set of cones over $G \circ D$ with apex c. Fix c and compute this hom-set through a chain of bijections, each natural in c.

Step 1: Cross the adjunction. The adjunction $F \dashv G$ gives

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \cong \mathrm{Hom}_{\mathbf{D}}(Fc, L)$$

natural in c (definition 4.1.1).

Step 2: Use that representables preserve limits. Since $L = \lim D$ in $\mathbf{D}$, the representable $\mathrm{Hom}_{\mathbf{D}}(Fc, -)$ preserves it (proposition 3.4.7):

$$\mathrm{Hom}_{\mathbf{D}}(Fc, L) \cong \lim_j \mathrm{Hom}_{\mathbf{D}}(Fc, Dj)$$

This is the step where the hypothesis that $\lim D$ exists is used: the limit-preservation of G is pulled back from the limit-preservation of a representable.

Step 3: Cross the adjunction back, inside the limit. Applying $F \dashv G$ objectwise in j, and using that these bijections are natural in j so they assemble into an isomorphism of the diagrams whose limits are being taken,

$$\lim_j \mathrm{Hom}_{\mathbf{D}}(Fc, Dj) \cong \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$$

Step 4: Reassemble. Composing the three bijections gives an isomorphism

$$\mathrm{Hom}_{\mathbf{C}}(c, GL) \cong \lim_j \mathrm{Hom}_{\mathbf{C}}(c, G(Dj))$$

natural in c. Tracing an element through the chain shows the composite is $f \mapsto (G\lambda_j \circ f)_j$, the comparison map above. By the Yoneda embedding (definition 2.3.1) a natural isomorphism of the representables $\mathrm{Hom}_{\mathbf{C}}(-, GL)$ and $\mathrm{Hom}_{\mathbf{C}}(-, \lim(G \circ D))$ comes from a unique isomorphism $GL \cong \lim(G \circ D)$, and by naturality this isomorphism is the comparison morphism. Hence G preserves the limit.

The dual statement follows from the duality principle (box 1.3.3): a left adjoint $F \colon \mathbf{C} \to \mathbf{D}$ is $F \dashv G$, so $F^{\mathrm{op}} \colon \mathbf{C}^{\mathrm{op}} \to \mathbf{D}^{\mathrm{op}}$ is a right adjoint (its left adjoint is G^{op}, as one checks by dualizing the hom-set bijection), and a colimit in $\mathbf{C}$ is a limit in $\mathbf{C}^{\mathrm{op}}$. The result just proved, applied to F^{op}, says F^{op} preserves limits in $\mathbf{C}^{\mathrm{op}}$, which is exactly the statement that F preserves colimits in $\mathbf{C}$, completing the proof.

RAPL is a source of both computations and obstructions. On the computational side it says a right adjoint may be evaluated on a limit by evaluating it factorwise: G commutes with products, pullbacks, equalizers, and all other limits. On the obstruction side it gives a fast way to prove a functor is *not* a right adjoint, by exhibiting one limit it fails to preserve. The tensor product $M \otimes_R -$ does not preserve limits (it fails to preserve infinite products in general), so it cannot be a right adjoint; and indeed it is a left adjoint, adjoint to $\mathrm{Hom}_R(M, -)$, and preserves colimits instead (example 4.2.2). A functor that preserves neither all limits nor all colimits can be neither a left nor a right adjoint, which is one reason adjointness is a strong condition.

The corollary the reader has already met is that a forgetful functor which happens to be a right adjoint computes limits on underlying sets.

Corollary 4.3.5: Limits on Underlying Sets

Let $U\colon \mathbf{D} \to \mathbf{Set}$ be a forgetful functor that has a left adjoint (a free functor $F \dashv U$). Then U preserves all limits that exist in $\mathbf{D}$: the underlying set of a limit in $\mathbf{D}$ is the corresponding limit of underlying sets, with its structure determined by that of the diagram.

***Proof*:**
Immediate from theorem 4.3.4 applied to $F \dashv U$, since U is the right adjoint, as required.

This recovers, and now explains, the pattern computed directly when limits were introduced (definition 3.2.1): the underlying set of a product of groups is the product of the underlying sets with coordinatewise multiplication, the underlying set of an equalizer is the equalizer of underlying sets, and the limit of a diagram of modules is computed on underlying sets. Each forgetful functor in question, from **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, or **Top** to **Set**, is a right adjoint (example 4.2.1), so RAPL forces its compatibility with limits, in the sense of preservation from definition 3.4.6.

Example 4.3.6: Products and Kernels of Abelian Groups
The forgetful functor $U\colon \mathbf{Ab} \to \mathbf{Set}$ has the free-abelian-group functor as left adjoint, so U preserves limits. Two instances make this concrete.

A product $\prod_i A_i$ in **Ab** has underlying set $\prod_i UA_i$, the cartesian product of the underlying sets, because U preserves the product. The group structure is coordinatewise, which is precisely the structure the limit universal property forces once the underlying set is fixed as the product.

An equalizer of two homomorphisms $f, g\colon A \to B$ is the subgroup $\{a \in A \big| f(a) = g(a)\}$; its underlying set is $\{x \in UA \big| Uf(x) = Ug(x)\}$, the equalizer of Uf, Ug in **Set**, again because U preserves the equalizer. Taking $g = 0$ specializes this to the kernel, whose underlying set is $\{a \in UA \big| Uf(a) = 0\}$. Both computations are consequences of RAPL, not separate verifications.

4.4 Adjoint Functor Theorems

The RAPL theorem (theorem 4.3.4) makes limit-preservation a necessary condition for a functor $G\colon \mathbf{D} \to \mathbf{C}$ to be a right adjoint. The adjoint functor theorems address the converse: when a limit-preserving G is guaranteed to have a left adjoint. Preservation of limits alone is not enough, because a proper class of candidate arrows can obstruct the construction. The extra ingredient is a smallness hypothesis. The General Adjoint Functor Theorem asks for a set of arrows that jointly dominates every arrow out of a fixed object, the solution-set condition; the Special Adjoint Functor Theorem replaces that condition with a global structural hypothesis on $\mathbf{C}$. We prove the general theorem in full and record the special one with its key ideas.

The whole chapter turns on one reformulation, established in section 4.1: a left adjoint to G is the same data as a universal arrow $\eta_c\colon c \to Gd_c$ from each object c of $\mathbf{C}$ to G. A universal arrow from c to G (definition 3.1.4) is exactly an initial object of the comma category (c/G), whose objects are pairs (d, f) with $f : c \to Gd$ and whose morphisms $(d, f) \to (d', f')$ are arrows $h : d \to d'$ in $\mathbf{D}$ with $Gh \circ f = f'$. Constructing a left adjoint therefore reduces to producing an initial object in each (c/G). The solution-set condition is precisely the smallness input that lets a completeness argument produce one.

Definition 4.4.1: Solution-Set Condition

Let $G : \mathbf{D} \to \mathbf{C}$ be a functor and c an object of $\mathbf{C}$. Then G satisfies the *solution-set condition* at c if there is a *set* of morphisms

$$\{ f_i : c \to Gd_i \}_{i \in I}$$

such that every morphism $f : c \to Gd$ factors as $f = Gh \circ f_i$ for some index i and some morphism $h : d_i \to d$ in $\mathbf{D}$. The functor G *satisfies the solution-set condition* if it does so at every object c of $\mathbf{C}$.

In the language of the comma category, the condition says the objects (d_i, f_i) form a set that is *weakly initial* in (c/G): every object receives at least one morphism from a member of the set. The word "weakly" marks that the arrow witnessing the factorization is neither required to be unique nor to come from a single chosen source. The force of the definition is entirely its smallness. In a locally small $\mathbf{D}$ the objects (d, f) of (c/G) already form a proper class, and there is no reason a class-sized family should admit a set-sized dominating subfamily; the solution-set condition asserts that one exists. That is the gap the theorem fills between the necessary condition (RAPL) and sufficiency.

Example 4.4.2: The Solution Set for Free Groups

Let $G : \mathbf{Grp} \to \mathbf{Set}$ be the forgetful functor and c a set. A single arrow already works: take the free group Fc on c with its insertion of generators $\eta_c : c \to G(Fc)$, and let $\{f_i\}$ be the one-element family $\{\eta_c\}$. Every function $f : c \to G(H)$ into the underlying set of a group H extends uniquely to a homomorphism $\bar{f} : Fc \to H$ with $G(\bar{f}) \circ \eta_c = f$, so $f = G(\bar{f}) \circ \eta_c$ exhibits the required factorization. Here the solution set has one element because a genuine universal arrow already exists; the theorem below runs the implication in the other direction, manufacturing such a universal arrow from any solution set, however large.

The example makes the shape of the argument visible. When a left adjoint is present, the singleton solution set is the universal arrow itself. The content of the General Adjoint Functor Theorem is that a solution set of *any* size, together with completeness of $\mathbf{D}$ and preservation of limits by G, suffices to assemble a universal arrow that was not given in advance. The assembly is a completeness argument inside the comma category, and its engine is a lemma about initial objects, isolated first.

The lemma is where a set-sized weakly initial family is compressed to a single genuine initial object. In a complete category one can form the product of the family; the product receives an arrow to every object, so it is again weakly initial, and now it is a single object. Passing from weakly initial to initial then requires killing the ambiguity in those arrows, which is what an equalizer of endomorphisms does.

Lemma 4.4.3: Initial Object from a Weakly Initial Set

Let $\mathbf{A}$ be a complete, locally small category. Suppose $\mathbf{A}$ has a *weakly initial set*: a set $\{a_i\}_{i\in I}$ of objects such that every object of $\mathbf{A}$ admits at least one morphism from some a_i. Then $\mathbf{A}$ has an initial object.

***Proof*:**
Step 1: Form a weakly initial object. Since I is a set and $\mathbf{A}$ is complete, the product $P = \prod_{i\in I} a_i$ exists, with projections $\pi_i\colon P \to a_i$. Given any object a, weak initiality supplies some i and a morphism $g\colon a_i \to a$; then $g \circ \pi_i\colon P \to a$ is a morphism from P to a. Thus P is a weakly initial *object*: every object receives at least one morphism from P.

Step 2: Equalize the endomorphisms of P. Local smallness makes $\mathbf{A}(P, P)$ a set, so the family of all endomorphisms of P is a set of parallel arrows $P \rightrightarrows P$. Completeness gives their equalizer

$$e\colon E \to P$$

the joint equalizer of $\{\, u\colon P \to P\,\}$, characterized by $u \circ e = e$ for every endomorphism u of P, and universal among arrows into P with this property. We claim E is initial.

Step 3: Existence of a morphism from E. For any object a, weak initiality of P gives a morphism $P \to a$, and precomposing with e yields a morphism $E \to a$. So E admits at least one morphism to every object.

Step 4: Uniqueness. Let $s, t\colon E \to a$ be two morphisms to a common object a; we must show $s = t$. Form the equalizer $k\colon K \to E$ of s and t, so $s \circ k = t \circ k$ and k is universal with this property. Since P is weakly initial there is a morphism $g\colon P \to K$, and composing produces an endomorphism

$$v = e \circ k \circ g\colon P \to P$$

Now e is the joint equalizer of all endomorphisms of P, and v is one such endomorphism, so $v \circ e = e$, that is $e \circ k \circ g \circ e = e$. The morphism e is a monomorphism, being an equalizer, so it is left-cancellable: from $e \circ (k \circ g \circ e) = e = e \circ \mathrm{id}_P$ we cancel e to get

$$k \circ (g \circ e) = \mathrm{id}_E$$

Thus k is a split epimorphism, with section $g \circ e$. But k is also a monomorphism, being an equalizer, and a split epimorphism that is monic is an isomorphism: composing $k \circ (g \circ e) = \mathrm{id}_E$ with k on the right gives $k \circ (g \circ e) \circ k = k = k \circ \mathrm{id}_K$, and cancelling the monic k on the left yields $(g \circ e) \circ k = \mathrm{id}_K$, so $g \circ e$ is a two-sided inverse of k. Since $s \circ k = t \circ k$ and k is an isomorphism, right-cancelling k gives $s = t$.

Combining Steps 3 and 4, every object of $\mathbf{A}$ admits exactly one morphism from E, so E is initial, completing the proof.

The lemma is the whole mechanism in miniature. The product collapses a set of objects into one weakly initial object; the equalizer of its endomorphisms then removes the freedom in the arrows out of it, leaving a single arrow to each target. The two hypotheses do disjoint work: completeness supplies the product and the equalizers, local smallness guarantees that the endomorphisms of P form a set so that their equalizer exists. Drop either and the argument fails. With the lemma in hand the theorem is a matter of transporting it into the comma category (c/G), which is where the smallness of the solution set becomes the weakly initial set the lemma requires.

The remaining input is that (c/G) inherits completeness from $\mathbf{D}$. This is exactly where the hypothesis that G preserves limits is spent: limits in (c/G) are computed in $\mathbf{D}$, and the cone over c that makes the limit an object of the comma category exists precisely because G carries the $\mathbf{D}$-limit to the corresponding $\mathbf{C}$-limit.

Theorem 4.4.4: General Adjoint Functor Theorem

Let $\mathbf{D}$ be a locally small, complete category and $G\colon \mathbf{D} \to \mathbf{C}$ a functor that preserves all small limits. Then G has a left adjoint if and only if G satisfies the solution-set condition (definition 4.4.1).

***Proof*:**
Necessity. Suppose G has a left adjoint $F \dashv G$, with unit η. For each object c of $\mathbf{C}$, the component $\eta_c\colon c \to G(Fc)$ is a universal arrow from c to G (theorem 4.1.5): every $f\colon c \to Gd$ factors as $f = Gh \circ \eta_c$ for a unique $h\colon Fc \to d$. The singleton family $\{\eta_c\}$ is then a solution set at c, so G satisfies the solution-set condition.

Sufficiency. Assume G preserves small limits and satisfies the solution-set condition. By theorem 4.1.5 it suffices to produce, for each object c of $\mathbf{C}$, a universal arrow from c to G, equivalently an initial object of the comma category (c/G) (definition 3.1.4). Fix c; we produce that initial object by applying lemma 4.4.3 to $\mathbf{A} = (c/G)$, which requires that (c/G) be locally small and complete and possess a weakly initial set.

Step 1: (c/G) is locally small. A morphism $(d, f) \to (d', f')$ in (c/G) is a morphism $h\colon d \to d'$ in $\mathbf{D}$ subject to the equation $Gh \circ f = f'$. Thus the hom-set of (c/G) is a subset of $\mathbf{D}(d, d')$, which is a set because $\mathbf{D}$ is locally small. So (c/G) is locally small.

Step 2: (c/G) is complete, using preservation of limits. Let $D\colon \mathbf{J} \to (c/G)$ be a small diagram, writing $D(j) = (d_j, f_j)$ with structure arrows $f_j\colon c \to Gd_j$; a morphism $j \to j'$ in $\mathbf{J}$ gives $\alpha_{jj'}\colon d_j \to d_{j'}$ with $G\alpha_{jj'} \circ f_j = f_{j'}$. The projection to $\mathbf{D}$ sends D to the diagram $j \mapsto d_j$, which has a limit $L = \lim_{\mathbf{J}} d_j$ in $\mathbf{D}$ by completeness, with limit cone $\lambda_j\colon L \to d_j$.

Because G preserves this limit, $G(L)$ with the cone $G\lambda_j\colon G(L) \to Gd_j$ is a limit of $j \mapsto Gd_j$ in $\mathbf{C}$. The arrows $f_j\colon c \to Gd_j$ form a cone over $j \mapsto Gd_j$ with apex c: for a morphism $j \to j'$ one has $G\alpha_{jj'} \circ f_j = f_{j'}$, which is exactly the compatibility condition for a cone. By the universal property of the limit $G(L)$, this cone factors through a unique morphism

$$f_L\colon c \to G(L)$$

satisfying $G\lambda_j \circ f_L = f_j$ for every j. The pair (L, f_L) is then an object of (c/G), and each $\lambda_j\colon L \to d_j$ satisfies $G\lambda_j \circ f_L = f_j$, so λ_j is a morphism $(L, f_L) \to (d_j, f_j)$ in (c/G); these assemble into a cone over D.

This cone is a limit cone in (c/G). Given any cone over D with apex (d, f), its legs are morphisms $\mu_j\colon (d, f) \to (d_j, f_j)$, hence morphisms $\mu_j\colon d \to d_j$ in $\mathbf{D}$ forming a cone over $j \mapsto d_j$; the limit L in $\mathbf{D}$ yields a unique $u\colon d \to L$ with $\lambda_j \circ u = \mu_j$. It remains to check u is a morphism $(d, f) \to (L, f_L)$, that is $Gu \circ f = f_L$. Both sides are morphisms $c \to G(L)$ into the limit $G(L)$; they agree after composing with each projection $G\lambda_j$, since $G\lambda_j \circ Gu \circ f = G(\lambda_j u) \circ f = G\mu_j \circ f = f_j$ and $G\lambda_j \circ f_L = f_j$. By the uniqueness clause of the limit $G(L)$, $Gu \circ f = f_L$. So u is the unique factorization in (c/G), and (L, f_L) is a limit of D. Hence (c/G) has all small limits: it is complete.

Step 3: the solution set is a weakly initial set for (c/G)*.* By hypothesis G satisfies the solution-set condition at c: there is a set $\{f_i : c \to Gd_i\}_{i \in I}$ such that every $f : c \to Gd$ factors as $f = Gh \circ f_i$ for some i and some $h : d_i \to d$. Read in (c/G), the objects $\{(d_i, f_i)\}_{i \in I}$ form a set, and the factorization $f = Gh \circ f_i$ says exactly that $h : (d_i, f_i) \to (d, f)$ is a morphism of (c/G). So every object (d, f) receives a morphism from some (d_i, f_i): the set $\{(d_i, f_i)\}$ is weakly initial.

Step 4: apply the lemma. By Steps 1-3 the comma category (c/G) is locally small, complete, and has a weakly initial set. Lemma 4.4.3 produces an initial object (d_c, η_c) of (c/G). An initial object of (c/G) is a universal arrow $\eta_c : c \to Gd_c$ from c to G (definition 3.1.4). This holds for every c, so by theorem 4.1.5 the assignment $c \mapsto d_c$ extends to a left adjoint $F \dashv G$, completing the proof.

The theorem isolates the two obstructions to being a right adjoint and names both. The soft obstruction is failure to preserve limits, ruled out by RAPL and assumed away here. The hard obstruction is size: a limit-preserving functor between well-behaved categories can still fail to have a left adjoint if the comma categories are too large to admit an initial object, and the solution-set condition is exactly the hypothesis that excludes this. When one has a candidate construction in hand, the theorem is often applied by exhibiting an explicit small solution set (frequently a single arrow, as in example 4.4.2) rather than by verifying an abstract smallness principle.

A recurring difficulty in practice is that verifying the solution-set condition can be as much work as constructing the adjoint by hand. The Special Adjoint Functor Theorem removes this burden under stronger global hypotheses on $\mathbf{C}$: a small cogenerating set does the work of every solution set at once. The cost is machinery, well-poweredness and cogenerators phrased through subobject lattices, that this book does not develop, so the theorem is stated with its key ideas rather than proved in full.

Theorem 4.4.5: Special Adjoint Functor Theorem

Let $\mathbf{D}$ be a locally small, complete, well-powered category with a small cogenerating set, and let $G : \mathbf{D} \to \mathbf{C}$ be a functor between locally small categories that preserves all small limits. Then G has a left adjoint.

***Proof Key ideas*:**
The full argument requires two notions this book does not develop: a category is *well-powered* when every object has only a set of subobjects (its subobjects form a set rather than a proper class), and a set $\{s_k\}$ of objects is *cogenerating* when any two distinct parallel morphisms $u \neq v : d \rightrightarrows d'$ are separated by some morphism $d' \to s_k$ that distinguishes them. These are hypotheses on the lattice of subobjects, whose systematic development belongs to a treatment of well-powered categories and their factorization systems; see Mac Lane, *Categories for the Working Mathematician*, Chapter V.8.

Granting them, the shape of the proof matches the general theorem, with the cogenerating set replacing the solution-set condition. Fix an object c of $\mathbf{C}$ and form, inside $\mathbf{D}$, the product $\prod_k s_k^{\mathbf{C}(c, Gs_k)}$ indexed by the cogenerating set and, for each factor, by the set of morphisms $c \to G(s_k)$. The candidate universal object is the smallest subobject of this product through which the canonical arrow from c factors; well-poweredness guarantees a smallest such subobject exists, since the subobjects form a set closed under the intersections that completeness provides, and the cogenerating property forces the resulting arrow to be universal. Preservation of limits by G is used, as before, to place the construction in the comma category (c/G). The upshot is that a small cogenerating set functions as a uniform solution set: it supplies, in one stroke, the smallness input that the general theorem requires separately at each object, completing the sketch.

The two theorems are the standard existence criteria for left adjoints, and their hypotheses trade off against each other. The general theorem requires little of the target categories but requires a solution set verified case by case; the special theorem front-loads strong structural hypotheses on $\mathbf{D}$, a small cogenerating set and well-poweredness, and then asks nothing further of the individual objects. Many categories of everyday interest, **Set**, **Grp**, **Ab**, **Top**, and $\mathbf{Mod}_R$, are complete and well-powered with a small cogenerating set, so the special theorem applies to limit-preserving functors out of them without any per-object verification.

A final application of the general theorem recurs across the series: producing a reflection onto a subcategory.

Example 4.4.6: A Reflective Subcategory via GAFT

Let $\mathbf{D}$ be a locally small, complete category and $\mathbf{C} \subseteq \mathbf{D}$ a full subcategory that is closed under small limits, meaning the inclusion $G\colon \mathbf{C} \hookrightarrow \mathbf{D}$ preserves and reflects small limits and every $\mathbf{D}$-limit of a diagram in $\mathbf{C}$ again lies in $\mathbf{C}$. Suppose in addition that $\mathbf{C}$ is complete and that G satisfies the solution-set condition: for each object d of $\mathbf{D}$ there is a set of morphisms $\{d \to c_i\}$ with c_i in $\mathbf{C}$ through which every morphism $d \to c$ into an object of $\mathbf{C}$ factors. Then theorem 4.4.4, applied to $G\colon \mathbf{C} \to \mathbf{D}$, produces a left adjoint $L \dashv G$: a *reflection* of $\mathbf{D}$ onto $\mathbf{C}$.

Concretely, take $\mathbf{D} = \mathbf{Grp}$ and $\mathbf{C} = \mathbf{Ab}$ the full subcategory of abelian groups, with $G\colon \mathbf{Ab} \hookrightarrow \mathbf{Grp}$ the inclusion. A limit of abelian groups computed in **Grp** is again abelian, so G preserves limits and **Ab** is closed under them; **Ab** is complete. For the solution-set condition at a group H, a single arrow suffices: the quotient map $q_H\colon H \to H^{\mathrm{ab}} = H/[H,H]$ onto the abelianization. Any homomorphism $H \to A$ into an abelian group A kills the commutator subgroup $[H,H]$, so it factors uniquely as $\bar{\varphi} \circ q_H$ with $\bar{\varphi}\colon H^{\mathrm{ab}} \to A$. The general theorem then returns the left adjoint $H \mapsto H^{\mathrm{ab}}$, recovering abelianization as the reflection of **Grp** onto **Ab**. As with example 4.4.2, the singleton solution set is present because the universal arrow already exists; the value of the theorem is that a solution set of any size would have sufficed.

5

Monads and their Algebras

A monad is the algebraic structure an adjunction leaves on one side: from $F \dashv G$ the composite GF inherits a unit and a multiplication, and this structure alone suffices to rebuild a category of algebras. This chapter defines *monads*, constructs the two canonical adjunctions that realize any monad, the Eilenberg-Moore and Kleisli categories, and asks the converse, *monadicity*: when is a forgetful functor the forgetful functor of a monad? Beck's theorem answers it, and explains why the algebraic categories over **Set** are exactly the monadic ones.

5.1 Monads from Adjunctions

Every adjunction $F \dashv G$ leaves a trace on the category where its two functors meet: the composite GF is an endofunctor, the unit is a natural transformation $\mathrm{id} \Rightarrow GF$, and the counit supplies a second natural transformation $GFGF \Rightarrow GF$. This section isolates that trace as a structure in its own right, the *monad*, defines it by the two laws it must obey, and shows that every adjunction produces one. The free-monoid, powerset, and error monads run alongside as the guiding examples.

Definition 5.1.1: Monad

A *monad* on a category C is an endofunctor $T\colon C \to C$ together with two natural transformations, the *unit*

$$\eta\colon \mathrm{id}_C \Rightarrow T$$

and the *multiplication*

$$\mu\colon T^2 \Rightarrow T$$

where T^2 denotes the composite TT, subject to two laws. The *associativity law* is the equality of natural transformations $T^3 \Rightarrow T$

$$\mu \circ T\mu = \mu \circ \mu T \tag{5.1}$$

and the *unit laws* are the equalities of natural transformations $T \Rightarrow T$

$$\mu \circ T\eta = \mathrm{id}_T = \mu \circ \eta T \tag{5.2}$$

Here $T\mu$ and μT are the whiskered composites: $T\mu$ has component $T(\mu_c)\colon T^3c \to T^2c$ at c, and μT has component $\mu_{Tc}\colon T^3c \to T^2c$; similarly $T\eta$ and ηT have components $T(\eta_c)$ and η_{Tc}. The two laws are the commutativity of the associativity square

$$\begin{tikzcd} T^3 \arrow[r, Rightarrow, "T\mu"] \arrow[d, Rightarrow, "\mu T"'] & T^2 \arrow[d, Rightarrow, "\mu"] \\ T^2 \arrow[r, Rightarrow, "\mu"'] & T \end{tikzcd}$$

and the two unit triangles

$$\begin{tikzcd} T \arrow[r, Rightarrow, "\eta T"] \arrow[dr, Rightarrow, "\mathrm{id}_T"'] & T^2 \arrow[d, Rightarrow, "\mu"] & T \arrow[l, Rightarrow, "T\eta"'] \arrow[dl, Rightarrow, "\mathrm{id}_T"] \\ & T & \end{tikzcd}$$

The pattern is that of a monoid: μ plays the role of a multiplication and η the role of a unit element, with (5.1) the associativity of the multiplication and (5.2) its two-sided unit law. The slogan is that a monad is a monoid in the category of endofunctors of C under composition, with the functor category $[C, C]$ (definition 1.5.4) supplying the ambient category and functor composition supplying the tensor product. The definition is the monoid axioms transported into that setting, with vertical composition of natural transformations for equality of maps and horizontal composition (whiskering) for the two ways of associating T^3.

The endofunctor alone carries no content; the unit and multiplication are what make a monad an algebraic object. Before the examples, the source of every such structure is an adjunction.

The chapter opening framed a monad as the structure an adjunction leaves on one side. That statement is now a proposition: the composite GF of an adjunction, equipped with the adjunction unit and the multiplication built from the counit, satisfies definition 5.1.1. The verification is a diagram chase in which every square commutes by naturality of the counit or by a triangle identity.

Proposition 5.1.2: The Monad of an Adjunction

Let $F\colon C \to D$ and $G\colon D \to C$ be functors with $F \dashv G$, unit $\eta\colon \mathrm{id}_C \Rightarrow GF$ and counit $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ (definition 4.1.1, definition 4.1.3). Then

$$T = GF\colon C \to C, \qquad \eta\colon \mathrm{id}_C \Rightarrow T, \qquad \mu = G\varepsilon F\colon T^2 \Rightarrow T$$

is a monad on C, where μ is the whiskered natural transformation with component $\mu_c = G\varepsilon_{Fc}\colon GFGFc \to GFc$ at c.

***Proof*:**
The unit of the monad is the adjunction unit, already a natural transformation $\mathrm{id}_C \Rightarrow GF = T$ (definition 4.1.3). The multiplication $\mu = G\varepsilon F$ is the whiskering of the natural transformation $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ by G on the left and F on the right, hence a natural transformation $GFGF \Rightarrow GF$, that is, $T^2 \Rightarrow T$. It remains to check the associativity law and the two unit laws.

Step 1: Associativity. Both $\mu \circ T\mu$ and $\mu \circ \mu T$ are natural transformations $T^3 = GFGFGF \Rightarrow GF = T$; the claim is that their components agree at every c. Writing out the components,

$$(\mu \circ T\mu)_c = G\varepsilon_{Fc} \circ GF(G\varepsilon_{Fc}) = G\varepsilon_{Fc} \circ GFG\varepsilon_{Fc}$$

and

$$(\mu \circ \mu T)_c = G\varepsilon_{Fc} \circ G\varepsilon_{FGFc}$$

Applying the functor G to the naturality square of ε reduces both to a common expression. Naturality of $\varepsilon\colon FG \Rightarrow \mathrm{id}_D$ at the morphism $\varepsilon_{Fc}\colon FGFc \to Fc$ in D gives the commuting square

$$\begin{tikzcd} FGFGFc \arrow[r, "\varepsilon_{FGFc}"] \arrow[d, "FG\varepsilon_{Fc}"'] & FGFc \arrow[d, "\varepsilon_{Fc}"] \\ FGFc \arrow[r, "\varepsilon_{Fc}"'] & Fc \end{tikzcd}$$

that is, $\varepsilon_{Fc} \circ FG\varepsilon_{Fc} = \varepsilon_{Fc} \circ \varepsilon_{FGFc}$ in D. Applying G to this equation yields $G\varepsilon_{Fc} \circ GFG\varepsilon_{Fc} = G\varepsilon_{Fc} \circ G\varepsilon_{FGFc}$, which is precisely $(\mu \circ T\mu)_c = (\mu \circ \mu T)_c$. As c was arbitrary, (5.1) holds. Associativity of the monad is thus the naturality of the counit, pushed through G.

Step 2: The unit laws. The component of $\mu \circ T\eta$ at c is

$$(\mu \circ T\eta)_c = G\varepsilon_{Fc} \circ GF(\eta_c) = G\varepsilon_{Fc} \circ GF\eta_c = G(\varepsilon_{Fc} \circ F\eta_c)$$

using that G is a functor. By the second triangle identity (theorem 4.3.1), $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$, so the component is $G(\mathrm{id}_{Fc}) = \mathrm{id}_{GFc} = (\mathrm{id}_T)_c$. Hence $\mu \circ T\eta = \mathrm{id}_T$. For the other unit law, the component of $\mu \circ \eta T$ at c is

$$(\mu \circ \eta T)_c = G\varepsilon_{Fc} \circ \eta_{GFc}$$

the composite $GFc \to GFGFc \to GFc$. This is exactly the first triangle identity (theorem 4.3.1) evaluated at the object Fc of D, namely $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ with $d = Fc$, giving $G\varepsilon_{Fc} \circ \eta_{GFc} = \mathrm{id}_{GFc}$. Hence $\mu \circ \eta T = \mathrm{id}_T$, and both equalities in (5.2) hold, completing the proof.

The proof shows the two monad laws are repackaged adjunction data: associativity is the naturality of the counit, and each unit law is one of the two triangle identities. This is the origin of every monad met in practice. A monad is not usually written down axiomatically and then studied; it arises as GF for a free-forgetful adjunction, and its multiplication is the counit describing how a free object built on a free object collapses. The examples make this concrete, and each recovers a familiar operation.

Example 5.1.3: The Free-Monoid, Powerset, and Error Monads
Three monads on **Set**, each the monad of a free-forgetful adjunction (proposition 5.1.2) or checked directly against definition 5.1.1.

1. *The free-monoid monad.* Let TX be the set of finite words $(x_1, \dots, x_n)$, $n \geq 0$, in the alphabet X, including the empty word; for $f\colon X \to Y$, Tf applies f letterwise. This is GF for the free-monoid adjunction $F \dashv G$ with $G\colon \mathbf{Mon} \to \mathbf{Set}$ the underlying-set functor and F the free-monoid functor ([4]). The unit $\eta_X\colon X \to TX$ sends x to the length-one word (x). The multiplication $\mu_X\colon T^2X \to TX$ takes a word of words and concatenates: a length-one word of words $\big((x_1, \dots, x_k)\big)$ flattens to $(x_1, \dots, x_k)$, and in general
$$\mu_X\big((w_1, \dots, w_m)\big) = w_1 \cdot w_2 \cdots w_m$$
the concatenation of the constituent words $w_i \in TX$. Associativity (5.1) is the associativity of concatenation: flattening a triple-nested word by first concatenating the inner groupings or first concatenating the outer groupings gives the same word. The unit laws (5.2) say that turning each letter into a length-one word and then flattening, or wrapping a whole word into a length-one word of words and then flattening, both return the original word.

2. *The powerset monad.* Let $T = P$ be the covariant powerset functor: PX is the set of subsets of X, and $Pf(S) = f(S)$ is the image of $S \subseteq X$ under $f\colon X \to Y$. The unit $\eta_X\colon X \to PX$ is the singleton map $x \mapsto \{x\}$. The multiplication $\mu_X\colon P^2X \to PX$ is the union: for a set $\mathcal{S}$ of subsets of X,
$$\mu_X(\mathcal{S}) = \bigcup_{S \in \mathcal{S}} S$$
Associativity is the equality of the two ways to flatten a set of sets of sets by union, which both give the union of all the innermost members. The unit laws read $\bigcup_{x \in S}\{x\} = S$ (union of the singletons of the elements of S) and $\bigcup\{S\} = S$ (union of the one-element family $\{S\}$). This is the monad of the adjunction between **Set** and the category of complete sup-lattices, whose algebras are taken up in the next section.

3. *The error monad.* Fix a one-point set $\{*\}$ and let $TX = X \sqcup \{*\}$, the disjoint union adjoining a single point $*$ read as an error value; for $f\colon X \to Y$, Tf acts as f on X and fixes $*$. The unit $\eta_X\colon X \to X \sqcup \{*\}$ is the inclusion of X. The multiplication $\mu_X\colon T^2X \to TX$, where $T^2X = X \sqcup \{*_1\} \sqcup \{*_2\}$ carries two error points ($*_1$ from the inner T, $*_2$ from the outer), acts as the identity on X and sends both $*_1$ and $*_2$ to the single $*$, merging the two error points. Associativity holds because on the X-part every map in sight is the identity and on the error part every composite lands on $*$; the unit laws hold because η adjoins an error point that μ immediately absorbs. This is the monad GF for the free-pointed-set adjunction, F adjoining a basepoint and G forgetting it.

The three examples share a shape: TX is a syntax of "X-terms" (words, subsets, values-or-error), the unit embeds each element as a trivial term, and the multiplication evaluates a term of terms into a single term. A monad is thus a theory of substitution, with μ the operation that carries out one round of it, and the two laws are exactly what make repeated substitution unambiguous. This reading anticipates the theory of algebras for a monad, developed in the next section, where an algebra supplies the missing evaluation of a term into an actual element.

Reversing every arrow in definition 5.1.1 produces the dual notion, obtained by applying the definition in the opposite category. A comonad packages the trace an adjunction casts on the *other* side, the composite FG with counit and comultiplication, and is the structure whose coalgebras encode "co-substitution" in the same way a monad's algebras encode substitution.

Definition 5.1.4: Comonad

A *comonad* on a category C is a monad on C^{op}: an endofunctor $T\colon C \to C$ with a *counit* $\varepsilon\colon T \Rightarrow \mathrm{id}_C$ and a *comultiplication* $\delta\colon T \Rightarrow T^2$ satisfying the coassociativity law $T\delta \circ \delta = \delta T \circ \delta$ and the counit laws $\varepsilon T \circ \delta = \mathrm{id}_T = T\varepsilon \circ \delta$, the diagrams of definition 5.1.1 with every arrow reversed. Dually to proposition 5.1.2, an adjunction $F \dashv G$ induces a comonad FG on D with counit ε and comultiplication $F\eta G$ (box 1.3.3).

Everything proved for monads holds for comonads by the duality principle, with the roles of unit and multiplication interchanged and arrows reversed. The comonad and its coalgebras are the opposite-category translation of the monad theory developed here and in the sections that follow, and are not pursued further.

5.2 Algebras for a Monad

Proposition 5.1.2 produced a monad from every adjunction. This section runs the construction the other way. Given a monad (T, η, μ) on $\mathbf{C}$ (definition 5.1.1), an *algebra* for it is an object equipped with an action of T compatible with η and μ; the algebras assemble into a category, the Eilenberg-Moore category $\mathbf{C}^T$, and the free-algebra construction supplies an adjunction whose induced monad is the original T. A monad thus records an algebraic theory, and $\mathbf{C}^T$ is its category of models.

The definition is the categorical distillation of what it means to be an algebra for an operation. If T is the free-monoid endofunctor on **Set**, then TA is the set of finite words in A; an action $a\colon TA \to A$ that respects the unit and multiplication is precisely a rule multiplying words, that is, a monoid structure. The two laws below are exactly associativity and unitality of that multiplication, transcribed into the language of T.

Definition 5.2.1: Algebra for a Monad

Let (T, η, μ) be a monad on a category $\mathbf{C}$. A *T-algebra* is a pair (A, a) consisting of an object A of $\mathbf{C}$ and a morphism $a\colon TA \to A$, called the *structure map*, making the two diagrams

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[dr, "\mathrm{id}_A"'] & TA \arrow[d, "a"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} T^2A \arrow[r, "\mu_A"] \arrow[d, "Ta"'] & TA \arrow[d, "a"] \\ TA \arrow[r, "a"'] & A \end{tikzcd}$$

commute, that is, $a \circ \eta_A = \mathrm{id}_A$ (the *unit law*) and $a \circ \mu_A = a \circ Ta$ (the *associativity law*).

A *morphism of T-algebras* $(A, a) \to (B, b)$ is a morphism $f\colon A \to B$ in $\mathbf{C}$ commuting with the structure maps, that is, making

$$\begin{tikzcd} TA \arrow[r, "Tf"] \arrow[d, "a"'] & TB \arrow[d, "b"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

commute: $f \circ a = b \circ Tf$.

The unit law forces the structure map to restore an element inserted by η_A without altering it: the free elements act as they must. The associativity law compares the two ways of collapsing a doubly-applied T down to A (first collapse the outer layer with μ_A, then act by a; or first act by a inside the outer T, then act again) and requires they agree. For the free-monoid monad these are the associativity and unit axioms of a monoid, as the next passage makes precise; the definition abstracts that pattern away from any one theory.

Example 5.2.2: The Trivial Algebra and the Free Algebra

Two algebras exist for every monad. First, (TA, μ_A) is a T-algebra for each object A: the unit law $\mu_A \circ \eta_{TA} = \mathrm{id}_{TA}$ is one of the monad unit laws, and the associativity law $\mu_A \circ \mu_{TA} = \mu_A \circ T\mu_A$ is the monad associativity law (definition 5.1.1). This is the *free T-algebra* on A. Second, for any algebra (A, a) the associativity law read as a commuting square makes the structure map $a\colon (TA, \mu_A) \to (A, a)$ itself a morphism of T-algebras.

The algebras and their morphisms form a category, and the underlying-object assignment forgets back to $\mathbf{C}$.

Definition 5.2.3: Eilenberg-Moore Category

Let (T, η, μ) be a monad on $\mathbf{C}$. The *Eilenberg-Moore category* $\mathbf{C}^T$ has T-algebras as objects and morphisms of T-algebras as arrows, with composition and identities computed in $\mathbf{C}$. The *forgetful functor*

$$U^T : \mathbf{C}^T \to \mathbf{C}, \qquad (A, a) \mapsto A, \qquad (f : (A, a) \to (B, b)) \mapsto (f : A \to B)$$

sends an algebra to its underlying object and an algebra morphism to its underlying morphism.

Composition is well-defined because a composite of algebra morphisms is again one: stacking the two commuting squares of definition 5.2.1 for $f : (A, a) \to (B, b)$ and $g : (B, b) \to (C, c)$ gives the square for gf, using $T(gf) = Tg \circ Tf$. The identity of $\mathbf{C}$ on A is an algebra endomorphism of (A, a). So $\mathbf{C}^T$ is a genuine category and U^T a functor; the content lies not in these formalities but in what U^T admits as a partner. The free-algebra assignment $A \mapsto (TA, \mu_A)$ of example 5.2.2 is the object part of its left adjoint.

The motivation for the adjunction is the recovery program. Proposition 5.1.2 showed an adjunction breeds a monad; the reverse question is whether every monad comes from an adjunction, and if so from which one. The Eilenberg-Moore construction answers yes and produces a canonical such adjunction, built entirely out of T itself.

Theorem 5.2.4: The Eilenberg-Moore Adjunction

Let (T, η, μ) be a monad on $\mathbf{C}$. The assignment

$$F^T : \mathbf{C} \to \mathbf{C}^T, \qquad c \mapsto (Tc, \mu_c), \qquad (g : c \to c') \mapsto (Tg : (Tc, \mu_c) \to (Tc', \mu_{c'}))$$

is a functor, the *free T-algebra functor*, and it is left adjoint to the forgetful functor U^T. The adjunction $F^T \dashv U^T$ induces (via proposition 5.1.2) the original monad (T, η, μ).

***Proof*:**
Step 1: F^T is a functor. On objects $F^T c = (Tc, \mu_c)$ is a T-algebra by example 5.2.2. On a morphism $g : c \to c'$, the arrow $Tg : Tc \to Tc'$ is an algebra morphism $(Tc, \mu_c) \to (Tc', \mu_{c'})$: the required square $Tg \circ \mu_c = \mu_{c'} \circ T(Tg)$ is naturality of $\mu : T^2 \Rightarrow T$ applied to g. Functoriality of F^T then follows from functoriality of T, since F^T acts as T on morphisms.

Step 2: The transposition bijection. Fix an object c of $\mathbf{C}$ and a T-algebra (A, a). Define

$$\Phi : \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (A, a)\big) \longrightarrow \mathrm{Hom}_{\mathbf{C}}(c, A), \qquad \Phi(f) = f \circ \eta_c$$

which precomposes an algebra map out of the free algebra with the unit component $\eta_c : c \to Tc$; this is a morphism of $\mathbf{C}$ because $U^T f = f$ and η_c live there. In the other direction define

$$\Psi : \mathrm{Hom}_{\mathbf{C}}(c, A) \longrightarrow \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (A, a)\big), \qquad \Psi(g) = a \circ Tg$$

Two claims must be checked: that $\Psi(g)$ is an algebra morphism, and that Φ and Ψ are mutually inverse.

Step 3: $\Psi(g)$ is an algebra morphism. Write $h = a \circ Tg : Tc \to A$. The algebra-morphism square for $h\colon (Tc,\mu_c) \to (A,a)$ requires $h \circ \mu_c = a \circ Th$. Compute the right-hand side:

$$a \circ Th = a \circ T(a \circ Tg) = a \circ Ta \circ T^2 g$$

Apply the associativity law $a \circ Ta = a \circ \mu_A$ of (A,a):

$$a \circ Ta \circ T^2 g = a \circ \mu_A \circ T^2 g$$

Naturality of $\mu\colon T^2 \Rightarrow T$ applied to g gives $\mu_A \circ T^2 g = Tg \circ \mu_c$, so

$$a \circ \mu_A \circ T^2 g = a \circ Tg \circ \mu_c = h \circ \mu_c$$

which is the identity required. Hence $\Psi(g)$ is a morphism $(Tc,\mu_c) \to (A,a)$ of T-algebras.

Step 4: Φ and Ψ are inverse. For $g\colon c \to A$,

$$\Phi(\Psi(g)) = (a \circ Tg) \circ \eta_c = a \circ Tg \circ \eta_c = a \circ \eta_A \circ g = g$$

where the third equality is naturality of $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow T$ applied to g (namely $Tg \circ \eta_c = \eta_A \circ g$) and the fourth is the unit law $a \circ \eta_A = \mathrm{id}_A$ of (A,a). Conversely, for an algebra morphism $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Psi(\Phi(f)) = a \circ T(f \circ \eta_c) = a \circ Tf \circ T\eta_c$$

Because f is an algebra morphism, $a \circ Tf = f \circ \mu_c$, so

$$a \circ Tf \circ T\eta_c = f \circ \mu_c \circ T\eta_c = f \circ \mathrm{id}_{Tc} = f$$

the middle step using the monad unit law $\mu_c \circ T\eta_c = \mathrm{id}_{Tc}$ (definition 5.1.1). Thus Φ and Ψ are mutually inverse bijections.

Step 5: Naturality. The bijection must be natural in c and in (A,a), so that it is an adjunction (definition 4.1.1). For naturality in the algebra variable, let $\varphi\colon (A,a) \to (B,b)$ be an algebra morphism. Postcomposition by φ acts on the left-hand hom-set (in $\mathbf{C}^T$) and by $U^T\varphi = \varphi$ on the right (in $\mathbf{C}$); for $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Phi(\varphi \circ f) = \varphi \circ f \circ \eta_c = \varphi \circ \Phi(f)$$

which is the naturality square for the target variable. For naturality in c, let $k\colon c' \to c$ be a morphism of $\mathbf{C}$. Precomposition acts by k on the right-hand hom-set and by $F^T k = Tk$ on the left; for $f\colon (Tc,\mu_c) \to (A,a)$,

$$\Phi(f \circ Tk) = f \circ Tk \circ \eta_{c'} = f \circ \eta_c \circ k = \Phi(f) \circ k$$

using naturality of η applied to k (namely $Tk \circ \eta_{c'} = \eta_c \circ k$). Both squares commute, so Φ is a natural isomorphism and $F^T \dashv U^T$.

Step 6: The induced monad is T. By proposition 5.1.2, the monad induced by $F^T \dashv U^T$ has endofunctor $U^T F^T$, unit the adjunction unit, and multiplication $U^T\varepsilon^T F^T$, where ε^T is the counit of $F^T \dashv U^T$. On objects, $U^T F^T c = U^T(Tc,\mu_c) = Tc$, and on morphisms $U^T F^T g = Tg$, so $U^T F^T = T$ as endofunctors. The adjunction unit is the transpose of the identity on $F^T c = (Tc,\mu_c)$, which by Step 2 is $\Phi(\mathrm{id}) = \mathrm{id}_{Tc} \circ \eta_c = \eta_c$; the induced unit is η. The counit component at (A,a) is the transpose of id_A under $F^T \dashv U^T$, that is $\varepsilon^T_{(A,a)} = \Psi(\mathrm{id}_A) = a \circ T\mathrm{id}_A = a$, the structure map itself. Hence the induced multiplication at c is $U^T\varepsilon^T_{F^T c} = \varepsilon^T_{(Tc,\mu_c)} = \mu_c$. Endofunctor, unit, and multiplication all agree with (T,η,μ), completing the proof.

The counit computation in Step 6 isolates a useful fact: the counit of $F^T \dashv U^T$ at an algebra (A, a) is its own structure map a, viewed as the algebra morphism $(TA, \mu_A) \to (A, a)$. This recovers the second observation of example 5.2.2 as the counit of the adjunction, and it says that every algebra is a canonical quotient of a free one, namely the free algebra on its underlying object, acted on by a. The theorem thus does two things at once: it realizes an arbitrary monad by an adjunction, closing the loop opened in section 5.1, and it does so through a category $\mathbf{C}^T$ assembled purely from T, with no auxiliary data.

The examples make the slogan "a monad is an algebraic theory, and $\mathbf{C}^T$ is its category of models" concrete. Each monad from example 5.1.3 has an Eilenberg-Moore category that is, up to equivalence, a familiar category of structured sets.

Example 5.2.5: Eilenberg-Moore Categories of Standard Monads

1. *The free-monoid monad.* Let T be the free-monoid monad on **Set**, so TA is the set of finite words $(a_1, \ldots, a_n)$ in A, with $\eta_A(a) = (a)$ the length-one word and μ_A concatenating a word of words. A T-algebra (A, a) has a structure map $a\colon TA \to A$ assigning an element to each word. Write $a_1 \cdots a_n$ for $a(a_1, \ldots, a_n)$. The unit law $a \circ \eta_A = \mathrm{id}_A$ says $a(a) = a$: single letters are unchanged. The associativity law $a \circ \mu_A = a \circ Ta$ says that evaluating a word of words either by first concatenating (via μ_A) or by first evaluating each inner word (via Ta) gives the same answer; taking the empty outer word gives a distinguished element $e = a()$ and taking a two-word split gives $(a_1 \cdots a_i)(a_{i+1} \cdots a_n) = a_1 \cdots a_n$. These are exactly the associativity and unit axioms of a monoid with unit e and product $xy = a(x, y)$. A morphism of T-algebras is a map preserving the product and unit, that is, a monoid homomorphism ([4]). Hence $\mathbf{Set}^T$ is equivalent to **Mon**, the category of monoids.

2. *The powerset monad.* Let $T = P$ be the covariant powerset monad on **Set**, with $\eta_A(a) = \{a\}$ and μ_A taking a union of a set of subsets. A P-algebra is a map $a\colon PA \to A$ sending each subset $S \subseteq A$ to an element $\bigvee S := a(S)$. The unit law $a(\{x\}) = x$ pins the singletons. Setting $x \le y$ iff $\bigvee\{x, y\} = y$ defines a partial order on A, and the associativity law $a \circ \mu_A = a \circ Pa$ forces $a(S)$ to be the least upper bound of S for every subset S, including the empty set (whose supremum is the bottom element $a(\varnothing)$). An algebra structure on A is thus a rule assigning a supremum to *every* subset, which is exactly a complete sup-lattice; a P-algebra morphism is a map preserving arbitrary suprema. So $\mathbf{Set}^P$ is the category of complete sup-lattices and sup-preserving maps.

3. *The maybe monad.* For the maybe monad $MA = A \sqcup \{*\}$ (example 5.1.3), an M-algebra $a\colon A \sqcup \{*\} \to A$ is determined by its value $a(*)$ together with the unit law fixing A; the associativity law imposes nothing further, so an M-algebra is a set with a chosen basepoint $a(*)$, and $\mathbf{Set}^M$ is the category of pointed sets.

In each case the structure map is the operation of the theory: word-multiplication for monoids, supremum for sup-lattices, basepoint selection for pointed sets. This is the sense in which a monad on **Set** encodes an algebraic theory whose operations are "T-many arguments in, one out", and $\mathbf{C}^T$ is its category of models with the theory-preserving maps. The converse question, which categories over **Set** arise this way, is the subject of the monadicity theorem later in the chapter.

One interpretive point closes the section. Theorem 5.2.4 realizes a given monad by an adjunction, but it is not the only such realization: $\mathbf{C}^T$ is the *terminal* one. Among all adjunctions $F \dashv U$ inducing the given monad, the Eilenberg-Moore adjunction receives a unique comparison functor from every other, making it the largest category of T-algebras compatible with the monad. The next section constructs the initial realization, the Kleisli category, together with the comparison functor between them; the two together bound every adjunction that induces T.

5.3 The Kleisli Category

Where the Eilenberg-Moore category $\mathbf{C}^T$ (section 5.2) realizes T from *every* T-algebra, this section builds the opposite extreme from the smallest possible supply, the free ones. The construction packages the free algebras into a category $\mathbf{C}_T$ using only η and μ, exhibits it as a second adjunction that induces T, and locates it against $\mathbf{C}^T$: $\mathbf{C}_T$ is initial and $\mathbf{C}^T$ terminal among all adjunctions inducing T.

The idea is to read a morphism $c \to Tc'$ not as a map landing in Tc' but as a map from c to c' that carries a T-shaped effect: a function returning a formal word, a subset, or a possibly-undefined value rather than a single element. Composing two such effectful maps requires flattening the doubled effect, and μ is exactly the flattening.

Definition 5.3.1: Kleisli Category

Let (T, η, μ) be a monad on a category $\mathbf{C}$ (definition 5.1.1). The *Kleisli category* $\mathbf{C}_T$ has the same objects as $\mathbf{C}$. A morphism $c \to c'$ in $\mathbf{C}_T$ is a $\mathbf{C}$-morphism $f\colon c \to Tc'$, called a *Kleisli arrow*. The identity on c is the unit component

$$\eta_c\colon c \to Tc$$

and the composite of $f\colon c \to Tc'$ with $g\colon c' \to Tc''$ (in that order, g after f) is the Kleisli arrow

$$g \odot f \;=\; \mu_{c''} \circ Tg \circ f \;:\; c \to Tc'' \tag{5.3}$$

where $\odot$ denotes composition in $\mathbf{C}_T$ to distinguish it from composition $\circ$ in $\mathbf{C}$.

The formula (5.3) reads left to right through the effect: apply f to reach Tc', apply Tg to push each intermediate value through g and land in TTc'', then apply $\mu_{c''}$ to collapse the doubled effect down to Tc''. That $\mathbf{C}_T$ is a category is not automatic; it is exactly the monad laws in disguise, and the verification shows why associativity and unitality of μ and η are the right axioms to impose.

Proposition 5.3.2: The Kleisli Category is a Category

With the composition and identities of definition 5.3.1, $\mathbf{C}_T$ is a category.

***Proof*:**
Write $\odot$ for Kleisli composition and $\circ$ for composition in $\mathbf{C}$; naturality of η and μ and the monad laws (definition 5.1.1) are used throughout.

Step 1: Left unit. Let $f\colon c \to Tc'$ be a Kleisli arrow, so its target identity is $\eta_{c'}\colon c' \to Tc'$. Then

$$\eta_{c'} \odot f = \mu_{c'} \circ T\eta_{c'} \circ f = (\mu_{c'} \circ T\eta_{c'}) \circ f = \mathrm{id}_{Tc'} \circ f = f$$

using the right unit law $\mu \circ T\eta = \mathrm{id}_T$ of the monad.

Step 2: Right unit. With $\eta_c : c \to Tc$ the identity on the source,

$$f \odot \eta_c = \mu_{c'} \circ Tf \circ \eta_c$$

Naturality of $\eta : \mathrm{id}_{\mathbf{C}} \Rightarrow T$ applied to $f : c \to Tc'$ gives $Tf \circ \eta_c = \eta_{Tc'} \circ f$, so

$$f \odot \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = (\mu_{c'} \circ \eta_{Tc'}) \circ f = \mathrm{id}_{Tc'} \circ f = f$$

now using the left unit law $\mu \circ \eta T = \mathrm{id}_T$.

Step 3: Associativity. Let $f : c \to Tc'$, $g : c' \to Tc''$, and $h : c'' \to Tc'''$. Expanding the two bracketings,

$$h \odot (g \odot f) = \mu_{c'''} \circ Th \circ (\mu_{c''} \circ Tg \circ f) \tag{5.4}$$
$$(h \odot g) \odot f = \mu_{c'''} \circ T(\mu_{c'''} \circ Th \circ g) \circ f \tag{5.5}$$

In (5.5), functoriality of T splits $T(h \odot g) = T\mu_{c'''} \circ TTh \circ Tg$, so the right side becomes $\mu_{c'''} \circ T\mu_{c'''} \circ TTh \circ Tg \circ f$. In (5.4), naturality of $\mu : TT \Rightarrow T$ applied to $h : c'' \to Tc'''$ gives $Th \circ \mu_{c''} = \mu_{Tc'''} \circ TTh$, so the left side becomes $\mu_{c'''} \circ \mu_{Tc'''} \circ TTh \circ Tg \circ f$. The two agree once $\mu_{c'''} \circ T\mu_{c'''} = \mu_{c'''} \circ \mu_{Tc'''}$, which is precisely the associativity law $\mu \circ T\mu = \mu \circ \mu T$ of the monad, evaluated at c'''. Hence (5.4) and (5.5) coincide, and Kleisli composition is associative, completing the proof.

The proof is the monad laws read backward: the two unit laws become the two identity axioms, and the single associativity square becomes associativity of composition. A monad is thus exactly the data needed to make Kleisli arrows compose, which is one sense in which a monad is an "algebraic theory" presented by its effects. A concrete instance makes the effects visible.

Example 5.3.3: The Kleisli Category of the Powerset Monad

Let $P : \mathbf{Set} \to \mathbf{Set}$ be the covariant powerset monad (example 5.1.3), with unit $\eta_X : X \to PX$ sending x to the singleton $\{x\}$ and multiplication $\mu_X : PPX \to PX$ sending a set of subsets to its union. A Kleisli arrow $f : X \to PY$ is a function assigning to each $x \in X$ a subset $f(x) \subseteq Y$, that is, a *relation* between X and Y. The Kleisli identity η_X is the diagonal relation $x \mapsto \{x\}$. Given $f : X \to PY$ and $g : Y \to PZ$, the Kleisli composite is

$$(g \odot f)(x) = \mu_Z\big((Pg)(f(x))\big) = \bigcup_{y \in f(x)} g(y)$$

which is exactly the composite relation: x is related to z if and only if some y has $x\ f\ y$ and $y\ g\ z$. Thus $\mathbf{Set}_P$ is the category $\mathbf{Rel}$ of sets and relations, with relational composition read off the monad structure. The maybe monad $X \mapsto X + 1$ (add a disjoint basepoint) gives instead the category of partial functions, and the free-monoid monad gives functions returning formal words.

Every Kleisli category comes with a canonical adjunction, and it induces the monad one started from. The left adjoint reinterprets an ordinary map as a pure Kleisli arrow, one with trivial effect, and the right adjoint sends an object to the free algebra Tc.

Theorem 5.3.4: The Kleisli Adjunction

Let (T, η, μ) be a monad on $\mathbf{C}$ with Kleisli category $\mathbf{C}_T$. Define functors

$$F_T \colon \mathbf{C} \to \mathbf{C}_T, \qquad U_T \colon \mathbf{C}_T \to \mathbf{C}$$

as follows. The functor F_T is the identity on objects, and sends $f \colon c \to c'$ in $\mathbf{C}$ to the Kleisli arrow $\eta_{c'} \circ f \colon c \to Tc'$. The functor U_T sends an object c to Tc, and a Kleisli arrow $f \colon c \to Tc'$ to the $\mathbf{C}$-morphism $\mu_{c'} \circ Tf \colon Tc \to Tc'$. Then $F_T \dashv U_T$, and the monad induced by this adjunction (proposition 5.1.2) is (T, η, μ).

***Proof*:**
Step 1: F_T and U_T are functors. For F_T: the identity on c in $\mathbf{C}$ goes to $\eta_c \circ \mathrm{id}_c = \eta_c$, the Kleisli identity. For composability, given $f \colon c \to c'$ and $g \colon c' \to c''$ in $\mathbf{C}$, the Kleisli composite of their images is

$$(\eta_{c''} \circ g) \odot (\eta_{c'} \circ f) = \mu_{c''} \circ T(\eta_{c''} \circ g) \circ \eta_{c'} \circ f$$

Naturality of $\eta \colon \mathrm{id}_{\mathbf{C}} \Rightarrow T$ at the morphism $\eta_{c''} \circ g \colon c' \to Tc''$ gives $T(\eta_{c''} \circ g) \circ \eta_{c'} = \eta_{Tc''} \circ \eta_{c''} \circ g$, whence

$$(\eta_{c''} \circ g) \odot (\eta_{c'} \circ f) = \mu_{c''} \circ \eta_{Tc''} \circ \eta_{c''} \circ g \circ f = \eta_{c''} \circ (g \circ f)$$

using $\mu_{c''} \circ \eta_{Tc''} = \mathrm{id}_{Tc''}$ (left unit law). This is $F_T(g \circ f)$, so F_T is a functor. For U_T: the identity η_c goes to $\mu_c \circ T\eta_c = \mathrm{id}_{Tc}$ (right unit law); and U_T preserves composition because, for $f \colon c \to Tc'$ and $g \colon c' \to Tc''$,

$$U_T(g \odot f) = \mu_{c''} \circ T(\mu_{c''} \circ Tg \circ f) = \mu_{c''} \circ T\mu_{c''} \circ TTg \circ Tf$$

while $U_T(g) \circ U_T(f) = \mu_{c''} \circ Tg \circ \mu_{c'} \circ Tf$. Naturality of μ at g gives $Tg \circ \mu_{c'} = \mu_{Tc''} \circ TTg$, and associativity $\mu_{c''} \circ \mu_{Tc''} = \mu_{c''} \circ T\mu_{c''}$ turns the second expression into the first. Thus U_T is a functor.

Step 2: The hom-set bijection. By construction, for objects c of $\mathbf{C}$ and c' of $\mathbf{C}_T$,

$$\mathrm{Hom}_{\mathbf{C}_T}(F_T c, c') = \mathrm{Hom}_{\mathbf{C}_T}(c, c') = \mathrm{Hom}_{\mathbf{C}}(c, Tc') = \mathrm{Hom}_{\mathbf{C}}(c, U_T c') \tag{5.6}$$

the middle equality being the definition of a Kleisli arrow and the last the definition of U_T on objects. Take $\Phi_{c,c'}$ to be this identification.

Step 3: Naturality of the bijection. Naturality in c requires that for $u \colon c_0 \to c$ in $\mathbf{C}$ and a Kleisli arrow $k \colon c \to c'$, precomposing in $\mathbf{C}_T$ with $F_T u$ matches precomposing in $\mathbf{C}$ with u under (5.6). The left side is $k \odot (\eta_c \circ u) = \mu_{c'} \circ Tk \circ \eta_c \circ u$; naturality of η at k gives $Tk \circ \eta_c = \eta_{Tc'} \circ k$, so this is $\mu_{c'} \circ \eta_{Tc'} \circ k \circ u = k \circ u$, the right side. Naturality in c' requires that for a Kleisli arrow $v \colon c' \to c''$, postcomposing in $\mathbf{C}_T$ with v matches postcomposing in $\mathbf{C}$ with $U_T v = \mu_{c''} \circ Tv$. The left side is $v \odot k = \mu_{c''} \circ Tv \circ k = (U_T v) \circ k$, the right side. Thus Φ is natural and $F_T \dashv U_T$.

Step 4: The induced monad is T. The composite $U_T F_T$ sends an object c to $U_T c = Tc$ and a morphism $f \colon c \to c'$ to $U_T(\eta_{c'} \circ f) = \mu_{c'} \circ T(\eta_{c'} \circ f) = \mu_{c'} \circ T\eta_{c'} \circ Tf = Tf$, using the right unit law. Hence $U_T F_T = T$ as an endofunctor. The unit of the adjunction is the transpose of the identity Kleisli arrow on $F_T c$: under (5.6) the identity $\mathrm{id}_{F_T c} = \eta_c \in \mathrm{Hom}_{\mathbf{C}_T}(c, c)$ corresponds to $\eta_c \in \mathrm{Hom}_{\mathbf{C}}(c, Tc)$, so the induced unit is η. The induced multiplication is $U_T \varepsilon F_T$ (proposition 5.1.2), where the counit $\varepsilon_{c'} \colon F_T U_T c' \to c'$ is the transpose of $\mathrm{id}_{U_T c'} = \mathrm{id}_{Tc'}$; under (5.6) at object Tc' this is the Kleisli arrow $\mathrm{id}_{Tc'} \colon Tc' \to Tc'$. Then $U_T \varepsilon_{c'} = \mu_{c'} \circ T\mathrm{id}_{Tc'} = \mu_{c'}$, so the induced multiplication at c' is $\mu_{c'}$. The induced monad is (T, η, μ), as we sought to show.

Together with theorem 5.2.4, the Kleisli adjunction proves the converse promised in section 5.1: every monad arises from an adjunction, and now from at least two. The left adjoint F_T names each object by its free algebra and each map by its pure lift; the counit at c is the identity Kleisli arrow on Tc, which under U_T is μ_c, so the counit *is* the multiplication seen through the underlying category. Two adjunctions inducing the same T raises the question of how they relate, and the answer places $\mathbf{C}_T$ inside $\mathbf{C}^T$.

The bridge is the comparison functor. Every free T-algebra (Tc, μ_c) lives in $\mathbf{C}^T$, and a Kleisli arrow induces a canonical map between free algebras; assembling these identifies $\mathbf{C}_T$ with the free-algebra part of $\mathbf{C}^T$.

Theorem 5.3.5: Kleisli as the Free Algebras

Let (T, η, μ) be a monad on $\mathbf{C}$. The assignment

$$K\colon \mathbf{C}_T \to \mathbf{C}^T$$

sending an object c to the free T-algebra (Tc, μ_c) and a Kleisli arrow $f\colon c \to Tc'$ to the $\mathbf{C}$-morphism $\mu_{c'} \circ Tf\colon Tc \to Tc'$ is a fully faithful functor. Its image is the full subcategory of $\mathbf{C}^T$ on the free algebras, so K identifies $\mathbf{C}_T$ with that subcategory. Moreover, among all adjunctions inducing T, the Kleisli adjunction is initial and the Eilenberg-Moore adjunction (theorem 5.2.4) is terminal.

***Proof*:**
Step 1: K lands in $\mathbf{C}^T$ and is a functor. That (Tc, μ_c) is a T-algebra (definition 5.2.1) is the content of the free-algebra construction: the unit law $\mu_c \circ \eta_{Tc} = \mathrm{id}_{Tc}$ and the associativity law $\mu_c \circ \mu_{Tc} = \mu_c \circ T\mu_c$ are two of the monad laws. For a Kleisli arrow $f\colon c \to Tc'$, the morphism $Kf = \mu_{c'} \circ Tf$ is an algebra map $(Tc, \mu_c) \to (Tc', \mu_{c'})$: the required square $Kf \circ \mu_c = \mu_{c'} \circ T(Kf)$ reads $\mu_{c'} \circ Tf \circ \mu_c = \mu_{c'} \circ T(\mu_{c'} \circ Tf)$, and both sides equal $\mu_{c'} \circ \mu_{Tc'} \circ TTf$ by naturality of μ at f (left) and functoriality of T together with associativity of μ (right). On objects and morphisms K agrees with U_T followed by the free-algebra data, so it preserves identities and composition exactly as U_T does (theorem 5.3.4, Step 1). Thus K is a functor.

Step 2: K is faithful. A Kleisli arrow $f\colon c \to Tc'$ can be recovered from $Kf = \mu_{c'} \circ Tf$ by precomposing with the unit:

$$Kf \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = f$$

using naturality of η at f and the left unit law $\mu \circ \eta T = \mathrm{id}$. Hence $Kf = Kg$ forces $f = Kf \circ \eta_c = Kg \circ \eta_c = g$, so K is injective on each hom-set.

Step 3: K is full. Let $h\colon (Tc, \mu_c) \to (Tc', \mu_{c'})$ be any algebra map, so $h \circ \mu_c = \mu_{c'} \circ Th$. Set $f = h \circ \eta_c\colon c \to Tc'$, a Kleisli arrow. Then

$$Kf = \mu_{c'} \circ Tf = \mu_{c'} \circ T(h \circ \eta_c) = \mu_{c'} \circ Th \circ T\eta_c$$

The algebra-map condition rewrites $\mu_{c'} \circ Th = h \circ \mu_c$, so

$$Kf = h \circ \mu_c \circ T\eta_c = h \circ \mathrm{id}_{Tc} = h$$

using the right unit law $\mu \circ T\eta = \mathrm{id}$. Thus every algebra map between free algebras is Kf for some Kleisli arrow f, and combined with Step 2, K is fully faithful.

Step 4: The image and the identification. By Steps 2 and 3, K is injective on objects up to the free-algebra assignment and a bijection on each hom-set $\mathrm{Hom}_{\mathbf{C}_T}(c, c') \to \mathrm{Hom}_{\mathbf{C}^T}\big((Tc, \mu_c), (Tc', \mu_{c'})\big)$. Its objects are exactly the free algebras (Tc, μ_c), and fullness says every $\mathbf{C}^T$-morphism between two free algebras lies in the image. Hence K is an isomorphism of $\mathbf{C}_T$ onto the full subcategory of $\mathbf{C}^T$ spanned by the free algebras.

Step 5: Initial and terminal placement. Fix the monad T and let $\mathbf{Adj}(T)$ be the category whose objects are adjunctions inducing T: a category $\mathbf{D}$ with $F \dashv G$, $F\colon \mathbf{C} \to \mathbf{D}$ and $G\colon \mathbf{D} \to \mathbf{C}$, such that $GF = T$ and the induced unit and multiplication (proposition 5.1.2) are η and μ. A morphism from $(\mathbf{D}, F, G)$ to $(\mathbf{D}', F', G')$ is a functor $R\colon \mathbf{D} \to \mathbf{D}'$ with $RF = F'$ and $G'R = G$.

Step 5a: The Kleisli adjunction is initial. Fix a target $(\mathbf{D}', F', G')$ with counit ε'; write $\Phi\colon \mathbf{D}'(F'c, F'c') \to \mathbf{C}(c, G'F'c')$ for the adjunction bijection of $F' \dashv G'$, which sends g to $G'g \circ \eta_c$ (definition 4.1.1), using that the induced unit of $F' \dashv G'$ is the monad unit η. A morphism $R\colon \mathbf{C}_T \to \mathbf{D}'$ must satisfy $RF_T = F'$ and $G'R = U_T$. Since F_T is the identity on objects, $Rc = RF_Tc = F'c$ on objects. For a Kleisli arrow $f\colon c \to Tc'$, note $Tc' = G'F'c'$, so f lies in $\mathbf{C}(c, G'F'c')$; the constraint $G'(Rf) = U_Tf = \mu_{c'} \circ Tf$ determines the transpose of Rf, because

$$\Phi(Rf) = G'(Rf) \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = \mu_{c'} \circ \eta_{Tc'} \circ f = f$$

using naturality of η at f and the left unit law $\mu \circ \eta T = \mathrm{id}$. As Φ is a bijection, Rf is forced to be $\Phi^{-1}(f) = \varepsilon'_{F'c'} \circ F'f\colon F'c \to F'c'$. Both the object and the morphism assignment are thus determined, so at most one R exists.

This assignment is a morphism of adjunctions. It is a functor: the Kleisli identity η_c goes to $\varepsilon'_{F'c} \circ F'\eta_c = \mathrm{id}_{F'c}$ by the triangle identity $\varepsilon'F' \circ F'\eta = \mathrm{id}_{F'}$, and for Kleisli arrows $f\colon c \to Tc'$, $g\colon c' \to Tc''$ the two morphisms $R(g \odot f)$ and $Rg \circ Rf$ have the same transpose under Φ: the computation above gives $\Phi(R(g \odot f)) = g \odot f$, while

$$\Phi(Rg \circ Rf) = G'(Rg) \circ G'(Rf) \circ \eta_c = U_Tg \circ (U_Tf \circ \eta_c) = U_Tg \circ f = \mu_{c''} \circ Tg \circ f = g \odot f$$

where $U_Tf \circ \eta_c = \mu_{c'} \circ Tf \circ \eta_c = f$ as before, so $R(g \odot f) = Rg \circ Rf$. The constraint $G'R = U_T$ holds by construction: $G'(Rf) = G'\varepsilon'_{F'c'} \circ G'F'f = \mu_{c'} \circ Tf = U_Tf$, using that the induced multiplication is $\mu = G'\varepsilon'F'$, so $\mu_{c'} = G'\varepsilon'_{F'c'}$. Finally $RF_T = F'$: on a $\mathbf{C}$-morphism $f\colon c \to c'$, the image $F_Tf = \eta_{c'} \circ f$ has $R(F_Tf) = \varepsilon'_{F'c'} \circ F'\eta_{c'} \circ F'f = F'f$ by the same triangle identity. Hence R exists and is unique, and $\mathbf{C}_T$ is initial.

Step 5b: The Eilenberg-Moore adjunction is terminal. Fix a source $(\mathbf{D}, F, G)$ with counit ε. Define $K^{\mathbf{D}}\colon \mathbf{D} \to \mathbf{C}^T$ on an object d by $K^{\mathbf{D}}d = (Gd, G\varepsilon_d)$ and on a morphism $p\colon d \to d'$ by $K^{\mathbf{D}}p = Gp$. That $(Gd, G\varepsilon_d)$ is a T-algebra (definition 5.2.1) uses the two adjunction facts. Its unit law $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ is the triangle identity $G\varepsilon \circ \eta G = \mathrm{id}_G$ at d. Its associativity law reads $G\varepsilon_d \circ \mu_{Gd} = G\varepsilon_d \circ T(G\varepsilon_d)$; since $\mu_{Gd} = G\varepsilon_{FGd}$ and $T(G\varepsilon_d) = GFG\varepsilon_d$, both sides are G applied to $\varepsilon_d \circ \varepsilon_{FGd}$ and $\varepsilon_d \circ FG\varepsilon_d$ respectively, equal by naturality of $\varepsilon\colon FG \Rightarrow \mathrm{id}$ at ε_d. For $p\colon d \to d'$, the morphism Gp is an algebra map $(Gd, G\varepsilon_d) \to (Gd', G\varepsilon_{d'})$: the condition $Gp \circ G\varepsilon_d = G\varepsilon_{d'} \circ T(Gp)$ is G applied to $p \circ \varepsilon_d = \varepsilon_{d'} \circ FGp$, again naturality of ε at p. Being G on morphisms, $K^{\mathbf{D}}$ preserves identities and composition, so it is a functor.

It is a morphism of adjunctions. On objects $U^TK^{\mathbf{D}}d = Gd$ and on morphisms $U^TK^{\mathbf{D}}p = Gp$, so $U^TK^{\mathbf{D}} = G$. For F: on an object c, $K^{\mathbf{D}}(Fc) = (GFc, G\varepsilon_{Fc}) = (Tc, \mu_c) = F^Tc$, using $\mu_c = G\varepsilon_{Fc}$; on a morphism f, $K^{\mathbf{D}}(Ff) = GFf = Tf = F^Tf$, so $K^{\mathbf{D}}F = F^T$.

For uniqueness, let $S\colon \mathbf{D} \to \mathbf{C}^T$ be any morphism of adjunctions, so $SF = F^T$ and $U^TS = G$. The second constraint forces the underlying object of Sd to be $U^T(Sd) = Gd$, so $Sd = (Gd, x_d)$

for some structure map $x_d\colon TGd \to Gd$; only x_d remains to be pinned. Apply S to the counit $\varepsilon_d\colon FGd \to d$: since $SF = F^T$, the source is $SFGd = F^T(Gd) = (TGd, \mu_{Gd})$, so $S\varepsilon_d$ is an algebra map $(TGd, \mu_{Gd}) \to (Gd, x_d)$ whose underlying $\mathbf{C}$-morphism is $U^T(S\varepsilon_d) = G\varepsilon_d$. Under the free-forgetful bijection of theorem 5.2.4, an algebra map out of the free algebra (TGd, μ_{Gd}) equals $x_d \circ T(-)$ applied to its restriction along η_{Gd}; that restriction is $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ by the triangle identity, so

$$G\varepsilon_d = x_d \circ T(\mathrm{id}_{Gd}) = x_d$$

Hence $x_d = G\varepsilon_d$ and $Sd = (Gd, G\varepsilon_d) = K^{\mathbf{D}}d$ on objects. On a morphism p, $U^T(Sp) = Gp = U^T(K^{\mathbf{D}}p)$, and U^T is faithful (a $\mathbf{C}^T$-morphism is determined by its underlying $\mathbf{C}$-morphism, definition 5.2.3), so $Sp = K^{\mathbf{D}}p$. Thus $S = K^{\mathbf{D}}$, and $\mathbf{C}^T$ is terminal.

Composing the two extremes, every $(\mathbf{D}, F, G)$ inducing T receives the unique morphism $R\colon \mathbf{C}_T \to \mathbf{D}$ from $\mathbf{C}_T$ and admits the unique morphism $K^{\mathbf{D}}\colon \mathbf{D} \to \mathbf{C}^T$ to $\mathbf{C}^T$, so the comparison factors as $\mathbf{C}_T \to \mathbf{D} \to \mathbf{C}^T$. Taking $\mathbf{D} = \mathbf{C}^T$ shows the composite $K^{\mathbf{C}^T} \circ R\colon \mathbf{C}_T \to \mathbf{C}^T$ is the unique morphism $\mathbf{C}_T \to \mathbf{C}^T$, which is the K of Steps 1 to 4. This establishes the initial and terminal placement, completing the proof.

The theorem exhibits $\mathbf{C}_T$ and $\mathbf{C}^T$ as the two ends of a spectrum. Every construction of T from an adjunction factors as $\mathbf{C}_T \to \mathbf{D} \to \mathbf{C}^T$: the Kleisli category adjoins nothing beyond the free algebras, the Eilenberg-Moore category adjoins every algebra, and any other realization sits between. When T is the free-monoid monad, $\mathbf{C}_T$ is the category whose objects are sets and whose maps $X \to Y$ are functions $X \to$ (words in Y), while $\mathbf{C}^T = \mathbf{Mon}$; the inclusion $\mathbf{C}_T \hookrightarrow \mathbf{Mon}$ is the free monoids sitting inside all monoids. Fullness on the free part is what makes $\mathbf{C}_T$ an object of study rather than a bookkeeping device: a Kleisli arrow and the algebra map it induces carry the same information, so one may compute with whichever is convenient. When the comparison functor $\mathbf{D} \to \mathbf{C}^T$ is not just unique but an equivalence, the category $\mathbf{D}$ is $\mathbf{C}^T$ up to equivalence, so the adjunction recovers the full algebraic content of T; whether a given forgetful functor has this property is the question of *monadicity*.

5.4 Monadicity

The previous sections travel one direction: every monad T on $\mathbf{C}$ produces a category of algebras $\mathbf{C}^T$, and the forgetful functor $U^T\colon \mathbf{C}^T \to \mathbf{C}$ carries the monad back (definition 5.2.3, theorem 5.2.4). This section runs the converse. Given a functor $G\colon \mathbf{D} \to \mathbf{C}$ with a left adjoint, G remembers a monad $T = GF$ on $\mathbf{C}$; the question is whether $\mathbf{D}$ is nothing more than the algebras for that monad. When it is, G is *monadic*: the objects of $\mathbf{D}$ are algebraic structures on their images, and G forgets exactly the operations that a T-algebra structure records. Beck's theorem gives the exact test.

Fix an adjunction $F \dashv G$ with $F\colon \mathbf{C} \to \mathbf{D}$, $G\colon \mathbf{D} \to \mathbf{C}$, unit η, and counit ε. By proposition 5.1.2 this induces the monad $T = GF$ on $\mathbf{C}$, with unit $\eta\colon \mathrm{id}_C \Rightarrow T$ and multiplication $\mu = G\varepsilon F$. Every object d of $\mathbf{D}$ then carries a canonical T-algebra structure on its image Gd: the counit component $\varepsilon_d\colon FGd \to d$ has image $G\varepsilon_d\colon TGd = GFGd \to Gd$, and the triangle identities and naturality of the counit make $(Gd, G\varepsilon_d)$ a T-algebra. This assignment is functorial, and it is the object of study.

Definition 5.4.1: Monadic Functor

Let $G\colon \mathbf{D} \to \mathbf{C}$ have a left adjoint $F \dashv G$, and let $T = GF$ be the induced monad on $\mathbf{C}$. The *comparison functor* is

$$K\colon \mathbf{D} \to \mathbf{C}^T, \qquad K(d) = (Gd,\, G\varepsilon_d), \qquad K(h\colon d \to d') = Gh$$

into the Eilenberg-Moore category of T. The functor G is *monadic* if it has a left adjoint and the comparison functor K is an equivalence of categories.

The comparison functor always exists and always commutes with the forgetful functors: $U^T K = G$, since $U^T(Gd, G\varepsilon_d) = Gd$. So K factors G through the algebras, presenting $\mathbf{D}$ as sitting over $\mathbf{C}^T$. Monadicity is the requirement that this factorization lose nothing, that K be an equivalence. When it is, $\mathbf{D}$ is, up to equivalence, precisely $\mathbf{C}^T$: every object of $\mathbf{D}$ is a T-algebra, every T-algebra comes from an object of $\mathbf{D}$, and morphisms match. The functor G is then a bona fide forgetful functor, and T is the complete record of the structure it forgets.

Before stating Beck's test, it helps to see why a naive expectation fails. One might hope that any G with a left adjoint is monadic. It is not: the free-forgetful adjunction between **Set** and **Top** has $GF = \mathrm{id}_{\mathbf{Set}}$ (the free topology on a set is discrete, whose underlying set is the set again), so T is the identity monad, $\mathbf{Set}^T \cong \mathbf{Set}$, and $K\colon \mathbf{Top} \to \mathbf{Set}$ is the forgetful functor, which is not an equivalence. A topology is not encoded by any operations on the underlying set. Beck's theorem isolates the exact structure whose presence certifies monadicity.

The certifying structure is a condition on how G interacts with certain coequalizers (definition 3.3.5). A parallel pair $f, g\colon d \rightrightarrows d'$ in $\mathbf{D}$ is a *G-split pair* if its image $Gf, Gg\colon Gd \rightrightarrows Gd'$ admits a split coequalizer in $\mathbf{C}$: a coequalizer $q\colon Gd' \to c$ together with sections $s\colon c \to Gd'$ and $t\colon Gd' \to Gd$ satisfying $qs = \mathrm{id}_c$, $Gf\,t = \mathrm{id}_{Gd'}$, and $Gg\,t = sq$. A split coequalizer is preserved by every functor, because the equations defining it are equational: applying any functor to q, s, t produces the split coequalizer of the image pair. This is the property that makes the condition usable, and it is why the coequalizers Beck asks for are exactly the G-split ones.

Theorem 5.4.2: Beck's Monadicity Theorem

A functor $G\colon \mathbf{D} \to \mathbf{C}$ is monadic if and only if

1. G has a left adjoint, and
2. G creates coequalizers of G-split pairs (definition 3.4.6): for every G-split pair $f, g\colon d \rightrightarrows d'$ in $\mathbf{D}$, the coequalizer cocone of Gf, Gg in $\mathbf{C}$ has a unique lift along G to a cocone on f, g in $\mathbf{D}$, and that lifted cocone is a coequalizer.

***Proof Key ideas*:**
The full proof is a long diagram bookkeeping (the *precise* and *crude* monadicity theorems in the literature); it is deferred to Mac Lane's *Categories for the Working Mathematician* (VI.7) and Riehl's *Category Theory in Context* (5.5), which is why only the ideas are recorded here (BOOK.md §6 sanctions this deferral). Three points carry the argument.

The canonical presentation of a T-algebra. Every T-algebra (A, a) is a coequalizer of free algebras.

The free algebras (T^2A, μ_{TA}) and (TA, μ_A) sit in a parallel pair

$$(T^2A,\ \mu_{TA}) \rightrightarrows (TA,\ \mu_A)$$

with legs μ_A and Ta, and (A, a) is its coequalizer in $\mathbf{C}^T$. Applying U^T gives the pair $\mu_A, Ta\colon T^2A \rightrightarrows TA$ in $\mathbf{C}$, which is split by η_{TA} and η_A: the algebra axioms $a \circ \eta_A = \mathrm{id}_A$ and $a \circ \mu_A = a \circ Ta$ supply the section and coequalizing equations, the remaining splitting data coming from the monad unit laws and the naturality of η. So every T-algebra is a U^T-split pair's coequalizer, and it is its own free presentation.

Constructing the inverse to K. The comparison functor K has a left adjoint $L\colon \mathbf{C}^T \to \mathbf{D}$ built by forming these coequalizers in $\mathbf{D}$. Given (A, a), transport its free presentation $\mu_A, Ta\colon T^2A \rightrightarrows TA$ through F to the parallel pair $FTA \rightrightarrows FA$ in $\mathbf{D}$ (the legs are $F\mu_A$ and the composite FTa followed by ε_{FA}, the transported free presentation); this pair is G-split, its splitting descending from the split coequalizer of μ_A, Ta established above. Condition (2) creates the coequalizer of this pair in $\mathbf{D}$, and $L(A, a)$ is defined to be that coequalizer. Creation makes L well-defined and functorial.

Why the G-split condition forces an equivalence. The unit and counit of $L \dashv K$ are the comparison maps between an object and the coequalizer of its canonical presentation. On the algebra side, $KL(A, a) \cong (A, a)$ because K preserves the created coequalizer (split coequalizers are absolute, preserved by every functor) and (A, a) is the coequalizer of its own presentation. On the $\mathbf{D}$ side, $LK(d) \cong d$ because d is the coequalizer of the G-split pair $\varepsilon_{FGd}, FG\varepsilon_d\colon FGFGd \rightrightarrows FGd$ (the presentation of d), and creation delivers exactly that object. Thus K and L are mutually inverse up to isomorphism, so K is an equivalence and G is monadic. Conversely, if G is monadic then G inherits condition (2) from U^T, which creates these coequalizers by the free-presentation computation above, completing the proof.

The theorem trades the global requirement "K is an equivalence" for a local, checkable one: create the coequalizers of G-split pairs. The mechanism is the canonical presentation. Every algebra is the coequalizer of free algebras built from it, and G knows the free objects because it has a left adjoint; the only thing G needs to supply on its own is these coequalizers, lifted uniquely from $\mathbf{C}$. Where G can build them, $\mathbf{D}$ reassembles from the free part exactly as the algebras do, and the two categories coincide. This is why the condition is about coequalizers and not general colimits: the presentation of an algebra is a single reflexive coequalizer, so that one shape suffices.

The payoff is that the standard forgetful functors of algebra pass the test, and this is the precise content of the slogan that groups, rings, and modules are algebraic structures.

Example 5.4.3: Groups, Rings, and Modules are Monadic over Set
The forgetful functors

$$U\colon \mathbf{Grp} \to \mathbf{Set}, \qquad U\colon \mathbf{Ring} \to \mathbf{Set}, \qquad U\colon \mathbf{Mod}_R \to \mathbf{Set}$$

are each monadic. Take $U\colon \mathbf{Grp} \to \mathbf{Set}$; the other two are identical in form.

The left adjoint is the free-group functor F, so condition (1) holds by the free-forgetful adjunction ([4]). The induced monad $T = UF$ sends a set X to the underlying set of the free group on X, and its algebras are groups: a T-algebra structure on a set is precisely a choice of multiplication, inverse, and identity satisfying the group axioms, recovered from the operations that the free-group monad encodes ([4] for the free group and the group axioms). So the comparison functor $K\colon \mathbf{Grp} \to \mathbf{Set}^T$ is already the identification of groups with T-algebras.

For condition (2), let $f, g\colon G \rightrightarrows H$ be group homomorphisms whose underlying maps Uf, Ug admit a split coequalizer of sets. The coequalizer of Uf, Ug in **Set** is the quotient of UH by the equivalence relation generated by $Uf(x) \sim Ug(x)$. The split hypothesis is what guarantees this set-level quotient is compatible with the operations, so a group structure descends onto it: under that hypothesis the set-level quotient coincides with UH modulo the normal subgroup generated by $\{f(x)g(x)^{-1}\}$, which is the coequalizer of f, g in **Grp**. So the quotient carries a unique group structure making q a homomorphism, and with that structure it is the coequalizer in **Grp**. The lift exists, is unique, and is a coequalizer, so U creates the coequalizer. Both conditions hold, and Beck's theorem gives monadicity. The rings and modules cases run the same way, with the relevant quotient (by an ideal, by a submodule) descending along the split coequalizer. This is what "algebraic over **Set**" means: the category is the algebras for a monad on **Set** that its forgetful functor remembers.

By contrast, $U\colon \mathbf{Top} \to \mathbf{Set}$ is not monadic. As noted above, the free topology on a set is the discrete one, whose underlying set is the original set, so the induced monad is the identity and $\mathbf{Set}^T \cong \mathbf{Set}$; the comparison functor is U itself, sending a space to its point-set and collapsing all topologies on a fixed set to one object. No monad on **Set** has topological spaces as its algebras, because a topology is not a system of finitary operations on the underlying set. Beck's theorem locates the failure precisely in condition (2): U does *not* create coequalizers of U-split pairs. Given a U-split pair with set-coequalizer $q\colon UY \to Q$, the lift is not unique. The final topology on Q (the quotient topology) does make q a coequalizer in **Top**, but every topology on Q coarser than the final one also makes q continuous, so the coequalizer cocone admits many lifts along U, only one of which is a coequalizer. Creation requires a *unique* lift, and uniqueness fails. This is the same multiplicity-of-topologies phenomenon that also breaks conservativity: U does not reflect isomorphisms, because a continuous bijection from a finer to a coarser topology has underlying map a set-isomorphism yet is not a homeomorphism, and a monadic functor is conservative. Both failures trace to one fact: many topologies sit over a single set, and U cannot tell them apart. The dividing line Beck's theorem draws is exactly the line between structure recorded by operations and structure, like a topology, that is not.

6

Kan Extensions

Given a functor $F\colon \mathbf{C} \to \mathbf{E}$ and a functor $K\colon \mathbf{C} \to \mathbf{D}$, a *Kan extension* extends F along K universally, in a left and a right variant. This is the last construction the book needs: limits, colimits, adjunctions, and the Yoneda lemma are all Kan extensions. This chapter defines the two Kan extensions as adjoints to restriction, computes them by a pointwise (co)limit formula, derives that slogan, and closes with the calculus of ends and coends and the density theorem, which writes every presheaf as a colimit of the representables the Yoneda embedding produced.

6.1 Left and Right Kan Extensions

A functor $F\colon \mathbf{C} \to \mathbf{E}$ records data indexed by the objects and morphisms of $\mathbf{C}$. Given a second functor $K\colon \mathbf{C} \to \mathbf{D}$, one may ask to transport that data along K: to produce a functor on $\mathbf{D}$ that agrees with F on the image of K, and does so in the best possible way. When K is an inclusion this is a genuine extension of F from a subcategory to the whole of $\mathbf{D}$; in general K need not be injective on objects, so the word "extension" is optimistic, but the universal formulation survives intact. Two such best approximations exist, one approximating F from the left and one from the right, and each is characterized by a universal property. This section defines them and shows that the two universal properties are precisely the two halves of a chain of adjunctions with restriction along K.

Throughout, fix $K\colon \mathbf{C} \to \mathbf{D}$. Restriction along K is the functor

$$K^*\colon [\mathbf{D}, \mathbf{E}] \to [\mathbf{C}, \mathbf{E}], \qquad H \mapsto HK$$

sending a functor $H\colon \mathbf{D} \to \mathbf{E}$ to the composite $HK\colon \mathbf{C} \to \mathbf{E}$ and a natural transformation $\theta\colon H \Rightarrow H'$ to its whiskering θK (the natural transformation with component θ_{Kc} at c). Left and right Kan extension will turn out to be the two adjoints of K^*.

Definition 6.1.1: Left and Right Kan Extension

Let $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$ be functors. A *left Kan extension* of F along K is a functor

$$\operatorname{Lan}_K F\colon \mathbf{D} \to \mathbf{E}$$

together with a natural transformation $\eta\colon F \Rightarrow (\operatorname{Lan}_K F)K$, the *unit*, that is universal from F to K^*: for every functor $G\colon \mathbf{D} \to \mathbf{E}$ and every natural transformation $\alpha\colon F \Rightarrow GK$, there is a unique natural transformation $\beta\colon \operatorname{Lan}_K F \Rightarrow G$ with

$$(\beta K) \circ \eta = \alpha$$

where βK is the whiskering of β by K.

Dually, a *right Kan extension* of F along K is a functor $\operatorname{Ran}_K F\colon \mathbf{D} \to \mathbf{E}$ with a natural transformation $\varepsilon\colon (\operatorname{Ran}_K F)K \Rightarrow F$, the *counit*, universal to F from K^*: for every $G\colon \mathbf{D} \to \mathbf{E}$ and every $\alpha\colon GK \Rightarrow F$, there is a unique $\beta\colon G \Rightarrow \operatorname{Ran}_K F$ with $\varepsilon \circ (\beta K) = \alpha$.

The two triangles record the same shape with the 2-cell pointing opposite ways. For the left Kan extension the unit fills the extension triangle

$$\begin{tikzcd} \mathbf{C} \arrow[r, "F"] \arrow[d, "K"'] \arrow[dr, phantom, "\Downarrow\eta"] & \mathbf{E} \\ \mathbf{D} \arrow[ur, "\operatorname{Lan}_K F"'] & \end{tikzcd} \qquad \begin{tikzcd} \mathbf{C} \arrow[r, "F"] \arrow[d, "K"'] \arrow[dr, phantom, "\Uparrow\varepsilon"] & \mathbf{E} \\ \mathbf{D} \arrow[ur, "\operatorname{Ran}_K F"'] & \end{tikzcd}$$

and the universal property says every competing pair (G, α) factors through $(\operatorname{Lan}_K F, \eta)$ by a unique 2-cell β. This is the defining shape of a universal arrow from F into the functor K^*, and universal objects are unique up to unique isomorphism. Applied here: if $(\operatorname{Lan}_K F, \eta)$ and (L', η') both left-Kan-extend F along K, the universal property of each produces a comparison, and the uniqueness clause forces the two comparisons to be mutually inverse. So the left Kan extension, when it exists, is determined up to a unique natural isomorphism compatible with the units; likewise the right Kan extension.

Existence, on the other hand, is not automatic: for a fixed K and F there may be no functor on $\mathbf{D}$ carrying a universal unit, and whether one exists depends on $\mathbf{E}$ having enough (co)limits. That question is settled in section 6.2, which constructs the extensions explicitly as (co)limits and pins down exactly when they exist. The present section treats the extensions where they do exist and extracts the structural consequence: the universal property of definition 6.1.1 is an adjunction.

Before the adjunction, one worked example, to see that "extension" is sometimes literal. It also isolates the mechanism, a coproduct that copies Fc once for each arrow $Kc \to d$, that section 6.2 will generalize into the pointwise formula.

Example 6.1.2: Extension Along a Coproduct Inclusion

Let $\mathbf{C}$ be the discrete category on one object $*$, so a functor $F\colon \mathbf{C} \to \mathbf{E}$ is the choice of a single object $e = F*$ of $\mathbf{E}$. Let $\mathbf{D}$ be the discrete category on a two-element set $\{0, 1\}$, and let $K\colon \mathbf{C} \to \mathbf{D}$ pick out the object 0. Assume $\mathbf{E}$ has an initial object and binary coproducts.

A functor $G\colon \mathbf{D} \to \mathbf{E}$ is a pair of objects $(G0, G1)$, and GK is the single object $G0$; a natural transformation $F \Rightarrow GK$ is a single morphism $e \to G0$. The left Kan extension is

$$(\mathrm{Lan}_K F)(0) = e, \qquad (\mathrm{Lan}_K F)(1) = 0_{\mathbf{E}}$$

where $0_{\mathbf{E}}$ is the initial object, with unit $\eta\colon e \to e$ the identity. Given $\alpha\colon e \to G0$, the unique $\beta\colon \mathrm{Lan}_K F \Rightarrow G$ satisfying $(\beta K) \circ \eta = \alpha$ has component $\beta_0 = \alpha$ at 0, forced by the equation, and component $\beta_1\colon 0_{\mathbf{E}} \to G1$ the unique map out of the initial object. So the extension copies F where K lands and fills the rest with the initial object, the "empty" value that imposes no constraint. The right Kan extension is the mirror image: $(\mathrm{Ran}_K F)(0) = e$ and $(\mathrm{Ran}_K F)(1) = 1_{\mathbf{E}}$, the terminal object.

The mechanism sharpens when K is not injective. Take $\mathbf{C}$ discrete on $\{a, b\}$ and $\mathbf{D} = \mathbf{1}$ the terminal category, so K sends both objects to the single object $*$. A functor F is a pair (Fa, Fb) of objects of $\mathbf{E}$; a natural transformation $F \Rightarrow GK$ is a pair of maps $Fa \to G*$, $Fb \to G*$ into the one object $G*$. Such a pair is exactly a map out of the coproduct $Fa \sqcup Fb$, so

$$(\mathrm{Lan}_K F)(*) = Fa \sqcup Fb$$

with unit the two coprojections. This is the colimit of F, computed by hand. That a left Kan extension along the functor to $\mathbf{1}$ is a colimit is the first of the identifications developed in section 6.3.

The example shows the universal property doing local work; the following theorem shows it is global. Read across all F at once, the correspondence $\alpha \leftrightarrow \beta$ of definition 6.1.1 is exactly the defining bijection of an adjunction between Lan_K and K^*.

Theorem 6.1.3: Kan Extensions as Adjoints to Restriction

Fix $K\colon \mathbf{C} \to \mathbf{D}$ and a category $\mathbf{E}$. Suppose the left Kan extension $\mathrm{Lan}_K F$ exists for every $F\colon \mathbf{C} \to \mathbf{E}$. Then $F \mapsto \mathrm{Lan}_K F$ is the object part of a functor $\mathrm{Lan}_K\colon [\mathbf{C}, \mathbf{E}] \to [\mathbf{D}, \mathbf{E}]$ that is left adjoint to restriction,

$$\mathrm{Lan}_K \dashv K^*$$

Dually, if $\mathrm{Ran}_K F$ exists for every F, then Ran_K is right adjoint to K^*, so that

$$\mathrm{Lan}_K \dashv K^* \dashv \mathrm{Ran}_K$$

The unit η of definition 6.1.1 is the unit of the first adjunction, and the counit ε is the counit of the second.

***Proof*:**
The two claims are dual (box 1.3.3): a right Kan extension of F along K is a left Kan extension of F^{op} along K^{op} into $\mathbf{E}^{\mathrm{op}}$, and Ran_K $\dashv$-ness for $\mathbf{E}$ is $\mathrm{Lan}_{K^{\mathrm{op}}}$ $\dashv$-ness for $\mathbf{E}^{\mathrm{op}}$. We prove the left case; the right case follows by replacing every category by its opposite and reversing every 2-cell.

Step 1: the hom-set bijection. Fix $F\colon \mathbf{C} \to \mathbf{E}$ and let $(\mathrm{Lan}_K F, \eta_F)$ be its left Kan extension. For any $G\colon \mathbf{D} \to \mathbf{E}$, the universal property of definition 6.1.1 states that the assignment

$$[\mathbf{D}, \mathbf{E}](\mathrm{Lan}_K F, G) \longrightarrow [\mathbf{C}, \mathbf{E}](F, K^*G), \qquad \beta \longmapsto (\beta K) \circ \eta_F \tag{6.1}$$

is a bijection: existence of a unique β for each α on the right is exactly the statement that this map is bijective, with inverse sending α to the induced β. Here $[\mathbf{D}, \mathbf{E}]$ and $[\mathbf{C}, \mathbf{E}]$ are the functor categories of definition 1.5.4 and the elements are natural transformations in the sense of definition 1.5.2.

Step 2: Lan_K *is a functor.* Given $\varphi\colon F \Rightarrow F'$ in $[\mathbf{C},\mathbf{E}]$, define $\mathrm{Lan}_K\,\varphi\colon \mathrm{Lan}_K\,F \Rightarrow \mathrm{Lan}_K\,F'$ to be the unique natural transformation β obtained by applying the universal property of $(\mathrm{Lan}_K\,F, \eta_F)$ to the competitor

$$F \xRightarrow{\varphi} F' \xRightarrow{\eta_{F'}} (\mathrm{Lan}_K\,F')K$$

that is, $\mathrm{Lan}_K\,\varphi$ is the unique 2-cell with $((\mathrm{Lan}_K\,\varphi)\,K) \circ \eta_F = \eta_{F'} \circ \varphi$. Uniqueness in the universal property gives functoriality directly: for $\varphi = \mathrm{id}_F$ the transformation $\mathrm{id}_{\mathrm{Lan}_K F}$ satisfies the defining equation, so $\mathrm{Lan}_K\,\mathrm{id}_F = \mathrm{id}$; and for $F \overset{\varphi}{\Rightarrow} F' \overset{\psi}{\Rightarrow} F''$ both $\mathrm{Lan}_K(\psi\circ\varphi)$ and $(\mathrm{Lan}_K\,\psi)\circ(\mathrm{Lan}_K\,\varphi)$ solve the universal problem for $\eta_{F''}\circ\psi\circ\varphi$, hence coincide.

Step 3: naturality of the bijection. An adjunction is a bijection (6.1) natural in both variables (definition 4.1.1). Naturality in G: for $\gamma\colon G \Rightarrow G'$ in $[\mathbf{D},\mathbf{E}]$ and $\beta\colon\ \mathrm{Lan}_K\,F \Rightarrow G$, post-composing then restricting gives $((\gamma\circ\beta)K)\circ\eta_F = (\gamma K)\circ(\beta K)\circ\eta_F$, since whiskering by K preserves vertical composition; this is exactly $K^*\gamma$ applied to the image of β, so the square commutes. Naturality in F: for $\varphi\colon F' \Rightarrow F$ and $\beta\colon\ \mathrm{Lan}_K\,F \Rightarrow G$, one must check

$$\big((\beta\circ\mathrm{Lan}_K\,\varphi)\,K\big)\circ\eta_{F'} = \big((\beta K)\circ\eta_F\big)\circ\varphi$$

The left side is $(\beta K)\circ\big((\mathrm{Lan}_K\,\varphi)\,K\big)\circ\eta_{F'}$, and by the defining equation of $\mathrm{Lan}_K\,\varphi$ from Step 2 the tail $\big((\mathrm{Lan}_K\,\varphi)\,K\big)\circ\eta_{F'}$ equals $\eta_F\circ\varphi$; substituting gives $(\beta K)\circ\eta_F\circ\varphi$, the right side. So (6.1) is natural in both variables, and $\mathrm{Lan}_K \dashv K^*$.

Step 4: the unit. Under the correspondence of definition 4.1.3, the unit of $\mathrm{Lan}_K \dashv K^*$ at F is the image of $\mathrm{id}_{\mathrm{Lan}_K F}$ under (6.1), namely $(\mathrm{id}\,K)\circ\eta_F = \eta_F$. So the unit of the adjunction is the unit η of definition 6.1.1, as claimed. Dualizing, the counit of $K^* \dashv \mathrm{Ran}_K$ is the counit ε, completing the proof.

The theorem reorganizes the definition rather than adding to it: the phrase "universal among functors restricting to something receiving F" is the phrase "left adjoint to restriction", once one reads the universal property as a natural hom-set bijection. This is the productive form of the definition. Adjoints are unique when they exist, which re-derives the uniqueness noted after definition 6.1.1; a right adjoint preserves limits (theorem 4.3.4) and, dually, a left adjoint preserves colimits, so Lan_K and Ran_K inherit exactness properties for free; and composites of Kan extensions along composable functors become composites of adjoints. The hypothesis, that the extension exist for *every* F, is what upgrades the object-level universal property into a functor and hence an adjunction. When only some F admit an extension there is no global adjoint, only the pointwise universal arrows of definition 6.1.1.

6.2 The Pointwise Formula

The adjoint characterization of theorem 6.1.3 shows that Kan extensions exist, but it does not say what they are: knowing that Lan_K is left adjoint to restriction gives no recipe for the value $(\mathrm{Lan}_K\,F)(d)$ at a particular object d of $\mathbf{D}$. This section supplies the recipe. When the target category $\mathbf{E}$ has enough colimits, the left Kan extension is computed one object at a time as a colimit, and dually the right Kan extension as a limit. The indexing shape of that colimit is a comma category, so the formula reduces the abstract extension to a concrete diagram assembled out of the values of F.

Fix functors $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$. For an object d of $\mathbf{D}$, the relevant indexing category is the comma category $(K \downarrow d)$ of definition 3.1.6: its objects are pairs (c, φ) with c an object of $\mathbf{C}$ and $\varphi\colon Kc \to d$ a morphism of $\mathbf{D}$, and a morphism $(c, \varphi) \to (c', \varphi')$ is a morphism $f\colon c \to c'$ of $\mathbf{C}$ with $\varphi' \circ Kf = \varphi$. Projecting to the first coordinate gives a functor

$$\Pi_d\colon (K \downarrow d) \to \mathbf{C}, \qquad (c, \varphi) \mapsto c, \quad f \mapsto f$$

and composing with F gives the diagram $F \circ \Pi_d\colon (K \downarrow d) \to \mathbf{E}$ whose colimit will be the value at d. The dual indexing category $(d \downarrow K)$ has objects $(c, \psi\colon d \to Kc)$ with the projection $\Pi^d\colon (d \downarrow K) \to \mathbf{C}$ defined the same way.

The colimit of $F \circ \Pi_d$ carries more structure than its value at a single d: as d varies it assembles into a functor, and that functor is the Kan extension. The next theorem proves this, and in doing so exhibits the universal 2-cell of definition 6.1.1 directly from the colimit injections.

Theorem 6.2.1: The Pointwise Formula for Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ and $F\colon \mathbf{C} \to \mathbf{E}$ be functors, and suppose that for every object d of $\mathbf{D}$ the colimit of $F \circ \Pi_d\colon (K \downarrow d) \to \mathbf{E}$ exists. Then the left Kan extension $\operatorname{Lan}_K F$ exists and is computed pointwise by

$$(\operatorname{Lan}_K F)(d) \cong \operatorname{colim}_{(K\downarrow d)} F \circ \Pi_d$$

Dually, if for every d the limit of $F \circ \Pi^d\colon (d \downarrow K) \to \mathbf{E}$ exists, then the right Kan extension $\operatorname{Ran}_K F$ exists and is computed pointwise by

$$(\operatorname{Ran}_K F)(d) \cong \lim_{(d\downarrow K)} F \circ \Pi^d$$

***Proof*:**
The right Kan extension case follows from the left by duality (box 1.3.3), so the proof concentrates on $\operatorname{Lan}_K F$. The argument has four steps: define the object function on $\mathbf{D}$, extend it to a functor $L\colon \mathbf{D} \to \mathbf{E}$, build a natural transformation $\eta\colon F \Rightarrow L \circ K$, and verify that (L, η) satisfies the universal property of the left Kan extension.

Step 1: The value at each object. For each object d of $\mathbf{D}$, write

$$Ld = \operatorname{colim}_{(K\downarrow d)} F \circ \Pi_d$$

which exists by hypothesis. The colimit comes with injection morphisms

$$\iota^d_{(c,\varphi)}\colon Fc \to Ld$$

one for each object (c, φ) of $(K \downarrow d)$, and these form a cocone: for a morphism $f\colon (c, \varphi) \to (c', \varphi')$ in $(K \downarrow d)$ one has $\iota^d_{(c',\varphi')} \circ Ff = \iota^d_{(c,\varphi)}$. The colimit is universal among such cocones: a family of morphisms $\lambda_{(c,\varphi)}\colon Fc \to X$ compatible with all the Ff factors uniquely through a single morphism $Ld \to X$.

Step 2: Functoriality in d. A morphism $g\colon d \to d'$ in $\mathbf{D}$ induces a functor

$$g_*\colon (K \downarrow d) \to (K \downarrow d'), \qquad (c, \varphi) \mapsto (c, g \circ \varphi)$$

which acts as the identity on the underlying morphisms of $\mathbf{C}$, so that $\Pi_{d'} \circ g_* = \Pi_d$ and hence $F \circ \Pi_{d'} \circ g_* = F \circ \Pi_d$. The injections for d' therefore restrict along g_* to a cocone on $F \circ \Pi_d$ with vertex Ld', whose leg at (c, φ) is $\iota^{d'}_{(c,g\circ\varphi)} \colon Fc \to Ld'$. By the universal property of Ld this cocone factors through a unique morphism

$$Lg \colon Ld \to Ld'$$

characterized by $Lg \circ \iota^{d}_{(c,\varphi)} = \iota^{d'}_{(c,g\circ\varphi)}$ for every (c, φ). Functoriality is now a uniqueness argument: both $L(\mathrm{id}_d)$ and id_{Ld} satisfy the defining equation for $g = \mathrm{id}_d$, so they agree, and for composable $g \colon d \to d'$, $h \colon d' \to d''$ both $L(h \circ g)$ and $Lh \circ Lg$ send $\iota^{d}_{(c,\varphi)}$ to $\iota^{d''}_{(c,(h\circ g)\circ\varphi)}$, so they agree by the uniqueness clause of the universal property. Thus $L \colon \mathbf{D} \to \mathbf{E}$ is a functor.

Step 3: The universal 2*-cell.* An object c of $\mathbf{C}$ gives a canonical object (c, id_{Kc}) of $(K \downarrow Kc)$, and the injection there defines a morphism

$$\eta_c = \iota^{Kc}_{(c,\mathrm{id}_{Kc})} \colon Fc \longrightarrow L(Kc)$$

To see that $\eta \colon F \Rightarrow L \circ K$ is natural, let $f \colon c \to c'$ be a morphism of $\mathbf{C}$. The map f is a morphism $(c, Kf) \to (c', \mathrm{id}_{Kc'})$ in $(K \downarrow Kc')$, since $\mathrm{id}_{Kc'} \circ Kf = Kf$, so the cocone condition for $L(Kc')$ gives

$$\iota^{Kc'}_{(c',\mathrm{id}_{Kc'})} \circ Ff = \iota^{Kc'}_{(c,Kf)}$$

On the other hand $Kf \colon Kc \to Kc'$ acts by Step 2 with $(Kf)_*(c, \mathrm{id}_{Kc}) = (c, Kf)$, so

$$L(Kf) \circ \eta_c = L(Kf) \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = \iota^{Kc'}_{(c,Kf)}$$

Comparing the two displays, $L(Kf) \circ \eta_c = \eta_{c'} \circ Ff$, which is exactly the naturality square for η.

Step 4: The universal property. It remains to show that (L, η) is initial among pairs (G, γ) consisting of a functor $G \colon \mathbf{D} \to \mathbf{E}$ and a natural transformation $\gamma \colon F \Rightarrow G \circ K$; this is the universal property of definition 6.1.1. Given such a pair, a natural transformation $\alpha \colon L \Rightarrow G$ satisfies $(\alpha K) \circ \eta = \gamma$ if and only if, for every c,

$$\alpha_{Kc} \circ \eta_c = \gamma_c$$

The claim is that there is exactly one such α. Fix an object d of $\mathbf{D}$. For each object (c, φ) of $(K \downarrow d)$ the morphism $\varphi \colon Kc \to d$ gives, by naturality of the candidate α against φ and the required equation at c, the composite

$$Gd \xleftarrow{G\varphi} G(Kc) \xleftarrow{\gamma_c} Fc$$

and the family $\{G\varphi \circ \gamma_c\}_{(c,\varphi)}$ is a cocone on $F \circ \Pi_d$ with vertex Gd: for a morphism $f \colon (c, \varphi) \to (c', \varphi')$ one has $\varphi' \circ Kf = \varphi$, so naturality of γ gives

$$(G\varphi' \circ \gamma_{c'}) \circ Ff = G\varphi' \circ G(Kf) \circ \gamma_c = G(\varphi' \circ Kf) \circ \gamma_c = G\varphi \circ \gamma_c$$

which is the cocone compatibility condition. By the universal property of the colimit Ld there is a unique morphism

$$\alpha_d \colon Ld \to Gd, \qquad \alpha_d \circ \iota^{d}_{(c,\varphi)} = G\varphi \circ \gamma_c$$

Taking $d = Kc$ and $(c, \varphi) = (c, \mathrm{id}_{Kc})$ recovers $\alpha_{Kc} \circ \eta_c = G(\mathrm{id}_{Kc}) \circ \gamma_c = \gamma_c$, so any α with components so defined satisfies the required equation; and conversely, for any such α, naturality against φ and the equation at c force

$$\alpha_d \circ \iota^{d}_{(c,\varphi)} = \alpha_d \circ L\varphi \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = G\varphi \circ \alpha_{Kc} \circ \iota^{Kc}_{(c,\mathrm{id}_{Kc})} = G\varphi \circ \gamma_c$$

which is exactly the defining equation, so α_d is the only possibility. Naturality of α against a morphism $g\colon d \to d'$ is checked on injections: using the defining equations for $\alpha_{d'}$, Lg, and α_d,

$$\alpha_{d'} \circ Lg \circ \iota^{d}_{(c,\varphi)} = \alpha_{d'} \circ \iota^{d'}_{(c,g\circ\varphi)} = G(g \circ \varphi) \circ \gamma_c = Gg \circ G\varphi \circ \gamma_c = Gg \circ \alpha_d \circ \iota^{d}_{(c,\varphi)}$$

Since the $\iota^{d}_{(c,\varphi)}$ are jointly epic (a colimit cocone is), $\alpha_{d'} \circ Lg = Gg \circ \alpha_d$, so α is natural. This exhibits (L, η) as initial in the category of pairs (G, γ), hence as the left Kan extension $\mathrm{Lan}_K F$ with universal 2-cell η, completing the proof.

The comma-category colimit deserves a plain reading. An object (c, φ) of $(K \downarrow d)$ is a way of mapping the image Kc into d; the value $(\mathrm{Lan}_K F)(d)$ glues together all the values Fc over all such approaches to d, identifying Fc with Fc' whenever a morphism of $\mathbf{C}$ carries one approach to another. The extension therefore builds the value at d out of "everything $\mathbf{C}$ knows about d through K", and does so in the most efficient way, which is what a colimit is. When K is the inclusion of a full subcategory and $\varphi = \mathrm{id}$ is available (that is, when $d = Kc$ lies in the image of K), the object (c, id_{Kc}) is terminal in $(K \downarrow d)$, the colimit collapses to Fc, and the extension extends F; the general formula measures how far the Kan extension must interpolate when no such terminal approach exists.

The hypothesis of theorem 6.2.1 is a supply of colimits indexed by the categories $(K \downarrow d)$. When $\mathbf{C}$ is small these indexing categories are small, so a cocomplete target guarantees every required colimit at once.

Corollary 6.2.2: Existence of Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ be a functor with $\mathbf{C}$ small. If $\mathbf{E}$ is cocomplete, then $\mathrm{Lan}_K F$ exists for every functor $F\colon \mathbf{C} \to \mathbf{E}$; if $\mathbf{E}$ is complete, then $\mathrm{Ran}_K F$ exists for every F. In either case the extension along K is defined for all functors simultaneously, so restriction $K^*\colon [\mathbf{D}, \mathbf{E}] \to [\mathbf{C}, \mathbf{E}]$ acquires the adjoints $\mathrm{Lan}_K \dashv K^* \dashv \mathrm{Ran}_K$ of theorem 6.1.3.

***Proof*:**
For each object d of $\mathbf{D}$ the comma category $(K \downarrow d)$ is small: its objects are pairs (c, φ) with c ranging over the small object collection of $\mathbf{C}$ and φ over the set $\mathbf{D}(Kc, d)$ (a set since $\mathbf{D}$ is locally small, as throughout), and its morphisms form a set for the same reason. A cocomplete category has a colimit for every functor out of a small category (definition 3.4.1), so $F \circ \Pi_d$ has a colimit for every d, and theorem 6.2.1 produces $\mathrm{Lan}_K F$. Since this holds for every F, restriction has a left adjoint on all of $[\mathbf{C}, \mathbf{E}]$, which is the content of the left half of theorem 6.1.3. The complete case is dual (box 1.3.3): $(d \downarrow K)$ is small by the same count, a complete $\mathbf{E}$ supplies the limits, and restriction gains its right adjoint Ran_K, completing the proof.

Most target categories met so far are cocomplete and complete: **Set**, **Grp**, **Ab**, **Ring**, $\mathbf{Mod}_R$, and **Top** all have all small limits and colimits, so along any functor out of a small category the Kan extensions into these categories always exist. The pointwise formula is thus not an existence caveat in practice but a computation: it turns the question "what is $\mathrm{Lan}_K F$" into the question "what is this colimit", which the concrete structure of $\mathbf{E}$ answers. The following example runs that computation to the end.

Example 6.2.3: Left Kan Extension along an Inclusion of Ordinals

Let $\mathbf{C}$ be the two-object discrete category $\{0,1\}$ (no morphisms beyond identities), let $\mathbf{D}$ be the poset $\{0 < 1 < 2\}$ regarded as a category, and let $K\colon \mathbf{C} \to \mathbf{D}$ send $0 \mapsto 0$ and $1 \mapsto 1$. Take $\mathbf{E} = \mathbf{Set}$ and let $F\colon \mathbf{C} \to \mathbf{Set}$ pick out two sets, $F0 = A$ and $F1 = B$. Since $\mathbf{C}$ is small and $\mathbf{Set}$ is cocomplete, corollary 6.2.2 guarantees $\mathrm{Lan}_K F$ exists, and theorem 6.2.1 computes it object by object.

Compute each comma category $(K \downarrow d)$. Its objects are pairs $(c, \varphi\colon Kc \to d)$, and in a poset there is at most one morphism between any two objects, so φ exists precisely when $Kc \leq d$.

- For $d = 0$: the only c with $Kc \leq 0$ is $c = 0$ (since $K0 = 0 \leq 0$ but $K1 = 1 \not\leq 0$), and φ is the unique arrow $0 \leq 0$. So $(K \downarrow 0)$ has a single object $(0, \varphi)$, and the colimit of $F \circ \Pi_0$ is just $F0 = A$.
- For $d = 1$: both $K0 = 0 \leq 1$ and $K1 = 1 \leq 1$, giving two objects $(0, \varphi_0)$ and $(1, \varphi_1)$. There are no morphisms between them ($\mathbf{C}$ is discrete), so $(K \downarrow 1)$ is the discrete two-object category and the colimit is the coproduct $F0 \coprod F1 = A \coprod B$, the disjoint union.
- For $d = 2$: again both $K0 = 0 \leq 2$ and $K1 = 1 \leq 2$, so $(K \downarrow 2)$ is discrete on two objects and the colimit is $A \coprod B$.

The value function is therefore $(\mathrm{Lan}_K F)(0) = A$ and $(\mathrm{Lan}_K F)(1) = (\mathrm{Lan}_K F)(2) = A \coprod B$. The functor structure is read off Step 2 of the proof of theorem 6.2.1: the morphism $0 \leq 1$ of $\mathbf{D}$ induces the coprojection $A \hookrightarrow A \coprod B$ into the first summand, while the morphism $h\colon 1 \leq 2$ induces the identity $A \coprod B \to A \coprod B$, since the functor $h_*\colon (K \downarrow 1) \to (K \downarrow 2)$ sends $(c, \varphi) \mapsto (c, h \circ \varphi)$, matching the two objects of each comma category by their $\mathbf{C}$-component, so the induced map on colimits carries each coprojection to itself. The unit $\eta\colon F \Rightarrow (\mathrm{Lan}_K F) \circ K$ is read off the colimit injections at the identity approaches $(0, \mathrm{id})$ and $(1, \mathrm{id})$: $\eta_0\colon A \to (\mathrm{Lan}_K F)(0) = A$ is the identity, an isomorphism, while $\eta_1\colon B \to (\mathrm{Lan}_K F)(1) = A \coprod B$ is the coprojection into the second summand, which is monic but not an isomorphism unless $A = \emptyset$; there $\eta_1\colon B \to \emptyset \coprod B$ is itself an isomorphism. So η is not invertible. This is consistent with the general principle that runs in one direction: when K is fully faithful the pointwise unit η is a natural isomorphism, so F is recovered on the nose. Here K is not full, since $\mathbf{C}(0,1) = \emptyset$ while $\mathbf{D}(K0, K1) = \{0 \leq 1\}$ is nonempty, and the direct computation shows η fails to be invertible. The converse fails in general: fullness is sufficient but not necessary for η to be an isomorphism, as the case $A = \emptyset$ of this same example shows, where η is a natural isomorphism while K remains non-full. The extension therefore fails to restrict back to F on the nose.

The picture matches the interpretive reading above. At objects in the image of K (namely 0 and 1) the extension records the data F prescribes, assembled by coproduct where K maps two objects below the same d; at the new object 2, which lies above both 0 and 1 but is not in the image of K, the extension is forced by the colimit to be the coproduct $A \coprod B$, the freest set receiving compatible maps from A and B. A left Kan extension fills in unseen values as freely as the diagram permits, which is the precise sense in which it is the left, rather than the right, adjoint to restriction.

6.3 All Concepts are Kan Extensions

The machinery of the previous two sections was built for one payoff: to show that the central constructions of this book are Kan extensions in disguise. Limits and colimits, adjoints, and the Yoneda lemma each reappear here as an instance of Lan or Ran, computed by the universal property of definition 6.1.1 or the pointwise formula of theorem 6.2.1. The identifications are short, so the section is a sequence of corollaries rather than a development of new theory.

The first identification is the cleanest. A functor $F\colon \mathbf{J} \to \mathbf{E}$ out of a small category has a colimit exactly when a certain Kan extension exists, and the two objects agree. The trick is to extend F not along an inclusion but along the collapse $\mathbf{J} \to \mathbf{1}$ to the terminal category, the category with one object $*$ and only its identity arrow.

Proposition 6.3.1: Limits and Colimits as Kan Extensions

Let $\mathbf{J}$ be a small category and $!\colon \mathbf{J} \to \mathbf{1}$ the unique functor to the terminal category. For $F\colon \mathbf{J} \to \mathbf{E}$:

1. the colimit of F exists if and only if $\mathrm{Lan}_! F$ exists, and then $(\mathrm{Lan}_! F)(*) \cong \operatorname{colim} F$;
2. the limit of F exists if and only if $\mathrm{Ran}_! F$ exists, and then $(\mathrm{Ran}_! F)(*) \cong \lim F$.

***Proof*:**
We prove (1); part (2) follows by the duality principle (box 1.3.3), reading the statement in $\mathbf{E}^{\mathrm{op}}$.

A functor $\mathbf{1} \to \mathbf{E}$ is the same datum as an object of $\mathbf{E}$: it picks out the image $G(*)$ of the unique object, and its action on the sole arrow is forced to be id. Write $e = G(*)$. Precomposition with $!$ sends such a functor to the constant functor $\Delta e\colon \mathbf{J} \to \mathbf{E}$ at e, since every object of $\mathbf{J}$ maps to $*$ and thence to e, and every arrow to id_e. A natural transformation $F \Rightarrow (\Delta e)$ assigns to each object j of $\mathbf{J}$ a map $\lambda_j\colon Fj \to e$, naturally in j: for every arrow $u\colon j \to j'$ the square

$$\begin{tikzcd} Fj \arrow[r, "Fu"] \arrow[dr, "\lambda_j"'] & Fj' \arrow[d, "\lambda_{j'}"] \\ & e \end{tikzcd}$$

commutes, that is, $\lambda_{j'} \circ Fu = \lambda_j$. This is precisely the data of a cocone under F with summit e (definition 3.3.1). So a natural transformation $F \Rightarrow (\Delta e) = G \circ {!}$ is the same thing as a cocone under F with summit e.

Now unwind definition 6.1.1. The left Kan extension $\mathrm{Lan}_! F$ is a functor $\mathbf{1} \to \mathbf{E}$, hence an object $L = (\mathrm{Lan}_! F)(*)$, equipped with a universal natural transformation $\eta\colon F \Rightarrow (\mathrm{Lan}_! F) \circ {!} = \Delta L$. By the previous paragraph η is a cocone under F with summit L. Universality of η says: every natural transformation $F \Rightarrow G \circ {!}$, equivalently every cocone (λ_j) under F with some summit e, factors uniquely as η followed by a natural transformation $(\mathrm{Lan}_! F) \Rightarrow G$; the latter is a single map $L \to e$ in $\mathbf{E}$. Uniqueness of the factoring map is exactly the statement that (L, η) is initial among cocones under F, which is the universal property of the colimit (definition 3.3.1, definition 3.1.1). Hence $L \cong \operatorname{colim} F$, and the objects exist together, completing the proof.

This is the pointwise formula (theorem 6.2.1) at the single object $*$. The comma category $(! \downarrow *)$ has an object of $\mathbf{J}$ for each way of mapping $!\,j = *$ to $*$ in $\mathbf{1}$, of which there is exactly one, so $(! \downarrow *) \cong \mathbf{J}$ and the projection to $\mathbf{J}$ is the identity; the colimit formula $(\mathrm{Lan}_! F)(*) \cong \mathrm{colim}_{(!\downarrow *)} F\pi$ collapses to $\mathrm{colim}_{\mathbf{J}} F$. The proposition says a (co)limit is the universal way to compress a diagram to a single object, and Kan extension along $\mathbf{J} \to \mathbf{1}$ is the universal compression. Every general Kan extension is then, by the pointwise formula, a (co)limit computed one object of the target at a time.

The second identification places adjunctions inside the same frame. An adjoint is determined by a universal property, and a universal property is what a Kan extension records; the content of the next result is that a right adjoint is literally the Kan extension of the identity along its left adjoint.

Proposition 6.3.2: Adjoints as Kan Extensions

Let $F\colon \mathbf{C} \to \mathbf{D}$ be left adjoint to $G\colon \mathbf{D} \to \mathbf{C}$, with unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ and counit $\varepsilon\colon F \circ G \Rightarrow \mathrm{id}_{\mathbf{D}}$. Then G is the left Kan extension of $\mathrm{id}_{\mathbf{C}}$ along F,

$$G \cong \mathrm{Lan}_F \mathrm{id}_{\mathbf{C}}$$

with the unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ serving as the universal cell. Dually, F is the right Kan extension of $\mathrm{id}_{\mathbf{D}}$ along G, $F \cong \mathrm{Ran}_G \mathrm{id}_{\mathbf{D}}$, with the counit as its universal cell.

***Proof*:**
We prove the first statement; the second is its dual under box 1.3.3, read in the opposite categories, exchanging the left and right adjoints. By definition 6.1.1, exhibiting G as $\mathrm{Lan}_F \mathrm{id}_{\mathbf{C}}$ means producing a natural transformation $u\colon \mathrm{id}_{\mathbf{C}} \Rightarrow G \circ F$ such that for every functor $H\colon \mathbf{D} \to \mathbf{C}$ the map

$$\Phi_H\colon [\mathbf{D}, \mathbf{C}](G, H) \longrightarrow [\mathbf{C}, \mathbf{C}](\mathrm{id}_{\mathbf{C}}, H \circ F), \qquad \theta \longmapsto (\theta F) \circ u$$

is a bijection natural in H, where $(\theta F) \circ u$ has components $\theta_{Fc} \circ u_c\colon c \to GFc \to HFc$. We take $u = \eta$, the unit, and produce an inverse to Φ_H from the adjunction $F \dashv G$ (definition 4.1.1).

Step 1: the inverse. Given $\beta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow H \circ F$ with components $\beta_c\colon c \to HFc$, define

$$\Psi_H(\beta)_d \;=\; H\varepsilon_d \circ \beta_{Gd}\colon\; Gd \xrightarrow{\beta_{Gd}} HFGd \xrightarrow{H\varepsilon_d} Hd$$

for each object d of $\mathbf{D}$, using the counit $\varepsilon_d\colon FGd \to d$. Naturality of β and of ε makes $\Psi_H(\beta)\colon G \Rightarrow H$ a natural transformation: for $g\colon d \to d'$ the square for $\Psi_H(\beta)$ commutes because β is natural at Gg and ε is natural at g. This defines a map Ψ_H in the reverse direction.

Step 2: Ψ_H inverts Φ_H. Take $\theta\colon G \Rightarrow H$ and compute $\Psi_H(\Phi_H(\theta))$. Its component at d is

$$H\varepsilon_d \circ (\Phi_H\theta)_{Gd} \;=\; H\varepsilon_d \circ \theta_{FGd} \circ \eta_{Gd}$$

Naturality of θ at $\varepsilon_d\colon FGd \to d$ gives $H\varepsilon_d \circ \theta_{FGd} = \theta_d \circ G\varepsilon_d$, so the component equals $\theta_d \circ G\varepsilon_d \circ \eta_{Gd}$. The triangle identity $G\varepsilon_d \circ \eta_{Gd} = \mathrm{id}_{Gd}$ (definition 4.1.3) collapses this to θ_d. Hence $\Psi_H \circ \Phi_H = \mathrm{id}$.

Conversely take $\beta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow HF$ and compute $\Phi_H(\Psi_H(\beta))$. Its component at c is

$$(\Psi_H\beta)_{Fc} \circ \eta_c \;=\; H\varepsilon_{Fc} \circ \beta_{GFc} \circ \eta_c$$

Naturality of β at $\eta_c\colon c \to GFc$ gives $\beta_{GFc} \circ \eta_c = HF\eta_c \circ \beta_c$, so the component equals $H\varepsilon_{Fc} \circ HF\eta_c \circ \beta_c = H(\varepsilon_{Fc} \circ F\eta_c) \circ \beta_c$. The other triangle identity $\varepsilon_{Fc} \circ F\eta_c = \mathrm{id}_{Fc}$ (definition 4.1.3) collapses this to β_c. Hence $\Phi_H \circ \Psi_H = \mathrm{id}$.

So Φ_H is a bijection, and it is natural in H because it is given by postcomposition with the fixed cell η. By definition 6.1.1 this exhibits (G, η) as the left Kan extension $\mathrm{Lan}_F \, \mathrm{id}_{\mathbf{C}}$, completing the proof.

An adjoint is a Kan extension. The unit $\eta\colon \mathrm{id}_{\mathbf{C}} \Rightarrow GF$, which one meets first as the insertion of generators of the free construction, is exactly the universal cell exhibiting the right adjoint G as $\mathrm{Lan}_F \, \mathrm{id}_{\mathbf{C}}$; the right Kan extension $\mathrm{Ran}_G \, \mathrm{id}_{\mathbf{D}} \cong F$ recovers the left adjoint dually, with the counit as its universal cell. The converse holds only under an extra hypothesis: if $\mathrm{Lan}_F \, \mathrm{id}_{\mathbf{C}}$ exists and is *preserved* by F, it is right adjoint to F; without the preservation clause a Kan extension of the identity need not arise from an adjunction.

The last identification returns to the Yoneda lemma. The Yoneda embedding $\mathbf{y}\colon \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$ (definition 2.3.1) is fully faithful (theorem 2.2.1), so restricting a functor along $\mathbf{y}$ loses nothing that the functor does on representables. The precise statement is that a functor out of the presheaf category which respects colimits is recovered as the left Kan extension of its restriction to representables.

Proposition 6.3.3: Yoneda via Kan Extension

Let $\mathbf{C}$ be small and $L\colon \mathbf{PSh}(\mathbf{C}) \to \mathbf{E}$ a colimit-preserving functor into a cocomplete category. Then L is the left Kan extension along $\mathbf{y}$ of its restriction to representables:

$$L \cong \mathrm{Lan}_{\mathbf{y}}(L \circ \mathbf{y})$$

In particular L is determined up to canonical isomorphism by the composite $L \circ \mathbf{y}\colon \mathbf{C} \to \mathbf{E}$.

***Proof*:**
Every presheaf is canonically a colimit of representables: for $P \in \mathbf{PSh}(\mathbf{C})$ there is a natural isomorphism $P \cong \mathrm{colim}_{(\mathbf{y}\downarrow P)} \, \mathbf{y}\pi$, the colimit of the representables mapping into P. This is the density theorem, proved in the next section; grant it here. Because L preserves colimits,

$$L(P) \cong L\big(\mathrm{colim}_{(\mathbf{y}\downarrow P)} \, \mathbf{y}\pi\big) \cong \mathrm{colim}_{(\mathbf{y}\downarrow P)} (L \circ \mathbf{y})\pi$$

The right-hand side is the pointwise colimit formula (theorem 6.2.1) for the left Kan extension of $L \circ \mathbf{y}$ along $\mathbf{y}$, evaluated at P. Hence $L(P) \cong (\mathrm{Lan}_{\mathbf{y}}(L \circ \mathbf{y}))(P)$, naturally in P, so $L \cong \mathrm{Lan}_{\mathbf{y}}(L \circ \mathbf{y})$. The composite $L \circ \mathbf{y}$ therefore determines L, as we sought to show.

The presheaf category is the free cocompletion of $\mathbf{C}$: a colimit-preserving functor out of $\mathbf{PSh}(\mathbf{C})$ is no more and no less than an arbitrary functor out of $\mathbf{C}$, extended by the universal formula. This is the sense in which the Yoneda lemma is a Kan extension. The density theorem that supplies the canonical presentation $P \cong \mathrm{colim} \, \mathbf{y}\pi$ is the subject of the next section, where the co-Yoneda identity $P \cong \int^c Pc \cdot \mathbf{y}(c)$ makes the same statement in the language of coends.

6.4 Ends, Coends, and Density

The pointwise formula (theorem 6.2.1) writes a Kan extension as a colimit over a comma category. That comma-category colimit has a more compact description once one has a piece of notation for integrating a functor of two variables that is contravariant in one and covariant in the other. This section develops that notation, the calculus of ends and coends, and applies it to presheaves. The results are the *density* theorem, which recovers every presheaf as a colimit of representables, and a one-line coend formula for the left Kan extension.

A functor $T\colon \mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{E}$ pairs each object c with itself along the diagonal, giving objects $T(c,c)$, but a morphism $f\colon c \to c'$ acts on T in two clashing directions: covariantly in the second slot, $T(c,f)\colon T(c,c) \to T(c,c')$, and contravariantly in the first, $T(f,c')\colon T(c',c') \to T(c,c')$. A single object of $\mathbf{E}$ that sees all the $T(c,c)$ at once must reconcile these two actions. The device that does so is a wedge.

Definition 6.4.1: End and Coend

Let $T\colon \mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{E}$ be a functor, with $\mathbf{C}$ small. A *wedge* from an object $e \in \mathbf{E}$ to T is a family of morphisms $w_c\colon e \to T(c,c)$, one for each object c, that is *dinatural*: for every morphism $f\colon c \to c'$ the square

$$\begin{tikzcd} e \arrow[r, "w_{c'}"] \arrow[d, "w_c"'] & T(c',c') \arrow[d, "T(f,c')"] \\ T(c,c) \arrow[r, "T(c,f)"'] & T(c,c') \end{tikzcd}$$

commutes, that is $T(c,f) \circ w_c = T(f,c') \circ w_{c'}$. The *end* of T, written $\int_c T(c,c)$, is a universal wedge: an object $\int_c T(c,c)$ with a wedge $\pi_c\colon \int_c T(c,c) \to T(c,c)$ such that every wedge $w_c\colon e \to T(c,c)$ factors as $w_c = \pi_c \circ u$ through a unique $u\colon e \to \int_c T(c,c)$.

Dually, a *cowedge* from T to an object e is a dinatural family $\iota_c\colon T(c,c) \to e$, meaning $\iota_c \circ T(f,c) = \iota_{c'} \circ T(c',f)$ for every $f\colon c \to c'$. The *coend* of T, written $\int^c T(c,c)$, is a universal cowedge: an object $\int^c T(c,c)$ with a cowedge $\iota_c\colon T(c,c) \to \int^c T(c,c)$ through which every cowedge factors uniquely.

The universal property is the abstract characterization; the computational handle is a (co)equalizer. A wedge $w_c\colon e \to T(c,c)$ is exactly a map $e \to \prod_c T(c,c)$ whose two composites into $\prod_{f\colon c\to c'} T(c,c')$ agree, one composite applying $T(c,f)$ in the codomain and the other $T(f,c')$. The end is therefore the equalizer

$$\int_c T(c,c) \longrightarrow \prod_c T(c,c) \rightrightarrows \prod_{f\colon c\to c'} T(c,c') \tag{6.2}$$

of these two maps (definition 3.2.4), so an end exists whenever $\mathbf{E}$ has the products indexed by the objects and the morphisms of $\mathbf{C}$ together with equalizers. Dualizing every arrow, the coend is the coequalizer

$$\coprod_{f\colon c\to c'} T(c',c) \rightrightarrows \coprod_c T(c,c) \longrightarrow \int^c T(c,c) \tag{6.3}$$

of the two maps sending the f-summand $T(c',c)$ into $T(c,c)$ by $T(f,c)$ and into $T(c',c')$ by $T(c',f)$ (definition 3.3.5). A coend is thus $\coprod_c T(c,c)$ with the two ways of moving along a morphism identified. This is the description used in every computation below.

Two pieces of notation package the copies of an object indexed by a set. For a set S and an object $e \in \mathbf{E}$, the *copower* $S \cdot e$ is the S-indexed coproduct $\coprod_S e$, and the *power* e^S is the S-indexed product $\prod_S e$. When $\mathbf{E} = \mathbf{Set}$ the copower $S \cdot e = S \times e$ and the power e^S is the set of functions $S \to e$.

These appear the moment a coend involves a hom-set: a coend $\int^c \mathbf{C}(c',c)\cdot Fc$ has each summand $\mathbf{C}(c',c)\cdot Fc$ a copower of Fc by a hom-set.

The end over a Hom-valued functor is the object that justifies the notation: it is the set of natural transformations.

Example 6.4.2: Natural Transformations as an End

Let $F, G\colon \mathbf{C}\to\mathbf{Set}$ be functors on a small category $\mathbf{C}$. The assignment $(c',c)\mapsto \mathbf{Set}(Fc',Gc)$ is a functor $\mathbf{C}^{\mathrm{op}}\times\mathbf{C}\to\mathbf{Set}$, contravariant in c' (precompose with Ff) and covariant in c (postcompose with Gf). A wedge from a one-point set to this functor is a choice, for each c, of an element $\alpha_c\in\mathbf{Set}(Fc,Gc)$, that is a map $\alpha_c\colon Fc\to Gc$; dinaturality at $f\colon c\to c'$ reads

$$Gf\circ\alpha_c=\alpha_{c'}\circ Ff$$

which is exactly the naturality square of α. By (6.2) the end is the equalizer of

$$\prod_c \mathbf{Set}(Fc,Gc)\rightrightarrows\prod_{f\colon c\to c'}\mathbf{Set}(Fc,Gc')$$

whose elements are the families (α_c) satisfying every naturality square, so

$$\int_c \mathbf{Set}(Fc,Gc)\;\cong\;\mathrm{Nat}(F,G)$$

The set of natural transformations is an end. This is the identity that makes the Yoneda lemma, and the density theorem below, coend computations rather than ad hoc arguments.

The dual computation is a coend that identifies elements rather than selecting them. For a functor $M\colon\mathbf{C}^{\mathrm{op}}\to\mathbf{Set}$ and a functor $N\colon\mathbf{C}\to\mathbf{Set}$ on a small category $\mathbf{C}$, the coend $\int^c Mc\times Nc$ is the set of pairs $(x\in Mc,\, y\in Nc)$ modulo the relation identifying $(x\cdot f, y)$ with $(x, f\cdot y)$ for $f\colon c\to c'$ and $x\in Mc'$, by (6.3) with $T(c',c)=Mc'\times Nc$. This is the tensor product of the two functors over $\mathbf{C}$: the same coequalizer shape, taken over a one-object category enriched in abelian groups (a ring R) with $\otimes$ in place of $\times$, produces the module tensor product $M\otimes_R N$ by identifying (xf,y) with (x,fy). The coend is the categorical tensor product, which is why the copower notation borrows the multiplicative sign.

That tensor reading underlies the density theorem. Every presheaf decomposes as a coend of representables, and the coend is a colimit, so every presheaf is a colimit of representables: a presheaf is assembled from the representables $\mathbf{y}(c)=\mathbf{C}(-,c)$ by gluing along $\mathbf{C}$'s own morphisms.

Theorem 6.4.3: Density (Co-Yoneda)

Let $\mathbf{C}$ be small and $F\colon\mathbf{C}^{\mathrm{op}}\to\mathbf{Set}$ a presheaf (definition 2.1.3). Then there is a canonical isomorphism

$$F\;\cong\;\int^c Fc\cdot\mathbf{y}(c) \tag{6.4}$$

in $\mathbf{PSh}(\mathbf{C})$, where $\mathbf{y}(c)=\mathbf{C}(-,c)$ is the representable presheaf (definition 2.1.1) under the Yoneda embedding $\mathbf{y}$ (definition 2.3.1) and $Fc\cdot\mathbf{y}(c)$ is the copower. Consequently every presheaf is a colimit of representables, and the Yoneda embedding $\mathbf{y}\colon\mathbf{C}\to\mathbf{PSh}(\mathbf{C})$ is dense.

***Proof*:**
The integrand $(c', c) \mapsto Fc' \cdot \mathbf{y}(c) = Fc' \times \mathbf{C}(-, c)$ is a functor $\mathbf{C}^{\mathrm{op}} \times \mathbf{C} \to \mathbf{PSh}(\mathbf{C})$, contravariant in the first slot through F and covariant in the second through $\mathbf{y}$; its coend is computed in $\mathbf{PSh}(\mathbf{C})$, where colimits are formed objectwise (definition 3.3.1). It suffices to produce, for each presheaf G, a bijection

$$\mathbf{PSh}(\mathbf{C})\Big(\int^{c} Fc \cdot \mathbf{y}(c),\, G\Big) \;\cong\; \mathbf{PSh}(\mathbf{C})(F, G)$$

natural in G: two objects of $\mathbf{PSh}(\mathbf{C})$ whose covariant hom-functors are naturally isomorphic are themselves isomorphic, this being the Yoneda lemma (theorem 2.2.1) applied in the opposite category $\mathbf{PSh}(\mathbf{C})^{\mathrm{op}}$, where the covariant hom-functors become the representables.

Step 1: Unwind the left-hand side to a wedge. By the universal property of the coend, a map $\int^{c} Fc \cdot \mathbf{y}(c) \to G$ is a cowedge, a dinatural family of maps $Fc \cdot \mathbf{y}(c) \to G$. A map out of a copower $Fc \cdot \mathbf{y}(c) = \coprod_{x \in Fc} \mathbf{y}(c)$ is a choice, for each $x \in Fc$, of a map $\mathbf{y}(c) \to G$ in $\mathbf{PSh}(\mathbf{C})$. Thus

$$\mathbf{PSh}(\mathbf{C})\big(Fc \cdot \mathbf{y}(c),\, G\big) \;\cong\; \prod_{x \in Fc} \mathbf{PSh}(\mathbf{C})\big(\mathbf{y}(c), G\big) \;\cong\; \mathbf{Set}\big(Fc,\; \mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G)\big)$$

Step 2: Apply Yoneda. The Yoneda lemma (theorem 2.2.1) gives a bijection $\mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cong Gc$, natural in c. Substituting,

$$\mathbf{PSh}(\mathbf{C})\big(Fc \cdot \mathbf{y}(c),\, G\big) \;\cong\; \mathbf{Set}(Fc, Gc)$$

and a cowedge into G becomes a wedge from the one-point set to the functor $(c', c) \mapsto \mathbf{Set}(Fc', Gc)$, the dinaturality conditions matching under the Yoneda bijection because that bijection is natural in c.

Step 3: Recognize the end. By example 6.4.2 the wedges from a point to $(c', c) \mapsto \mathbf{Set}(Fc', Gc)$ are exactly the natural transformations $F \Rightarrow G$:

$$\int_{c} \mathbf{Set}(Fc, Gc) \;\cong\; \mathrm{Nat}(F, G) = \mathbf{PSh}(\mathbf{C})(F, G)$$

Composing the three bijections gives $\mathbf{PSh}(\mathbf{C})\big(\int^{c} Fc \cdot \mathbf{y}(c), G\big) \cong \mathbf{PSh}(\mathbf{C})(F, G)$, natural in G. By the Yoneda lemma applied in $\mathbf{PSh}(\mathbf{C})$, the two presheaves are isomorphic, and the isomorphism is the canonical one sending $x \in Fc$ to the pair (x, id_c) in the copower, as we sought to show.

For the final clause, the coend (6.4) is the coequalizer (6.3), hence a colimit, of a diagram of representables. So F is a colimit of representables, and since F was arbitrary the representables generate $\mathbf{PSh}(\mathbf{C})$ under colimits, which is the statement that $\mathbf{y}$ is dense, completing the proof.

The density theorem is the promised converse to the Yoneda embedding. Yoneda embeds $\mathbf{C}$ fully faithfully into $\mathbf{PSh}(\mathbf{C})$; density says the image is not lost inside a larger category but generates it, every presheaf being reachable from the representables by colimits. Together they identify $\mathbf{PSh}(\mathbf{C})$ as the *free cocompletion* of $\mathbf{C}$: freely adjoining colimits to $\mathbf{C}$ produces exactly the presheaf category, with $\mathbf{y}$ the universal inclusion. The isomorphism (6.4) makes this concrete: a presheaf F is rebuilt from the sets Fc of its elements and the representables that receive them.

The concrete content is a formula for the elements of a presheaf and how they patch, seen most on a poset.

Example 6.4.4: Density over a Poset
Let $\mathbf{C}$ be a poset, viewed as a category with a single arrow $c \to c'$ when $c \le c'$, and let $F\colon \mathbf{C}^{\mathrm{op}} \to \mathbf{Set}$ be a presheaf (an order-reversing assignment of sets with restriction maps $Fc' \to Fc$ for $c \le c'$). The representable $\mathbf{y}(c) = \mathbf{C}(-, c)$ sends b to the one-point set if $b \le c$ and to the empty set otherwise, so $\mathbf{y}(c)$ is the presheaf represented by the down-set of c. The coend (6.4) identifies, for each $x \in Fc$ and each $b \le c$, the "germ" of x at b with the actual restriction $x|_b \in Fb$: the copower $Fc \cdot \mathbf{y}(c)$ contributes one copy of the down-set of c for every section $x \in Fc$, and the coequalizer glues the copy attached to x to the copy attached to $x|_b$ along b. The result reassembles F from its sections and their restrictions, the poset analogue of recovering a sheaf from its stalks, that is from its local data together with the compatibilities that let the pieces be glued.

The same coend calculus rewrites the pointwise formula. The colimit over the comma category $(K \downarrow d)$ in theorem 6.2.1 is a coend once the comma category is unwound, and the coend form is the one that composes with the density theorem and generalizes to the enriched setting.

Proposition 6.4.5: Coend Formula for Kan Extensions

Let $K\colon \mathbf{C} \to \mathbf{D}$ with $\mathbf{C}$ small, $F\colon \mathbf{C} \to \mathbf{E}$, and suppose the relevant (co)limits exist in $\mathbf{E}$. Then the left Kan extension is the coend

$$(\mathrm{Lan}_K\, F)(d) \;\cong\; \int^{c} \mathbf{D}(Kc, d) \cdot Fc \tag{6.5}$$

and the right Kan extension is the end

$$(\mathrm{Ran}_K\, F)(d) \;\cong\; \int_{c} Fc^{\mathbf{D}(d,Kc)} \tag{6.6}$$

where $\mathbf{D}(Kc, d) \cdot Fc$ is the copower and $Fc^{\mathbf{D}(d,Kc)}$ the power.

***Proof*:**
Fix $d \in \mathbf{D}$. By theorem 6.2.1 the value $(\mathrm{Lan}_K\, F)(d)$ is the colimit of the composite $(K \downarrow d) \xrightarrow{\pi} \mathbf{C} \xrightarrow{F} \mathbf{E}$, where the comma category $(K \downarrow d)$ has objects the pairs (c, g) with $g\colon Kc \to d$ and morphisms $(c, g) \to (c', g')$ the maps $h\colon c \to c'$ with $g' \circ Kh = g$. The claim is that this colimit is computed by the coequalizer (6.3) for the integrand $T(c', c) = \mathbf{D}(Kc', d) \cdot Fc$.

Step 1: Match the coproducts. The coproduct $\coprod_c T(c, c) = \coprod_c \mathbf{D}(Kc, d) \cdot Fc$ is $\coprod_c \coprod_{g\colon Kc \to d} Fc$, one copy of Fc for each pair (c, g), that is one copy of Fc for each object of $(K \downarrow d)$. This is the coproduct over the objects of $(K \downarrow d)$ of $F\pi$, the vertex coproduct in the coequalizer presentation of the colimit $\mathrm{colim}_{(K\downarrow d)} F\pi$ (definition 3.3.1).

Step 2: Match the relations. The parallel pair in (6.3) identifies, for each $h\colon c \to c'$ in $\mathbf{C}$, the summand $T(c', c) = \mathbf{D}(Kc', d) \cdot Fc$ under the two maps $T(h, c)$ and $T(c', h)$. Concretely a summand indexed by $h\colon c \to c'$ and $g'\colon Kc' \to d$ is sent, on one side, to the copy of Fc at $(c, g' \circ Kh)$ acting by the identity on Fc, and on the other to the copy of Fc' at (c', g') acting by $Fh\colon Fc \to Fc'$. This is exactly the relation imposed in the colimit of $F\pi$: a morphism

> $h\colon (c, g' \circ Kh) \to (c', g')$ of $(K \downarrow d)$ identifies the image of Fc in the vertex with the image of Fc' precomposed by Fh. The two coequalizers therefore present the same quotient, giving (6.5).
>
> *Step 2 (dual): the right extension.* Applying the duality principle (box 1.3.3) to the argument just given, $(\mathrm{Ran}_K F)(d)$ is the limit over $(d \downarrow K)$ of $F\pi$, presented by the equalizer (6.2) for the integrand $(c', c) \mapsto Fc^{\mathbf{D}(d,Kc')}$. The vertex product $\prod_c Fc^{\mathbf{D}(d,Kc)}$ is one copy of Fc for each map $d \to Kc$, that is one copy for each object of $(d \downarrow K)$, and the equalizer imposes the morphisms of $(d \downarrow K)$ exactly as in the left case with all arrows reversed. This gives (6.6), completing the proof.

Read against density, the coend formula (6.5) says the left Kan extension along K is what one gets by resolving F into representables and then relabeling their indices along K: the coend integrates Fc against the hom-set $\mathbf{D}(Kc, d)$, the copower weighting each Fc by the ways Kc maps to d. Taking $K = \mathbf{y}$ the Yoneda embedding and $F = \mathbf{y}$ itself, (6.5) computes $(\mathrm{Lan}_{\mathbf{y}}\, \mathbf{y})(G) \cong \int^c \mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cdot \mathbf{y}(c) \cong \int^c Gc \cdot \mathbf{y}(c)$ at a presheaf G, using $\mathbf{PSh}(\mathbf{C})(\mathbf{y}(c), G) \cong Gc$; this is the density isomorphism (6.4) for G. Density is the case of the coend formula where the extension is along Yoneda. The end and coend are thus not a separate topic appended to Kan extensions but the notation in which the pointwise formula, the Yoneda lemma, and density are one computation.

7

Monoidal Categories

A *monoidal category* carries a tensor product $\otimes$, a binary operation on objects that is associative and unital up to coherent isomorphism. It is the structure behind every tensor product in mathematics, and behind a category acting on itself. This chapter defines monoidal categories and the coherence theorem that tames their associativity isomorphisms, adds the symmetry and closedness most examples carry, and identifies the monoids that live inside a monoidal category, whose instance among endofunctors is the monad of chapter 5.

7.1 Monoidal Categories and Coherence

A tensor product multiplies objects. On R-modules it is $\otimes_R$; on sets it is the cartesian product; on the endofunctors of a category it is composition. Each is associative and unital, but only up to a natural isomorphism, and those isomorphisms are themselves data that must fit together. This section isolates that data as the notion of a monoidal category, states the two axioms (pentagon and triangle) that control it, and proves the coherence theorem, which shows the axioms suffice to make every reassociation unambiguous. We write $\otimes$ for the tensor product and I for the unit object throughout.

The associativity isomorphism $(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z)$ is called the *associator*, and the isomorphisms $I \otimes X \cong X$ and $X \otimes I \cong X$ the *left* and *right unitors*. Requiring these to be natural in their arguments is the first constraint; the pentagon and triangle axioms are the second, and they are what distinguishes a coherent choice of associativity from an arbitrary one.

Definition 7.1.1: Monoidal Category

A *monoidal category* is a category C equipped with

- a functor $\otimes\colon C \times C \to C$, the *tensor product*;
- an object I of C, the *unit object*;
- a natural isomorphism α, the *associator*, with components

$$\alpha_{X,Y,Z}\colon (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$$

- natural isomorphisms λ and ρ, the *left* and *right unitors*, with components

$$\lambda_X\colon I \otimes X \to X, \qquad \rho_X\colon X \otimes I \to X$$

subject to two axioms. The *pentagon axiom* requires that the diagram

$$\begin{tikzcd}
 & (W \otimes X) \otimes (Y \otimes Z) \arrow[dr, "\alpha_{W,X,Y\otimes Z}"] & \\
\big((W \otimes X) \otimes Y\big) \otimes Z \arrow[ur, "\alpha_{W\otimes X,Y,Z}"] \arrow[d, "\alpha_{W,X,Y}\otimes \mathrm{id}_Z"'] & & W \otimes \big(X \otimes (Y \otimes Z)\big) \\
\big(W \otimes (X \otimes Y)\big) \otimes Z \arrow[rr, "\alpha_{W,X\otimes Y,Z}"'] & & W \otimes \big((X \otimes Y) \otimes Z\big) \arrow[u, "\mathrm{id}_W\otimes \alpha_{X,Y,Z}"']
\end{tikzcd}$$

commutes for all objects W, X, Y, Z, and the *triangle axiom* requires that the diagram

$$\begin{tikzcd}
(X \otimes I) \otimes Y \arrow[rr, "\alpha_{X,I,Y}"] \arrow[dr, "\rho_X\otimes \mathrm{id}_Y"'] & & X \otimes (I \otimes Y) \arrow[dl, "\mathrm{id}_X\otimes \lambda_Y"] \\
 & X \otimes Y &
\end{tikzcd}$$

commutes for all objects X, Y.

The pentagon governs a fourfold tensor product. There are five ways to fully parenthesize $W \otimes X \otimes Y \otimes Z$, and the associator supplies an isomorphism between adjacent parenthesizations; the pentagon says the two paths around its perimeter, from $\big((W \otimes X) \otimes Y\big) \otimes Z$ to $W \otimes \big(X \otimes (Y \otimes Z)\big)$, agree. The triangle relates the associator to the unitors, forcing the two ways of cancelling a unit factor wedged between X and Y to coincide. These two small diagrams are the entire input; the coherence theorem below shows they already force every larger diagram of associators and unitors to commute.

When the associator and unitors are not just coherent but are identity morphisms, the tensor product is strictly associative and strictly unital, and the bookkeeping vanishes.

Definition 7.1.2: Strict Monoidal Category

A monoidal category is *strict* if the associator α and the unitors λ, ρ are all identity morphisms, so that $(X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$, $I \otimes X = X$, and $X \otimes I = X$ on the nose.

In a strict monoidal category the pentagon and triangle axioms hold trivially, being equalities of identity morphisms, and one may write iterated tensor products $X_1 \otimes \cdots \otimes X_n$ without parentheses. Most monoidal categories arising from a tensor construction are not strict, since $(X \otimes Y) \otimes Z$ and $X \otimes (Y \otimes Z)$ are built by different constructions, isomorphic but not equal. The endofunctor example below is a case where strictness does hold.

The unit object is determined by the structure. It is not extra data to be pinned down by a further choice, but is fixed up to a unique compatible isomorphism, in the same way a terminal object is (definition 3.1.1).

Proposition 7.1.3: Uniqueness of the Unit Object

Let $(C, \otimes, I, \alpha, \lambda, \rho)$ be a monoidal category, and suppose I' is another object admitting isomorphisms $\lambda'_X : I' \otimes X \to X$ and $\rho'_X : X \otimes I' \to X$ natural in X and satisfying the triangle axiom with α. Then there is a unique isomorphism $u : I \to I'$ satisfying $u \circ \lambda_I = \lambda'_{I'} \circ (u \otimes u)$; it is the canonical comparison of the two unit structures.

The compatibility condition packages the two units into a single square: it says u intertwines the multiplication $\lambda_I : I \otimes I \to I$ that the unit induces on itself with the corresponding multiplication $\lambda'_{I'} : I' \otimes I' \to I'$ on I'. Three standard facts about the unitors drive the proof. Two are recorded once and for all in Mac Lane, *Categories for the Working Mathematician*, Chapter VII: the left and right unitors agree at the unit, $\lambda_I = \rho_I$ (and likewise $\lambda'_{I'} = \rho'_{I'}$), so the induced multiplication is unambiguous; and the two unit structures are compared by the left-unit coherence identity $\lambda'_X \circ (u \otimes \mathrm{id}_X) = \lambda_X$, a consequence of the pentagon and triangle axioms. The third is elementary: tensoring by an object whose unitors are isomorphisms is an equivalence, hence *faithful*, which lets one cancel a factor $(-) \otimes \mathrm{id}_I$ from an equation.

***Proof Key ideas*:**
Write $v = \lambda_I = \rho_I : I \otimes I \to I$ and $v' = \lambda'_{I'} = \rho'_{I'} : I' \otimes I' \to I'$ for the two induced multiplications, using $\lambda_I = \rho_I$ (and its primed analogue) cited above.

Existence. Set

$$u = \rho_{I'} \circ (\lambda'_I)^{-1} : I \to I' \otimes I \to I',$$

an isomorphism as a composite of isomorphisms. The compatibility square $u \circ v = v' \circ (u \otimes u)$ reduces to a single unitor identity. Indeed, write $u \otimes u = (u \otimes \mathrm{id}_{I'}) \circ (\mathrm{id}_I \otimes u)$; naturality of λ at $u : I \to I'$ gives $u \circ \lambda_I = \lambda_{I'} \circ (\mathrm{id}_I \otimes u)$, so $u \circ v = u \circ \lambda_I$ and $v' \circ (u \otimes u) = \lambda'_{I'} \circ (u \otimes \mathrm{id}_{I'}) \circ (\mathrm{id}_I \otimes u)$ agree if and only if, after cancelling the isomorphism $\mathrm{id}_I \otimes u$,

$$\lambda'_{I'} \circ (u \otimes \mathrm{id}_{I'}) = \lambda_{I'}$$

This is the instance at $X = I'$ of the left-unit coherence identity $\lambda'_X \circ (u \otimes \mathrm{id}_X) = \lambda_X$, which relates the two unit structures through α and holds in any monoidal category by the pentagon and triangle axioms; it is a companion to the coherence theorem (theorem 7.1.5), deferred with it to Mac Lane, Chapter VII, which is why the existence half is given in outline. Granting it, the square commutes and u is a compatible comparison.

Uniqueness. Suppose $f: I \to I'$ is another isomorphism with $f \circ \upsilon = \upsilon' \circ (f \otimes f)$. Both u and f are then isomorphisms $I \to I'$ of the two multiplications υ, υ', so $g = u^{-1} \circ f: I \to I$ is an isomorphism satisfying $g \circ \upsilon = \upsilon \circ (g \otimes g)$, an automorphism of the multiplication υ. Naturality of λ at g gives $g \circ \lambda_I = \lambda_I \circ (\mathrm{id}_I \otimes g)$, that is $g \circ \upsilon = \upsilon \circ (\mathrm{id}_I \otimes g)$; comparing with $g \circ \upsilon = \upsilon \circ (g \otimes g)$ and cancelling the isomorphism υ yields $g \otimes g = \mathrm{id}_I \otimes g$. By functoriality of $\otimes$, $g \otimes g = (g \otimes \mathrm{id}_I) \circ (\mathrm{id}_I \otimes g)$, so the equation reads $(g \otimes \mathrm{id}_I) \circ (\mathrm{id}_I \otimes g) = \mathrm{id}_I \otimes g$; precomposing with the inverse of the isomorphism $\mathrm{id}_I \otimes g$ gives $g \otimes \mathrm{id}_I = \mathrm{id}_{I \otimes I} = \mathrm{id}_I \otimes \mathrm{id}_I$, whence $g = \mathrm{id}_I$ by faithfulness of $(-) \otimes I$. Thus $f = u$, and the unit object is unique up to a unique compatible isomorphism, as we sought to show.

The proposition is the reason one speaks of *the* unit object: any two choices are canonically identified, so the ambiguity carries no content. This is the monoidal analogue of the uniqueness of a terminal object (definition 3.1.1), and it is what allows I to be treated as a fixed piece of the structure rather than a variable one.

The standing cast supplies the primary examples. Each is a category already in use, now viewed as carrying a tensor product.

Example 7.1.4: Monoidal Structures on the Standing Cast

Four monoidal categories, each read against the data and axioms of definition 7.1.1.

1. *Sets under cartesian product.* The category **Set** is monoidal with $\otimes = \times$ the cartesian product (definition 3.2.3) and $I = 1$ any singleton, terminal in **Set** (definition 3.1.1). The associator $\alpha_{X,Y,Z}: (X \times Y) \times Z \to X \times (Y \times Z)$ sends $((x, y), z)$ to $(x, (y, z))$, and the unitors $\lambda_X : 1 \times X \to X$, $\rho_X : X \times 1 \to X$ delete the singleton coordinate. These are bijections, natural in their arguments, and the pentagon and triangle are the evident equalities of tuple regroupings. This is not strict: $(X \times Y) \times Z$ is a set of pairs whose first entry is a pair, not equal to $X \times (Y \times Z)$.

2. *Vector spaces under tensor product.* The category $\mathbf{Vect}_k$ of vector spaces over a field k is monoidal with $\otimes = \otimes_k$ the tensor product over k ([4]) and $I = k$ the ground field as a one-dimensional space. The associator is the canonical isomorphism $(U \otimes V) \otimes W \to U \otimes (V \otimes W)$ sending $(u \otimes v) \otimes w$ to $u \otimes (v \otimes w)$, and the unitors $k \otimes V \to V$, $V \otimes k \to V$ send $c \otimes v$ and $v \otimes c$ to cv. The pentagon and triangle hold because both sides of each act identically on the generating simple tensors. Again not strict.

3. *Modules under tensor product.* The same construction gives a monoidal structure on $\mathbf{Mod}_R$ for a commutative ring R, with $\otimes = \otimes_R$ and $I = R$ ([4]); the associator and unitors are the standard isomorphisms of module tensor products, and $\mathbf{Vect}_k$ is the special case $R = k$.

4. *Endofunctors under composition.* For any category C, the functor category $[C, C]$ (definition 1.5.4) of endofunctors is monoidal with $\otimes = \circ$ functor composition and $I = \mathrm{id}_C$ the identity functor. Composition of functors is associative on the nose, $(F \circ G) \circ H = F \circ (G \circ H)$, and the identity functor is a strict two-sided unit, $\mathrm{id}_C \circ F = F = F \circ \mathrm{id}_C$, so the associator and unitors are identity natural transformations. Thus $([C, C], \circ, \mathrm{id}_C)$ is a *strict* monoidal category (definition 7.1.2).

The endofunctor example is the one to keep in mind. It is where strictness appears without contrivance, composition being an operation that is equal, not just isomorphic, when reassociated, and it is the setting in which the monad (T, η, μ) met earlier will reappear as a monoid object. The strictification argument in the coherence proof is modelled on exactly this category, replacing each object by its translation endofunctor. The first three examples, by contrast, are the reason coherence is needed at all: their associators and unitors are nontrivial isomorphisms, and without the pentagon and triangle there would be no guarantee that a long chain of reassociations returns an unambiguous answer.

More generally, any category with finite products is monoidal under $(\times, 1)$, with $\times$ the binary product (definition 3.2.3) and 1 a terminal object (definition 3.1.1) as unit; the first example is the case $C = \mathbf{Set}$. This is the *cartesian monoidal* structure, and it is available on **Grp**, **Top**, **Cat**, and every other member of the standing cast with finite products.

The pentagon and triangle are two small diagrams. It is not obvious that they suffice: a diagram built from a dozen associators and unitors offers many paths between its corners, and nothing so far forces those paths to agree. Mac Lane's coherence theorem is the guarantee that they do. It has two equivalent formulations, one internal (every formal diagram commutes) and one structural (every monoidal category is equivalent to a strict one), and the second is the route to proving the first.

Theorem 7.1.5: Mac Lane's Coherence Theorem

Let C be a monoidal category. Then:

1. *Coherence.* Every diagram in C whose edges are built by tensoring identities, associators $\alpha^{\pm 1}$, and unitors $\lambda^{\pm 1}$, $\rho^{\pm 1}$ commutes. Any two morphisms between the same pair of iterated tensor products built from these constituents are equal.

2. *Strictification.* There is a strict monoidal category C^{st} and a monoidal equivalence $C \simeq C^{\mathrm{st}}$.

The full proof of part (1) is a combinatorial argument about a word problem: one shows that any two formal composites of associators and unitors with the same source and target are provably equal from the pentagon and triangle, and the complete verification is long. It is given in Mac Lane, *Categories for the Working Mathematician*, Chapter VII. What follows establishes part (2) in full and shows how it yields part (1), which is the conceptual content.

***Proof Key ideas*:**
The strictification construction is given in full; the word-problem proof of part (1) from part (2) is sketched, with the complete combinatorial argument cited to Mac Lane, Chapter VII, as noted above.

Step 1: The strict category of lists. Let C^{st} be the category whose objects are finite lists $\underline{X} = (X_1, \ldots, X_n)$ of objects of C (including the empty list, of length 0), and whose morphisms

$$C^{\mathrm{st}}(\underline{X}, \underline{Y}) = \mathrm{Hom}_C\big(\overline{X}, \overline{Y}\big)$$

are the C-morphisms between the tensorings $\overline{X} = ((X_1 \otimes X_2) \otimes \cdots) \otimes X_n$ associated to the left in a fixed order (the empty list tensoring to I), with composition and identities taken in C. Concatenation of lists, $\underline{X} \otimes \underline{Y} = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$, is an associative operation on objects with strict unit the empty list, and on morphisms it is defined by transporting the tensor of the underlying C-morphisms across the associators that compare $\overline{\underline{X} \otimes \underline{Y}}$ with $\overline{X} \otimes \overline{Y}$; the pentagon axiom makes this transport well-defined and functorial. Thus C^{st} is a *strict* monoidal category (definition 7.1.2): concatenation is associative and unital on the nose. This is the list model of the endofunctor picture, where a length-one list (X) plays the role of the translation $X \otimes -$ and concatenation the role of composition, but built so that the bookkeeping isomorphisms live inside the definition of the tensor rather than obstructing strictness.

Step 2: L is a monoidal equivalence. Define $L\colon C \to C^{\mathrm{st}}$ on objects by $L(X) = (X)$, the length-one list, and on morphisms by $L(f) = f$, viewing $f\colon X \to Y$ as a morphism $\overline{(X)} = X \to Y = \overline{(Y)}$ in C. Because $C^{\mathrm{st}}((X),(Y)) = \mathrm{Hom}_C(X,Y)$ by definition, L is *fully faithful* with no further argument, its action on each hom-set being the identity map of $\mathrm{Hom}_C(X,Y)$. It is essentially surjective: every list $\underline{X}$ is isomorphic in C^{st} to the length-one list $(\overline{\underline{X}})$, the isomorphism being the identity of $\overline{\underline{X}}$ regarded as a morphism $\underline{X} \to (\overline{\underline{X}})$. A functor that is fully faithful and essentially surjective is an equivalence (proposition 1.5.6), so L is an equivalence of categories. Finally L is *strong monoidal*: the structure isomorphism $L(X) \otimes L(Y) = (X,Y) \to (X \otimes Y) = L(X \otimes Y)$ is the identity of $X \otimes Y$ regarded as a morphism $\overline{(X,Y)} = X \otimes Y \to X \otimes Y$, and the unit comparison $I_{C^{\mathrm{st}}} = () \to (I)$ is id_I, since both $\overline{()}$ and $\overline{(I)}$ equal I; the pentagon and triangle axioms of C are exactly the coherence identities these comparisons must satisfy. Hence L is a monoidal equivalence $C \simeq C^{\mathrm{st}}$ with strict target, proving part (2).

Step 3: Coherence from strictification. Given a formal diagram in C built from associators and unitors, apply the equivalence $L\colon C \simeq C^{\mathrm{st}}$. Since L is *strong* (not strict) monoidal, it does not send an associator to an identity; rather, the strong-monoidal coherence axioms and the strictness of the target force $L(\alpha_{X,Y,Z})$ to equal a fixed composite of the structure isomorphisms $\varphi_{A,B}\colon L(A)\otimes L(B) \cong L(A\otimes B)$ of Step 2 (each of which, in the list model, is an identity morphism of C regarded across a change of tensoring), and likewise $L(\lambda_X)$, $L(\rho_X)$ to fixed composites of the $\varphi_{A,B}$ and the unit comparison. Because concatenation in C^{st} is strictly associative and unital, these image composites depend only on the source and target parenthesizations, not on the path taken: any two formal composites of associators and unitors sharing a source and target map under L to the *same* morphism of C^{st}. As L is faithful, they are already equal in C, so the formal diagram commutes. The content that Step 3 defers is exactly the path-independence just used: that the structure isomorphisms assemble into a well-defined map depending only on the endpoints is the word-problem statement, that the free monoidal category on the objects involved has at most one morphism between any two parenthesizations. This is the combinatorial argument deferred to Mac Lane, Chapter VII; granting it, every formal diagram of associators and unitors commutes, completing the proof of part (1).

Coherence licenses a controlled sloppiness. Because every formal diagram of associators and unitors commutes, one may drop parentheses from iterated tensor products and suppress the unit isomorphisms entirely, writing $X \otimes Y \otimes Z$ and $I \otimes X = X$ as though the category were strict. No ambiguity results: whichever way a suppressed reassociation is reinstated, part (1) guarantees the same morphism. In practice one works in a monoidal category as if it were strict, invoking coherence to justify the omitted brackets, and the strictification of part (2) is what makes this rigorous rather than convenient. Every monoidal category met later in this chapter is handled this way.

7.2 Braided and Symmetric Monoidal Categories

A monoidal structure (definition 7.1.1) relates $X \otimes Y$ and $Y \otimes X$ by nothing at all; in $(\mathbf{Vect}_k, \otimes, k)$ they are visibly isomorphic by the swap $x \otimes y \mapsto y \otimes x$, coherently with the tensor. This section isolates that extra datum. A *braiding* is a natural isomorphism $\beta_{X,Y}\colon X \otimes Y \to Y \otimes X$ compatible with the associator, and a braiding is *symmetric* when swapping twice returns to the start. The compatibility is governed not by one axiom but by two hexagons, one for each way of moving a factor past a tensor product.

The compatibility a braiding must satisfy is forced by coherence. Given three objects, $X \otimes (Y \otimes Z)$ can be rewritten as $(Y \otimes Z) \otimes X$ in two ways: apply the braiding to the whole factor $Y \otimes Z$ at once, or slide X past Y and then past Z one factor at a time, reassociating as needed. A braiding is required to make these two routes agree. Each route is a hexagon of associators and braidings, and there are two of them because X may be moved to the right (past $Y \otimes Z$) or an object may be moved to the left (past $X \otimes Y$); the second hexagon is the analogue of the first for the reverse braiding.

Definition 7.2.1: Braided Monoidal Category

A *braided monoidal category* is a monoidal category $(\mathbf{C}, \otimes, I, \alpha, \lambda, \rho)$ (definition 7.1.1) equipped with a natural isomorphism

$$\beta_{X,Y} \colon X \otimes Y \to Y \otimes X$$

called the *braiding*, natural in X and Y (definition 1.5.2), such that the two hexagon diagrams commute for all objects X, Y, Z. The first hexagon

$$\begin{tikzcd}
 & (X \otimes Y) \otimes Z \arrow[r, "\beta_{X,Y} \otimes \mathrm{id}_Z"] & (Y \otimes X) \otimes Z \arrow[dr, "\alpha"] & \\
X \otimes (Y \otimes Z) \arrow[ur, "\alpha^{-1}"] \arrow[dr, "\beta_{X,Y \otimes Z}"'] & & & Y \otimes (X \otimes Z) \arrow[dl, "\mathrm{id}_Y \otimes \beta_{X,Z}"] \\
 & (Y \otimes Z) \otimes X \arrow[r, "\alpha"'] & Y \otimes (Z \otimes X) &
\end{tikzcd}$$

expresses the compatibility of β with α when the object X is braided past the tensor product $Y \otimes Z$. The second hexagon

$$\begin{tikzcd}
 & X \otimes (Y \otimes Z) \arrow[r, "\mathrm{id}_X \otimes \beta_{Y,Z}"] & X \otimes (Z \otimes Y) \arrow[dr, "\alpha^{-1}"] & \\
(X \otimes Y) \otimes Z \arrow[ur, "\alpha"] \arrow[dr, "\beta_{X \otimes Y,Z}"'] & & & (X \otimes Z) \otimes Y \arrow[dl, "\beta_{X,Z} \otimes \mathrm{id}_Y"] \\
 & Z \otimes (X \otimes Y) \arrow[r, "\alpha^{-1}"'] & (Z \otimes X) \otimes Y &
\end{tikzcd}$$

expresses the same compatibility when Z is braided past $X \otimes Y$.

Naturality of β means that for morphisms $f \colon X \to X'$ and $g \colon Y \to Y'$, the square

$$\begin{tikzcd}
X \otimes Y \arrow[r, "\beta_{X,Y}"] \arrow[d, "f \otimes g"'] & Y \otimes X \arrow[d, "g \otimes f"] \\
X' \otimes Y' \arrow[r, "\beta_{X',Y'}"'] & Y' \otimes X'
\end{tikzcd}$$

commutes: braiding and then acting by $g \otimes f$ agrees with acting by $f \otimes g$ and then braiding. The two hexagons are the coherence conditions for the braiding against the associator; together with Mac Lane's coherence theorem (theorem 7.1.5) for the underlying monoidal structure, they guarantee that every diagram built from associators, unitors, and braidings that one might expect to commute does commute, provided its two sides braid the factors into the same permutation by isotopic routes. The proviso is the point: unlike the associator and unitors, the braiding is not forced to be its own inverse, so the order in which factors are braided can matter.

In a strict monoidal category (definition 7.1.2) the associators are identities and each hexagon collapses to a triangle. The first becomes

$$\beta_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes \beta_{X,Z}) \circ (\beta_{X,Y} \otimes \mathrm{id}_Z)$$

and the second becomes

$$\beta_{X\otimes Y,Z} = (\beta_{X,Z} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes \beta_{Y,Z})$$

so that a braiding past a tensor product is determined by braiding past each factor in turn. These are the relations one draws as strands crossing: $\beta_{X,Y}$ is the strand carrying X passing over the strand carrying Y, and the hexagons say that a strand passing over two others equals its passing over them one at a time.

The extra condition that makes a braiding into a symmetry is the one the swap map satisfies: undoing the crossing recovers the identity.

Definition 7.2.2: Symmetric Monoidal Category

A braided monoidal category (definition 7.2.1) is *symmetric* if its braiding satisfies

$$\beta_{Y,X} \circ \beta_{X,Y} = \mathrm{id}_{X\otimes Y}$$

for all objects X and Y. A braiding with this property is called a *symmetry*.

For a symmetry, $\beta_{Y,X} = \beta_{X,Y}^{-1}$, so the two hexagons of definition 7.2.1 become equivalent: each is obtained from the other by inverting the braidings, and symmetry identifies a braiding with its inverse. A symmetric monoidal category therefore carries a single hexagon axiom, and the strand carrying X over the strand carrying Y is indistinguishable from X under Y. The distinction between over and under crossings is what a general braiding retains and a symmetry discards.

The condition $\beta_{Y,X}\beta_{X,Y} = \mathrm{id}$ is the categorical form of a familiar identity: it says the swap is an involution. Symmetry is the monoidal counterpart of commutativity. Just as a commutative operation lets a product $a_1a_2\cdots a_n$ be formed without regard to the order of its factors, a symmetric braiding lets a tensor $X_1 \otimes \cdots \otimes X_n$ be reindexed by any permutation, with any two reindexings related by a canonical isomorphism independent of how the permutation is factored into transpositions. This is coherence for the symmetric group: the symmetric group S_n acts on the n-fold tensor, and the action is well-defined precisely because $\beta_{Y,X}\beta_{X,Y} = \mathrm{id}$ makes a transposition square to the identity. In a braided category that is not symmetric the braid group B_n acts in place of S_n, and B_n is infinite, so the order of braiding is remembered.

Example 7.2.3: The Standing Cast as Symmetric Monoidal Categories

The commutative examples of example 7.1.4, $(\mathbf{Vect}_k, \otimes, k)$ and $(\mathbf{Set}, \times, 1)$, each carry a symmetry; the third example there, the endofunctor category $([\mathbf{C}, \mathbf{C}], \circ, \mathrm{id})$, is deliberately excluded, since functor composition is non-commutative ($F \circ G$ is not naturally isomorphic to $G \circ F$ in general) and so admits no braiding, a point that returns when a monad is identified as a monoid object in that category. The vector-space case extends to modules over any commutative ring.

1. In $(\mathbf{Vect}_k, \otimes, k)$ the braiding is the swap

$$\beta_{V,W} \colon V \otimes W \to W \otimes V, \qquad v \otimes w \mapsto w \otimes v$$

extended k-bilinearly. This is a well-defined linear isomorphism because $(v, w) \mapsto w \otimes v$ is k-bilinear, and its inverse is $\beta_{W,V}$. Applying it twice returns $v \otimes w \mapsto w \otimes v \mapsto v \otimes w$, so $\beta_{W,V}\beta_{V,W} = \mathrm{id}_{V\otimes W}$ and the structure is symmetric. The hexagon axioms hold because both routes send $u \otimes v \otimes w$ (with the associators identifying the two bracketings) to the same reordered pure tensor.

2. For a commutative ring R, the category $(\mathbf{Mod}_R, \otimes_R, R)$ is symmetric with the same swap $m \otimes n \mapsto n \otimes m$. Well-definedness uses commutativity of R: for $r \in R$ the assignment must respect $rm \otimes n = m \otimes rn$, and the target $n \otimes rm = rn \otimes m$ agrees only because r is central, which it is. The tensor product itself is the tensor product of R-modules cited from [4]; here it is repackaged as a symmetric monoidal structure. The case $R = k$ a field recovers the previous item.

3. In $(\mathbf{Set}, \times, 1)$ the braiding is the swap of ordered pairs

$$\beta_{A,B} \colon A \times B \to B \times A, \qquad (a, b) \mapsto (b, a)$$

which is its own inverse: applying it twice returns $(a, b) \mapsto (b, a) \mapsto (a, b)$, so it is a symmetry. Every category with finite products carries this symmetry on its cartesian monoidal structure $(\mathbf{C}, \times, 1)$ (the structure of definition 3.2.3): the braiding $\beta_{X,Y} \colon X \times Y \to Y \times X$ is the unique map whose projections to Y and X are the projections of $X \times Y$ in the other order, and its square is the identity because both maps have the same projections. A cartesian monoidal category is thus symmetric with no further choice.

Braidings that are not symmetric exist. The category of finite-dimensional representations of a quantum group $U_q(\mathfrak{g})$ carries a braiding, built from the R-matrix solving the Yang-Baxter equation, for which $\beta_{Y,X}\beta_{X,Y} \neq \mathrm{id}$: braiding past twice is a nontrivial automorphism, and the braid group B_n acts faithfully on tensor powers. This is the source of the quantum knot invariants, the Jones polynomial among them, where an over-crossing and an under-crossing of strands must be assigned different operators precisely because β and β^{-1} differ. These braided-not-symmetric categories are the natural home of that theory and are developed in the literature on quantum groups; the present book stays with the symmetric case, where the standing cast lives.

The three examples share a mechanism. In each, the braiding is a swap of coordinates, its square is the identity, and the hexagons hold because reassociating and swapping commute on pure tensors or pairs. The symmetry is not an added structure to be checked case by case so much as a reflection of the fact that the underlying product, of vector spaces, of modules, of sets, is commutative up to canonical isomorphism. Symmetric monoidal categories are the ambient setting in which the algebra of tensor products behaves like the algebra of a commutative operation.

7.3 Closed Monoidal Categories

The tensor product $\otimes$ combines two objects into one. A closed structure supplies the inverse move: an object $[X, Y]$ inside the category whose maps out of any A are exactly the maps $A \otimes X \to Y$. This section makes that require precise as an adjunction (definition 4.1.1), identifies the two canonical maps it carries, and reads off the closure of the two examples the adjunction chapter already met in another guise.

The condition to impose is that tensoring with a fixed object have a right adjoint. Naming that right adjoint $[X,-]$ turns the informal "object of maps $X \to Y$" into a functor internal to the category.

Definition 7.3.1: Closed Monoidal Category

A monoidal category $(\mathbf{C}, \otimes, I)$ (definition 7.1.1) is *closed* if for every object X the functor

$$-\otimes X : \mathbf{C} \to \mathbf{C}$$

has a right adjoint (definition 4.1.1), written $[X,-] : \mathbf{C} \to \mathbf{C}$ and called the *internal hom*. Explicitly, for each X there is a bijection

$$\mathbf{C}(A \otimes X, Y) \;\cong\; \mathbf{C}(A, [X,Y]) \tag{7.1}$$

natural in A and in Y. The object $[X,Y]$ is the *internal hom* from X to Y.

The word "closed" names the fact that the hom-set $\mathbf{C}(X,Y)$, an object of **Set** external to $\mathbf{C}$, has been promoted to an object $[X,Y]$ of $\mathbf{C}$ itself: the category is closed under forming its own hom-objects. (7.1) is the defining feature. It says a map from A into the internal hom $[X,Y]$ is the same datum as a map from $A \otimes X$ to Y, so $[X,Y]$ represents the functor $A \mapsto \mathbf{C}(A \otimes X, Y)$. The external hom is recovered by taking $A = I$: since $I \otimes X \cong X$ through the left unitor, (7.1) gives $\mathbf{C}(X,Y) \cong \mathbf{C}(I, [X,Y])$, so the ordinary maps $X \to Y$ are the "points" $I \to [X,Y]$ of the internal hom.

The definition asks $-\otimes X$ to have a right adjoint, tensoring on the right. In a symmetric monoidal category (definition 7.2.2) the braiding gives a natural isomorphism $A \otimes X \cong X \otimes A$, so a right adjoint to $-\otimes X$ is at once a right adjoint to $X \otimes -$: left and right closedness coincide, and one speaks simply of a closed symmetric monoidal category.[1]

The adjunction packaged by (7.1) carries a unit and a counit (definition 4.1.3). Named and unwound in the present notation, they are the two structural maps a closed category always provides: an evaluation and a currying map.

[1]Without symmetry the two conditions may differ, and a category can be right closed without being left closed. Every example below is symmetric, so the distinction does not arise.

Theorem 7.3.2: The Tensor-Hom Adjunction

Let $(\mathbf{C}, \otimes, I)$ be a closed monoidal category (definition 7.3.1), and fix an object X. The adjunction $- \otimes X \dashv [X, -]$ has:

1. counit the *evaluation* morphism, natural in Y,
$$\mathrm{ev}_{X,Y} \colon [X, Y] \otimes X \to Y$$
the transpose of $\mathrm{id}_{[X,Y]}$;
2. unit the *coevaluation* (or *currying*) morphism, natural in A,
$$\mathrm{coev}_{A,X} \colon A \to [X, A \otimes X]$$
the transpose of $\mathrm{id}_{A\otimes X}$.

The bijection (7.1) sends $f \colon A \otimes X \to Y$ to $[X, f] \circ \mathrm{coev}_{A,X} \colon A \to [X, Y]$, and is natural in A and in Y.

***Proof*:**
By definition 7.3.1 the functors $- \otimes X$ and $[X, -]$ are adjoint, so the general theory of section 4.1 applies with $F = - \otimes X$ and $G = [X, -]$; the content of the theorem is to name the unit and counit in this setting and confirm the transpose formula.

Step 1: The counit is evaluation. By definition 4.1.3 the counit of $- \otimes X \dashv [X, -]$ has component at Y the transpose of $\mathrm{id}_{[X,Y]}$ under the inverse of (7.1). Setting $A = [X, Y]$, that bijection reads $\mathbf{C}([X, Y] \otimes X, Y) \cong \mathbf{C}([X, Y], [X, Y])$, and the transpose of $\mathrm{id}_{[X,Y]}$ is a morphism
$$\mathrm{ev}_{X,Y} \colon [X, Y] \otimes X \to Y$$
which is the counit, hence natural in Y. It evaluates the internal hom: for $\mathbf{C} = \mathbf{Set}$ with $[X, Y] = \mathbf{Set}(X, Y)$, ev is the map $(g, x) \mapsto g(x)$.

Step 2: The unit is coevaluation. Dually, the unit has component at A the transpose of $\mathrm{id}_{A\otimes X}$ under (7.1) with $Y = A \otimes X$: the bijection $\mathbf{C}(A \otimes X, A \otimes X) \cong \mathbf{C}(A, [X, A \otimes X])$ sends $\mathrm{id}_{A\otimes X}$ to a morphism
$$\mathrm{coev}_{A,X} \colon A \to [X, A \otimes X]$$
the unit, natural in A. In **Set** it sends a to the function $x \mapsto (a, x)$, the map that curries the second coordinate.

Step 3: The transpose formula. Proposition 4.1.4 computes the transpose of a morphism through the unit and counit: for $F \dashv G$ with unit η, the transpose of $f \colon FA \to Y$ is $Gf \circ \eta_A$. With $F = - \otimes X$, $G = [X, -]$, $\eta_A = \mathrm{coev}_{A,X}$, and $Gf = [X, f]$, the transpose of $f \colon A \otimes X \to Y$ is
$$A \xrightarrow{\mathrm{coev}_{A,X}} [X, A \otimes X] \xrightarrow{[X,f]} [X, Y] \tag{7.2}$$
which is the asserted formula: curry the identity, then apply the internal hom to f.

Step 4: Naturality. Naturality of (7.1) in A and Y is naturality of the adjunction isomorphism $\Phi \colon \mathbf{C}(F-, -) \cong \mathbf{C}(-, G-)$ of definition 4.1.1, which holds by the definition of an adjunction. Concretely, for $u \colon A' \to A$ the square relating $\Phi_{A,Y}$ and $\Phi_{A',Y}$ commutes because (7.2) is built from the natural transformation coev and the functor $[X, -]$; the Y-variable is dual, using ev. Thus the bijection is natural in both variables, completing the proof.

The evaluation and coevaluation maps are the internal forms of two operations on ordinary functions. Evaluation $[X,Y] \otimes X \to Y$ applies a map to an argument; coevaluation $A \to [X, A \otimes X]$ holds an element of A and waits for an argument in X, producing the pair. The triangle identities of theorem 4.1.5 record that currying and then evaluating returns the original map, which is the content of (7.2) run forward and back. This is the tensor-hom adjunction of example 4.2.2, now read as structure carried by a single category rather than a bijection between two external hom-sets: the module Hom-tensor identity $\mathrm{Hom}_R(A \otimes_R X, Y) \cong \mathrm{Hom}_R(A, \mathrm{Hom}_R(X,Y))$ and the currying isomorphism $\mathbf{Set}(A \times X, Y) \cong \mathbf{Set}(A, Y^X)$ are exactly (7.1) for $\mathbf{Mod}_R$ and for $\mathbf{Set}$. What the adjunction chapter presented as two adjoint functors, the closed structure presents as the category's ability to internalize its own hom.

When the monoidal structure is the cartesian one built from finite products (definition 3.2.3), closedness has a classical name and the internal hom is the exponential object.

Definition 7.3.3: Cartesian Closed Category

A category $\mathbf{C}$ with finite products is *cartesian closed* if its cartesian monoidal structure $(\mathbf{C}, \times, 1)$ is closed (definition 7.3.1): for every object X the functor $- \times X$ has a right adjoint $(-)^X : \mathbf{C} \to \mathbf{C}$, the *exponential*. The object $Y^X = [X, Y]$ is the *exponential object*, and the adjunction reads

$$\mathbf{C}(A \times X, Y) \;\cong\; \mathbf{C}(A, Y^X) \tag{7.3}$$

natural in A and Y.

Cartesian closedness is the special case where the tensor is the product and the unit is the terminal object 1. The bijection (7.3) is currying in its familiar form: a map of two variables $A \times X \to Y$ is the same as a map $A \to Y^X$ into the object of maps $X \to Y$. The evaluation counit $Y^X \times X \to Y$ applies a function to an argument, and taking $A = 1$ recovers the global elements of Y^X as the maps $X \to Y$, so the exponential is a genuine internal object of functions.

Example 7.3.4: Set and Vect as Closed Categories

1. $\mathbf{Set}$ *is cartesian closed.* The cartesian monoidal structure is $(\mathbf{Set}, \times, 1)$ with 1 a one-point set. For sets X and Y the exponential is the set of all functions,

$$Y^X = \mathbf{Set}(X, Y)$$

and currying is the bijection

$$\mathbf{Set}(A \times X, Y) \;\cong\; \mathbf{Set}(A, Y^X)$$

sending $f : A \times X \to Y$ to $a \mapsto (x \mapsto f(a, x))$. This is a bijection because a function of two arguments is determined by, and determines, its curried form; it is natural in A and Y because currying commutes with precomposition on A and postcomposition on Y. Thus $- \times X \dashv (-)^X$, and the evaluation counit $Y^X \times X \to Y$ is $(g, x) \mapsto g(x)$.

2. $\mathbf{Vect}_k$ *is closed monoidal, not cartesian.* The relevant monoidal structure on vector spaces over k is $(\mathbf{Vect}_k, \otimes_k, k)$, the tensor product with unit the ground field k, not the cartesian

product. It is closed with internal hom the space of linear maps,

$$[X, Y] = \mathrm{Hom}_k(X, Y)$$

itself a k-vector space. The bijection

$$\mathrm{Hom}_k(A \otimes_k X, Y) \;\cong\; \mathrm{Hom}_k(A, \mathrm{Hom}_k(X, Y))$$

is the tensor-hom adjunction over k (example 4.2.2): a bilinear map $A \times X \to Y$ corresponds to a linear map $A \otimes_k X \to Y$ and, curried, to a linear map $A \to \mathrm{Hom}_k(X, Y)$. The evaluation counit $\mathrm{Hom}_k(X, Y) \otimes_k X \to Y$ is $g \otimes x \mapsto g(x)$. The cartesian product $X \times Y$ (the direct sum, since finite products and coproducts coincide here) does *not* make $\mathbf{Vect}_k$ cartesian closed: the exponential would need $\mathbf{Vect}_k(A \times X, Y) \cong \mathbf{Vect}_k(A, Y^X)$, but the left side is not additive in A while the representable right side is. Since $A \times X = A \oplus X$, the left side is $\mathbf{Vect}_k(A \oplus X, Y) \cong \mathbf{Vect}_k(A, Y) \times \mathbf{Vect}_k(X, Y)$, which carries the constant summand $\mathbf{Vect}_k(X, Y)$: at $A = 0$ it returns $\mathbf{Vect}_k(X, Y)$, not a one-point set, so with $X = Y = k$ it fails to send the zero space to a terminal set. A representable functor $\mathbf{Vect}_k(-, Y^X)$ does send 0 to a point, so the two cannot be naturally isomorphic and no cartesian exponential exists. Closedness here is a feature of $\otimes_k$, not of $\times$.

The two examples separate the cartesian case from the general one. In **Set** the tensor is the product, so the internal hom is the exponential and the category is cartesian closed. In $\mathbf{Vect}_k$ the closure lives on the tensor product, and the cartesian product carries no matching exponential; the category is closed monoidal but not cartesian closed. The pattern recurs: the tensor-hom adjunction of example 4.2.2 for $\mathbf{Mod}_R$ makes each module category closed monoidal under $\otimes_R$, while a topos such as **Set** or a presheaf category is cartesian closed under $\times$.

Cartesian closed categories are the categorical semantics of the typed lambda calculus: types are objects, terms are morphisms, the product $\times$ interprets pairing, the exponential Y^X interprets the function type $X \to Y$, and the evaluation and currying maps interpret function application and lambda abstraction, with the adjunction (7.3) the statement that a function of two arguments and its curried form are interchangeable.[2]

7.4 Monoids, Comonoids, and Duality

The definition of a monoid transcribes verbatim into any monoidal category once the product is a functor $\otimes$ rather than a set-theoretic multiplication, and the reward is that the same three axioms name at once monoids, rings, algebras, and, in the endofunctor category of example 7.1.4, monads. This section makes that transcription, dualizes it to comonoids, and closes with a taste of the finiteness condition, dualizability, that lets an object have a linear dual.

Fix a monoidal category $(\mathbf{C}, \otimes, I)$ with associator α and unitors λ, ρ (definition 7.1.1). A monoid in **Set** is a set M together with a multiplication $M \times M \to M$ and a chosen unit element, which is the same as a map $1 \to M$ from the terminal set. Replacing $\times$ by $\otimes$ and the terminal set by the monoidal unit I gives the general definition.

[2]This correspondence, that cartesian closed categories model simply typed lambda calculus, is due to Lambek; a standard reference is Lambek and Scott, *Introduction to Higher Order Categorical Logic*.

Definition 7.4.1: Monoid Object

A *monoid object* in a monoidal category $(\mathbf{C}, \otimes, I)$ is an object M of $\mathbf{C}$ together with two morphisms

$$\mu\colon M \otimes M \to M \qquad \text{and} \qquad \eta\colon I \to M$$

the *multiplication* and the *unit*, such that the associativity diagram

$$\begin{tikzcd}
(M \otimes M) \otimes M \arrow[r, "\alpha_{M,M,M}"] \arrow[d, "\mu\otimes\mathrm{id}_M"'] & M \otimes (M \otimes M) \arrow[r, "\mathrm{id}_M\otimes\mu"] & M \otimes M \arrow[d, "\mu"] \\
M \otimes M \arrow[rr, "\mu"'] & & M
\end{tikzcd}$$

and the two unit diagrams

$$\begin{tikzcd}
I \otimes M \arrow[r, "\eta\otimes\mathrm{id}_M"] \arrow[dr, "\lambda_M"'] & M \otimes M \arrow[d, "\mu"] & M \otimes I \arrow[l, "\mathrm{id}_M\otimes\eta"'] \arrow[dl, "\rho_M"] \\
& M &
\end{tikzcd}$$

commute.

The associativity square says the two ways of multiplying a triple $M \otimes M \otimes M$ agree once the parenthesization is corrected by the associator, and the two triangles say that inserting the unit on either side and multiplying is the same as forgetting the unit factor via λ or ρ. When $\mathbf{C}$ is strict the associator and unitors are identities and the diagrams reduce to the familiar equations $\mu(\mu \otimes \mathrm{id}) = \mu(\mathrm{id} \otimes \mu)$, $\mu(\eta \otimes \mathrm{id}) = \mathrm{id} = \mu(\mathrm{id} \otimes \eta)$; theorem 7.1.5 guarantees no generality is lost by picturing the strict case. A morphism of monoid objects $(M, \mu, \eta) \to (M', \mu', \eta')$ is a map $f\colon M \to M'$ with $f\mu = \mu'(f \otimes f)$ and $f\eta = \eta'$; monoid objects and their morphisms form a category $\mathrm{Mon}(\mathbf{C})$.

Reversing every arrow gives the dual notion. Rather than a multiplication that fuses two copies of M into one, a comonoid carries a comultiplication that splits one copy into two, and in place of a unit picking out an element it has a counit that measures an element against the ground object I.

Definition 7.4.2: Comonoid Object

A *comonoid object* in $(\mathbf{C}, \otimes, I)$ is a monoid object in the opposite monoidal category $(\mathbf{C}^{\mathrm{op}}, \otimes, I)$: an object C with a *comultiplication* $\delta\colon C \to C \otimes C$ and a *counit* $\varepsilon\colon C \to I$ making the coassociativity diagram

$$\begin{tikzcd} C \arrow[r, "\delta"] \arrow[d, "\delta"'] & C \otimes C \arrow[d, "\mathrm{id}_C \otimes \delta"] \\ C \otimes C \arrow[r, "\delta \otimes \mathrm{id}_C"'] & C \otimes C \otimes C \end{tikzcd}$$

and the two counit diagrams

$$\begin{tikzcd} & C \arrow[dl, "\lambda_C^{-1}"'] \arrow[d, "\delta"] \arrow[dr, "\rho_C^{-1}"] & \\ I \otimes C & C \otimes C \arrow[l, "\varepsilon \otimes \mathrm{id}_C"] \arrow[r, "\mathrm{id}_C \otimes \varepsilon"'] & C \otimes I \end{tikzcd}$$

commute (associators suppressed in the strict picture).

By the duality principle (box 1.3.3), every statement about monoid objects has a mirror statement about comonoid objects, obtained by reversing the arrows in $\mathbf{C}$; there is nothing to prove separately. Comonoids are the less familiar of the pair only because the everyday examples, sets and modules, carry no interesting comultiplication under $\times$ or $\otimes$: the diagonal $C \to C \otimes C$ is forced. They become substantial in categories where $\otimes$ is genuinely non-cartesian, where a comonoid is the categorical form of a coalgebra, the structure dual to an algebra that organizes, for instance, the representation theory of a group through its group algebra.

The point of the definition is that a single set of axioms, read in different monoidal categories, returns structures that look unrelated on their face. The following example collects the payoff.

Example 7.4.3: Monoids, Rings, Algebras, and Monads

A monoid object is a monoid, a ring, an algebra, or a monad according to where it is read.

1. *In* $(\mathbf{Set}, \times, 1)$. A monoid object is exactly a monoid in the ordinary sense. The map $\mu\colon M \times M \to M$ is a binary operation, $\eta\colon 1 \to M$ picks out an element $e = \eta(*)$, the associativity square is $\mu(\mu(a, b), c) = \mu(a, \mu(b, c))$, and the unit triangles say $ea = a = ae$. The cartesian structure $(\mathbf{C}, \times, 1)$ of definition 3.2.3 is what makes this the definition of a monoid; the same reading in **Top** or **Grp** gives topological monoids and monoids internal to **Grp**.

2. *In* $(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$. A monoid object is exactly a ring. An abelian group R with a $\mathbb{Z}$-bilinear multiplication $\mu\colon R \otimes_{\mathbb{Z}} R \to R$ and a unit $\eta\colon \mathbb{Z} \to R$, $\eta(1) = 1_R$, is precisely a ring: bilinearity is built into the tensor product, associativity and unitality are the two diagrams. Replacing $\mathbb{Z}$ by a commutative ring k and **Ab** by $(\mathbf{Mod}_k, \otimes_k, k)$ gives a k-algebra (the multiplication and unit of k-modules cited to [4]). Thus rings and algebras are monoid objects, one monoidal category up from monoids.

3. *In* $([\mathbf{C}, \mathbf{C}], \circ, \mathrm{id})$. A monoid object in the strict monoidal category of endofunctors of $\mathbf{C}$, with $\otimes$ the composition of functors and unit object the identity functor (example 7.1.4, definition 1.5.4), is exactly a monad on $\mathbf{C}$. An object of $[\mathbf{C}, \mathbf{C}]$ is an endofunctor T; the

multiplication $\mu\colon T \circ T \to T$ and unit $\eta\colon \mathrm{id} \to T$ are natural transformations; and the associativity and unit diagrams of definition 7.4.1 are, symbol for symbol, the associativity and unit laws of a monad (T, η, μ) from definition 5.1.1. This is the identity

a monad is a monoid in the category of endofunctors

stated as the theorem it is: the two definitions are not analogous, they are the same definition read in two monoidal categories.

The third reading is the reason the notion of monoid object is worth isolating. Chapter 5 built monads by hand, checking the associativity and unit squares of natural transformations directly; definition 7.4.1 now shows those squares were never special to endofunctors. Any construction valid for monoids in an arbitrary monoidal category applies at once to monads: the category of monoid objects, the free monoid on an object, morphisms of monoids. A monad morphism, for instance, is nothing but a morphism of monoid objects in $[\mathbf{C}, \mathbf{C}]$.

A monoid in $(\mathbf{Set}, \times, 1)$ becomes a group exactly when every element has an inverse, and the inverse is itself a morphism $i\colon G \to G$ subject to a diagram. This upgrades the monoid-object definition in a cartesian monoidal category to a group object, the notion that lets "group" be interpreted inside geometry.

Example 7.4.4: Group Objects

Let $(\mathbf{C}, \times, 1)$ be a category with finite products (definition 3.2.3), regarded as a cartesian monoidal category. A *group object* is a monoid object (G, μ, η) together with an *inverse* morphism $i\colon G \to G$ such that

$$\begin{tikzcd} G \arrow[r, "(\mathrm{id}_G, i)"] \arrow[d, "e"'] & G \times G \arrow[d, "\mu"] & G \arrow[l, "(i, \mathrm{id}_G)"'] \arrow[d, "e"] \\ 1 \arrow[r, "\eta"'] & G & 1 \arrow[l, "\eta"] \end{tikzcd}$$

commutes, where $e\colon G \to 1$ is the unique map to the terminal object and (id_G, i), (i, id_G) are the maps into the product with the indicated components. The two squares say $\mu(g, i(g)) = \eta e(g) = \mu(i(g), g)$: multiplying an element by its inverse on either side returns the unit. In $(\mathbf{Set}, \times, 1)$ a group object is a group. In $(\mathbf{Top}, \times, 1)$ it is a topological group, a group whose multiplication and inversion are continuous; in the category $\mathbf{Man}$ of smooth manifolds with the product, it is a Lie group, a group whose operations are smooth, the central object of the differential-geometric theory in [9].

Group objects show that "group" is not a set-theoretic notion but a diagrammatic one: the same diagrams, interpreted in **Top** or **Man**, define topological and Lie groups without a word changing. Comultiplication reverses the picture once more, and a group object with a compatible comonoid structure in a symmetric monoidal category (definition 7.2.2) is a Hopf algebra, the structure underlying quantum groups; the symmetry is what lets the compatibility between μ and δ be stated.

The remaining topic is a finiteness property of the object itself, orthogonal to the algebraic structure above. In $\mathbf{Vect}_k$ a finite-dimensional space V has a dual V^*, and the pairing $V^* \otimes V \to k$ together with the "identity element" $k \to V \otimes V^*$ obey two triangle identities that reconstruct V from V^* and conversely. Abstracting those two identities gives dualizability.

Definition 7.4.5: Dualizable Object

An object X of a monoidal category $(\mathbf{C}, \otimes, I)$ is *(left) dualizable* if there is an object $X^\vee$, the *dual*, together with morphisms

$$\mathrm{ev}\colon X^\vee \otimes X \to I \qquad \text{and} \qquad \mathrm{coev}\colon I \to X \otimes X^\vee$$

the *evaluation* and *coevaluation*, such that the two *zigzag identities*

$$\begin{tikzcd} X \arrow[r, "\mathrm{coev}\otimes\mathrm{id}_X"] \arrow[dr, "\mathrm{id}_X"'] & X \otimes X^\vee \otimes X \arrow[d, "\mathrm{id}_X\otimes\mathrm{ev}"] \\ & X \end{tikzcd} \qquad \begin{tikzcd} X^\vee \arrow[r, "\mathrm{id}\otimes\mathrm{coev}"] \arrow[dr, "\mathrm{id}"'] & X^\vee \otimes X \otimes X^\vee \arrow[d, "\mathrm{ev}\otimes\mathrm{id}"] \\ & X^\vee \end{tikzcd}$$

commute (associators and unitors suppressed in the strict picture). A monoidal category in which every object is dualizable is called *rigid*.

The two triangles are the categorical form of the linear-algebra identity that expresses id_V through a basis and its dual basis. Reading them in $\mathbf{Vect}_k$ makes the abstraction concrete, and pins down exactly which spaces qualify.

Example 7.4.6: Dualizable Vector Spaces

In $(\mathbf{Vect}_k, \otimes_k, k)$ a vector space V is dualizable if and only if it is finite-dimensional, with dual $V^\vee = V^* = \mathrm{Hom}_k(V, k)$. The evaluation is the pairing $\mathrm{ev}(\varphi \otimes v) = \varphi(v)$, and the coevaluation is

$$\mathrm{coev}\colon k \to V \otimes V^*, \qquad 1 \mapsto \sum_i e_i \otimes e_i^*$$

for a basis (e_i) with dual basis (e_i^*); the element $\sum_i e_i \otimes e_i^*$ is independent of the basis, being the image of id_V under the isomorphism $\mathrm{End}_k(V) \cong V \otimes V^*$ that holds precisely when V is finite-dimensional. The first zigzag sends $v \mapsto \sum_i e_i \otimes e_i^* \otimes v \mapsto \sum_i e_i\, e_i^*(v) = v$, so it is id_V; the second is the symmetric computation. When V is infinite-dimensional no coevaluation exists: a map $k \to V \otimes V^*$ has image an element of $V \otimes V^*$, necessarily a finite sum of simple tensors, and no such finite sum can satisfy the first zigzag for every v. Thus dualizability is exactly the finiteness that makes $V \cong V^{**}$ canonically.

Dualizable objects are where the linear-algebraic notion of trace survives into a symmetric monoidal category (definition 7.2.2): for an endomorphism $f\colon X \to X$ of a dualizable object one forms the composite $I \xrightarrow{\mathrm{coev}} X \otimes X^\vee \xrightarrow{f\otimes\mathrm{id}} X \otimes X^\vee \xrightarrow{\beta_{X,X^\vee}} X^\vee \otimes X \xrightarrow{\mathrm{ev}} I$, an endomorphism of I that specializes to the ordinary trace in $\mathbf{Vect}_k$. The symmetry β is what supplies the swap $X \otimes X^\vee \to X^\vee \otimes X$; in a bare monoidal category no such canonical map exists, so the trace needs at least a braiding. Rigid symmetric monoidal categories, in which every object is dualizable, are the setting for Tannakian duality, which reconstructs a group from its rigid tensor category of representations; both developments belong to the next stage of the theory and are not pursued here.

8

Enriched and 2-Categories

Two facts about the previous chapters were left unremarked. The hom-*sets* of a category are objects of **Set**, but nothing forced that choice; and **Cat** carried natural transformations, a second layer of morphism an ordinary category lacks. This chapter follows both threads. *Enriched category theory* replaces hom-sets by objects of a monoidal category, recovering additive, topological, and simplicial categories as the base varies. *Two-category theory* takes the morphisms between morphisms seriously, with **Cat** as the guiding example, and recovers adjunctions and monads as statements internal to a 2-category.

The two threads are developed to different depths. The enriched theory is built in full through section 8.3: tensors and cotensors, the enriched functor category, the enriched Yoneda lemma with its proof, and weighted limits, which subsume ordinary limits and both of the first two. The 2-categorical theory is developed only as far as the enriched theory already carries it, since a strict 2-category is the **Cat**-enriched case of definition 8.1.1 and the structures whose coherence is genuine, rather than provable, are the subject of the sequel. The chapter, and the book, close at the edge of the subject, where the strict equalities weaken to coherent isomorphism, pointing beyond it to higher category theory.

8.1 Enriched Categories

The hom-set of a category is a set, and nothing more. Yet the standing cast carries hom-sets with extra structure that the plain definition discards: the morphisms between two abelian groups form an abelian group under pointwise addition, the maps between two topological spaces form a topological space, and between two categories the functors form a category (with natural transformations as its own morphisms). *Enriched category theory* takes this extra structure seriously. It replaces the hom-*set* $\mathbf{C}(x, y)$ by a hom-*object* living in a fixed monoidal category $\mathbf{V}$ (definition 7.1.1), whose tensor product supplies the composition and whose unit supplies the identities. The one requirement is that $\mathbf{V}$ be monoidal: composition pairs two hom-objects, and $\otimes$ is exactly the operation for pairing objects of $\mathbf{V}$.

Definition 8.1.1: Enriched Category

Let $(\mathbf{V}, \otimes, I)$ be a monoidal category. A *category enriched over* $\mathbf{V}$, or a $\mathbf{V}$*-category*, $\mathbf{C}$ consists of

- a collection of *objects* $\mathrm{Ob}(\mathbf{C})$;
- for each ordered pair (x, y) of objects, a *hom-object* $\mathbf{C}(x, y)$ of $\mathbf{V}$;
- for each ordered triple (x, y, z), a *composition* morphism of $\mathbf{V}$

$$\circ_{xyz} \colon \mathbf{C}(y,z) \otimes \mathbf{C}(x,y) \to \mathbf{C}(x,z)$$

- for each object x, an *identity* morphism of $\mathbf{V}$

$$j_x \colon I \to \mathbf{C}(x,x)$$

subject to the associativity and unit laws, which are the following diagrams in $\mathbf{V}$. Associativity is the commuting square (writing α for the associator of $\mathbf{V}$)

$$\begin{tikzcd}
\big(\mathbf{C}(z,w) \otimes \mathbf{C}(y,z)\big) \otimes \mathbf{C}(x,y) \arrow[r, "\alpha"] \arrow[d, "\circ\otimes\mathrm{id}"'] & \mathbf{C}(z,w) \otimes \big(\mathbf{C}(y,z) \otimes \mathbf{C}(x,y)\big) \arrow[d, "\mathrm{id}\otimes\circ"] \\
\mathbf{C}(y,w) \otimes \mathbf{C}(x,y) \arrow[d, "\circ"'] & \mathbf{C}(z,w) \otimes \mathbf{C}(x,z) \arrow[d, "\circ"] \\
\mathbf{C}(x,w) \arrow[r, equal] & \mathbf{C}(x,w)
\end{tikzcd}$$

and the two unit laws are the commuting triangles (writing λ, ρ for the left and right unitors of $\mathbf{V}$)

$$\begin{tikzcd}
I \otimes \mathbf{C}(x,y) \arrow[r, "j_y\otimes\mathrm{id}"] \arrow[dr, "\lambda"'] & \mathbf{C}(y,y) \otimes \mathbf{C}(x,y) \arrow[d, "\circ"] & \mathbf{C}(x,y) \otimes I \arrow[l, "\mathrm{id}\otimes j_x"'] \arrow[dl, "\rho"] \\
& \mathbf{C}(x,y) &
\end{tikzcd}$$

The two diagrams are the ordinary associativity and unit laws with every equation of *elements* replaced by a commuting diagram of *morphisms in* $\mathbf{V}$, and every reassociation or unit-insertion routed through the coherence isomorphisms α, λ, ρ that a monoidal category supplies for exactly this purpose. Nowhere does the definition speak of an element of $\mathbf{C}(x, y)$: $\mathbf{C}(x, y)$ is an object of $\mathbf{V}$, and $\mathbf{V}$ need not be a category of structured sets. Composition is not a function applied to a pair of morphisms but a single arrow out of a tensor product, and the identity at x is not a chosen element but an arrow $I \to \mathbf{C}(x, x)$ selecting it in whatever sense $\mathbf{V}$ allows. This is the shift the definition forces: from *elements and functions* to *objects and morphisms of* $\mathbf{V}$.

An ordinary category is the special case $\mathbf{V} = (\mathbf{Set}, \times, 1)$. Then $\mathbf{C}(x, y)$ is a hom-set, the composition arrow $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is the usual composition function $(g, f) \mapsto gf$, and the identity arrow $j_x\colon 1 \to \mathbf{C}(x,x)$ from the one-point set picks out id_x. The associativity and unit diagrams reduce to $h(gf) = (hg)f$ and $f\,\mathrm{id}_x = f = \mathrm{id}_y f$. Enrichment loses nothing and is a genuine generalization: it remembers structure on the hom-objects that the passage to underlying sets throws away.

That passage is itself a construction. Given a $\mathbf{V}$-category, one recovers an ordinary category by asking $\mathbf{V}$ for the elements of each hom-object, where "element" means a morphism out of the monoidal unit.

Definition 8.1.2: Underlying Ordinary Category

Let $\mathbf{C}$ be a $\mathbf{V}$-category. Its *underlying ordinary category* $\mathbf{C}_0$ has the same objects as $\mathbf{C}$, and hom-sets

$$\mathrm{Hom}_{\mathbf{C}_0}(x,y) = \mathbf{V}\big(I, \mathbf{C}(x,y)\big)$$

the set of $\mathbf{V}$-morphisms from the unit I to the hom-object $\mathbf{C}(x,y)$. The identity of x in $\mathbf{C}_0$ is the enriched identity $j_x\colon I \to \mathbf{C}(x,x)$. The composite in $\mathbf{C}_0$ of $f\colon I \to \mathbf{C}(x,y)$ and $g\colon I \to \mathbf{C}(y,z)$ is

$$I \xrightarrow{\lambda^{-1}} I \otimes I \xrightarrow{g \otimes f} \mathbf{C}(y,z) \otimes \mathbf{C}(x,y) \xrightarrow{\circ} \mathbf{C}(x,z)$$

where $\lambda^{-1}\colon I \to I \otimes I$ is the inverse left unitor.

The associativity and unit laws of $\mathbf{C}_0$ are not assumed; they are *consequences* of the enriched laws of definition 8.1.1. Composing three elements in $\mathbf{C}_0$ produces two bracketings of the same triple tensor, and the enriched associativity square, together with the coherence (theorem 7.1.5) that makes all instances of α, λ, ρ agree, identifies them; the enriched unit triangles give the identity laws in $\mathbf{C}_0$. The underlying category is what the $\mathbf{V}$-category induces on $\mathbf{Set}$ by probing each hom-object with the unit. When $\mathbf{V} = \mathbf{Set}$ it changes nothing, since $\mathbf{Set}(1, S) \cong S$: an ordinary category is its own underlying category. The interesting cases are those where $\mathbf{C}_0$ forgets real structure, and the enriched data is strictly richer than the ordinary category it sits over.

Example 8.1.3: Enrichment over the Standing Cast

Each choice of monoidal base $\mathbf{V}$ names a familiar species of category.

1. $\mathbf{Set}$-*enriched categories are ordinary categories.* With $\mathbf{V} = (\mathbf{Set}, \times, 1)$ the data of definition 8.1.1 is precisely objects, hom-sets, a composition function, and chosen identities obeying associativity and unit laws. This is the definition of a locally small category, and the underlying category $\mathbf{C}_0$ is $\mathbf{C}$ itself.

2. $\mathbf{Ab}$-*enriched categories are preadditive categories.* Take $\mathbf{V} = (\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$, the tensor product of abelian groups with unit $\mathbb{Z}$. Each hom-object $\mathbf{C}(x,y)$ is then an abelian group, and composition is a morphism of abelian groups

$$\mathbf{C}(y,z) \otimes_{\mathbb{Z}} \mathbf{C}(x,y) \to \mathbf{C}(x,z)$$

which is exactly a $\mathbb{Z}$-bilinear map $\mathbf{C}(y,z) \times \mathbf{C}(x,y) \to \mathbf{C}(x,z)$, by the universal property of $\otimes_{\mathbb{Z}}$. Composition is therefore biadditive: $h(g+g') = hg + hg'$ and $(h+h')g = hg + h'g$. A category with abelian-group hom-sets and biadditive composition is a *preadditive category*, and the underlying category recovers the hom-set as $\mathbf{Ab}(\mathbb{Z}, \mathbf{C}(x,y)) \cong \mathbf{C}(x,y)$. This is the enrichment that homological algebra rests on: an additive category is a preadditive category with a zero object and biproducts, and **Ab**-enrichment is the first axiom (see [4] for the module-theoretic origin of $\otimes_{\mathbb{Z}}$).

3. **Top**-*enriched categories topologize their hom-sets.* With $\mathbf{V} = (\mathbf{Top}, \times, *)$ each $\mathbf{C}(x, y)$ is a topological space and composition $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is continuous. The identity $* \to \mathbf{C}(x, x)$ from the one-point space picks out id_x. The underlying category has hom-sets $\mathbf{Top}(*, \mathbf{C}(x, y)) \cong \mathbf{C}(x, y)$ as sets, discarding the topology; the enriched structure remembers, for instance, when two maps are homotopic through a continuous path of morphisms.

4. **Cat**-*enriched categories are strict* 2-*categories.* Take $\mathbf{V} = (\mathbf{Cat}, \times, \mathbf{1})$, categories under product with the terminal category $\mathbf{1}$ as unit. Each hom-object $\mathbf{C}(x, y)$ is now a *category*: its objects are the morphisms $x \to y$, and its own morphisms are a second layer of structure between them. Composition is a functor $\mathbf{C}(y, z) \times \mathbf{C}(x, y) \to \mathbf{C}(x, z)$. This is exactly a strict 2-category, the subject of section 8.4: the prototype is **Cat** itself, whose hom-categories are the functor categories $[\mathbf{C}, \mathbf{D}]$ (definition 1.5.4) with natural transformations as the second layer.

5. *Lawvere metric spaces.* Let $\mathbf{V} = ([0, \infty], +, 0)$ be the extended nonnegative reals, viewed as a category by the order $\geq$ (an arrow $a \to b$ exactly when $a \geq b$), monoidal under addition with unit 0. A category enriched over this $\mathbf{V}$ has objects, and for each pair a number $\mathbf{C}(x, y) \in [0, \infty]$ to be read as a *distance*. The composition arrow $\mathbf{C}(y, z) + \mathbf{C}(x, y) \to \mathbf{C}(x, z)$ is, in this order category, precisely the inequality

$$\mathbf{C}(x, z) \leq \mathbf{C}(y, z) + \mathbf{C}(x, y)$$

the triangle inequality, and the identity $0 \to \mathbf{C}(x, x)$ is the inequality $\mathbf{C}(x, x) \leq 0$, forcing $\mathbf{C}(x, x) = 0$. Such a structure is a *Lawvere metric space*: a metric space that drops symmetry and the separation axiom (distinct points may sit at distance 0) and permits ∞. Composition of morphisms and the triangle inequality are the same axiom seen through two monoidal bases.

The five bases run from the trivial to the exotic and read off five theories at a stroke. **Set** recovers the ordinary notion; **Ab** recovers the preadditive categories of homological algebra; **Cat** recovers strict 2-categories; and the order category $[0, \infty]$ shows that enrichment need not be over structured sets at all, turning the triangle inequality into an instance of composition. The last example is the sharpest test of the definition, since $[0, \infty]$ has no underlying sets in the naive sense: what matters is only that it is monoidal, and that alone is enough to carry categorical composition. The lesson is that "category" is not tied to **Set**; it is tied to whatever monoidal **V** one composes in.

8.2 Enriched Functors and Naturality

A **V**-category (definition 8.1.1) carries its structure in **V**: a functor between two such categories must act on hom-objects by morphisms of **V**, and a natural transformation must have components that live in **V** rather than in Hom-sets. This section installs the two notions, shows how a monoidal functor transports the whole theory from one base to another, and records the enriched form of the Yoneda lemma.

An ordinary functor sends each hom-set $\mathrm{Hom}(x, y)$ to $\mathrm{Hom}(Fx, Fy)$ by post- and pre-composition, and the two functoriality axioms say this map respects composition and identities. The enriched version replaces those set functions with $\mathbf{V}$-morphisms and rewrites the two axioms as commuting diagrams in $\mathbf{V}$.

Definition 8.2.1: Enriched Functor

Let $\mathbf{V}$ be a monoidal category and let $\mathbf{C}$, $\mathbf{D}$ be $\mathbf{V}$-categories. A $\mathbf{V}$-*functor* (or *enriched functor*) $F\colon \mathbf{C} \to \mathbf{D}$ consists of

- a map $x \mapsto Fx$ on objects, and
- for each pair of objects x, y of $\mathbf{C}$, a $\mathbf{V}$-morphism

$$F_{x,y}\colon \mathbf{C}(x,y) \to \mathbf{D}(Fx, Fy)$$

such that the following two diagrams in $\mathbf{V}$ commute. Writing $\circ$ for the composition morphisms of $\mathbf{C}$ and $\mathbf{D}$ and $j_x\colon I \to \mathbf{C}(x,x)$ for the identity morphisms, compatibility with composition is the commutativity of

$$\begin{tikzcd} \mathbf{C}(y,z) \otimes \mathbf{C}(x,y) \arrow[r, "F_{y,z} \otimes F_{x,y}"] \arrow[d, "\circ"'] & \mathbf{D}(Fy,Fz) \otimes \mathbf{D}(Fx,Fy) \arrow[d, "\circ"] \\ \mathbf{C}(x,z) \arrow[r, "F_{x,z}"'] & \mathbf{D}(Fx,Fz) \end{tikzcd}$$

and compatibility with identities is the commutativity of

$$\begin{tikzcd} I \arrow[r, "j_x"] \arrow[dr, "j_{Fx}"'] & \mathbf{C}(x,x) \arrow[d, "F_{x,x}"] \\ & \mathbf{D}(Fx,Fx) \end{tikzcd}$$

Reading the squares in $\mathbf{V} = \mathbf{Set}$ recovers the ordinary axioms: the first says $F(g \circ f) = Fg \circ Ff$ and the second $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$. The content is that these hold at the level of $\mathbf{V}$, so the functor is compatible with whatever extra structure the hom-objects carry.

Example 8.2.2: Additive Functors as Ab-Functors

Let $\mathbf{V} = \mathbf{Ab}$, so that $\mathbf{C}$ and $\mathbf{D}$ are preadditive categories (example 8.1.3): each Hom-set is an abelian group and composition is bilinear. An $\mathbf{Ab}$-functor $F\colon \mathbf{C} \to \mathbf{D}$ is an ordinary functor whose action on morphisms

$$F_{x,y}\colon \mathbf{C}(x,y) \to \mathbf{D}(Fx, Fy)$$

is a group homomorphism for every pair x, y, that is, $F(f+g) = Ff + Fg$. These are exactly the *additive functors*. The composition square is the statement that F is bilinear-compatible, and the identity triangle that $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$; both are automatic for a functor once additivity on hom-groups is imposed. For $\mathbf{V} = \mathbf{Cat}$, an enriched functor is a strict 2-functor, sending objects, 1-cells, and 2-cells compatibly (this is developed in section 8.4).

An enriched functor moves objects and hom-objects; a natural transformation must compare two such functors at each object. In the unenriched setting a component is a morphism $\alpha_x\colon Fx \to Gx$, and morphisms out of x are elements of a hom-set. In $\mathbf{V}$ there are no elements; a "point" of the hom-object $\mathbf{D}(Fx, Gx)$ is a $\mathbf{V}$-morphism $I \to \mathbf{D}(Fx, Gx)$ out of the unit, matching the underlying-category convention $\mathrm{Hom}(x, y) = \mathbf{V}(I, \mathbf{C}(x, y))$ from definition 8.1.2. A component is therefore such a morphism, and naturality becomes a single diagram in $\mathbf{V}$.

Definition 8.2.3: Enriched Natural Transformation

Let $F, G\colon \mathbf{C} \to \mathbf{D}$ be $\mathbf{V}$-functors. A $\mathbf{V}$-*natural transformation* (or *enriched natural transformation*) $\alpha\colon F \Rightarrow G$ assigns to each object x of $\mathbf{C}$ a $\mathbf{V}$-morphism

$$\alpha_x\colon I \to \mathbf{D}(Fx, Gx)$$

(the *component* at x) such that for every pair x, y the following diagram in $\mathbf{V}$ commutes:

$$\begin{tikzcd}
\mathbf{C}(x,y) \arrow[r, "\rho^{-1}"] \arrow[d, "\lambda^{-1}"'] & \mathbf{C}(x,y) \otimes I \arrow[r, "G_{x,y} \otimes \alpha_x"] & \mathbf{D}(Gx,Gy) \otimes \mathbf{D}(Fx,Gx) \arrow[d, "\circ"] \\
I \otimes \mathbf{C}(x,y) \arrow[r, "\alpha_y \otimes F_{x,y}"'] & \mathbf{D}(Fy,Gy) \otimes \mathbf{D}(Fx,Fy) \arrow[r, "\circ"'] & \mathbf{D}(Fx,Gy)
\end{tikzcd}$$

Here λ and ρ are the unitors of $\mathbf{V}$ (definition 7.1.1), inserting the unit I so that a component may be composed against a hom-object.

The two legs of the diagram are the enriched versions of the two paths around the ordinary naturality square. The top-right leg applies G to a morphism $f\colon x \to y$ and pre-composes with α_x, producing the path $Gf \cdot \alpha_x$; the bottom-left leg applies F and post-composes with α_y, producing $\alpha_y \cdot Ff$. Equality of the legs is precisely $\alpha_y \circ Ff = Gf \circ \alpha_x$ written internally to $\mathbf{V}$. In $\mathbf{V} = \mathbf{Set}$ each component $\alpha_x\colon I = \{*\} \to \mathbf{D}(Fx, Gx)$ names an element $\alpha_x \in \mathrm{Hom}(Fx, Gx)$, and the diagram collapses to the familiar commuting square.

Example 8.2.4: Additive Natural Transformations

For $\mathbf{V} = \mathbf{Ab}$, a component $\alpha_x\colon \mathbb{Z} \to \mathbf{D}(Fx, Gx)$ is a group homomorphism out of $\mathbb{Z}$, hence pinned down by the single element $\alpha_x(1) \in \mathrm{Hom}(Fx, Gx)$: an ordinary component morphism. The enriched naturality diagram then says nothing more than ordinary naturality, because both composition maps of $\mathbf{Ab}$ are $\mathbb{Z}$-bilinear and evaluating at $1 \otimes 1$ returns the set-level square. An $\mathbf{Ab}$-natural transformation between additive functors is thus an ordinary natural transformation, with no further condition to check.

The definitions of $\mathbf{V}$-category, $\mathbf{V}$-functor, and $\mathbf{V}$-natural transformation depend on $\mathbf{V}$ only through its monoidal structure. A functor between monoidal categories that preserves that structure should therefore transport enriched categories along it. The correct hypothesis is weaker than strict preservation of $\otimes$: it is enough that $\mathbf{V}$-tensors map to $\mathbf{W}$-tensors up to a coherent comparison morphism, which is exactly a lax monoidal functor. That comparison is what carries composition across the base.

Proposition 8.2.5: Change of Enriching Base

Let $(\Phi, \varphi, \varphi_0)\colon \mathbf{V} \to \mathbf{W}$ be a lax monoidal functor: Φ is a functor equipped with a natural comparison $\varphi_{a,b}\colon \Phi a \otimes \Phi b \to \Phi(a \otimes b)$ and a unit comparison $\varphi_0\colon I_{\mathbf{W}} \to \Phi(I_{\mathbf{V}})$, satisfying the associativity and unit coherence laws. Then Φ induces a functor

$$\Phi_*\colon \mathbf{V}\text{-}\mathbf{Cat} \to \mathbf{W}\text{-}\mathbf{Cat}$$

called *change of base*, sending a $\mathbf{V}$-category $\mathbf{C}$ to the $\mathbf{W}$-category $\Phi_*\mathbf{C}$ with the same objects and hom-objects $(\Phi_*\mathbf{C})(x, y) = \Phi\big(\mathbf{C}(x, y)\big)$.

***Proof*:**
Fix a lax monoidal functor $(\Phi, \varphi, \varphi_0)$. The construction proceeds in two stages: first the object $\Phi_*\mathbf{C}$, then the action of Φ_* on enriched functors, checking the axioms at each stage.

Step 1: The category $\Phi_*\mathbf{C}$. Take the objects of $\mathbf{C}$ and the hom-objects $\Phi(\mathbf{C}(x, y))$ in $\mathbf{W}$. The composition morphism of $\Phi_*\mathbf{C}$ is the composite

$$\Phi(\mathbf{C}(y, z)) \otimes \Phi(\mathbf{C}(x, y)) \xrightarrow{\varphi} \Phi\big(\mathbf{C}(y, z) \otimes \mathbf{C}(x, y)\big) \xrightarrow{\Phi(\circ)} \Phi(\mathbf{C}(x, z))$$

which first uses the comparison φ to gather the two hom-objects inside Φ and then applies Φ to the composition of $\mathbf{C}$. The identity morphism is

$$I_{\mathbf{W}} \xrightarrow{\varphi_0} \Phi(I_{\mathbf{V}}) \xrightarrow{\Phi(j_x)} \Phi(\mathbf{C}(x, x))$$

using the unit comparison followed by Φ applied to the identity of $\mathbf{C}$. Associativity of composition in $\Phi_*\mathbf{C}$ follows from associativity in $\mathbf{C}$ (apply Φ to its associativity square) glued to the associativity coherence of the lax structure φ; the two unit laws follow likewise from the unit laws in $\mathbf{C}$ glued to the unit coherence of φ_0. Each verification is a diagram in $\mathbf{W}$ built from a Φ-image of a $\mathbf{C}$-axiom and one lax-coherence cell sharing a common edge: the coherence cell rebrackets the φ-comparisons to expose Φ applied to the tensor of the hom-objects, and the Φ-image of the $\mathbf{C}$-axiom equates the two composites there, so the two cells paste along that edge into the required square. Thus $\Phi_*\mathbf{C}$ is a $\mathbf{W}$-category.

Step 2: The action on functors and naturality. Given a $\mathbf{V}$-functor $F\colon \mathbf{C} \to \mathbf{D}$, define Φ_*F to agree with F on objects and to act on hom-objects by

$$(\Phi_*F)_{x,y} = \Phi(F_{x,y})\colon \Phi(\mathbf{C}(x, y)) \to \Phi(\mathbf{D}(Fx, Fy))$$

Its composition square is Φ applied to the composition square of F, with the naturality of φ used to slide $\Phi(F_{y,z}) \otimes \Phi(F_{x,y})$ past the two copies of the comparison; the identity triangle is Φ applied to that of F, precomposed with φ_0. Both commute, so Φ_*F is a $\mathbf{W}$-functor. Functoriality of Φ_*, preservation of enriched composition and identities of functors, is immediate, since Φ_* acts as Φ on hom-objects and Φ is a functor. This gives the assignment on $\mathbf{V}\text{-}\mathbf{Cat}$, completing the proof.

Change of base is the mechanism by which enriched category theory specializes to ordinary category theory, and by which one kind of enrichment passes to another. The main instance takes $\mathbf{W} = \mathbf{Set}$.

Example 8.2.6: The Underlying Category as Change of Base
Let $\mathbf{V}$ be monoidal. The representable functor

$$\mathbf{V}(I,-)\colon \mathbf{V} \to \mathbf{Set}, \qquad a \mapsto \mathbf{V}(I,a)$$

is lax monoidal: its comparison sends a pair $(u\colon I \to a,\ v\colon I \to b)$ to the composite $I \xrightarrow{\lambda^{-1}} I \otimes I \xrightarrow{u \otimes v} a \otimes b$, and its unit comparison $\{*\} \to \mathbf{V}(I,I)$ picks out id_I. Change of base along it,

$$\big(\mathbf{V}(I,-)\big)_*\colon \mathbf{V}\text{-}\mathbf{Cat} \to \mathbf{Set}\text{-}\mathbf{Cat} = \mathbf{Cat}$$

sends a $\mathbf{V}$-category $\mathbf{C}$ to a category with hom-sets $\mathbf{V}(I,\mathbf{C}(x,y))$, exactly the underlying category $\mathbf{C}_0$ of definition 8.1.2. The construction of $\mathbf{C}_0$ is thus not an ad hoc device but the $\mathbf{W} = \mathbf{Set}$ case of this proposition. Concretely, for $\mathbf{V} = \mathbf{Ab}$ the functor $\mathbf{Ab}(\mathbb{Z},-)$ is the forgetful functor $\mathbf{Ab} \to \mathbf{Set}$, and change of base along it strips a preadditive category down to its underlying ordinary category, forgetting the abelian-group structure on the hom-sets while keeping the same objects and morphisms.

A $\mathbf{V}$-natural transformation, as definition 8.2.3 defines it, is a family of morphisms, and families form a set. Every other hom in this chapter has been an object of $\mathbf{V}$, and the collection of $\mathbf{V}$-natural transformations ought to be one too. Supplying that object is the first task of section 8.3, and it is what the enriched Yoneda lemma needs before it can even be stated.

8.3 Enriched Functor Categories and Weighted Limits

Two constructions stand between the definitions of the previous section and a working enriched theory. The first is the *end*, a limit in $\mathbf{V}$ that cuts the product of the hom-objects $\mathbf{D}(Fc, Gc)$ down to the part respecting naturality; it is what promotes the $\mathbf{V}$-natural transformations from a set to an object of $\mathbf{V}$, and with it the $\mathbf{V}$-functors $\mathbf{C} \to \mathbf{D}$ become a $\mathbf{V}$-category $[\mathbf{C}, \mathbf{D}]$. The second is the *tensor* and *cotensor*, by which an object of $\mathbf{V}$ acts on an object of a $\mathbf{V}$-category. Together they make the enriched Yoneda lemma provable, and they yield one definition, the weighted limit, that has ordinary limits, tensors and cotensors as three instances.

Throughout this section $\mathbf{V}$ is symmetric monoidal closed (definition 7.2.2, definition 7.3.1) with all small limits and colimits, and $\mathbf{C}$ is small. Closedness supplies the internal hom $[a,b]$; symmetry is what allows $\mathbf{C}^{\mathrm{op}}$ to be formed as a $\mathbf{V}$-category and what orients the second of the two morphisms in definition 8.3.3; completeness supplies the products and equalizers that the end is built from, and cocompleteness the coproducts and coequalizers its dual needs. The bracket $[a,b]$ for the internal hom and $[\mathbf{C}, \mathbf{D}]$ for the functor category are distinguished by their arguments, objects in the first case and categories in the second.

Ordinary category theory copies an object along a set: the copower $S \cdot e$ of section 6.4 is S-many copies of e, characterized by $\mathrm{Hom}(S \cdot e, d) \cong \mathbf{Set}(S, \mathrm{Hom}(e,d))$. Enriched over $\mathbf{V}$ the indexing set becomes an object of $\mathbf{V}$ and the hom-sets become hom-objects, and the same two universal properties read in $\mathbf{V}$ define the two constructions below.

Definition 8.3.1: Tensor and Cotensor

Let $\mathbf{C}$ be a $\mathbf{V}$-category, c an object of $\mathbf{C}$, and v an object of $\mathbf{V}$. The *tensor* of c by v is an object $v \odot c$ of $\mathbf{C}$ together with an isomorphism

$$\mathbf{C}(v \odot c,\ d)\ \cong\ \big[v,\ \mathbf{C}(c,d)\big]$$

of objects of $\mathbf{V}$, natural in d. The *cotensor* of c by v is an object c^{v} of $\mathbf{C}$ together with an isomorphism

$$\mathbf{C}(d,\ c^{v})\ \cong\ \big[v,\ \mathbf{C}(d,c)\big]$$

natural in d. The category $\mathbf{C}$ is *tensored* if $v \odot c$ exists for every v and c, and *cotensored* if c^{v} always exists.

Each is defined by a representing property, so each is unique up to a unique isomorphism compatible with the displayed natural isomorphism, by the argument of theorem 2.2.1 applied to the underlying category. The tensor is v-indexed copies of c and the cotensor is v-indexed families in c, with "v-indexed" interpreted by $\mathbf{V}$ rather than by a set. Note which variable each is natural in: the tensor represents a functor of the target and the cotensor a functor of the source, which is why the tensor is a colimit-like construction and the cotensor a limit-like one.

Example 8.3.2: Tensors and Cotensors over the Standing Bases

1. $\mathbf{V}$ *over itself.* The base $\mathbf{V}$ is a $\mathbf{V}$-category with hom-objects $[a,b]$, and there $v \odot a = v \otimes a$ and $a^{v} = [v,a]$. The first is the internal form of (7.1): the required isomorphism $[v \otimes a, b] \cong [v,[a,b]]$ holds because, for every u, the adjunction gives $\mathbf{V}(u,[v \otimes a,b]) \cong \mathbf{V}(u \otimes v \otimes a, b) \cong \mathbf{V}(u,[v,[a,b]])$ naturally in u, and theorem 2.2.1 turns a natural isomorphism of representables into one of the representing objects. The second follows from the first by symmetry of $\otimes$.

2. $\mathbf{V} = \mathbf{Set}$. For an ordinary category $\mathbf{C}$ the defining isomorphism reads $\mathrm{Hom}(S \odot c, d) \cong \mathbf{Set}(S, \mathrm{Hom}(c,d))$, and an S-indexed family of maps $c \to d$ is a single map out of the coproduct. So $S \odot c = \coprod_S c$ and, dually, $c^S = \prod_S c$: the tensor and cotensor are the copower and power of section 6.4, and a category is tensored and cotensored over $\mathbf{Set}$ exactly when it has all small coproducts and products.

3. $\mathbf{V} = \mathbf{Ab}$. For $\mathbf{C} = \mathbf{Ab}$ itself, $A \odot B = A \otimes_{\mathbb{Z}} B$ and $B^{A} = \mathrm{Hom}_{\mathbb{Z}}(A,B)$, by (1), since $\mathbf{Ab}$ is closed with internal hom $\mathrm{Hom}_{\mathbb{Z}}$. For $\mathbf{C}$ the R-modules, the tensor of a module M by an abelian group A is $A \otimes_{\mathbb{Z}} M$ with R acting through M, because a homomorphism $A \to \mathrm{Hom}_R(M,N)$ is the same as an R-linear map $A \otimes_{\mathbb{Z}} M \to N$.

The three bases show the pattern: over $\mathbf{Set}$ the tensor is a coproduct of copies, over $\mathbf{Ab}$ it is a tensor product, and over $\mathbf{V}$ itself it is $\otimes$. In every case the operation is the one that lets an object of the base act on an object of the enriched category, and that action is what definition 8.3.7 will weight a diagram by.

The second construction is the one the previous section stopped short of. A $\mathbf{V}$-natural transformation $F \Rightarrow G$ is a family of morphisms $I \to \mathbf{D}(Fc, Gc)$ satisfying the naturality condition of definition 8.2.3. Replacing "family of morphisms out of I" by "object of $\mathbf{V}$ mapping to each $\mathbf{D}(Fc,Gc)$", and imposing naturality as an equation between two morphisms into a product, turns that collection into a limit in $\mathbf{V}$.

Definition 8.3.3: The Object of Enriched Natural Transformations

Let $F, G\colon \mathbf{C} \to \mathbf{D}$ be $\mathbf{V}$-functors. The *object of* $\mathbf{V}$*-natural transformations* from F to G, also called the *end* of $c \mapsto \mathbf{D}(Fc, Gc)$ and written

$$[\mathbf{C},\mathbf{D}](F,G) \;=\; \int_c \mathbf{D}(Fc,Gc)$$

is the equalizer in $\mathbf{V}$ (definition 3.2.4) of the two morphisms

$$\prod_c \mathbf{D}(Fc,Gc) \;\rightrightarrows\; \prod_{c,c'} \big[\,\mathbf{C}(c,c'),\ \mathbf{D}(Fc,Gc')\,\big]$$

whose components at a pair (c, c') are the transposes under (7.1) of

$$\textstyle\prod_b \mathbf{D}(Fb,Gb)\otimes\mathbf{C}(c,c') \xrightarrow{\pi_{c'}\otimes F_{c,c'}} \mathbf{D}(Fc',Gc')\otimes\mathbf{D}(Fc,Fc') \xrightarrow{\circ} \mathbf{D}(Fc,Gc')$$

and

$$\textstyle\prod_b \mathbf{D}(Fb,Gb)\otimes\mathbf{C}(c,c') \xrightarrow{\pi_c\otimes G_{c,c'}} \mathbf{D}(Fc,Gc)\otimes\mathbf{D}(Gc,Gc') \xrightarrow{\sigma} \mathbf{D}(Gc,Gc')\otimes\mathbf{D}(Fc,Gc) \xrightarrow{\circ} \mathbf{D}(Fc,Gc')$$

where π_c is a product projection and σ the symmetry of $\mathbf{V}$. The products are indexed by the objects and the ordered pairs of objects of the small category $\mathbf{C}$, and both they and the equalizer exist because $\mathbf{V}$ is complete.

The two morphisms are the two routes from a component at one object to a value at another: the first carries the component at c' backwards along F, the second carries the component at c forwards along G, and a family is natural exactly when the two agree. Writing α_c for a component, the first is $\alpha_{c'} \circ F(f)$ and the second $G(f) \circ \alpha_c$ in the notation of an ordinary category; the symmetry σ appears in the second because $\circ$ takes its arguments in the order (later, earlier) while the projection produces them in the order (earlier, later).

Probing the equalizer with the unit recovers the unenriched notion. The functor $\mathbf{V}(I, -)$ is representable, hence preserves limits, so $\mathbf{V}\big(I, [\mathbf{C},\mathbf{D}](F,G)\big)$ is the equalizer of the corresponding pair of functions, which is precisely the set of families $\alpha_c\colon I \to \mathbf{D}(Fc, Gc)$ satisfying definition 8.2.3. The new object therefore refines the old set rather than replacing it.

Proposition 8.3.4: The Enriched Functor Category

Let $\mathbf{C}$ be a small $\mathbf{V}$-category and $\mathbf{D}$ a $\mathbf{V}$-category. The $\mathbf{V}$-functors $\mathbf{C} \to \mathbf{D}$ are the objects of a $\mathbf{V}$-category $[\mathbf{C},\mathbf{D}]$ whose hom-objects are those of definition 8.3.3. Its underlying ordinary category (definition 8.1.2) is the category of $\mathbf{V}$-functors and $\mathbf{V}$-natural transformations, and for $\mathbf{V} = \mathbf{Set}$ it is the functor category of definition 1.5.4.

***Proof*:**
Write $E(F,G) = [\mathbf{C},\mathbf{D}](F,G)$ and let $\pi_c\colon E(F,G) \to \mathbf{D}(Fc,Gc)$ denote the composite of the equalizer inclusion with the projection at c.

Step 1: Composition. The componentwise composites

$$E(G,H) \otimes E(F,G) \xrightarrow{\pi_c \otimes \pi_c} \mathbf{D}(Gc,Hc) \otimes \mathbf{D}(Fc,Gc) \xrightarrow{\circ} \mathbf{D}(Fc,Hc)$$

are the components of a morphism into $\prod_c \mathbf{D}(Fc,Hc)$. It factors through the equalizer $E(F,H)$, and this is the one point needing argument. Transposing, the claim is that the two routes from $E(G,H) \otimes E(F,G) \otimes \mathbf{C}(c,c')$ to $\mathbf{D}(Fc,Hc')$ agree. Take the route carrying the composite component at c' backwards along F. Since $E(G,H)$ and $E(F,G)$ are equalizers, their own defining equations apply: the β-factor may be moved from "at c', transported along G" to "at c, transported along G" and then the α-factor from "at c' along F" to "at c along H". Concretely, insert $\mathbf{C}(c,c')$ between the two factors using symmetry, apply the equalizer equation of $E(F,G)$ to replace $\alpha_{c'} \circ F$ by $G \circ \alpha_c$, then apply the equalizer equation of $E(G,H)$ to replace $\beta_{c'} \circ G$ by $H \circ \beta_c$. The two applications are along different tensor factors, so they commute, and associativity of $\circ$ in $\mathbf{D}$ closes the square onto the second route. Hence the composite lands in $E(F,H)$.

Step 2: Identities. The morphisms $j_{Fc}\colon I \to \mathbf{D}(Fc,Fc)$ assemble into $I \to \prod_c \mathbf{D}(Fc,Fc)$, and this factors through $E(F,F)$: the two routes out of $I \otimes \mathbf{C}(c,c')$ are $\mathrm{id}_{Fc'} \circ F(f)$ and $F(f) \circ \mathrm{id}_{Fc}$, equal by the unit laws of $\mathbf{D}$. This gives $j_F\colon I \to E(F,F)$.

Step 3: The axioms. Associativity and the unit laws for the composition of Step 1 are equations between morphisms into $E(F,K)$, and the projections π_c out of an equalizer of a product are jointly monic, so it suffices to check them after each π_c. There they read as associativity and the unit laws of $\mathbf{D}$ at the object Fc, which hold by definition 8.1.1. So $[\mathbf{C},\mathbf{D}]$ is a $\mathbf{V}$-category.

Step 4: The underlying category and the **Set** *case.* By definition 8.1.2 the underlying category has hom-sets $\mathbf{V}(I, E(F,G))$, identified above with the $\mathbf{V}$-natural transformations $F \Rightarrow G$, and Steps 1 and 2 restrict to their usual composition and identities. For $\mathbf{V} = \mathbf{Set}$ the internal hom is the hom-set, the product is the cartesian product and the equalizer the subset on which the two functions agree, so $E(F,G)$ is the set of natural transformations and $[\mathbf{C},\mathbf{D}]$ is definition 1.5.4.

The construction is the ordinary functor category with two substitutions: the product of hom-sets becomes a product in $\mathbf{V}$, and the naturality condition, stated in the ordinary case as an equation holding at each morphism, becomes the equalizer that imposes it. Everything else, composition and identities, is inherited componentwise from $\mathbf{D}$, which is why Step 3 reduces to axioms already available.

One base makes both constructions concrete at once.

Example 8.3.5: Modules over a Ring as an Enriched Functor Category

Let R be a ring and let $\mathbf{R}$ be the $\mathbf{Ab}$-category with a single object $*$ and hom-object $\mathbf{R}(*,*) = R$, its composition the multiplication of R and its identity $j_*\colon \mathbb{Z} \to R$ the map sending 1 to 1_R. This is definition 8.1.1 over $\mathbf{V} = (\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z})$, and the associativity and unit diagrams are the ring axioms.

An $\mathbf{Ab}$-functor $F\colon \mathbf{R} \to \mathbf{Ab}$ is an abelian group $M = F(*)$ together with a homomorphism $R \to \mathrm{Hom}_{\mathbb{Z}}(M,M)$ respecting multiplication and the unit, which is exactly a left R-module.

For two such, $F \leftrightarrow M$ and $G \leftrightarrow N$, definition 8.3.3 has a single-factor product $\mathrm{Hom}_{\mathbb{Z}}(M,N)$ and equalizes the two maps $\mathrm{Hom}_{\mathbb{Z}}(M,N) \to \mathrm{Hom}_{\mathbb{Z}}\big(R, \mathrm{Hom}_{\mathbb{Z}}(M,N)\big)$ sending f to $r \mapsto f \circ (r \cdot -)$ and to $r \mapsto (r \cdot -) \circ f$. Their equalizer is the subgroup of f with $f(rm) = rf(m)$, so

$$[\mathbf{R},\mathbf{Ab}](F,G) \;=\; \mathrm{Hom}_R(M,N)$$

and proposition 8.3.4 says the left R-modules form an **Ab**-category whose hom-objects are the R-linear maps. The **Ab**-enrichment of R-modules that homological algebra takes as given is thus an instance of the enriched functor category.

With $[\mathbf{C}, \mathbf{V}]$ available, the enriched Yoneda lemma can be stated and proved. The ordinary statement (theorem 2.2.1) identified the natural transformations out of a representable with the elements of Fx; the enriched one identifies an object of $\mathbf{V}$ with an object of $\mathbf{V}$, and the proof is the same evaluation-at-the-identity argument written in $\mathbf{V}$.

Theorem 8.3.6: Enriched Yoneda Lemma

Let $\mathbf{C}$ be a small $\mathbf{V}$-category and $F\colon \mathbf{C} \to \mathbf{V}$ a $\mathbf{V}$-functor into the base. For each object x of $\mathbf{C}$ there is an isomorphism

$$[\mathbf{C}, \mathbf{V}]\big(\mathbf{C}(x, -),\, F\big) \;\cong\; Fx$$

of objects of $\mathbf{V}$, natural in x and in F. For $\mathbf{V} = \mathbf{Set}$ it is the bijection $\mathrm{Nat}(\mathrm{Hom}(x, -), F) \cong Fx$ of theorem 2.2.1.

***Proof*:**
Write $E = [\mathbf{C}, \mathbf{V}]\big(\mathbf{C}(x, -), F\big) = \int_y \big[\mathbf{C}(x, y), Fy\big]$, with $\pi_y\colon E \to [\mathbf{C}(x, y), Fy]$ the map of definition 8.3.3. Two morphisms are constructed and shown to be mutually inverse.

Step 1: The map $\psi\colon E \to Fx$. Take the component at $y = x$ and evaluate at the identity. The identity $j_x\colon I \to \mathbf{C}(x, x)$ induces $[\,j_x, Fx\,]\colon [\mathbf{C}(x, x), Fx] \to [I, Fx]$, and $[I, Fx] \cong Fx$ since I is the unit. Set $\psi = \big([\,j_x, Fx\,] \circ \pi_x\big)$ followed by that isomorphism.

Step 2: The map $\varphi\colon Fx \to E$. The $\mathbf{V}$-functor F supplies $F_{x,y}\colon \mathbf{C}(x, y) \to [Fx, Fy]$, whose transpose under (7.1) is an action $\mathbf{C}(x, y) \otimes Fx \to Fy$. Transposing that in the other variable gives $\varphi_y\colon Fx \to [\mathbf{C}(x, y), Fy]$ for each y. These form a wedge: the equalizer condition of definition 8.3.3 at (y, z) is, after transposition, the assertion that acting by a morphism of $\mathbf{C}(y, z)$ after acting by one of $\mathbf{C}(x, y)$ agrees with acting by their composite, which is the composition axiom of the $\mathbf{V}$-functor F. So the φ_y factor through E, defining φ.

Step 3: $\psi\varphi = \mathrm{id}_{Fx}$. Composing, $\psi\varphi$ is the transpose of $\mathbf{C}(x, x) \otimes Fx \to Fx$ precomposed with $j_x \otimes \mathrm{id}$, that is, the action of the identity. The unit axiom of the $\mathbf{V}$-functor F says $F_{x,x} \circ j_x = j_{Fx}$, and j_{Fx} is the transpose of the left unitor, so the action of the identity is λ and $\psi\varphi = \mathrm{id}$.

Step 4: $\varphi\psi = \mathrm{id}_E$. The projections π_z are jointly monic, so it suffices that $\pi_z\varphi\psi = \pi_z$ for every z. Transpose to morphisms $E \otimes \mathbf{C}(x, z) \to Fz$. Now take the equalizer equation of definition 8.3.3 at the pair (x, z) and transpose it to an equation between morphisms $E \otimes \mathbf{C}(x, z) \otimes \mathbf{C}(x, x) \to Fz$: one side applies π_x and then acts by $\mathbf{C}(x, z)$, the other composes $\mathbf{C}(x, z) \otimes \mathbf{C}(x, x) \to \mathbf{C}(x, z)$ and then applies π_z. Precompose both with $\mathrm{id}_E \otimes \mathrm{id} \otimes j_x$ and use the unit law of $\mathbf{C}$. The left side becomes the transpose of $\pi_z\varphi\psi$, since ψ is exactly π_x followed by evaluation at j_x and φ is the action; the right side becomes the transpose of π_z. Hence $\pi_z\varphi\psi = \pi_z$ for all z, and $\varphi\psi = \mathrm{id}_E$.

Step 5: Naturality. Naturality in F holds because ψ was built from π_x and j_x, neither of which involves F, so a $\mathbf{V}$-natural transformation $F \Rightarrow F'$ commutes with it by construction. Naturality in x follows from Step 4 read for the two objects together with the composition axiom of $\mathbf{C}$, in the same way that Step 2 used the composition axiom of F.

Nothing in the argument is new relative to theorem 2.2.1: Step 1 is "evaluate at id_x", Step 2 is "transport along the action", and Steps 3 and 4 are the two round trips. What the enrichment changes is that each step is a morphism of $\mathbf{V}$ rather than a function on elements, and the naturality condition that was a family of equations is now the equalizer through which the maps must factor. Setting $\mathbf{V} = \mathbf{Set}$ collapses every internal object to a set and returns the original proof verbatim.

The last definition of the chapter's enriched material unifies the two constructions above. An ordinary limit is a universal cone, and a cone over F with vertex d is a natural transformation from the constant functor at a point into $\mathrm{Hom}(d, F-)$. Nothing requires the constant functor: any $\mathbf{V}$-functor W may take its place, and the resulting notion is strictly more general because W may assign different objects of $\mathbf{V}$ to different objects of $\mathbf{C}$.

Definition 8.3.7: Weighted Limit and Weighted Colimit

Let $\mathbf{C}$ be a small $\mathbf{V}$-category, $\mathbf{D}$ a $\mathbf{V}$-category and $F\colon \mathbf{C} \to \mathbf{D}$ a $\mathbf{V}$-functor. For a *weight* $W\colon \mathbf{C} \to \mathbf{V}$, the *limit of F weighted by W* is an object $\lim^W F$ of $\mathbf{D}$ with an isomorphism

$$\mathbf{D}\big(d,\ \lim^W F\big) \ \cong\ [\mathbf{C}, \mathbf{V}]\big(W,\ \mathbf{D}(d, F-)\big)$$

of objects of $\mathbf{V}$, natural in d. For a weight $W\colon \mathbf{C}^{\mathrm{op}} \to \mathbf{V}$, the *colimit of F weighted by W*, written $W \star F$, is an object with

$$\mathbf{D}\big(W \star F,\ d\big) \ \cong\ [\mathbf{C}^{\mathrm{op}}, \mathbf{V}]\big(W,\ \mathbf{D}(F-, d)\big)$$

natural in d. Here $\mathbf{C}^{\mathrm{op}}$ is the $\mathbf{V}$-category with the same objects, hom-objects $\mathbf{C}^{\mathrm{op}}(c, c') = \mathbf{C}(c', c)$, and composition that of $\mathbf{C}$ precomposed with the symmetry σ.

Both are representing objects, so both are unique up to canonical isomorphism when they exist. The weight is the diagram's shape data: where an ordinary cone offers one morphism $d \to Fc$ for each c, a W-weighted cone offers an object Wc worth of them, and the naturality built into $[\mathbf{C}, \mathbf{V}]$ requires those to be compatible along the morphisms of $\mathbf{C}$. The definition of $\mathbf{C}^{\mathrm{op}}$ is where symmetry of $\mathbf{V}$ is needed, since reversing composition means feeding $\circ$ its two arguments in the opposite order.

Example 8.3.8: Three Weighted Limits

1. *Ordinary limits are the constant weight.* Take $\mathbf{V} = \mathbf{Set}$ and W the constant functor at the one-point set. Then $[\mathbf{C}, \mathbf{Set}]\big(W, \mathrm{Hom}(d, F-)\big)$ is the set of natural transformations from the constant functor into $\mathrm{Hom}(d, F-)$, which is the set of cones over F with vertex d. The defining isomorphism becomes $\mathrm{Hom}(d, \lim^W F) \cong \{\text{cones with vertex } d\}$, the universal property of the ordinary limit. Definition 3.2.1 is thus the special case in which every object of $\mathbf{C}$ carries the same one-point weight.

2. *Tensors and cotensors are weighted (co)limits.* Let $\mathbf{I}$ be the unit $\mathbf{V}$-category, one object $*$ with $\mathbf{I}(*, *) = I$. A $\mathbf{V}$-functor $F\colon \mathbf{I} \to \mathbf{D}$ is an object c of $\mathbf{D}$, and a weight $W\colon \mathbf{I} \to \mathbf{V}$ is an object v of $\mathbf{V}$. Unwinding definition 8.3.7, the products in definition 8.3.3 have one factor and the equalizer is trivial, so the defining isomorphism reads $\mathbf{D}(d, \lim^W F) \cong [v, \mathbf{D}(d, c)]$, which is definition 8.3.1: $\lim^W F = c^v$ and dually $W \star F = v \odot c$.

3. *Tensor product of modules.* In the setting of example 8.3.5, a weight $W\colon \mathbf{R}^{\mathrm{op}} \to \mathbf{Ab}$ is a right R-module and F a left R-module M. For an abelian group A, the group $\mathrm{Hom}_{\mathbb{Z}}(M, A)$ is a right R-module by $(f \cdot r)(m) = f(rm)$, and the defining isomorphism reads

$$\mathrm{Hom}_{\mathbb{Z}}(W \star F,\ A) \ \cong\ \mathrm{Hom}_{R^{\mathrm{op}}}\big(W,\ \mathrm{Hom}_{\mathbb{Z}}(M, A)\big)$$

the right-hand side being the right R-module maps, by example 8.3.5 applied to $\mathbf{R}^{\mathrm{op}}$. That is the universal property of $W \otimes_R M$, so $W \star F = W \otimes_R M$: the tensor product over a ring is a weighted colimit.

Three constructions that arose separately, the limit of a diagram, the power of an object and the tensor product of two modules, are one construction read over three bases with three weights. The generality is used again once homotopy enters: a homotopy limit is a weighted limit whose weight is built from simplicial sets, and the two-sided bar construction computing it is a weighted colimit. Both are developed in *Higher Category Theory* [1], which takes definition 8.3.7 as given.

8.4 Strict 2-Categories

The enriching base of the preceding sections has been an arbitrary monoidal category $\mathbf{V}$. Take $\mathbf{V} = \mathbf{Cat}$, and a $\mathbf{V}$-category acquires an extra layer of morphisms: each hom-object is now itself a category, so between two arrows there sit arrows-of-arrows. This is the setting in which the natural transformation lives on the same footing as the functor, and in which the adjunction of definition 4.1.1 and the monad of definition 5.1.1 become structures one can name on any object at all. This section makes the extra layer explicit, gives it its own composition laws, and identifies it with the **Cat**-enriched case of definition 8.1.1.

Definition 8.4.1: Strict 2-Category

A *strict 2-category* $\mathbf{K}$ consists of

- a class of 0-*cells* (or *objects*) $A, B, C, \ldots$;
- for each ordered pair A, B of 0-cells, a class of 1-*cells* $f\colon A \to B$;
- for each parallel pair $f, g\colon A \to B$ of 1-cells, a class of 2-*cells* $\alpha\colon f \Rightarrow g$,

together with two ways of composing 2-cells. *Vertical composition* takes 2-cells sharing a 1-cell boundary,

$$\alpha\colon f \Rightarrow g, \qquad \beta\colon g \Rightarrow h,$$

to a 2-cell $\beta \cdot \alpha\colon f \Rightarrow h$; *horizontal composition* takes 2-cells sharing a 0-cell boundary,

$$\alpha\colon f \Rightarrow g\ (A \to B), \qquad \alpha'\colon f' \Rightarrow g'\ (B \to C),$$

to a 2-cell $\alpha' \circ \alpha\colon f'f \Rightarrow g'g\ (A \to C)$. Both compositions are strictly associative and unital: vertical composition has, for each 1-cell f, an identity 2-cell $1_f\colon f \Rightarrow f$; horizontal composition has, for each 0-cell A, an identity 2-cell 1_{id_A} on the identity 1-cell id_A. The two compositions are compatible through the *interchange law*: whenever the composites are defined,

$$(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha) \tag{8.1}$$

The two compositions are pictured most cleanly by drawing 1-cells as arrows and 2-cells as double arrows between them. Vertical composition stacks two 2-cells inside a single hom-class $\mathbf{K}(A, B)$:

$$\begin{tikzcd} A \arrow[r, bend left=60, "f"{name=F}] \arrow[r, "g"{name=G, description}] \arrow[r, bend right=60, "h"'{name=H}] & B \arrow[Rightarrow, from=F, to=G, "\alpha"] \arrow[Rightarrow, from=G, to=H, "\beta"] \end{tikzcd}$$

Horizontal composition pastes two 2-cells along a shared 0-cell B:

$$\begin{tikzcd} A \arrow[r, bend left=60, "f"{name=F}] \arrow[r, bend right=60, "g"'{name=G}] & B \arrow[r, bend left=60, "f'"{name=F2}] \arrow[r, bend right=60, "g'"'{name=G2}] & C \arrow[Rightarrow, from=F, to=G, "\alpha"] \arrow[Rightarrow, from=F2, to=G2, "\alpha'"] \end{tikzcd}$$

The interchange law is the assertion that the four-fold pasting diagram

$$\begin{tikzcd} A \arrow[r, bend left=60, "f"{name=F}] \arrow[r, "g"{name=G, description}] \arrow[r, bend right=60, "h"'{name=H}] & B \arrow[r, bend left=60, "f'"{name=F2}] \arrow[r, "g'"{name=G2, description}] \arrow[r, bend right=60, "h'"'{name=H2}] & C \arrow[Rightarrow, from=F, to=G, "\alpha"] \arrow[Rightarrow, from=G, to=H, "\beta"] \arrow[Rightarrow, from=F2, to=G2, "\alpha'"] \arrow[Rightarrow, from=G2, to=H2, "\beta'"] \end{tikzcd}$$

evaluates to a single 2-cell $f'f \Rightarrow h'h$, no matter whether one first composes vertically within each column and then horizontally across, or first composes horizontally within each row and then vertically down. Equation (8.1) says the two readings agree. This is the one axiom that couples the two operations; associativity and unitality govern each in isolation.

The definition looks like a bookkeeping exercise until one reads it through the lens of section 8.1. Assemble, for each pair A, B, the 1-cells $A \to B$ as objects and the 2-cells between them as morphisms; vertical composition makes this data a category $\mathbf{K}(A, B)$, its identities the 1_f. Horizontal composition of 1-cells and of 2-cells then assembles into functors $\mathbf{K}(B, C) \times \mathbf{K}(A, B) \to \mathbf{K}(A, C)$, and the interchange law is exactly the statement that horizontal composition respects the categorical structure of each factor, that is, that it is a functor on the product and not only a function on objects. This is the hinge of the section.

Proposition 8.4.2: 2-Categories are Cat-enriched Categories

A strict 2-category is exactly a category enriched over $(\mathbf{Cat}, \times, \mathbf{1})$ in the sense of definition 8.1.1: the hom-object $\mathbf{K}(A, B)$ is the category whose objects are the 1-cells $A \to B$ and whose morphisms are the 2-cells between them.

***Proof*:**
Fix the correspondence between the two packages of data and check that the axioms match term by term.

From a 2-category to enrichment. Given a strict 2-category $\mathbf{K}$, for each pair A, B let $\mathbf{K}(A, B)$ be the category with the 1-cells $A \to B$ as objects, the 2-cells as morphisms, and vertical composition as composition; associativity and unitality of vertical composition are precisely the category axioms, with 1_f the identity on f. Horizontal composition assigns to a pair of objects (f', f) the object $f'f$ and to a pair of 2-cells (α', α) the 2-cell $\alpha' \circ \alpha$. That this assignment is a functor

$$c_{A,B,C} : \mathbf{K}(B, C) \times \mathbf{K}(A, B) \to \mathbf{K}(A, C)$$

is the content of the interchange law (8.1): it sends the identity morphism $(1_{f'}, 1_f)$ to $1_{f'f}$, which is unitality of horizontal composition, and it respects composition of morphisms in the product category, where composition is componentwise vertical, so functoriality is the equation $(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha)$ of (8.1). The identity 1-cell id_A together with 1_{id_A} is a functor $\mathbf{1} \to \mathbf{K}(A, A)$ from the terminal category, the unit map of the enrichment. Associativity and unitality of horizontal composition of 1-cells become the associativity and unit diagrams required of an enriched category.

From enrichment to a 2-category. Conversely, a **Cat**-enriched category supplies for each pair a category $\mathbf{K}(A, B)$: its objects are declared the 1-cells, its morphisms the 2-cells, its composition the vertical composition. The composition functors $c_{A,B,C}$ act on objects to give horizontal composition of 1-cells and on morphisms to give horizontal composition of 2-cells; functoriality of $c_{A,B,C}$ is the interchange law, and the enriched associativity and unit axioms are the associativity and unitality of horizontal composition. The two passages are mutually inverse, matching each datum and each axiom, so the two notions coincide, as we sought to show.

The proposition is the reason the tensor product $\times$ on **Cat** mattered. An ordinary category is enriched over $(\mathbf{Set}, \times)$; promoting the enriching base one level, from sets to categories, adds exactly one dimension of morphism and produces the strict 2-category. The interchange law is not an extra decoration but the naturality already built into enrichment, read in the base **Cat**. Every general fact about $\mathbf{V}$-categories from section 8.1 specializes: a 2-category has an underlying ordinary category $\mathbf{K}_0$ (definition 8.1.2), obtained by discarding the 2-cells and keeping only the 0-cells and 1-cells, since $\mathbf{Cat}(\mathbf{1}, \mathbf{K}(A, B))$ is the set of objects of $\mathbf{K}(A, B)$, namely the 1-cells $A \to B$.

The definition of a 2-category was abstracted from one example, in which all three layers of cell are already familiar and the two compositions of 2-cells have standing names.

Example 8.4.3: Cat as a 2-Category

Let **Cat** denote the 2-category whose

- 0-cells are (small) categories $\mathbf{C}, \mathbf{D}, \ldots$;
- 1-cells $F : \mathbf{C} \to \mathbf{D}$ are functors (definition 1.4.1);
- 2-cells $\alpha : F \Rightarrow G$ are natural transformations (definition 1.5.2) between parallel functors.

The hom-object $\mathbf{Cat}(\mathbf{C}, \mathbf{D})$ is the functor category $[\mathbf{C}, \mathbf{D}]$ (definition 1.5.4). Vertical composition is the composition of natural transformations: for $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$, the transformation $\beta \cdot \alpha$ has component $(\beta \cdot \alpha)_c = \beta_c \circ \alpha_c$ at each object c. Horizontal composition is the Godement product: for $\alpha : F \Rightarrow G$ between functors $\mathbf{C} \to \mathbf{D}$ and $\alpha' : F' \Rightarrow G'$ between functors $\mathbf{D} \to \mathbf{E}$, the component of $\alpha' \circ \alpha$ at $c \in \mathbf{C}$ is the common value of the two paths around the naturality square of α' at α_c:

$$(\alpha' \circ \alpha)_c = G'(\alpha_c) \circ \alpha'_{Fc} = \alpha'_{Gc} \circ F'(\alpha_c) \tag{8.2}$$

The interchange law is the middle-four exchange for natural transformations.

The two expressions in (8.2) are equal precisely because $\alpha' \colon F' \Rightarrow G'$ is natural: applied to the arrow $\alpha_c \colon Fc \to Gc$ of $\mathbf{D}$, naturality of α' is the commuting square

$$\begin{tikzcd} F'Fc \arrow[r, "\alpha'_{Fc}"] \arrow[d, "F'(\alpha_c)"'] & G'Fc \arrow[d, "G'(\alpha_c)"] \\ F'Gc \arrow[r, "\alpha'_{Gc}"'] & G'Gc \end{tikzcd}$$

whose two directed paths from $F'Fc$ to $G'Gc$ are the two formulas for $(\alpha' \circ \alpha)_c$. That **Cat** satisfies the axioms of definition 8.4.1 is the content of proposition 8.4.2 together with the fact that $[\mathbf{C}, \mathbf{D}]$ is a category (definition 1.5.4) and that these composites are natural. The interchange law $(\beta' \cdot \alpha') \circ (\beta \cdot \alpha) = (\beta' \circ \beta) \cdot (\alpha' \circ \alpha)$ is verified componentwise from (8.2) by a direct diagram chase; it is the classical middle-four exchange, so named because the four inner terms of the two sides pair off.

A frequent special case of horizontal composition fixes one of the two 2-cells to be an identity. This is common enough, and geometrically clean enough, to deserve its own name.

Definition 8.4.4: Whiskering

Let $\alpha \colon f \Rightarrow g$ be a 2-cell between 1-cells $f, g \colon A \to B$ in a 2-category $\mathbf{K}$. Given a 1-cell $h \colon B \to C$, the *left whiskering* of α by h is the horizontal composite of α with the identity 2-cell on h,

$$h * \alpha \coloneqq 1_h \circ \alpha \colon hf \Rightarrow hg,$$

a 2-cell between 1-cells $A \to C$. Given a 1-cell $k \colon Z \to A$, the *right whiskering* of α by k is

$$\alpha * k \coloneqq \alpha \circ 1_k \colon fk \Rightarrow gk,$$

a 2-cell between 1-cells $Z \to B$.

Pictorially, whiskering appends a 1-cell to one side of the 2-cell α without introducing a second nondegenerate 2-cell:

$$\begin{tikzcd} A \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & B \arrow[r, "h"] & C \end{tikzcd} \qquad \begin{tikzcd} Z \arrow[r, "k"] & A \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & B \end{tikzcd}$$

In **Cat**, where $\alpha \colon F \Rightarrow G$ is a natural transformation and H a functor, left whiskering $H * \alpha$ has component $(H * \alpha)_c = H(\alpha_c)$ at each object c, obtained by applying H to each component of α; right whiskering $\alpha * K$ has component $(\alpha * K)_z = \alpha_{Kz}$, obtained by reindexing α along K. Every horizontal composite factors through whiskering: for $\alpha \colon F \Rightarrow G$ and $\alpha' \colon F' \Rightarrow G'$ as in (8.2),

$$\alpha' \circ \alpha = (G' * \alpha) \cdot (\alpha' * F) = (\alpha' * G) \cdot (F' * \alpha),$$

the two factorizations being the two paths of the naturality square, so the two whiskerings together with vertical composition recover the full Godement product. Whiskering is therefore the elementary operation from which horizontal composition is built, and it is the notation in which the constructions below are cleanest.

With the two compositions and whiskering in hand, one can transcribe the central notions of the core theory into any 2-category, referring only to 0-, 1-, and 2-cells and never to elements. The payoff is that an adjunction (definition 4.1.1) and a monad (definition 5.1.1) become structures an object can carry in any $\mathbf{K}$, of which the definitions of the earlier chapters are the $\mathbf{Cat}$ case.

Proposition 8.4.5: Adjunctions and Monads in a 2-Category

Let $\mathbf{K}$ be a 2-category.

1. An *adjunction* in $\mathbf{K}$ consists of 1-cells $f\colon A \to B$ and $g\colon B \to A$ and 2-cells $\eta\colon \mathrm{id}_A \Rightarrow gf$ and $\varepsilon\colon fg \Rightarrow \mathrm{id}_B$ satisfying the triangle identities
$$(\varepsilon * f)\cdot(f * \eta) = 1_f, \qquad (g * \varepsilon)\cdot(\eta * g) = 1_g \tag{8.3}$$
2. A *monad* on an object A consists of a 1-cell $t\colon A \to A$ and 2-cells $\eta\colon \mathrm{id}_A \Rightarrow t$ and $\mu\colon tt \Rightarrow t$ satisfying the associativity and unit laws
$$\mu\cdot(\mu * t) = \mu\cdot(t * \mu), \qquad \mu\cdot(\eta * t) = 1_t = \mu\cdot(t * \eta) \tag{8.4}$$
3. Every adjunction in $\mathbf{K}$ induces a monad on A, with underlying 1-cell $t = gf$, unit the adjunction unit η, and multiplication $\mu = g * \varepsilon * f\colon gfgf \Rightarrow gf$.

In $\mathbf{K} = \mathbf{Cat}$, an adjunction is an adjunction of functors in the sense of definition 4.1.1 with unit and counit as in definition 4.1.3, a monad is a monad in the sense of definition 5.1.1, and the induced monad of part (3) is the monad an adjunction of functors induces on its source.

***Proof*:**
Parts (1) and (2) are definitions internal to $\mathbf{K}$; the content is part (3), that the data they package assemble into a monad, and that in $\mathbf{Cat}$ everything reduces to the earlier chapters.

Step 1: The induced multiplication and unit. From an adjunction $(f, g, \eta, \varepsilon)$ set $t = gf$. Left whiskering by g and right whiskering by f turn the counit $\varepsilon\colon fg \Rightarrow \mathrm{id}_B$ into a 2-cell

$$\mu = g * \varepsilon * f\colon gfgf \Rightarrow gf,$$

that is, $\mu\colon tt \Rightarrow t$, where $g*\varepsilon*f$ abbreviates the whiskering $g*(\varepsilon*f) = (g*\varepsilon)*f$ (associativity of horizontal composition makes the two readings equal). The unit of the monad is the adjunction unit $\eta\colon \mathrm{id}_A \Rightarrow t$.

Step 2: The unit laws. The two unit laws in (8.4) are the two triangle identities (8.3), whiskered into place. Whisker the first triangle identity $(\varepsilon * f)\cdot(f * \eta) = 1_f$ on the left by g: using functoriality of whiskering with respect to vertical composition,

$$(g * \varepsilon * f)\cdot(gf * \eta) = 1_{gf},$$

and since $t*\eta = gf*\eta$ and $\mu = g*\varepsilon*f$, this reads $\mu\cdot(t*\eta) = 1_t$. Whiskering the second triangle identity $(g * \varepsilon)\cdot(\eta * g) = 1_g$ on the right by f gives, by the same computation, $\mu\cdot(\eta * t) = 1_t$. Together these are the unit laws.

Step 3: Associativity. Both sides of $\mu\cdot(\mu*t) = \mu\cdot(t*\mu)$ are 2-cells $ttt \Rightarrow t$, that is, $gfgfgf \Rightarrow gf$. Expand using $\mu = g*\varepsilon*f$ and $t = gf$: the left side, $\mu\cdot(\mu * gf)$, whiskers the outer copy of ε

over the inner one on the left, while the right side, $\mu \cdot (gf * \mu)$, whiskers on the right; the two inner whiskerings apply ε at the two different copies of the pair fg in $fgfg$. Both reduce to a single whiskered double application of the counit,

$$g * (\varepsilon \circ \varepsilon) * f : gfgfgf \Rightarrow gf,$$

where $\varepsilon \circ \varepsilon : fgfg \Rightarrow \mathrm{id}_B$ is the horizontal composite of ε with itself. That this horizontal composite is unambiguous, that whiskering the outer ε over the inner one on the left equals whiskering it on the right, is precisely the interchange law (8.1) applied to the two copies of ε: $(\varepsilon \circ 1_{\mathrm{id}_B}) \cdot (1_{fg} \circ \varepsilon) = (\varepsilon \circ \varepsilon) = (1_{\mathrm{id}_B} \circ \varepsilon) \cdot (\varepsilon \circ 1_{fg})$. Hence the two sides agree and (t, η, μ) satisfies (8.4), a monad on A.

Step 4: The **Cat** *case.* Take $\mathbf{K} = \mathbf{Cat}$. A 1-cell $f : A \to B$ is a functor, a 2-cell is a natural transformation, and the triangle identities (8.3), read componentwise via the whiskering formulas $(H * \alpha)_c = H(\alpha_c)$ and $(\alpha * K)_z = \alpha_{Kz}$ of definition 8.4.4, are exactly the triangle identities of definition 4.1.3: at an object a the first reads $\varepsilon_{fa} \circ f(\eta_a) = \mathrm{id}_{fa}$. So an adjunction in **Cat** is an adjunction of functors (definition 4.1.1). Likewise $t = gf$ is the composite endofunctor, and the multiplication $\mu = g * \varepsilon * f$ has component $\mu_a = g(\varepsilon_{fa})$ at $a \in A$, which is the multiplication of the monad an adjunction of functors induces (definition 5.1.1). The construction of part (3) used only 2-cells, whiskering, and the interchange law, which is why the derivation for **Cat** was, without saying so, purely 2-categorical. This completes the proof.

The proposition is the abstract residue of the monad theory of the core chapters. The construction of a monad from an adjunction (definition 5.1.1) never touched an element of an object; it composed the counit with functors on the left and right and invoked naturality, and those are exactly whiskering and the interchange law. Reread through proposition 8.4.5, that construction proved a fact about **Cat** that holds verbatim in any 2-category: the endomorphism gf of the source of any adjunction carries a canonical monad structure. The same transcription runs the other way for the Eilenberg-Moore and Kleisli constructions (section 5.2, section 5.3), which recover an adjunction from a monad by building new objects, and it is the systematic version of this dictionary, one layer of categorical structure at a time, that the sequel takes up.

8.5 Toward Weak and Higher Categories

A strict 2-category asks the composition of 1-cells to be strictly associative and strictly unital, exactly as in an ordinary category. In the archetype **Cat** this holds on the nose. But many of the 2-dimensional structures that arise in practice do not: composing 1-cells there produces objects that are associative and unital only *up to* a specified invertible 2-cell, and one cannot wish those 2-cells away without discarding the mathematics that forced them. The passage from the strict laws of definition 8.4.1 to laws-up-to-coherent-isomorphism is the last structural step this book takes, and it is the door to the sequel.

The relaxation is governed by exactly the coherence discipline of chapter 7. There the associator and unitors of a monoidal category were required to satisfy the pentagon and triangle axioms, and Mac Lane's coherence theorem (theorem 7.1.5) guaranteed that every formal diagram then commutes. A bicategory imports that discipline one dimension up: the associativity and unit *equalities* of 1-cell composition become invertible 2-cells subject to a pentagon and a triangle, and the same kind of coherence theorem results.

Definition 8.5.1: Bicategory

A *bicategory* $\mathbf{B}$ consists of

- a collection of *0-cells* (objects) $A, B, C, \ldots$;
- for each ordered pair (A, B) a category $\mathbf{B}(A, B)$, whose objects are the *1-cells* $f : A \to B$ and whose morphisms are the *2-cells* $\alpha : f \Rightarrow g$ (composition within $\mathbf{B}(A, B)$ is *vertical* composition of 2-cells);
- for each triple (A, B, C) a *horizontal composition* functor
$$\circ : \mathbf{B}(B, C) \times \mathbf{B}(A, B) \to \mathbf{B}(A, C)$$
and for each A an identity 1-cell $\mathrm{id}_A \in \mathbf{B}(A, A)$;
- for all composable 1-cells f, g, h a natural *associator* 2-cell
$$a_{h,g,f} : (h \circ g) \circ f \;\Rightarrow\; h \circ (g \circ f)$$
and for each $f : A \to B$ natural *left* and *right unitor* 2-cells
$$\lambda_f : \mathrm{id}_B \circ f \Rightarrow f, \qquad \rho_f : f \circ \mathrm{id}_A \Rightarrow f$$
each of which is invertible,

subject to two coherence axioms: the *pentagon*, requiring that the two ways of reassociating a fourfold composite $((k \circ h) \circ g) \circ f$ to $k \circ (h \circ (g \circ f))$ agree, that is
$$(\mathrm{id}_k \circ a_{h,g,f}) \cdot a_{k,h\circ g,f} \cdot (a_{k,h,g} \circ \mathrm{id}_f) \;=\; a_{k,h,g\circ f} \cdot a_{k\circ h,g,f}$$
(where $\cdot$ is vertical composition and $\circ$ here denotes whiskering, definition 8.4.4); and the *triangle*, requiring for composable $g : B \to C$, $f : A \to B$ that
$$(\mathrm{id}_g \circ \lambda_f) \cdot a_{g,\mathrm{id}_B,f} \;=\; \rho_g \circ \mathrm{id}_f$$
A *strict 2-category* (definition 8.4.1) is the special case in which a, λ, and ρ are all identity 2-cells.

The two axioms are the pentagon and triangle of theorem 7.1.5, transplanted from the 1-cells of a monoidal category to the 1-cells of $\mathbf{B}$. This is not an analogy imposed after the fact: a bicategory with a single 0-cell is precisely a monoidal category, its one hom-category $\mathbf{B}(A, A)$ being the underlying category, horizontal composition the tensor, id_A the unit, and a, λ, ρ the monoidal associator and unitors. A bicategory is thus a monoidal category with many objects, and the coherence data is the same data seen one dimension higher.

The force of the pentagon and triangle is again a coherence theorem, which frees the working mathematician from tracking associators and unitors explicitly.

Theorem 8.5.2: Coherence for Bicategories

Every bicategory is biequivalent to a strict 2-category. Consequently, in any bicategory every diagram of 2-cells built from associators, unitors, their inverses, whiskerings, and identities commutes.

***Proof Key ideas*:**
The statement, and the notion of biequivalence it uses, belong to the weak 2-dimensional theory developed in the sequel; the full argument is beyond the light treatment here. The mechanism mirrors the strictification of a monoidal category behind theorem 7.1.5: one embeds $\mathbf{B}$ into a strict 2-category of $\mathbf{B}$-representations by a bicategorical Yoneda construction, sending each 0-cell to a strict object whose 1-cells compose on the nose, and checks the embedding is a biequivalence. A complete proof is given in the sequel ([1]) and in Mac Lane's and Bénabou's original treatments, to which the details are deferred.

Coherence has a practical consequence worth stating plainly: a bicategory may be manipulated as though it were strict. Parenthesization of a composite of 1-cells can be suppressed, and any two reassociations of the same string are identified by a canonical invertible 2-cell. The weakness is real, but coherence renders it invisible in computation, precisely as in the monoidal case.

The examples that make bicategories necessary are the ones where the associator fails to be an identity. Two are canonical.

Example 8.5.3: The Bicategory of Bimodules
Bimodules. Let $\mathbf{Bimod}$ have as 0-cells the (unital, associative) rings. A 1-cell $R \to S$ is an (S, R)-bimodule ${}_SM_R$, that is, an abelian group carrying commuting left S- and right R-actions ([4]). A 2-cell ${}_SM_R \Rightarrow {}_SN_R$ is a bimodule homomorphism. Vertical composition is composition of bimodule maps, so $\mathbf{Bimod}(R, S)$ is the category of (S, R)-bimodules.

Horizontal composition is the tensor product over the middle ring: the composite of $R \xrightarrow{M} S$ and $S \xrightarrow{N} T$ is the (T, R)-bimodule

$$N \circ M \;=\; N \otimes_S M$$

The identity 1-cell on R is the bimodule ${}_RR_R$, since $R \otimes_R M \cong M$ and $M \otimes_R R \cong M$ naturally.

Here associativity is not an equality. For bimodules ${}_SM_R$, ${}_TN_S$, ${}_UP_T$ the two triple tensor products

$$(P \otimes_T N) \otimes_S M \qquad \text{and} \qquad P \otimes_T (N \otimes_S M)$$

are distinct abelian groups: their elements are cosets in different quotients of $P \otimes N \otimes M$. They are canonically isomorphic by $(p \otimes n) \otimes m \mapsto p \otimes (n \otimes m)$, and this isomorphism is the associator $a_{P,N,M}$; it is invertible but not the identity. The unitors are the canonical maps $R \otimes_R M \cong M$ and $M \otimes_R R \cong M$. That these satisfy the pentagon and triangle is the statement that the standard tensor-product isomorphisms cohere, which they do because each is built from the universal property of $\otimes$. Thus $\mathbf{Bimod}$ is a bicategory that is not a strict 2-category, and no reasonable change of definition makes it one: the tensor product of bimodules is associative up to canonical isomorphism and no more.

The second canonical example needs only a category with pullbacks, and it shows the same phenomenon geometrically rather than algebraically.

Example 8.5.4: The Bicategory of Spans

Let $\mathbf{C}$ be a category with pullbacks. A *span* from A to B is a diagram $A \leftarrow S \rightarrow B$, an object S with a leg to each of A and B; this is a 1-cell $A \rightarrow B$. A 2-cell between spans $A \leftarrow S \rightarrow B$ and $A \leftarrow S' \rightarrow B$ is a morphism $S \rightarrow S'$ commuting with both legs.

Horizontal composition composes spans by pullback. Given $A \leftarrow S \rightarrow B$ and $B \leftarrow T \rightarrow C$, form the pullback $S \times_B T$ of the two legs over B (definition 3.2.6):

$$\begin{tikzcd}
 & & S \times_B T \arrow[dl] \arrow[dr] & & \\
 & S \arrow[dl] \arrow[dr] & & T \arrow[dl] \arrow[dr] & \\
A & & B & & C
\end{tikzcd}$$

The composite span $A \leftarrow S \times_B T \rightarrow C$ takes the outer legs. The identity 1-cell on A is the span $A \xleftarrow{\mathrm{id}} A \xrightarrow{\mathrm{id}} A$.

Associativity again holds only up to isomorphism. Composing three spans in the two possible orders forms two iterated pullbacks, $(S \times_B T) \times_C U$ and $S \times_B (T \times_C U)$; these are canonically isomorphic because both compute the limit of the same diagram, but the isomorphism is supplied by the universal property of the pullback and is not an identity. It is the associator, and the pentagon it satisfies is exactly the coherence of iterated pullback limits. So $\mathbf{Span}(\mathbf{C})$ is a bicategory, strict only in the degenerate cases where pullbacks can be chosen strictly functorially.

In both examples the weakness is not a defect of the definition to be engineered away but a feature of the mathematics: tensor product and pullback are operations defined by universal properties, and a universal property determines its output only up to canonical isomorphism. Any 2-dimensional structure organized by such operations is a bicategory and not, on the nose, a strict 2-category. The equalities of definition 8.4.1 were an idealization; the isomorphisms of definition 8.5.1 are what the objects satisfy.

This is where strictness gives way to coherence, and where the present book ends. Everything developed here has been strict 1-category theory. Composition of morphisms is strictly associative and strictly unital; a diagram commutes or it does not; two objects are the same when they are equal or, at the outermost, when they are isomorphic. Sameness has meant equality or isomorphism, and nothing weaker. The structured categories of the last two chapters have pushed against that boundary, introducing associators and unitors as data, but coherence has kept them tame: by theorem 7.1.5 and theorem 8.5.2 the strict theory recovers the weak one up to equivalence, and the strict picture remains adequate.

The subject changes character when equality weakens past this point. When the associativity of composition is asked to hold only up to a coherent *homotopy*, and that homotopy up to a further homotopy, and so on through every dimension, one is no longer doing strict 1-category theory but *higher category theory*. There the objects of study are model categories, quasi-categories, and ∞-categories, in which morphisms compose only up to coherent higher cells and the coherence data is infinite rather than a single pentagon. The theorems of this book do not disappear beyond it; they reappear in homotopy-coherent form. Limits and colimits become homotopy limits and colimits, taken up to coherent equivalence rather than on the nose. Adjunctions, monads, and the Yoneda lemma each have ∞-categorical statements that specialize back to the strict ones proved here. The duality principle of box 1.3.3 survives verbatim, opposite ∞-categories and all.

The sequel is *Higher Category Theory* ([1]), which takes up simplicial sets and nerves, Quillen's model categories, the quasi-category model of ∞-categories after Joyal, Lurie, and Riehl, homotopy (co)limits, and the stable and derived structures that connect back to homological algebra. It is the natural continuation for a reader who now recognizes a universal property on sight and wants to see what becomes of these constructions when equality is replaced throughout by coherent equivalence. The strict theory built here is the ground from which that generalization is read.

What Next?

The natural continuation is the one chapter 8 pointed to. A 2-category keeps strict equalities between composites; weaken them to coherent isomorphisms, and then to coherent homotopies, and one arrives at the weak and higher categories in which modern homotopy theory and derived geometry are written. *Higher Category Theory* [1] takes up that thread, through model categories, simplicial methods, and quasi-categories, where the limits, adjunctions, and Yoneda lemma of this book reappear in homotopy-coherent form. Lurie's *Higher Topos Theory* is the standard reference for the destination.

A different extension keeps categories strict but adds structure to their hom-sets. An *abelian* category axiomatizes the categories of modules and sheaves, with kernels, cokernels, and exact sequences; the series volume on homological algebra develops it, together with the derived functors Ext and Tor and the spectral sequences that compute them. This book leaves abelian categories to that setting, where the motivation from exactness is at hand.

Two further directions turn the machinery on itself. *Topos theory* studies categories that behave like the category of sets, uniting sheaf theory with a categorical account of logic in which a topos carries its own internal language; the companion volume is *Topos Theory* [3], and Mac Lane and Moerdijk's *Sheaves in Geometry and Logic* is the standard entry point. And the categorical viewpoint is by now inseparable from the rest of the series: the functor-of-points reading of a scheme, the fundamental group as the automorphism group of a fiber functor, and moduli problems presented as functors all recur in *Moduli, Deformation Theory, and Stacks* [2], where representability is a categorical question.

For deepening the present material rather than moving past it, the three books credited in the introduction remain the companions: Riehl for the modern viewpoint, Mac Lane for the standard account, and Leinster for a second, lighter pass.

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